Sharp local cyclicity of the complete quadratic graphic I_4^1

Haibo Lu
Shanghai Institute of Technology, Shanghai 201418, People's Republic of China
Corresponding author: luhaibo1985@gmail.com
ORCID: 0009-0000-2717-5968

18 October 2026

Abstract

The complete Dumortier–Roussarie–Rousseau graphic I_4^1 consists of two parabolic branches joining a finite saddle–node to a multiplicity-four vertex at infinity. We prove that its local cyclicity is exactly three in the full quadratic coefficient germ. One coefficient neighborhood and one whole-graphic Hausdorff collar give the upper bound, and every smaller neighborhood and collar contain one field with three hyperbolic complete cycles. We compare the curvature of the completed physical return with that of a cubic entry–exit action, whose quadratic coefficient cancels from the weighted curvature. Relative estimates near reversibility, separate root scales, and the ratio of the two physical tail gains preserve the comparison through root coalescence. At degree loss, comparison of the actual small-power section maps gives a two-zero bound. A finite coefficient cover yields one uniform count. The lower construction uses two coefficients to match a vanishing complete gain to a steep action transition and preserve three simple roots in one field. Whole-orbit convergence and a diagonal limit give attainment at the endpoint. In a fixed analytic slice, the ratio $p = \alpha_1/\sqrt{-\alpha_0}$ of the translated axis coefficients, defined for $\alpha_0 < 0$, tends to zero from above along three-cycle sequences approaching the organizing field and graphic. The bound is uniform down to $p = 0$, where at most one cycle occurs, and is sharp below every positive cutoff on p.

Keywords: finite cyclicity; quadratic vector field; DRR graphic; saddle–node; Dulac map; entry–exit asymptotics.

2020 Mathematics Subject Classification: 34C07 (primary); 34C23, 37G15 (secondary).

1 Introduction

Determining how many limit cycles bifurcate from a quadratic graphic under variation of all quadratic coefficients is a local part of the Dumortier–Roussarie–Rousseau finiteness programme [3, Introduction, author copy pp. 1–2]. We study its graphic I_4^1, whose two parabolic branches join a finite saddle–node to a nilpotent saddle–node at infinity [4, Proposition 4.2.2 and Figure 8, journal pp. 111–113].

For the present graphic the axis polynomial is cubic. Its incident roots can coalesce at different rates, and other coefficient limits reduce its degree. The cyclicity problem therefore asks for a bound on the return through both vertices throughout these configurations. We obtain the sharp bound three: one coefficient neighborhood and one Hausdorff collar admit at most three cycles, and arbitrarily small neighborhoods and collars admit three hyperbolic cycles in one field.

The return-map approach connects singular-passage geometry with zero counting. Saddle–node normal forms provide finitely smooth transition coordinates [3]; transition maps give

finite cyclicity for elementary quadratic graphics surrounding a focus or center [2]. For graphics through nilpotent singularities, finite cyclicity has been obtained by controlling the regular transition derivatives [12, Theorems 5.4 and 6.3, journal pp. 372 and 388–389] and nonlinear transitions through a finite saddle–node [11, Theorem 3.8, author copy p. 23]. In the center case, division in the center ideal permits a displacement–Rolle argument [10, Sections 5.1 and 5.4–6, author copy pp. 19 and 23–25]. Sharp counts have been obtained for a local codimension-four nilpotent saddle–node family [5, Theorem 1.1, peer-reviewed author version p. 4] and for hyperbolic hemicycles under quadratic perturbations [9, Definition 1.4 and Theorem B, published pp. 3 and 7]. For slow–fast canard cycles containing one remote hyperbolic saddle with nonunit hyperbolicity ratio, whose slow dynamics is regular or has only finitely many singularities, all in the interior of the slow segment, Huzak proves an upper bound of three [6, Definition 1 and Theorem 2.2, peer-reviewed author version pp. 7–8].

At the infinity passage of I_4^1, entry–exit theory describes how an orbit follows the axis and reaches its outgoing section. De Maesschalck and Schecter establish the entry–exit relation and its smooth parameter dependence [1, Theorem 1.1 and Corollary 1.2]. Huzak and Kristiansen develop this description for degenerate turning points, including the $n = 1$ geometry associated with I_4^1 [7, printed p. 3 and Theorem 6.1, printed p. 25]. Their strict $n = 1$ transition theorem assumes a uniformly negative axis polynomial on its selected chart [7, Theorem 2.4, printed pp. 6–7]. These local transition results provide entry–exit coordinates and regularity. The present argument also needs the other root and degree-loss charts, the finite saddle–node closing, and one count for the complete physical return. The neighboring fixed-parameter graphic I_2^1 is studied in [8]. At the I_4^1 endpoint its finite-passage multiplier vanishes; the lower construction below realizes three cycles internally before taking this endpoint limit.

The organizing field in the original classification coordinates is

$$F_* : \quad \dot{x} = \lambda_* x - y + x^2, \qquad \dot{y} = x + \lambda_* y + 2\lambda_* x^2 + xy, \qquad \lambda_* = \frac{1}{\sqrt{3}}. \tag{1.1}$$

Its invariant parabola is

$$y = \pi_*(x) := \frac{x^2}{2} - \lambda_* x - \frac{\lambda_*^2 + 1}{2}. \tag{1.2}$$

Write $\iota_*(x, y) = (x, y, 1)/\sqrt{1 + x^2 + y^2}$, let $p_* = (-2\lambda_*, 2/3)$, $q_* = (0, 1, 0)$, and let $\Gamma_- = \iota_*\{(x, \pi_*(x)) : x < -2\lambda_*\}$ and $\Gamma_+ = \iota_*\{(x, \pi_*(x)) : x > -2\lambda_*\}$. The complete marked graphic is

$$\Gamma_{I4} = \Gamma_- \cup \Gamma_+, \qquad q_* \longrightarrow p_* \longrightarrow q_*. \tag{1.3}$$

The two occurrences of q_* are one marked projective vertex.

Let $\mathcal{Q}_2$ be the twelve-dimensional affine space of real planar vector fields of degree at most two, and let d_{H} be the Hausdorff distance for a fixed smooth metric on the closed upper hemisphere.

Theorem 1.1 (Exact local cyclicity of I_4^1)**.** *There are a neighborhood $\mathcal{V}$ of F_* in $\mathcal{Q}_2$ and $\varepsilon > 0$ such that every $F \in \mathcal{V}$ has at most three distinct isolated periodic orbits γ with*

$$d_{\mathrm{H}}\big(\iota_*(\gamma), \Gamma_{I4}\big) < \varepsilon.$$

Conversely, for every neighborhood $\mathcal{W}$ of F_* and every $\delta > 0$, some $F \in \mathcal{W}$ has three geometrically distinct hyperbolic periodic orbits satisfying this inequality with δ in place of ε. Thus

$$\mathrm{Cycl}_{\mathcal{Q}_2}(\Gamma_{I4}, F_*) = 3.$$

Both assertions concern the complete two-branch marked graphic and the full twelve-coefficient quadratic germ.

1.1 The mechanism of the proof

The finite saddle–node supplies a small multiplier $m > 0$. Including the two regular closing maps, cycles are zeros of one displacement in an affine physical input:

$$F_{\theta,m}(x) = x - m\Pi_{\theta,m}(x) = 0. \tag{1.4}$$

The cubic entry–exit action reveals the key cancellation: its positively weighted curvature contains the linear and cubic coefficients, while the quadratic coefficient cancels. The product of the two matched axis distances decreases with the incoming quadratic label, so every curvature zero has the same crossing orientation. Section 4.2 derives the identity and transfers this orientation to the completed physical map. With a positive displacement derivative on the upper continuation, Rolle's theorem then gives at most three displacement zeros.

Uniformity requires more than convergence to the action. The two nonlinear section maps contribute to physical curvature, and their individual gains can vanish at different rates. We retain both gains and their actual section images. Near reversibility, division by the coefficient distance controls the curvature as its leading term vanishes. Separate root-collar estimates handle independently approaching roots; matching actual inward heights also controls the original ghost tails beyond the growing-action ranges. On the upper continuation, a bounded complete derivative gives $F' > 1/2$ after m is chosen small. At degree loss, a comparison of the two actual small-power section maps instead gives the same upward orientation at every critical point of the displacement, and hence at most two zeros on that whole coefficient fibre.

These estimates count one connected interval of actual trajectories. We fix the finite coefficient cover and overlap reserves, then a positive unscaled height cap, and finally independent coefficient and multiplier bounds. Geometric capture places every nearby whole cycle in the interval already counted. Fixed positive ratio intervals have a stronger two-cycle bound, forcing three-cycle parameters to concentrate at $p \to 0^+$ in the slice defined below.

The lower construction is independent of this upper count. First fix $c > 1$ near one. The complete organizing gain is $\exp(-4\pi/\sqrt{c-1})$; choose the action-model parameter $\epsilon > 0$ so that this equals $2\epsilon^{5/2}$. The central transition has slope of order $\epsilon^{-5/2}$, compensating for the vanishing gain. Two coefficients then calibrate the offset and the complete gain in one field, and one small scale preserves all three simple roots and their physical hyperbolicity. Their complete orbits converge to both branches of the organizing parabola. A diagonal choice as $c \downarrow 1$ gives the endpoint lower bound. The analytic affine and positive-time slice transports the conclusions to all twelve quadratic coefficients.

1.2 The organizing family and coordinate model

Put

$$A_0 = \begin{pmatrix} \sqrt{3}/2 & 0 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \qquad \rho = \frac{2}{\sqrt{3}}.$$

The affine change

$$\begin{pmatrix} X \\ Y \end{pmatrix} = A_0 \begin{pmatrix} x \\ y \end{pmatrix}, \qquad t_{\mathrm{can}} = \rho t \tag{1.5}$$

has positive time factor and sends (1.1) to

$$H_0 : \qquad \dot{X} = X - Y + X^2, \qquad \dot{Y} = X + X^2 + XY. \tag{1.6}$$

The parabola becomes $Y = (X^2 - 1)/2$, the finite vertex is $p = (-1, 0)$, and the point at infinity is $q = (0, 1, 0)$. For a quadratic field F in the original coordinates, its field in the canonical coordinates is

$$\mathscr{C}_0(F)(Z) = \rho^{-1} A_0 F(A_0^{-1} Z), \qquad Z = (X, Y)^{\mathsf{T}}. \tag{1.7}$$

The map $\mathscr{C}_0 : \mathcal{Q}_2 \to \mathcal{Q}_2$ is a linear isomorphism, with inverse

$$\mathscr{C}_0^{-1}(H)(u) = \rho A_0^{-1} H(A_0 u).$$

In particular, it sends the full twelve-coefficient germ at F_* onto the full twelve-coefficient germ at H_0.

Let

$$\iota_0(X, Y) = \frac{(X, Y, 1)}{\sqrt{1 + X^2 + Y^2}}$$

be the canonical affine embedding, and let S be the projective extension of A_0 to the closed upper hemisphere. Thus

$$S(U, V, W) = \frac{\left(A_0\binom{U}{V}, W\right)}{\sqrt{\left\|A_0\binom{U}{V}\right\|^2 + W^2}}, \qquad S \circ \iota_* = \iota_0 \circ A_0. \tag{1.8}$$

This is a smooth projective diffeomorphism preserving the upper hemisphere. Set

$$\Gamma := S(\Gamma_{I4}). \tag{1.9}$$

It is the complete two-branch marked graphic of H_0, with the same positive-time branch order because $\rho > 0$. For the fixed smooth metric used in Theorem 1.1, compactness gives a constant $1 \le L_0 < \infty$ such that

$$d_{\mathrm{H}}(SK, SK') \le L_0 d_{\mathrm{H}}(K, K') \tag{1.10}$$

for all nonempty compact subsets K, K' of the hemisphere. All calculations below are made for H_0 and Γ; the final proof pulls the resulting coefficient neighborhood back by (1.7) and uses (1.10) to transfer the Hausdorff collar.

The canonical model also lies in the organizing family

$$H_c : \quad \dot{X} = X - Y + X^2, \qquad \dot{Y} = cX + X^2 + XY, \qquad \Gamma_c = \overline{\iota_0\{Y = (X^2 - c)/2\}}, \tag{1.11}$$

where the closure is taken in the closed upper Poincare hemisphere and $\iota_0(X, Y) = (X, Y, 1)/\sqrt{1 + X^2 + Y^2}$. Along the invariant parabola,

$$\dot{X}\big|_{\Gamma_c} = \frac{(X + 1)^2 + c - 1}{2}. \tag{1.12}$$

For every fixed $c > 1$, the finite point $X = -1$ is crossed with positive speed. At $c = 1$, the same point becomes a finite saddle–node. The two infinite ends remain the two marked occurrences of one projective vertex; no equatorial arc is added. This is the geometric change from the fixed I_2^1 problem to the I_4^1 endpoint.

In the original coordinates, direct substitution and $3\lambda_*^2 = 1$ give

$$\begin{aligned} F_*(y - \pi_*(x)) &= \dot{y} - \pi_*'(x)\dot{x} = 2x(y - \pi_*(x)), \\ \dot{x}|_{y=\pi_*(x)} &= \lambda_* x - \pi_*(x) + x^2 = \frac{1}{2}(x + 2\lambda_*)^2. \end{aligned} \tag{1.13}$$

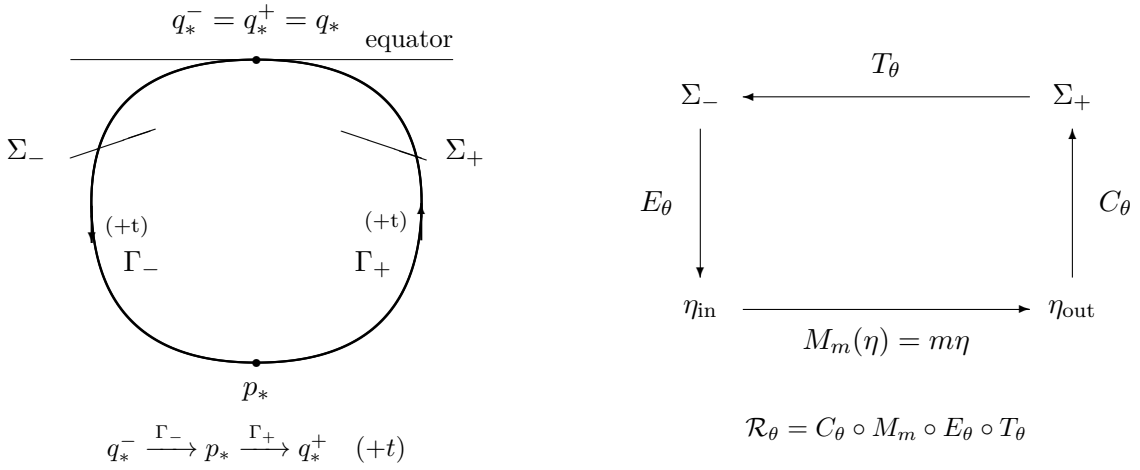
Writing $u = x + 2\lambda_*$ and $w = y - \pi_*(x)$, the complete system is

$$\dot{u} = \frac{u^2}{2} - w, \qquad \dot{w} = 2(u - 2\lambda_*)w. \tag{1.14}$$

Its linearization at $(u, w) = (0, 0)$ is

$$\begin{pmatrix} 0 & -1 \\ 0 & -4\lambda_* \end{pmatrix}.$$

Thus it has one zero and one nonzero eigenvalue, while the restriction to $w = 0$ has the quadratic degeneracy $\dot{u} = u^2/2$. The finite point is therefore a saddle–node. The multiplicity of the other marked vertex is checked separately in the infinity chart of Section 2. The lower bound used here is constructed internally in Theorem 6.2. The complete organizing multiplier $\exp(-4\pi/\sqrt{c-1})$ vanishes as $c \downarrow 1$; at the endpoint the regular finite passage becomes a saddle–node. We therefore derive the upper estimate at $c = 1$ with the finite multiplier separated from the actual infinity passage.



(a) The complete marked graphic

(b) The positive-time return composition

Figure 1: The complete marked graphic and the return used to count its nearby cycles. In panel (a), Γ_- runs from q_*^- to p_* and Γ_+ from p_* to q_*^+ in positive time. The equality $q_*^- = q_*^+ = q_*$ records two occurrences in the return word of one repeated projective vertex, not two points. The branches have a common tangent there, and no equatorial arc belongs to Γ_{I4}. Panel (b) shows the complete composition of positive-time first-hit maps on their common through-side domain: the infinity transition T_θ, the regular entry map E_θ, the finite saddle–node multiplier $M_m(\eta) = m\eta$, and the regular closing map C_θ. Its arrows denote maps, not additional trajectories. All shapes and distances are schematic.

1.3 The small positive ratio and its sharpness

The analytic slice in Lemma 2.1 is fixed once for the full quadratic germ. For the same field F, write $\theta_F = \theta(\mathscr{C}_0(F))$, let $P(v) = \mu_1 + \mu_2 v - v^3$, $R(v) = \mu_3 + v + v^2$, and use the root $\gamma(F)$ of R near zero. Set

$$\alpha_0(F) = P(\gamma(F)), \qquad \alpha_1(F) = P'(\gamma(F)), \qquad s(F) = \sqrt{-\alpha_0(F)}, \quad p(F) = \alpha_1(F)/s(F), \tag{1.15}$$

where s, p are used for $\alpha_0 < 0$. The ratio therefore has one specified normalization throughout the upper and lower arguments.

Theorem 1.2 (Uniform small positive ratio and sharpness)**.** *There are an ordinary open neighborhood $\mathcal{V}_{\mathrm{s}}$ of F_* in $\mathcal{Q}_2$, constants $\varepsilon_{\mathrm{s}} > 0$ and $p_{\mathrm{s}} > 0$, such that every $F \in \mathcal{V}_{\mathrm{s}}$ with*

$$\alpha_0(F) < 0, \qquad 0 \le p(F) \le p_{\mathrm{s}}$$

has at most three distinct isolated periodic orbits whose complete images have Hausdorff distance less than ε_{s} from Γ_{I4}. On its connected admitted physical return interval, the corresponding isolated displacement zeros have total multiplicity at most three. On the face $p = 0$ the bound is one.

For every neighborhood $\mathcal{W}$ of F_*, every $\delta > 0$ and every $\eta > 0$, one field $F \in \mathcal{W}$ has three geometrically distinct hyperbolic complete periodic orbits at Hausdorff distance less than δ from Γ_{I4} and satisfies

$$\alpha_0(F) < 0 < \alpha_1(F), \qquad 0 < p(F) < \min\{p_{\mathrm{s}}, \eta\}.$$

The upper bound retains the independently varying raw coefficients; no restriction on γ/s, relative root distances, or the actual finite multiplier compared with a source radius is added.

Proposition 6.10 locates the constructed triples more precisely: $p \to 0^+$ and $\gamma/s \to +\infty$, while the two positive roots remain separated in the positive-gamma chart. Thus the sharpness family is retained by the unrestricted small-ratio estimate. Section 7 proves the theorem from the full regional estimates below.

1.4 Organization

Section 2 constructs the physical return and slice. Sections 3–4 give the counting calculus and cubic mechanism. Section 5 assembles the regional estimates, uniform budgets, capture and concentration. Section 6 constructs the simultaneous lower bound with its model, passage, tail and coefficient proofs; the small-ratio and full-germ proofs follow. The remaining regional transition proofs and coefficient dictionaries are developed in the appendices. Two illustrative ghost paths are provided in the optional supplement.

Contents

1 Introduction **1**

2 The full quadratic germ and its physical return **7**

3 Counting on one physical interval **13**

4 Action curvature and coefficient geometry **16**

5 Uniform regional returns and the three-cycle upper bound **26**

6 Three hyperbolic cycles approaching the marked graphic **41**

7 The sharp small positive ratio sector **52**

A Scalar transitions and physical section calculus **53**

B Strong projection and physical section heights **61**

C Normalized transitions on the parabolic tails **63**

D Degenerating saddle transitions: action coordinates **68**

E Small characteristic ratios, finite middle passages, and fixed exit **73**

F The compact positive-ratio passage **81**

G Gamma-dominant and compact cubic passages **91**

H Bounded action, degree loss, and remote roots **119**

I Nonnegative opening: determinant and trace faces **157**

J Quadratic inward and outward section estimates **168**

K Coefficient dictionaries and passage domains **190**

L Notation and reading index **194**

2 The full quadratic germ and its physical return

We first separate the affine and positive-time changes from the five essential coefficients. We then construct the return through both singular points, including the two regular closing maps. The resulting displacement is written in an affine section height, which will be used throughout the zero count.

2.1 Reduction to five parameters

The following five-parameter family is transverse to the seven affine and positive-time directions:

$$H_\theta: \quad \begin{cases} \dot X = AX - Y + X^2 + (\mu_2+\mu_3)XY + \mu_1 Y^2, \\ \dot Y = CX + X^2 + XY + \mu_3 Y^2, \end{cases} \qquad \theta = (A, C, \mu_1, \mu_2, \mu_3) \tag{2.1}$$

near $\theta_0 = (1,1,0,0,0)$.

Lemma 2.1. *There are neighborhoods $\mathcal{U} \subset \mathcal{Q}_2$ of H_0 and $\mathcal{W}$ of $(I,0,1,\theta_0)$ such that every $H \in \mathcal{U}$ has a unique representation with parameters in $\mathcal{W}$,*

$$H(z) = a_{\mathrm{t}} L^{-1} H_\theta(Lz + c), \qquad (L, c, a_{\mathrm{t}}, \theta) \text{ near } (I,0,1,\theta_0), \tag{2.2}$$

with $a_{\mathrm{t}} > 0$. The resulting maps $H \mapsto \mathfrak{A}(H) := (L, c, a_{\mathrm{t}})$ and $H \mapsto \theta(H)$ are analytic on $\mathcal{U}$, and satisfy $\mathfrak{A}(H_0) = (I,0,1)$ and $\theta(H_0) = \theta_0$.

Proof. The derivative of the coefficient map has columns

$$DH_0 h - Dh\, H_0, \quad h = (1,0), (0,1), (X,0), (Y,0), (0,X), (0,Y),$$

followed by H_0 and the five vectors

$$(X,0), \quad (0,X), \quad (Y^2,0), \quad (XY,0), \quad (XY,Y^2).$$

In monomial order $(1, X, Y, X^2, XY, Y^2)$ in each component, the determinant is 3. The analytic inverse function theorem proves the claim; positivity of a_{t} persists near 1. □

For $\mathfrak{A} = (L, c, a_{\mathrm{t}})$, denote by $T_{\mathfrak{A}}$ the extension of $z \mapsto Lz + c$ to the closed hemisphere, obtained by normalizing $(X, Y, W) \mapsto (L(X,Y)^{\mathsf{T}} + cW, W)$. After shrinking $\mathcal{U}$ to have compact closure in the inverse-function neighborhood, these extensions and their inverses have a common Lipschitz constant $1 \le L_* < \infty$. Moreover,

$$d_{\mathrm{H}}\big(T_{\mathfrak{A}(H)}\Gamma, \Gamma\big) \longrightarrow 0 \qquad \text{as } H \longrightarrow H_0, \tag{2.3}$$

because $\mathfrak{A}(H) \to (I,0,1)$.

2.2 The two singular points

In the chart at the positive Y-direction, use $v = X/Y,\ z = 1/Y$. Multiplying the vector field by $z > 0$ preserves the direction of its orbits in the finite plane and gives

$$\dot{v} = D_\theta(v,z) = P_\theta(v) + Q_\theta(v)z, \qquad \dot{z} = -zG_\theta(v,z) = -z\{R_\theta(v) + S_\theta(v)z\}, \tag{2.4}$$

where

$$\begin{aligned} &P(v) = \mu_1 + \mu_2 v - v^3, \quad Q(v) = -1 + Av - Cv^2, \\ &R(v) = \mu_3 + v + v^2, \qquad S(v) = Cv. \end{aligned} \tag{2.5}$$

The physical side is $z > 0$; the equator $z = 0$ is invariant. We always use the positive-time chart field on this side. At the organizing parameter $\theta_0 = (1,1,0,0,0)$, the chart is

$$\dot{v} = -v^3 - (1 - v + v^2)z, \qquad \dot{z} = -z(v + v^2 + vz). \tag{2.6}$$

The linearization is $\left(\begin{smallmatrix} 0 & -1 \\ 0 & 0 \end{smallmatrix}\right)$. Eliminating the analytic unit $1 - v + v^2$ gives the nullcline $z = -v^3/(1 - v + v^2)$, on which

$$\dot{z}|_{\dot{v}=0} = \frac{v^4}{(1 - v + v^2)^2}. \tag{2.7}$$

Here $(1 + v)(1 - v + v^2) = 1 + v^3$; the reduced zero has order four, the multiplicity of the nilpotent singularity at infinity. The organizing arc has equation

$$z_\Gamma(v) = \sqrt{1 + v^2} - 1 \tag{2.8}$$

and is traversed with decreasing v; its expansion $z_\Gamma(v) = v^2/2 + O(v^4)$ gives the common equatorial tangent. Choose a small fixed $\delta > 0$ and the sections $\Sigma_\pm = \{v = \pm\delta\}$, with small height intervals around this arc. Take the height intervals small enough that $-D$ has a positive lower bound on their closures. On the fixed small chart, a coefficient shrink gives

$$Q < -q_0 < 0, \qquad G_v = 1 + 2v + Cz > g_0 > 0, \qquad P_\theta(\delta) < 0, \quad G_\theta(-\delta, z) < 0. \tag{2.9}$$

The small zero of the transverse eigenvalue and the first two equatorial coefficients there are

$$\gamma = \frac{-1 + \sqrt{1 - 4\mu_3}}{2}, \qquad \alpha_0 = P(\gamma), \qquad \alpha_1 = P'(\gamma). \tag{2.10}$$

The Jacobian with respect to (μ_3, μ_1, μ_2) at θ_0 is -1, so these, together with A, C, are parameter coordinates. The translated polynomials and their rescalings are introduced in Section 4.1. The five raw coefficients $A - 1, C - 1, \gamma, \alpha_0, \alpha_1$ vary independently.

For the finite saddle-node, set $u = X + 1,\ W = Y + u$. At θ_0,

$$\dot{u} = u^2 - W, \qquad \dot{W} = u^2 + (u - 2)W.$$

The zero set of the second component is an analytic graph $W = \psi(u, \theta)$, used here only to locate equilibria. The first component on this graph has a Weierstrass factor

$$e(u,\theta)(u^2 + b_1(\theta)u + b_0(\theta)), \qquad e(0,\theta_0) = 1/2.$$

Indeed, at θ_0 it equals $u^2(1 - u)/(2 - u)$. Differentiation gives

$$b_0 = C - A + O(|\theta - \theta_0|^2), \qquad b_1 = A - 1 + \mu_2 + \mu_3 + O(|\theta - \theta_0|^2).$$

Consequently $K = b_0 - b_1^2/4$ is a discriminant coordinate with $dK = dC - dA$ at the organizing field. We also write $K_{\text{fin}} = K$ for this finite-passage discriminant.

The organizing analytic germ has eigenvalues $0, -2$ and a nonzero quadratic weak coefficient $1/2$. Thus its multiplicity is two, the case $\mu = 1$ of the saddle–node normal form. Fix a finite joint differentiability order $N \geq 12$ before restricting the coefficients. The ordinary section maps and their inverses will be used through phase order three, while division by the discriminant costs at most one parameter derivative. The intrinsic root-collar kernels use phase order five, axis order six and selected mixed order seven; these estimates and the strong projection estimates are obtained from the analytic chart field. Choose the order for the normal-form theorem now, increasing it if necessary before choosing any neighborhood. This finite surplus accommodates coefficient differentiation and Hadamard division, so a jointly smooth normalizer of all orders is unnecessary. The finite-smooth multiparameter unfolding normal form [3, Theorem 3 and formula (2.2), October 2000 author copy p. 6] gives, by a joint C^N orbital equivalence preserving positive time,

$$\xi' = \varepsilon_f + \frac{\xi^2}{1 + b\xi}, \qquad \eta' = -\eta. \tag{2.11}$$

Here the parameter in the cited theorem is the complete vector $\theta - \theta_0$. For $\mu = 1$, its polynomial $P(\theta, \xi)$ is the constant ε_f. The reduction divides by the nonzero strong rate, which is positive at the organizing parameter and remains so nearby; it thus gives the stated positive-time equivalence. The theorem imposes no rank condition on the unfolding coefficients. The linear calibration $t_{\mathrm{nf}} = 2t$, $\xi = (X + 1 - Y)/8 + O(|(X + 1, Y)|^2)$ gives $\varepsilon_f = (C - A)/16 + O(|\theta - \theta_0|^2)$. The double-equilibrium loci in the two descriptions coincide. In parameter coordinates (K, ϑ), Hadamard division gives

$$\varepsilon_f(K, \vartheta) = Kh(K, \vartheta), \qquad h(K, \vartheta) = \int_0^1 \partial_K \varepsilon_f(tK, \vartheta)\, dt, \qquad h(\theta_0) = 1/16.$$

Continuity makes h a positive unit on a smaller coefficient neighborhood.

For fixed internal sections $\xi = \pm\rho$, with $1 + b\xi > 1/2$, the passage exists when $K > 0$ and is exactly

$$\eta_{\mathrm{out}} = m_{\mathrm{fin}} \eta_{\mathrm{in}}, \quad m_{\mathrm{fin}} = e^{-T_f}, \quad T_f = \int_{-\rho}^{\rho} \frac{d\xi}{\varepsilon_f + \xi^2/(1 + b\xi)} = \frac{\pi}{\sqrt{\varepsilon_f}} + O(1). \tag{2.12}$$

We write $m = m_{\mathrm{fin}}$ in the completed return below. Completing the square in $\xi^2 + b\varepsilon_f \xi + \varepsilon_f$ proves the last estimate. Differentiating its arctangent and logarithmic expressions also shows that each fixed finite parameter derivative of m is bounded by $CK^{-M} e^{-c/\sqrt{K}}$. Thus m extends flatly by zero to $K = 0$, to the chosen differentiability order. If $K \leq 0$, an orbit entering the selected channel approaches the saddle-node or the left equilibrium and cannot exit through $\xi = \rho$. The strong sides of this box point inward.

2.3 The exact displacement

Let $T_\theta : \Sigma_+ \to \Sigma_-$ be the infinity transition when it exists in positive time. The rest of the return consists of two regular transitions and the finite passage (2.12); see Figure 1.

Lemma 2.2. *There are uniformly regular increasing maps C_θ, E_θ on fixed section intervals such that the finite-side return is $q \mapsto C_\theta(mE_\theta(q))$. Their derivatives and inverse derivatives are bounded through the chosen finite order on buffered section intervals. If $a_\theta = C_\theta(0)$, $f_\theta = C_\theta - a_\theta$, and $d_\theta = E_\theta$, then in the affine coordinate $x = z_{\mathrm{in}} - a_\theta$ the cycle equation is*

$$\Delta_\theta(x) = x - m\Pi_\theta(x) = 0, \qquad \Pi_\theta(x) = \mathcal{C}_{\theta,m}(T_\theta(a_\theta + x)), \tag{2.13}$$

where

$$\mathcal{C}_{\theta,m}(q) = d_\theta(q) \int_0^1 f'_\theta(tmd_\theta(q))\, dt. \tag{2.14}$$

In the output coordinate t, the equivalent equation is

$$S_\theta(t) - a_\theta - f_\theta(mt) = 0, \qquad S_\theta = (d_\theta \circ T_\theta)^{-1}. \tag{2.15}$$

Proof. The parameter-dependent flow-box theorem on the two compact regular segments gives positively oriented E_θ, C_θ, with bounded derivatives and reciprocal slopes on smaller fixed intervals. The integral in (2.14) extends $f_\theta(md_\theta(q))/m$ to $m = 0$, with positive slope $f'_\theta(md_\theta)d'_\theta$. Substitution proves both equations and preserves isolated zeros and their multiplicities. □

At a zero, boundedness of $\mathcal{C}_{\theta,m}$ gives $|x| \le Cm$. In the inverse equation, the nonlinear closing term has an additional factor m^2:

$$f_\theta(mt) = mf'_\theta(0)t + m^2t^2 \int_0^1 (1-u)f''_\theta(umt)\,du. \tag{2.16}$$

The affine input keeps this small quadratic closing term explicit in the later derivative comparisons.

The separate value-and-slope normalization (C.7) is applied to the actual input height and completed output height. It retains their two centers and gains; only the input normalization is affine in the physical incoming height.

2.4 Order and uniform transverse sections

We first bound the through trajectories from above by an invariant graph. Choose $z_{\mathrm{r}} = \kappa\delta$, with $\delta \ll \kappa \ll 1$. On $z_{\mathrm{r}}/2 \le z \le 2z_{\mathrm{r}}$, a coefficient shrink gives $|P| \le z_{\mathrm{r}}/8,\ Q \le -1/2,\ |G| \le C\delta$. The orbit through (δ, z_{r}) consequently has $D \le -z_{\mathrm{r}}/8$ and $|dz/dv| \le C\delta$. It crosses the chart inside this band, by a first-exit argument. Its graph $\overline{z}_\theta(v)$ is the upper boundary; all selected section windows are strictly below it.

Lemma 2.3. *The heights on Σ_+ which reach Σ_- in the chart below this upper boundary form an interval*

$$\mathcal{T}_\theta = (\Theta_\theta, \overline{z}_\theta(\delta)]. \tag{2.17}$$

Every such orbit has $D < 0$. Both T_θ and

$$\sigma_\theta(h) = \min_{|v|\le\delta}\{-D_\theta(v, z_\theta(v;h))\} \tag{2.18}$$

are strictly increasing functions of the input h.

Proof. Every physical input has $D < 0$ on the right section by (2.9). On the nullclines,

$$\dot{D}|_{D=0} = -zQG, \qquad \dot{G}|_{G=0} = G_v D. \tag{2.19}$$

A first crossing from $D < 0$ to $D > 0$ requires $G > 0$; equality would be an equilibrium. The quadrant $D > 0, G > 0$ is forward invariant by these identities, so this orbit cannot subsequently reach the left section. A first terminal nullcline hit on $v = -\delta$ is also impossible: there $\dot{D} < 0$ by (2.9). A through orbit therefore has $D < 0$ everywhere and solves

$$z_v = -zG/D. \tag{2.20}$$

Scalar uniqueness strictly orders these graphs. A graph above a through graph has smaller D, since $Q < 0$, and cannot cross either $D = 0$ or the invariant upper boundary. It also passes through; openness proves the interval assertion and ordering proves that T increases. The positive difference of the two $-D$-values has a positive minimum on the compact v-interval, so σ increases. □

The lower endpoint is a stopped limiting passage. A *gap* is the positive height difference from it on a specified transverse cut, distinct from the central height.

Lemma 2.4. *There are $0 < r < \delta$ and $d_0, c_r > 0$, independent of the through orbit and of the sufficiently small coefficient, such that every through orbit with endpoints in the selected outer windows and $\sigma_\theta < d_0$ crosses $v = r$ and $v = -r$ once, in order. On $r \le |v| \le \delta$ it has $-D \ge c_r$. The two outer transitions consequently have uniformly bounded derivatives and inverse derivatives of any prescribed finite order on slightly larger fixed domains.*

Proof. The incidence set $\mathcal{E}_\theta = \{D = 0,\ zG = 0\}$ at θ_0 meets the small closed chart only at the origin. Compactness puts the whole nearby incidence set in $|v| < r/3$. Choose a neighborhood $\mathcal{O}$ of it in $|v| < 2r/3$. The endpoint lower bounds persist on the selected outer windows. At an interior minimum of $-D$ on a through graph,

$$D\,D_v - QzG = 0. \tag{2.21}$$

Transversality excludes a minimum near the fixed outer sections. As $\sigma_\theta \to 0$, (2.21) gives $|zG| \le C\sigma_\theta$; hence the minimizing point must enter $\mathcal{O}$. This proves the first assertion for some d_0. If the outer speed had no positive lower bound, a limiting orbit would have a point with $D = 0$, $D \le 0$ on both time sides, and thus $\dot{D} = 0$. By (2.19) this point lies in $\mathcal{E}$, a contradiction. The decreasing coordinate v now gives all ordered hits. The hitting-time theorem on these uniformly transverse compact slabs proves the derivative bounds. □

Fix nested section windows, whose interiors contain the corresponding intersections of the organizing graphic, and speed bounds

$$K_\pm \Subset K_\pm^\sharp \Subset \Sigma_\pm, \qquad 0 < d_{\mathrm{in}} < d_{\mathrm{out}} < d_0. \tag{2.22}$$

The inner and outer sets specified by

$$h \in K_+, \quad T_\theta(h) \in K_-, \quad \sigma_\theta(h) \le d_{\mathrm{in}},$$

and respectively by the interiors of $K_\pm^\sharp$ and $\sigma_\theta(h) < d_{\mathrm{out}}$, are intervals, denoted by $J_\theta^{\mathrm{in}} \subset J_\theta^{\mathrm{out}}$. All intervening transverse sections have nested windows as well. The central height

$$h_\theta(h) = z_\theta(\gamma(\theta); h) \tag{2.23}$$

is positive and strictly increasing on J_θ^{out}. For any fixed raw height cap $h_* > 0$, define the affine physical receiver

$$I_\theta(h_*) = \{x :\ a_\theta + x \in J_\theta^{\mathrm{out}},\ 0 < h_\theta(a_\theta + x) < h_*\}. \tag{2.24}$$

It is an interval, possibly empty. Thus the whole interval between any finite collection of its inputs satisfies the same input, inverse-output and speed restrictions. These geometric restrictions define the domain before any zero estimate is applied, and keep it on the through side of the stopped boundary.

The estimates in the following sections choose a positive cap h_{out} in (2.24) with strict buffers. Fix this cap and the nested windows before the final independent raw coefficient restrictions and the multiplier bound. Subsequent restrictions retain these caps and section bounds. After these choices, the following capture lemma places all nearby complete cycles below a smaller cap $0 < h_{\mathrm{in}} < h_{\mathrm{out}}$.

Lemma 2.5. *For every fixed set of these admissible nested windows and speed bounds in* (2.22)*, and every $h_{\mathrm{in}} > 0$, there are a coefficient neighborhood $\mathcal{V}_0$ of θ_0 and $\varepsilon_0 > 0$ such that, for every $\theta \in \mathcal{V}_0$, every periodic orbit C of H_θ with $d_{\mathrm{H}}(\iota_0(C), \Gamma) < \varepsilon_0$ follows the selected positive-time circuit and has $K > 0$. It meets the return section once in its prime period. Its infinity arc has input in J_θ^{in}, all selected transverse hits in the inner windows, and $0 < z(\gamma) < h_{\mathrm{in}}$.*

Proof. Surround the two singularities by small blocks and the two compact regular branches by flow boxes, with nested transverse windows. In the finite normal-form box take weak faces $\xi = -\rho$ and $\xi = \rho$, with transverse windows about the two organizing hits. The weak speed at these fixed faces is positive after a coefficient shrink. Thus they are respectively entrance and exit faces. The strong faces $\eta = \pm\eta_0$ point inward, since $\eta' = -\eta$, and are chosen away from the organizing arc. Remove from the two weak faces the parts outside the transverse windows. These parts and the strong faces are compact and disjoint from Γ.

The infinity block has transverse faces $v = \delta$ and $v = -\delta$, the invariant equator below, and the invariant graph $\overline{z}_\theta$ above. At the two selected face windows the speed D is uniformly negative. Its upper face and the portions of the transverse faces outside these windows stay a positive distance from Γ. The two regular flow boxes join the finite exit to the right infinity entrance and the left infinity exit to the finite entrance, in positive time. Their lateral faces have the same positive distance property. Choose the Hausdorff radius below all these boundary distances, which persist for nearby coefficients. A close cycle must visit both regular branch boxes, because it approximates the complete graphic. It can leave each visited box only through the selected next face; the equator cannot be crossed by uniqueness. These entrances and exits therefore force the finite channel, one regular branch, $\Sigma_+ \to \Sigma_-$, and the other regular branch in this order. The finite channel has no exit for $K \le 0$, so $K > 0$; Lemma 2.3 gives $D < 0$ on the infinity passage.

Every intersection of the cycle with the selected connected segment of Σ_+ lies in its inner window and has the same transverse orientation. A prime periodic orbit of a planar vector field is a Jordan curve. Along a connected transverse segment its intersections alternate between entering and leaving its bounded complementary component. Hence these equally oriented intersections consist of one point. The same circuit then contains one infinity arc, and its height is one isolated zero of (2.13) when the cycle is isolated.

If the uniform inner bounds failed, take $\theta_n \to \theta_0$ and C_n with $d_{\mathrm{H}}(\iota_0(C_n), \Gamma) \to 0$. Let A_n be the unique infinity-block subarc selected above and A_0 the complete organizing arc in that block. Every Hausdorff limit of A_n is contained in A_0. Conversely every interior point of A_0 is approached by $\iota_0(C_n)$, and the selected circuit forces the approaching points to be on A_n; closure includes the endpoints. Thus $d_{\mathrm{H}}(A_n, A_0) \to 0$. For a common modulus of continuity ω,

$$\left| \min_{A_n}(-D_{\theta_n}) - \min_{A_0}(-D_{\theta_0}) \right| \le \|D_{\theta_n} - D_{\theta_0}\|_\infty + \omega(d_{\mathrm{H}}(A_n, A_0)).$$

Since the minimum on A_0 is zero, $\sigma_{\theta_n}(A_n) \to 0$. The hits of $v = \gamma(\theta_n)$ converge to $(0, 0)$, the unique hit of $v = 0$ on A_0, and all transverse hits enter the inner windows by continuity. This contradiction supplies one coefficient neighborhood and one Hausdorff radius. □

2.5 Transferring a slice bound

The same geometric transfer will be used for the full coefficient cover and for its ratio-restricted parts.

Lemma 2.6. *Let $\mathcal{U}_5$ be a neighborhood of θ_0, let $\mathcal{B}$ be any subset of the slice parameter space, and fix $\varepsilon_5 > 0$ and an integer $N_{\mathrm{cyc}} \ge 0$. Suppose every $\theta \in \mathcal{U}_5 \cap \mathcal{B}$ has at most N_{cyc} distinct isolated cycles at whole-graphic Hausdorff distance less than ε_5 from Γ. There are one open neighborhood $\mathcal{V}$ of F_* in $\mathcal{Q}_2$ and one $\varepsilon > 0$ such that every $F \in \mathcal{V}$ with $\theta(\mathscr{C}_0(F)) \in \mathcal{B}$ has at most N_{cyc} distinct isolated periodic orbits C satisfying $d_{\mathrm{H}}(\iota_*(C), \Gamma_{I4}) < \varepsilon$. The transport preserves the complete marking, positive time and return-zero multiplicities. The maps and their inverses also preserve whole-orbit Hausdorff convergence to the marked graphics as the fields tend to F_*.*

Proof. For $H = \mathscr{C}_0(F)$ use the analytic decomposition in Lemma 2.1 for this field. Enlarge L_0 to bound both S and S^{-1}, and use the common bound L_* for $T_{\mathfrak{A}(H)}$ and its inverse. Continuity gives an open neighborhood $\widehat{\mathcal{V}}$ of H_0 on which

$$\theta(H) \in \mathcal{U}_5, \qquad \eta(H) := d_{\mathrm{H}}(T_{\mathfrak{A}(H)}\Gamma, \Gamma) < \varepsilon_5/2. \tag{2.25}$$

Set

$$\mathcal{V} = \mathscr{C}_0^{-1}(\widehat{\mathcal{V}}), \tag{2.26}$$

an ordinary open neighborhood in all twelve coefficients, and take

$$\varepsilon = \frac{\varepsilon_5}{2L_*L_0}. \tag{2.27}$$

The fixed change and $w = LZ + c$ send a cycle C of this same field to a cycle $\widetilde{C}$ of $H_{\theta(H)}$, with

$$\iota_0(\widetilde{C}) = T_{\mathfrak{A}(H)} S(\iota_*(C)).$$

Writing $d_{\mathrm{original}} = d_{\mathrm{H}}(\iota_*(C), \Gamma_{I4})$ and $d_{\mathrm{slice}} = d_{\mathrm{H}}(\iota_0(\widetilde{C}), \Gamma)$, we have

$$d_{\mathrm{slice}} \le L_*L_0 d_{\mathrm{original}} + \eta(H) < \varepsilon_5, \tag{2.28}$$

$$d_{\mathrm{original}} \le L_0L_*(d_{\mathrm{slice}} + \eta(H)). \tag{2.29}$$

The affine diffeomorphisms and positive time factors biject cycles, preserving distinctness, isolation, the two branches and their marked order. The induced transverse return maps are locally conjugate, so zero multiplicities are preserved as well. The first inequality applies the slice bound; both inequalities and $\eta(H) \to 0$ give the convergence assertion as the fields tend to the organizing field. □

3 Counting on one physical interval

The upper bound follows from two elementary counting criteria on a connected interval. One uses the orientation of displacement critical points; the other uses the orientation of curvature zeros. Each criterion counts the whole physical interval, including the regions between the captured roots.

3.1 A common coordinate and the counting criteria

Fix the incoming raw height coordinate in Lemma 2.2 before normalizing the tails. Absorb the regular maps between the fixed raw ports and the chosen sections into the actual adapters. All regional estimates will then concern the displacement

$$F(x) = x - m\Pi(x), \qquad m > 0. \tag{3.1}$$

Every local estimate below includes the divided adapter (2.14). Its derivatives remain bounded, with positive first derivative, as $m \downarrow 0$. Apply the local tail estimates with these adapters and normalize each map by its exact center and first variation. The relation with the physical map is the identity

$$\Pi(a_+ + c_+w) = a_- + c_-\widehat{\Pi}(w), \qquad c_\pm > 0. \tag{3.2}$$

Thus the signs of second and third derivatives transfer by the positive factors c_-/c_+^2 and c_-/c_+^3. The affine relation is essential here: independently chosen nonlinear coordinates need not preserve curvature. Comparisons obtained in an inverse coordinate are therefore transferred to the direct displacement before counting its critical points.

All carriers below are intervals of actual through trajectories in the nested section windows. Scalar first variation orders them by the central height. The preimage of an output interval under an increasing passage map is an interval. Consequently the interval between any finite set of captured roots stays in the same enlarged physical domain. The estimates used between roots therefore apply at nonroot points as well.

Lemma 3.1. *Let F be C^3 on a connected carrier I. If every critical point satisfies $F'' > 0$, then F has at most one critical point, which is simple and upward, and at most two isolated zeros, counted with multiplicity. More generally, let $J \subseteq I$ be a connected interval, possibly empty or all of I. If F' has at most j zeros on J, counted with multiplicity, where $0 \le j \le 2$, and $F' > 0$ on $I \setminus J$, then F' has at most j zeros on I, and F has at most $j+1$ isolated zeros there, counted with multiplicity. For $J = \varnothing$, take $j = 0$: there is no critical point and at most one simple displacement zero.*

Proof. In the first case every zero of F' is a simple upward crossing. Two such crossings would require an intervening zero with another orientation, so there is at most one. In the second case no critical point lies outside J, whether its complement is empty or lies on one or both sides. Rolle's theorem with multiplicities proves both assertions. □

Lemma 3.2. *Let $F = x - m\Pi$, $m > 0$, be C^3 on a connected carrier I, and let $J \subseteq I$ be a connected interval, possibly empty or all of I. Suppose that every zero of Π'' on J is simple and has the same crossing orientation, and that $F' > 0$ on $I \setminus J$. Then Π'' has at most one zero on J, F' has at most two zeros on I, counted with multiplicity, and F has at most three isolated displacement zeros on I, counted with multiplicity. If $J = \varnothing$, there is no critical point and at most one simple displacement zero.*

Proof. For nonempty J, two simple curvature zeros of the same orientation would require an intervening zero with another orientation, so there is at most one. Since $F'' = -m\Pi''$, Rolle's theorem with multiplicities gives at most two zeros of F' on J. Apply Lemma 3.1 on I. For empty J, $F' > 0$ throughout I. Positive affine changes in (3.2) preserve the curvature orientation, even if their scale ratios degenerate. □

The regional constructions specify the actual intervals I, J and establish the hypotheses at every point of those intervals. For each coefficient fibre, apply Lemma 3.1 or Lemma 3.2 to its whole carrier. The later coefficient cover takes the maximum of these bounds; its overlap choices are independent of the phase input.

3.2 Ordinary adapters and divided closing maps

The next two lemmas control the regular maps appended to a singular passage. They use ordinary third derivatives of those maps and allow the singular passage's first gain to tend to zero.

Lemma 3.3. *Let the ordinary adapters below have uniform C^3 norms and positive first derivatives bounded away from zero on their original buffered images. All parameters and gains are frozen under the phase derivation D. The following identities and bounds are uniform in those parameters.*

For an actual raw output T in such an image, put $\ell = \log \psi'$. Then

$$\begin{aligned} D \log \frac{(\psi \circ T)'}{T'} &= \ell'(T) DT, \\ D^2 \log \frac{(\psi \circ T)'}{T'} &= \ell''(T)(DT)^2 + \ell'(T) D^2 T. \end{aligned} \tag{3.3}$$

Here $T' > 0$ is the exact raw first gain; the formulas impose no lower bound on it. If the full translated output and its two absolute phase derivatives are bounded, and mT stays in the fixed closing buffer, then

$$\sum_{j=0}^{2} \Big(|D^j(f(mT) - f(0))| + |D^j(\log f'(mT) - \log f'(0))| \Big) \le Cm. \tag{3.4}$$

The constant depends on these absolute output bounds, not on a normalized raw gain or a ratio of gains.

For $f(0) = 0$, the divided closing map $C_m(q) = f(md(q))/m$, extended at zero by $C_0 = f'(0)d$, has jets

$$\begin{aligned} C_m' &= f'(md)d', \\ C_m'' &= mf''(md)(d')^2 + f'(md)d'', \\ C_m''' &= m^2 f'''(md)(d')^3 + 3mf''(md)d'd'' + f'(md)d'''. \end{aligned} \tag{3.5}$$

They are uniformly bounded and the first is a positive unit on each prescribed compact q-interval with buffered md-image. Finally, for a coefficient-only center z_0, $0 < d \le 1$, and a fixed compact phase interval whose $z_0 + dy$-image stays buffered,

$$N_d(y) = \frac{\psi(z_0 + dy) - \psi(z_0)}{d\psi'(z_0)}, \qquad N_0(y) = y, \qquad \|N_d - \mathrm{id}\|_{C^3} \le Cd. \tag{3.6}$$

This last bound concerns phase derivatives, not derivatives of the center or scale with respect to a parameter.

Proof. The chain rule proves (3.3) and (3.5); the integral formula $C_m = d\int_0^1 f'(tmd)\,dt$ gives the extension at zero. The ordinary C^3 bounds and positive slopes then give (3.4) by the mean value theorem. Taylor's integral formula gives (3.6), whose second and third rows contain factors d and d^2, respectively. □

Lemma 3.4. *Let M be an intrinsic core map with uniformly bounded C^4 norm on a fixed buffered phase interval. Let A, B be ordinary C^3 adapters close to the identity, with uniformly positive slopes, on buffers containing all compositions below. On a common compact interval for A^{-1} and $M \circ A^{-1}$,*

$$\|B \circ M \circ A^{-1} - M\|_{C^3} \le C(\|A - \mathrm{id}\|_{C^3} + \|B - \mathrm{id}\|_{C^3}). \tag{3.7}$$

The fourth derivative is required only of the intrinsic core.

Proof. The inverse formulas through order three and the uniform slope bound give $\|A^{-1} - \mathrm{id}\|_{C^3} \le C\|A - \mathrm{id}\|_{C^3}$. In the third chain-rule row for $M \circ A^{-1} - M$, the translated factor $M'''(A^{-1}) - M'''$ is bounded by $\|M''''\|_\infty \|A^{-1} - \mathrm{id}\|_\infty$; the other rows need fewer core derivatives. The ordinary composition of $B - \mathrm{id}$ with $M \circ A^{-1}$ is bounded in C^3 without translating any fourth derivative of B. These estimates prove (3.7). □

3.3 Weighted curvature of a completed passage

To use an action calculation in the physical input, we must include the curvature contributed by each section map. The following identity keeps these contributions separate and allows a positive weight adapted to the coefficient chart.

Lemma 3.5 (Weighted curvature of a completed passage)**.** *Let $u, v > 0$ be radial coordinates, with $P_\pm, B_\pm > 0$ on the two strict through intervals, and suppose*

$$v_u = \frac{uB_+(u)P_-(v)}{vB_-(v)P_+(u)}(1 + \varrho) > 0, \quad S(-u^2/2) = -v^2/2, \quad \Pi = \phi_- \circ S \circ \phi_+^{-1},$$

Here $P_\pm, B_\pm$ are C^2, $\varrho = \varrho(u,v)$ is C^2 on the product of the two radial intervals, and $\phi_\pm$ are the separately normalized actual C^3 section maps with positive derivatives. Put $\mathcal{D}_+ = P_+\partial_u/(uB_+)$, $\mathcal{D}_- = P_-\partial_v/(vB_-)$, and $R_B = B_+/B_-$. For any smooth positive weight $W = W(u,v)$, one has exactly

$$\begin{aligned}\phi'_+\frac{P_+}{W}\frac{\Pi''}{\Pi'} = {} & \frac{P'_+/u - R_B(1+\varrho)P'_-/v}{W} - \frac{P_+}{uW}\frac{d}{du}\log R_B \\ & - \frac{B_+}{W}\{\mathcal{D}_+ + (1+\varrho)\mathcal{D}_-\}\log(1+\varrho) \\ & - \frac{B_+(1+\varrho)}{W}\mathcal{D}_-\log\phi'_- + \frac{P_+}{u^2W}(u\partial_u)\log\phi'_+. \end{aligned} \tag{3.8}$$

Primes on each $\phi_\pm$ use its own quadratic label; primes on $P_\pm$ use the displayed radial variable. The derivative d/du follows the matched graph. In particular both section terms remain in the curvature of the actual affine input.

Proof. One has $S' = R_B P_-(1+\varrho)/P_+$. Logarithmically differentiate this identity using $d/db_+ = -u^{-1}d/du$, and substitute in

$$\phi'_+\frac{\Pi''}{\Pi'} = \frac{S''}{S'} + S'\frac{\phi''_-}{\phi'_-} - \frac{\phi''_+}{\phi'_+}.$$

Multiplication by P_+/W gives the formula. Its coefficients need not extend without weights to a stopped endpoint. □

4 Action curvature and coefficient geometry

We first define the coefficient coordinates and passage scales. The representative cubic calculation then explains how the limiting action controls the curvature of the completed physical return.

4.1 Coefficient geometry and the regime decision map

Different relative sizes of the five coefficients in (2.1) lead to different passage scales. We introduce these scales here and summarize the resulting cases in Figure 2. The corresponding bounds are proved in the later face propositions. Before counting zeros, every phase chart is transported to the physical sections of (2.13).

Translate the zero of the transverse eigenvalue to the origin and put

$$\xi = v - \gamma, \qquad \alpha_0 = P(\gamma), \qquad \alpha_1 = P'(\gamma), \qquad q_\gamma = 1 + 2\gamma, \qquad \ell(v) = 1 - Av + Cv^2. \tag{4.1}$$

Then (2.4) becomes

$$\begin{aligned}\dot\xi &= \alpha_0 + \alpha_1\xi - 3\gamma\xi^2 - \xi^3 - \ell(\gamma+\xi)z, \\ \dot z &= -z\{q_\gamma\xi + \xi^2 + C(\gamma+\xi)z\}.\end{aligned} \tag{4.2}$$

Thus $(A, C, \gamma, \alpha_0, \alpha_1)$ are analytic coordinates on the five-parameter slice near the organizing field. After one common coefficient shrink, q_γ and ℓ are positive units on the fixed raw rectangle, and the regular transition maps between fixed sections and their inverses have uniform bounds through every derivative order fixed in advance.

For negative opening write

$$\alpha_0 = -s^2, \qquad p = \frac{\alpha_1}{s}, \qquad v = \gamma + sX, \qquad z = s^2Y. \tag{4.3}$$

This gives the source field (C.1) with parameters $(a, p, \gamma, s) = (-1, p, \gamma, s)$. Two exact compactifications cover its small-p coefficient ends. For $\gamma \neq 0$, set

$$g = |\gamma|, \qquad k = \frac{s}{g^{3/2}}, \qquad \pi = \frac{p}{\sqrt{g}}, \qquad \mathfrak{q} = k\pi. \tag{4.4}$$

Here g is always a coefficient radius, never a section height. In the cubic chart set

$$t = s^{1/3}, \qquad U = \frac{p}{t}, \qquad G = \frac{\gamma}{t^2}, \qquad \tau_{\mathrm{cub}} = t^2. \tag{4.5}$$

We write τ_{cub} for this cubic scale throughout. The far-root, two-saddle and degree-loss arguments each use an unadorned τ for their local scale; any relation such as $\tau = p^2$ is stated where that scale is introduced.

θ in the five-parameter slice $\longrightarrow (\gamma, \alpha_0, \alpha_1; A, C)$

$K \leq 0$ or no through passage	$\longrightarrow$stopped/no-passage endpoint; it is not a periodic-orbit zero
$K > 0$, high central height	$\longrightarrow$affine high-height estimate, Corollary 5.1
$\alpha_0 \geq 0$, low height	$\longrightarrow$determinant chart or signed trace chart, Lemma I.4
$\alpha_0 < 0,\ p \leq 0$	$\longrightarrow$raw Jacobian contraction, Lemma H.15
$\alpha_0 < 0,\ 0 < p \ll 1$	$\longrightarrow$gamma-dominant, cubic, degree-loss, or remote-root face, Lemma 5.9
$\alpha_0 < 0,\ p \asymp 1$	$\longrightarrow$compact positive-ratio passage, Lemma F.1
$\alpha_0 < 0,\ p \gg 1$	$\longrightarrow$signed-trace terminal block plus compact annuli, Lemma 5.11

Each row $\longrightarrow$ zeros of the same exact displacement $x - m\Pi(x)$ on one coefficient fibre $\longrightarrow$ one bound on each whole phase fibre, then the maximum across coefficient alternatives $\longrightarrow$ one common coefficient neighborhood and one common Hausdorff collar $\longrightarrow$ lift from five to twelve coefficients.

Figure 2: Global coefficient–phase decision map. The arrows specify the logical routing, not additional coordinate changes. Strict overlaps between adjacent phase and coefficient blocks are recorded in Table 1.

For later reference, the symbols with global scope are $\theta = (A, C, \mu_1, \mu_2, \mu_3)$, $(\gamma, \alpha_0, \alpha_1)$, the finite multiplier m, and the actual input of (2.13). Symbols such as $g, k, \pi, \mathfrak{q}$, t, U, G, and the degree-loss exponents are local to the chart in which they are defined. Actual phase heights carry labels such as h, h_L, V, $\widehat{V}$, W, or Θ; they are never identified with a coefficient radius or with a stopped trace gap.

4.2 The cubic action and the physical curvature

The compact cubic block illustrates the curvature mechanism behind the three-zero bound. At the scale (4.5), its axis polynomial is

$$P(\xi) = 1 - U\xi + 3G\xi^2 + \xi^3, \qquad U \geq 0,$$

with U, G in a fixed compact set. On a strict through interval, let $u > 0$ and $-r < 0$ be the incoming and outgoing axis points. The limiting action equation is

$$\int_0^u \frac{\xi}{P(\xi)}\, d\xi = \int_0^r \frac{\xi}{P(-\xi)}\, d\xi, \qquad b = -u^2/2, \qquad S(b) = -r^2/2. \tag{4.6}$$

On the traversed segment $P > 0$. Differentiating gives $r_u = uP(-r)/(rP(u)) > 0$ and

$$S' = \frac{P(-r)}{P(u)}, \qquad \frac{P(u)}{u+r}\frac{S''}{S'} = 3 - \frac{U}{ur}, \qquad (ur)_b = -\frac{r}{u} - \frac{uP(-r)}{rP(u)} < 0. \tag{4.7}$$

These are the cubic specialization of (D.22)–(D.23). The term containing G cancels. At a curvature zero, $U = 3ur$, so the derivative of the middle expression with respect to b is strictly negative. Thus every curvature zero is a downward crossing, and there can be at most one on a connected interval.

The actual input is an affine raw section coordinate. To compare its curvature with the action formula, write the two separate outward maps as

$$\Gamma_\pm = a_\pm + c_\pm \phi_\pm, \qquad c_\pm > 0, \qquad \widehat{\Pi} = \phi_- \circ S_{\mathrm{act}} \circ \phi_+^{-1}.$$

Here S_{act} is the actual matched transition in the two normal section labels. The centers and gains are retained exactly. In these coordinates the second-derivative chain rule is

$$\phi_+' \frac{\widehat{\Pi}''}{\widehat{\Pi}'} = \frac{S_{\mathrm{act}}''}{S_{\mathrm{act}}'} + S_{\mathrm{act}}' \frac{\phi_-''}{\phi_-'} - \frac{\phi_+''}{\phi_+'}. \tag{4.8}$$

Each section derivative uses its own quadratic label. The actual gains do not enter this identity. Returning to the physical affine input multiplies its second and third derivatives by c_-/c_+^2 and c_-/c_+^3, so their signs are preserved.

On its bounded normal-section interval, Proposition 4.1 proves the weighted identity

$$\phi_+'(b) \frac{P(u)}{u+r} \frac{\widehat{\Pi}''}{\widehat{\Pi}'} = 3 - \frac{U}{ur} + E(u,r), \qquad \|E\|_{C^1_{u,r}} \le C\tau_{\mathrm{cub}}(1 + |\log \tau_{\mathrm{cub}}|). \tag{4.9}$$

It includes both section terms in (4.8). Near a possible zero the principal derivative with respect to u has a margin $c(1 + r_u)$, whereas the error derivative along the matched graph is at most $C\tau_{\mathrm{cub}}(1 + |\log \tau_{\mathrm{cub}}|)(1 + r_u)$. The physical input $\phi_+(-u^2/2)$ decreases with u. Consequently the actual curvature retains the same downward crossing. The bounded-action comparison overlaps this description on the same map. The rest of the admitted interval has bounded physical first derivative; choosing m last makes $F' > 1/2$ there. Lemma 3.2 then gives three zeros on the whole interval.

Two estimates make this calculation compatible with singular coefficient limits. First, the exterior continuations carry the common first-variation factor $\tau_{\mathrm{cub}}^{2-\lambda_G}$, where $\lambda_G = 12\tau_{\mathrm{cub}}G/(1 + 2\tau_{\mathrm{cub}}G)$. The common power cancels in the completed first derivative; the actual normalized section maps have ordinary C^3 bounds. The general raw cancellation is displayed in (C.24). Second, near an axis root the normal height has an action exponential. The proof bounds derivatives on the product of the two separate section domains, using this exponential to pay the losses in each root distance. Only then is the estimate restricted to the actual matching graph. This controls the two root distances independently and includes the dependence on the moving entrance.

The other coefficient faces use the same physical chain rule with different positive weights, as recorded in Lemma 3.5. On the reversible face the principal curvature vanishes. There the exact first integral permits division by the coefficient distance before taking the limit, giving the relative estimate (D.12). The coefficient cover selects one of these estimates for each whole coefficient fibre.

4.3 The complete physical return in the compact cubic block

Fix the passive coefficient set and the margins in the raw sections. Let $z_* > 0$ be the unscaled height cap supplied by Corollary 5.1; its high-height estimate is used below after the cubic low-height analysis.

Proposition 4.1 (Compact cubic coefficients)**.** *Fix $U_0, G_0 < \infty$, the original passive compact set and buffered sections. For $0 \le U \le U_0$, $|G| \le G_0$, $p = tU, \gamma = t^2 G, s = t^3$, put $\tau = \tau_{\mathrm{cub}} = t^2$.*

Choose a positive unscaled raw cap $0 < h_c < z_*$ before the final coefficient and multiplier bounds. On the entire original interval

$$I_\theta = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < H(x) = z_\theta(\gamma; a_\theta + x) < h_c\}, \qquad F(x) = x - m\Pi_{\theta,m}(x),$$

the connected lower curvature receiver satisfies $\Pi'' = 0 \Rightarrow \Pi''' < 0$. Its bounded and growing descriptions have actual resection overlap. The remaining admitted ghost raw tail has $0 < \Pi' \le C(d_0)\tau^3$; its upper core and signed high complement have bounded complete derivative and strictly overlap. After the common multiplier shrink the same F' has at most two zeros, and F has at most three isolated zeros, counted with multiplicity, on the whole coefficient fibre. All original window, inverse-image, speed, stop and height clips and intervening nonroot inputs are retained. Real-root and fold endpoints are one-sided; the stopped fold continuation is excluded.

Proof. We construct normal sections on which the action controls curvature, then continue their estimates to the original raw sections. Near the roots, the two section variables must be estimated independently before they are matched. The ghost tail and upper continuation require only a bound on the complete first derivative. These estimates will cover one connected physical interval to which the counting lemma applies.

In raw coordinates $v - \gamma = \tau\xi, z = \tau^2\eta$, the exact field is

$$\dot\xi = -D, \quad \dot\eta = -\eta H, \qquad D = \tau P + Q\eta, \quad H = \xi B + C\tau^2(G+\xi)\eta, \tag{4.10}$$

$$P = 1 - U\xi + 3G\xi^2 + \xi^3, \quad B = 1 + \tau(2G+\xi), \quad Q = 1 - A\tau(G+\xi) + C\tau^2(G+\xi)^2. \tag{4.11}$$

The unique simple negative root is $-c$, by Descartes' rule, and

$$P(\xi) = (\xi + c)[(\xi - \alpha)^2 - \Delta], \quad c - 2\alpha = 3G, \quad c(\alpha^2 - \Delta) = 1, \quad 2\alpha c = U + c^{-1}. \tag{4.12}$$

The four positive quantities $c, c^{-1}, \alpha, \alpha^{-1}$ are units. A triple root would force $P = (\xi+1)^3, U = -3$, outside this block.

Choose ℓ_0 so the original positive input window exceeds the first-lip height at its raw port with a fixed margin. The actual lips $\eta = \tau\ell_0 P$, with abscissae $u, -r$, obey

$$\frac{d}{dt}\log\frac{\eta}{P} = -\xi B + \tau P'(1 + Q\ell_0) - C\tau^3(G+\xi)\ell_0 P. \tag{4.13}$$

Beyond a fixed positive threshold the signs are negative on the positive lip and positive on the negative lip; their first hits are unique. At $a = \tau u$ the raw height is

$$z = \ell_0[a^3 + 3G\tau a^2 - U\tau^2 a + \tau^3]. \tag{4.14}$$

Choose $0 < d_{\mathrm{in}} < d_0 < \delta$, then $\ell_0\delta \ll 1$, and finally $\tau \ll d_{\mathrm{in}}$. Comparison with the organizing arc and Lemma 2.5 place captured low-height cycles at

$$0 < a = \tau u < d_{\mathrm{in}}. \tag{4.15}$$

The analytic carrier still includes all its original through inputs. To continue them to the raw sections, set $v - \gamma = \sigma q$, $\sigma = \pm 1$, and $\mathcal{W} = z/q^2$. The graph equation becomes

$$q\mathcal{W}_q = \mathcal{W}\frac{B_\sigma(q) + \sigma C v_\sigma(q) q\mathcal{W}}{\ell(v_\sigma(q))\mathcal{W} + M(\sigma q)/q^2} - 2\mathcal{W}. \tag{4.16}$$

Here

$$v_\sigma(q) = \gamma + \sigma q, \qquad B_\sigma(q) = 1 + 2\gamma + \sigma q, \qquad M(x) = x^3 + 3\gamma x^2 - spx + s^2,$$

and ℓ is defined in (4.1). At fixed relative distance from a lip this is a small perturbation of $q\mathcal{W}_q = 1 - 2\mathcal{W}$; constants around $1/2$ give a strict invariant strip to both original raw sections. The negative continuation crosses its axis root at positive height. The original windows and scalar order supply one through component.

Normal sections and independent root distances. In the bounded normal-section calculation below, p denotes the positive axis footpoint; the fixed coefficient $p = tU$ is represented by U in (4.11). Choose $v > 0$ with $P \geq 3/4$ on $[-4v, 4v]$. The limiting action exit for $p = v$ lies in $[v/2, 2v]$, because both densities are $s(1+O(v))$. Opposite strict action brackets and Corollary D.2 give an actual comparison orbit, independently of the return equation, with both footpoints in fixed subintervals of $(0, 3v)$. Fix normal boxes and independent weak cuts p_w, r_w below these footpoints. Scalar order selects the lower receiver, whose footpoints have fixed positive lower bounds. The strong projections and normal linearization give

$$\dot{p} = -\tau P(p), \quad \dot{\zeta} = -pB(p)\zeta, \quad \zeta_{\text{section}} = u_+(p),$$
$$J_+(p) = \int_{p_w}^{p} \frac{\xi B}{P}\, d\xi, \qquad J_-(r) = \int_{r_w}^{r} \frac{sB(-s)}{P(-s)}\, ds.$$

The projections have integrable kernels $C(1+T)^N e^{-(c_*-C\tau)T}$. To retain the higher rows, write the attracting field as $V = (h - qy, -ny - ky^2)$, where $h = O_{C^6}(\tau)$, $n \geq c_* > 0$, and q, n, k have bounded finite derivatives. Apply the variational induction in the proof of Lemma B.1 to this attracting field. The normal forcing always contains a height or a lower normal variation, while pure tangential forcing has an $O(\tau)$ coefficient. Integrating the normal row first gives the same integrable Volterra kernel. At phase order five it yields tangential rows $C(1+T)^N e^{C\tau T}$ and normal rows $C(1+T)^N e^{-(c_*-C\tau)T}$, using axis-flow order six. A selected coefficient derivative requires mixed order seven of the explicit coefficients and adds only polynomial time factors to this majorant. Taylor division thus gives the required C^4 units $u_\pm, u_\pm^{-1}$. The normal gap persists at the fold; these coordinates are defined on the actual normal sections.

The action exponential must absorb derivative losses as either endpoint approaches a root. We estimate those losses in five overlapping charts. For $\Delta = s_0^2$, $a_0 = \alpha - s_0$, $X = a_0 - p$, use $(X, s_0/X)$ and $(s_0, X/s_0)$; the singular action integral is

$$\frac{1}{2s_0} \log \frac{\ell(X + 2s_0)}{X(\ell + 2s_0)}, \qquad s_0 = 0: \quad X^{-1} - \ell^{-1}.$$

For $\Delta = -\kappa^2$, use the left, central and right charts $(X, \kappa/X)$, $p = \alpha + \kappa z, |z| \leq Z_0$, and $(\widetilde{X}, \kappa/\widetilde{X})$. Their arctangent action is $\kappa^{-1}[\arctan((p-\alpha)/\kappa) + \arctan(\ell/\kappa)]$. Restoring the smooth positive density gives lower bounds $c/X - C$, $cs_0^{-1}(1 + |\log(X/s_0)|) - C$, or $c_{Z_0}/\kappa - C$. Every required ordinary phase derivative loses only fixed powers of these independent variables. For example the real inner loss is bounded by $s_0^{-N} z^{-N} e^{-c(1+|\log z|)/(2\tau s_0)}$, uniformly after $c/(2\tau_0 s_{0,\max}) > N + 1$. Thus $e^{-J_+/(2\tau)}$ pays all root-distance losses, without comparing the two gaps.

Matching through the weak core. To compare the two actions, we need a core map defined all the way to zero height. We construct that map on the whole chord below each admitted orbit, before choosing an action cutoff. In raw height the fixed weak-core equation is

$$\eta_\xi = f_\tau(\xi, \eta) = \frac{\eta(a + c_1\eta)}{T + Q\eta}, \qquad a = \xi B, \quad c_1 = C\tau^2(G + \xi), \quad T = \tau P.$$

Here $P \geq p_c > 0$, $Q \geq q_c > 0$, and the coefficients are bounded on a fixed enlarged weak slab. Direct differentiation gives

$$\partial_\eta f_\tau = \frac{aT + c_1\eta(2T + Q\eta)}{(T + Q\eta)^2}, \qquad |\partial_\eta f_\tau| \leq \frac{|a|}{T} + \frac{|c_1|}{Q} \leq \frac{C}{\tau}, \qquad \eta \geq 0.$$

The constants use neither endpoint root distance.

Retain the actual intrinsic weak cuts from the projection construction above. Write their raw parametrizations as $\iota_\pm(w) = (\psi_\pm(w), h_\pm(w))$, where $w = \zeta_\pm$, $\psi_+(0) = p_w$, $\psi_-(0) = -r_w$,

and $h_\pm(0) = 0$. The projection and normal linearization above, constructed by the backward integrals of Lemma B.1 with the displayed higher rows, give uniformly C^4 inverse charts; $h'_\pm$ and the chart Jacobians $\Delta_\pm = \det D_{(v,\eta)}(\pi_\pm, \zeta_\pm)$, in the respective radii $v = \xi, -\xi$, are positive units. Their axis values are one. Use reversed physical time in the negative radial chart, so that both attracting projected equations are $\dot\pi_\pm = -\tau P(\pm\pi_\pm)$. The sections are $\pi_+ = p_w$, $\pi_- = r_w$, while ψ_- records the raw abscissa $-v$. Their exact projected axis equations give the transverse factors

$$A_+(w) := h'_+(w) - f_\tau(\iota_+(w))\psi'_+(w) = \frac{\tau P(p_w)}{D(\iota_+(w))\Delta_+(w)},$$
$$A_-(w) := h'_-(w) - f_\tau(\iota_-(w))\psi'_-(w) = \frac{\tau P(-r_w)}{D(\iota_-(w))\Delta_-(w)}.$$

The reflection on the negative side reverses the abscissa and the attracting time orientation, leaving both factors positive. Since the weak buffers have fixed bounded height, $c\tau \le A_\pm \le C$.

Fix any already admitted through orbit of the lower receiver, with weak heights $z > 0$ and $w > 0$. Its core segment, the axis and the two weak-section arcs from zero to these heights bound a strip. At either cut the positive-time derivative of $\xi - \psi_\pm(h_\pm^{-1}(\eta))$ is $-DA_\pm/h'_\pm < 0$. The strip lies to the left of the positive arc and to the right of the negative arc, making these respectively incoming and outgoing. Scalar uniqueness prevents a trajectory from crossing the axis or this admitted orbit, and $-\dot\xi \ge \tau p_c$ gives a finite passage. Thus every point of $0 \le z_0 \le z$ reaches the negative arc inside this strip. The same actual intrinsic map M_τ is consequently defined on the entire chord, with $M_\tau(0) = 0$ and $0 \le M_\tau(z_0) \le w$. Uniqueness identifies these chord maps with each other and with the local map of the weak-core flow.

Let V be the raw initial-height variation between the two actual endpoint abscissae. Its equation $V_\xi = (\partial_\eta f_\tau)V$ on a slab of fixed length gives $e^{-K/\tau} \le V \le e^{K/\tau}$. Differentiating the actual endpoint equation yields

$$M'_\tau(z_0) = V\,\frac{A_+(z_0)}{A_-(M_\tau(z_0))} > 0.$$

Absorbing the fixed powers of τ in the exponential constant,

$$e^{-K/\tau} \le M'_\tau(z_0) \le e^{K/\tau}, \qquad \frac{M_\tau(z)}{z} = \int_0^1 M'_\tau(tz)\,dt.$$

The matching of these same intrinsic heights is $w_- = M_\tau(w_+)$, $w_\pm = u_\pm e^{-J_\pm/\tau}$. The positive C^4 units $u_\pm, u_\pm^{-1}$ therefore give

$$J_+ - J_- = \tau \log\frac{M_\tau(w_+)}{w_+} + \tau\log\frac{u_+}{u_-}, \qquad |J_+ - J_-| \le K + C\tau \le K_0.$$

Thus the actions differ by a bounded amount on every admitted through orbit in the original independent section domains. The bound is obtained before choosing an action cutoff and uses no return equation.

On the fixed weak core $D \ge \tau p_c + q_c\eta$, the first four normalized variational equations have bounds $e^{K_j/\tau}$. Factor the map just constructed at zero height as $M_\tau(z) = m_\tau z Q_\tau(z)$, $Q_\tau(0) = 1, m_\tau^{\pm1} \le e^{K/\tau}$. In particular, the exact quotient is

$$Q_\tau(z) = m_\tau^{-1}\int_0^1 M'_\tau(vz)\,dv.$$

Four core variational equations bound its three ordinary derivatives. Enlarge the exponential constant to include this quotient, m_τ^{-1}, and the actual input and output chord reserves; then

choose J_* larger than twice that constant with a fixed buffer. Thus $w_+ = u_+(p)e^{-J_+/\tau}$ lies on the actual core chord and its image lies in the negative weak image. Since $u_-(r)e^{-J_-/\tau} \downarrow 0$ as $r \uparrow c$, it has a unique actual inverse. Taking logarithms on the product of separate section domains gives

$$\begin{aligned} J_+ - J_- &= C_\theta + \tau R(p,r), \qquad C_\theta = \tau \log m_\tau, \\ R &= \log u_+ - \log u_- + \log Q_\tau(w_+), \\ |C_\theta| &\le K_0, \qquad \|R\|_{C^3_{p,r}} \le C(1 + |\log\tau|). \end{aligned} \tag{4.17}$$

Split the positive action exponential to absorb the core bounds and the losses in the five root charts. Differentiation in the negative section variable acts directly on u_-; this avoids differentiating an exponentially large inverse. Since the heights are intrinsic normal coordinates, there is no moving initial $\log P$ term. On the actual matched graph, with $P_+ = P(p), P_- = P(-r)$,

$$r_p = \frac{pB_+P_-}{rB_-P_+}(1+\rho), \qquad 1+\rho = \frac{1-\tau P_+R_p/(pB_+)}{1+\tau P_-R_r/(rB_-)}, \qquad \|\rho\|_{C^2_{p,r}} \le \epsilon_\tau := C\tau(1+|\log\tau|). \tag{4.18}$$

The polynomial multipliers and their derivatives are bounded; all remaining denominators are units, including when P_+, P_- vanish independently.

Curvature in the physical input. Let $\Gamma_\pm(b) = a_\pm + c_\pm\phi_\pm(b)$, $c_\pm > 0$, be these actual outward embeddings into the incoming raw height h and divided output, with $b = -p^2/2, S(b) = -r^2/2$. The fixed normal-height equation $d(\eta - \xi^2/2)/d\log\eta = \tau\Phi$ has four bounded height rows and clock $O(1+|\log\tau|)$, independently of the axis gap. For the exterior continuation retain $\lambda_G = 12\tau G/(1+2\tau G)$. At each frozen bounded base $q > 0$,

$$\int_1^{\delta/(\tau q)} \rho^{-(2-\lambda_G)}[\tau q^{-2}\rho^{-2} + \tau U q^{-1}\rho^{-1} + \tau q\rho](1+|\log\rho|^2)\frac{d\rho}{\rho} \le C\tau(1+q). \tag{4.19}$$

The actual relative hierarchy yields $\phi_\pm = \mathrm{id} + O_{C^3}(\epsilon_\tau)$, including inverse maps on their actual images. At an ordinary observer with exact small gain $a_\tau \asymp \tau^{2-O(\tau)}$, the normalized second and third rows are $a_\tau\chi''/\chi'_*, a_\tau^2\chi'''/\chi'_*$. Only its ordinary C^3 bounds are used. Outgoing and incoming cocycles at every cut cancel before normalization; the output closing retains $f'(md)d'$ and its ordinary third chain rule. Thus $\widehat{\Pi} = \phi_- \circ S \circ \phi_+^{-1}$ is the actual completed map up to its two retained positive affine gains. The exact chain rule gives

$$\phi'_+(b)\frac{P(p)}{p+r}\frac{\widehat{\Pi}''}{\widehat{\Pi}'} = 3 - \frac{U}{pr} + E(p,r), \qquad \|E\|_{C^1_{p,r}} \le \epsilon_\tau. \tag{4.20}$$

To display the error, put $W = p + r$, $K_{\mathrm{mat}} = B_+(1+\rho)/B_-$, and $L_\pm = \phi''_\pm/\phi'_\pm$, each in its own quadratic label. Then

$$\begin{aligned} E = &-\frac{(K_{\mathrm{mat}}-1)P'_-}{rW} - \frac{P_+}{pW}\partial_p \log K_{\mathrm{mat}} - \frac{K_{\mathrm{mat}}P_-}{rW}\partial_r \log K_{\mathrm{mat}} \\ &+ \frac{K_{\mathrm{mat}}P_-}{W}L_- - \frac{P_+}{W}L_+. \end{aligned}$$

Here P'_- uses the radial variable r. Since $K_{\mathrm{mat}} - 1 = O_{C^2}(\epsilon_\tau)$ and $L_\pm = O_{C^1}(\epsilon_\tau)$, every term has the asserted C^1 bound on the product of the two normal-section domains: $P_\pm$ occur as multipliers and the remaining denominators are units. Near a zero, $U \ge U_* > 0$, and restriction to the matched graph gives

$$\begin{aligned} \frac{d}{dp}\left(3 - \frac{U}{pr}\right) &= \frac{U}{pr}\left(\frac{1}{p} + \frac{r_p}{r}\right) \ge c(1+r_p), \\ \left|\frac{d}{dp}E(p,r(p))\right| &\le \epsilon_\tau(1+r_p). \end{aligned}$$

Since $w_p = (\phi_+(-p^2/2))_p < 0$, every curvature zero is simple and downward in the physical input.

On $J_+ \le J_{\rm rf} > J_* + 1$, both actions are bounded. The five singular integrals give a fixed positive polynomial margin on the whole traversed segment and its buffered exits. Corollary D.2, with cubic coefficient one, gives actual absolute C^3 comparison there. The limiting map has $S_0' = P_-/P_+$, $S_0''/S_0' = (p+r)(3 - U/(pr))/P_+$, and $(pr)_b = -r/p - pP_-/(rP_+) < 0$. At $U = 3pr$ its S_0''' has a uniform negative margin; away from that set the curvature has a nonzero margin. This transfers the same crossing orientation to the bounded-action region. That comparison strictly overlaps the high-action estimate for the same map.

On ghost fibres choose M above all roots and bottlenecks, large for Lemma G.4. Its actual growing receiver $u \ge 2M, \tau u \le d_0$ has matched $r_0 \le r < c$ and (G.25), with error $C\tau(1+u)(1+|\log(\tau(1+u))|)^4$. Choose $M > 2U_0/r_0$, then d_0 with $Cd_0(1+|\log d_0|)^4 < 1/4$, then τ. Its curvature is positive throughout this growing range. To establish actual overlap, take a buffered band $[2M - \beta, 3M + \beta]$, $0 < \beta < M/2$, in the fixed normal boxes. The lip projection is $k_\tau(u) = u + O_{C^3}(\tau)$, its linearized height is $v_{\rm lip} = \tau\ell_0 P(u) V_\tau(u)$, with V_τ a C^3 unit. The same-orbit invariant gives

$$\int_{k_\tau(u)}^{p_\tau(u)} \frac{qB(q)}{P(q)}\,dq = \tau \log \frac{u_+(p_\tau(u))}{v_{\rm lip}(u)}, \qquad \|p_\tau - \mathrm{id}\|_{C^3} \le C\tau(1+|\log\tau|), \quad p_\tau' > 0. \tag{4.21}$$

The divergent bottleneck prefix cancels in this local integral. Its p-derivative after subtracting $\tau u_+'/u_+$ is a unit, so the implicit equation gives all three rows. The closed band $[2M+\beta, 3M-\beta]$ has $p_\tau < 3M, u > 2M$; the inverse image of $p = 3M$ also has $u > 2M$, leaving no gap. These actual orbits retain the original windows: their bounded normal labels have raw gains $\tau^{2-\lambda_G}$ times units, their reference images lie inside the original passive organizing-arc buffers, and the raw corridor continues them across the negative root at positive height. Scalar order keeps the band in the comparison component. Resection identifies orbits; each curvature estimate uses its own actual adapters.

The remaining raw ghost tail. On a ghost fibre, through trajectories can continue beyond the cutoff $\tau u = d_0$ of the growing estimate. The first-lip abscissa differs from the minimum speed σ_θ; small central height may coexist with $a = \tau u > d_0$. Put $a_b = d_0/2$, and let $x_* = \delta - \gamma$ be the actual positive raw port. For every admitted $a_b \le a \le x_*$, at fixed coefficients set

$$M_\tau = x^3 + 3\tau G x^2 - \tau^2 U x + \tau^3, \qquad B_0 = 1 + 2\tau G + x,$$
$$Q_0 = 1 - A(\tau G + x) + C(\tau G + x)^2, \quad \zeta_0 = C(\tau G + x), \quad D_0 = M_\tau + Q_0 z.$$

The true graph is $z_x = z(xB_0 + \zeta_0 z)/D_0$, $z(a) = \ell_0 M_\tau(a)$. On $[a, x_*]$, $M_\tau \ge c(d_0) > 0$ and Q_0 is a positive unit. The graph satisfies $z_x \le Cx + Cz$, so its positive initial height and Gronwall give $z \le C\delta^2$, below the fixed upper chart boundary. Its numerator is positive after the coefficient shrink, and hence $z \ge \ell_0 M_\tau(a) > 0$. This constructs the regular outward passage, including its auxiliary zero-flight endpoint. Append the observer only when the image lies in its original buffer; the auxiliary endpoint needs no observer value when its image is outside that buffer. Use (A.6), with $\sigma = \tau$, $K_{\rm lip} = K$, $I_0 = I$, and $x_T = x_*$:

$$\Gamma_{+,a}' = -a\chi'(z_{\rm port}) \frac{z_{\rm port}}{D_0(x_*, z_{\rm port})} B_0(a) K_{\rm lip}(a) e^{I_0(a)}. \tag{4.22}$$

The original lip transversality makes $K_{\rm lip}$ a positive unit. All admitted factors have bounds and reciprocals depending on d_0, so $c(d_0) \le -\Gamma_{+,a}' \le C(d_0)$, or $c(d_0)\tau^2 \le \Gamma_{+,b_+}' \le C(d_0)\tau^2$ for $b_+ = -a^2/(2\tau^2)$.

We match this outward map to the negative branch by their actual central heights. Differentiate the positive inward passage directly. For its height $\eta_0^+(u)$, $W = \eta/(\tau P)$, $H_+ = P'/(\xi B)$, $\varpi = C\tau^3(G+\xi)P/(\xi B)$,

$$\partial_u \log \eta_0^+ = -\frac{uB_+}{\tau P_+}\Lambda_+, \quad \Lambda_+ = \frac{K_{\mathrm{lip}}(a)}{1+Q_+\ell_0}\exp\int_u^0 \frac{\xi B}{\tau P}\frac{W(\varpi - Q)}{(1+QW)^2}\,d\xi. \tag{4.23}$$

In the action clock $dq = -\xi B\,d\xi/(\tau P)$,

$$W_q = W\left[-\frac{1+\varpi W}{1+QW} + \tau H_+\right], \qquad W(0) = \ell_0.$$

On the original raw chart $Q - \varpi > 0$; the braces increase with W, and at ℓ_0 equal $-K_{\mathrm{lip}}/(1+Q\ell_0) \le -c_1$. Thus $W \le \ell_0 e^{-c_1 q}$ down to a fixed small scaled cut δ_c, on the entire original raw range. The integral in (4.23) becomes the bounded integral $\int W(Q-\varpi)/(1+QW)^2\,dq$. For the negative branch put $P_-(v) = P(-v) = (c-v)L_-(v)$, $B_-(v) = B(-v)$, $Q_-(v) = Q(-v)$, with the exact unit L_- used in Lemma G.4. Its inverse inward graph is

$$\eta_v = f_-(v,\eta) = \frac{\eta\{vB_- + \zeta_-\eta\}}{\tau P_- + Q_-\eta}, \qquad \zeta_- = -C\tau^2(G-v), \qquad \eta(r) = \tau\ell_0 P_-(r).$$

On $\delta_c \le v \le r < c$, set $H_- = P_-'/(vB_-)$ and $\varpi_- = \tau P_-\zeta_-/(vB_-)$. The compact coefficient block gives bounded H_-, $\varpi_- = O(\tau^3)$, and positive-unit Q_-, even at the simple root. The reflected ratio W_- has the same action-clock equation and strict decay. Its complete moving-initial factor is

$$K_- = 1 - \tau H_-(1+\ell_0 Q_-) + \ell_0\varpi_- > c_K > 0.$$

Choose $2\delta_c < r_0$. A fixed positive band below every positive raw-tail label and the fixed negative band $[\delta_c, r_0]$ supply action margins $A_+, A_- > 0$. Thus both inward graphs reach their central slab with heights $\eta_i \le C\tau e^{-c_1 A_{\mathrm{in}}/\tau}$, $A_{\mathrm{in}} = \min(A_+, A_-)$. On that slab $P_\pm \ge p_c > 0$. In the decreasing radial coordinate each graph satisfies $\eta_t \le C\tau\eta^2$; first exit keeps $0 < \eta \le 2\eta_i$ and gives an actual finite continuation to $\xi = 0$. For f_+ use the positive graph, with $\zeta_+ = C\tau^2(G+v)$. The exact rational difference is

$$f_{\pm,\eta} - f_\pm/\eta = \frac{\eta\{\zeta_\pm\tau P_\pm - n_\pm Q_\pm\}}{(\tau P_\pm + Q_\pm\eta)^2}, \qquad n_\pm = vB_\pm.$$

Its central integral is bounded by $C\eta_i/\tau^2$. Together with the bounded noncentral ratio integral this proves the positive Λ_+ bound and, independently, the reflected identity

$$\begin{aligned} -\partial_r \log \eta_0^-(r) &= \frac{rB_-}{\tau P_-}\Lambda_-, \\ \Lambda_- = \frac{K_-(r)}{1+\ell_0 Q_-(r)}\exp\int_r^0 (f_{-,\eta} - f_-/\eta)\,dv, &\qquad 0 < c_\Lambda \le \Lambda_\pm, \Lambda_\pm^{-1} \le C_\Lambda. \end{aligned} \tag{4.24}$$

No root factor has been inverted in these error bounds.

The ratio memory and the central linear cocycle give $\eta_0^-(r) = \tau\ell_0 P_-(r)e^{-A_-(r)/\tau}E_-(r)$, where $E_-, E_-^{-1} \le C$ and $A_-(r) = \int_0^r vB_-/P_-\,dv$. The simple root makes $A_-(r) \to \infty$ as $r \uparrow c$. Hence the actual negative centre map is strictly decreasing with range

$$\eta_0^-([r_0, c)) = (0, \eta_0^-(r_0)]. \tag{4.25}$$

Start with an admitted shot u_{ref} in the fixed resection band, whose negative match has $r_{\mathrm{ref}} \ge r_0$. For small τ, $\tau u_{\mathrm{ref}} < a_b$. The weighted inward construction, before either port observer is attached, is strictly decreasing up to $u_b = a_b/\tau$, since $\tau u \le a_b < d_0$. Thus $\eta_0^+(u_b) < \eta_0^-(r_{\mathrm{ref}})$,

and the full negative image gives a unique $r_b \geq r_{\mathrm{ref}}$. This auxiliary centre match does not require the formal a_b port image to lie in an observer window. Positive centre-height monotonicity and this full negative range then give every original admitted $a \geq a_b$ its unique negative match. Scalar uniqueness at the same $\xi = 0$ identifies the joined graph with the original through orbit; its negative output and closing images therefore retain their original buffers. Matching them gives $S' = B_+P_-\Lambda_+/(B_-P_+\Lambda_-)$. In the root collar the negative output embedding keeps its own endpoint and exterior gains:

$$\Gamma'_{-,b_-} = d_c\theta_c X^{\beta_-}K_c(X, X^{\beta_-}), \quad X = c - r,$$
$$d_c = \tau^{2-\lambda_G}U_c, \quad \theta_c = \tau^{\beta_-}, \quad \beta_- = \tau P'(-c)/(cB(-c)) > 0.$$

Here U_c, K_c are bounded positive units. On the fixed complement the ordinary negative exterior calculation gives $\Gamma'_{-,b_-} = \tau^{2-\lambda_G}$ times a bounded positive unit. These estimates overlap on a fixed collar band and give $0 < \Gamma'_{-,b_-} \leq C\tau^2$ on the entire $r_0 \leq r < c$, since $\tau^{-\lambda_G}$ is a unit. The closing contributes its actual bounded positive first derivative. Now $P_+(u) = \tau^{-3}M_\tau(a) \geq c(d_0)\tau^{-3}$, while $P_- \leq C$. Therefore

$$0 < \Pi' = \frac{\Gamma'_{-,b_-}}{\Gamma'_{+,b_+}}\frac{B_+P_-}{B_-P_+}\frac{\Lambda_+}{\Lambda_-} \leq C(d_0)\tau^3. \tag{4.26}$$

The two absolute gains are retained in this ratio. The estimate covers $a_b \leq a \leq d_0$ and every original admitted $a > d_0$, subject to all the original window, speed and height restrictions. It therefore continues the derivative bound beyond the growing estimate's cutoff.

The upper continuation. For the upper complement continue the fixed comparison outward to $\xi = \pm 4v$. The fixed-normal fast invariants give positive-unit heights there; scalar order preserves these lower bounds for all larger inputs, and the field bounds them above through a fixed $\eta_{\max} > 0$. With $f = \eta H/D$, specialization of (A.5) gives

$$f_\eta = \frac{d}{d\xi}\log\frac{\eta}{D} + \frac{\tau P' + (Q' + C\tau^2(G + \xi))\eta}{D}. \tag{4.27}$$

Endpoint quotients are units, and $D \geq \tau p_c + q_c\eta$ bounds the integral even at exponentially small central height. The two outward first variations have the same exact factor $\tau^{2-\lambda_G}$; their ratio and closing give $0 < \Pi' \leq C_{\mathrm{core}}$. The comparison orbit and a fixed action neighborhood lie in both the lower and core regions. The comparison height η_{comp} is exponentially small. The signed threshold is $H_H = \tau^3K_H(1 + \tau U^2)$, $\eta_H = H_H/\tau^2$. After all fixed cuts and caps, choose τ_0 so $\eta_{\mathrm{comp}} < \eta_H < \eta_{\max}/2$ and $\tau^2\eta_{\max} < h_c/2$. Core and signed high then strictly overlap on the original height axis.

Covering and counting the whole interval. The through lemma and increasing H, σ_θ make I_θ connected. The comparison is admitted: its raw outward images retain the original buffers, its central speed is $O(\tau^3 + \tau^2\eta_{\mathrm{comp}}) < d_{\mathrm{out}}$, and $H < h_c$. At its transverse positive lip every lower input has $W < \ell_0$. On a ghost fibre the fixed lower height of the original positive window and the original lip reserve give $W > \ell_0$ at the raw port. Every input in I_θ already has an actual through trajectory; scalar order below the comparison gives $W < \ell_0$ at its positive lip. Thus a hit exists. Every hit has negative transverse derivative, so it is unique. At the hit the radial derivative of W is positive, whereas its input-height derivative is positive by scalar variation. The implicit hit equation therefore gives $a_x < 0$ in the original affine input, before any return-zero argument. On real/fold fibres a through orbit crosses the incident inner root at positive height, with $W \to \infty$ on its inner side, giving the same unique hit. An outer-component lip hit would force $\eta \leq \tau\ell_0 P \to 0$ at the next root and a stopped passage.

Thus all lower through inputs have one of the actual descriptions above. The bounded and growing ones have no gap by (4.21); the growing one reaches $a = d_0$, and (4.26) continues every admitted tail. The whole original carrier has the exact allocation

$$\begin{aligned} I_{\mathrm{tail}} &= \{x \in I_\theta : x \le x_{\mathrm{comp}},\ \Delta < 0,\ a(x) > d_0\},\\ I_{\mathrm{lower}} &= \{x \in I_\theta : x \le x_{\mathrm{comp}}\} \setminus I_{\mathrm{tail}},\\ I_{\mathrm{core}} &= \{x \in I_\theta : x > x_{\mathrm{comp}},\ H(x) \le H_H\},\\ I_{\mathrm{high}} &= \{x \in I_\theta : H(x) > H_H\}, \qquad I_\theta = I_{\mathrm{tail}} \cup I_{\mathrm{lower}} \cup I_{\mathrm{core}} \cup I_{\mathrm{high}}. \end{aligned} \tag{4.28}$$

They are intervals or empty clips, since $a(x)$ decreases. Equality $a = d_0$ and the comparison belong to the lower piece; $H = H_H$ belongs to the core. Closed action dividers are retained, and all estimate packages have the strict overlaps above. Natural stopped endpoints and the infinite-time fold branch are excluded. Choose m last with $m \max(C_{\mathrm{core}}, C_H, C(d_0)\tau_0^3) < 1/2$. Then $F' > 1/2$ on tail, core and high. On the lower receiver the exact two-gain composition gives

$$F = a_+ - a_\theta - ma_- + c_+ w - mc_- \widehat{\Pi}(w), \qquad F'' = -\frac{mc_-}{c_+^2}\widehat{\Pi}'', \quad F''' = -\frac{mc_-}{c_+^3}\widehat{\Pi}'''.$$

With $J = I_{\mathrm{lower}}$, Lemma 3.2 applies on the connected I_θ: the positive factors preserve the lower downward curvature crossings, and $F' > 1/2$ on $I_\theta \setminus J$. It gives at most one curvature zero on J, at most two zeros of F', counted with multiplicity, and at most three isolated zeros of the same physical F on I_θ, counted with multiplicity. All choices depend only on the fixed coefficient block and section margins. The estimates allow the two gains and the two root gaps to vary independently. □

5 Uniform regional returns and the three-cycle upper bound

The regional estimates apply to the completed physical map. We fix the coefficient subcover and its reserves, choose a common raw-height cap, and only then restrict the independent coefficients and finite multiplier. Each coefficient alternative counts the whole returning interval; geometric capture completes the upper bound.

5.1 From the action to the physical return

The passage estimates below concern the completed map in the physical input of Section 3. The ordinary section calculus of Lemma 3.3 and the curvature identity of Lemma 3.5 retain both nonlinear section maps and their independent gains. We use four comparisons.

On strict passages, Proposition C.3 gives ordinary phase C^3 convergence. The common absolute tail factor cancels in (C.24); after the regular closing this bounds the upper continuation. Near reversibility, Theorem D.1 instead divides the error by the coefficient distance before comparison. Its root-free margin and entry/exit buffers are part of its hypotheses.

For coalescing roots, Proposition 4.1 estimates the two root distances independently before matching their actual heights. Its bounded and growing descriptions overlap on the same map; the remaining original ghost tail is controlled by (4.26). The quadratic counterpart is Proposition G.6. In that case the signed density during the negative-root dwell is essential. Proposition 5.15 identifies both tail estimates with the common physical receiver.

At degree loss, the two saddle scales remain independent. The actual section comparison in Proposition H.13, the growing-range matching and the fixed-exit construction of Proposition E.5 cover one returning interval. Proposition H.14 counts its critical points. The exact dictionaries of Appendix K.1 place whole coefficient fibres in these ranges. The coefficient cover and raw budgets below fix their overlap reserves before the common height cap is chosen. The

following bounds apply to the same completed displacement on its actual connected through interval. The appendices prove the transition and section estimates used in each region, with the one-sided endpoint estimates and continuations needed to count the full physical carrier.

5.2 The signed high-height interval

Corollary 5.1 (Signed high-height interval)**.** *There exist $K, z_*, \gamma_*, C_H > 0$ such that, for all sufficiently small raw coefficients and either sign of α_0, the complete transition on the actual fixed sections satisfies $0 < \Pi' \le C_H$ throughout the interval of through trajectories*

$$K(|\alpha_0| + \alpha_1^2) < H := z(\gamma) < z_*, \qquad |\gamma| < \gamma_*. \tag{5.1}$$

After the common choice $mC_H < 1/2$, the complete displacement has $\Delta' > 1/2$ and hence at most one isolated zero, counted with multiplicity, on this entire interval and on each of its truncations. The assertion includes $\alpha_0 = \alpha_1 = 0$. At a positive boundary height $H = K(|\alpha_0| + \alpha_1^2) < z_*$, with the same coefficient bounds, the derivative bound still holds; this equality face is assigned to the low-height region.

Proof. Fix the coefficients and a through trajectory in (5.1). Freeze $r_0 = \sqrt{H}$ and put

$$a = \frac{\alpha_0}{H}, \qquad p = \frac{\alpha_1}{\sqrt{H}}, \qquad s = r_0, \qquad v = \gamma + r_0 X, \quad z = r_0^2 Y.$$

The reference orbit has $Y(0) = 1$, and

$$|a| < K^{-1}, \qquad |p| < K^{-1/2}.$$

Choose K so large that the closed block with the corresponding nonstrict bounds lies inside the signed compact strict family of Proposition C.4. Then choose z_* and γ_* small enough for its raw-tail estimates. Its complete derivative bound applies with the frozen scale r_0, including both signs of a and $a = p = 0$. The phase-independent constant bounds the derivative of the same physical map, with r_0 frozen during differentiation.

Scalar first variation makes H increasing with that input, so (5.1) and its truncations are intervals. Choose $mC_H < 1/2$ last; then $\Delta' > 1/2$ throughout. The closed block includes the positive equality face, assigned to low height. For double-zero coefficients every positive height is high. □

5.3 The compact positive-ratio whole-carrier bound

Proposition 5.2. *For each fixed $0 < p_0 < P < \infty$, the complete captured carrier with $a = -1$ and $p_0 \le p \le P$ has at most two isolated displacement roots, counted with multiplicity.*

Proof. Use the one connected actual receiver and its open preimages from Lemma F.1. Keep the physical affine input x throughout. On $I_E \cup I_M$, $\Pi_{xx} < 0$, so for the physical multiplier $m > 0$, $\Delta'' = -m\Pi'' > 0$. On I_H, the complete derivative bound of Lemma F.11 gives $\Delta' > 1/2$ after the single final choice $m < \min\{m_{\mathrm{ad}}, 1/(2C_H)\}$. Every critical point has $\Delta'' > 0$. Lemma 3.1 gives the whole-carrier multiplicity bound. □

5.4 The gamma-dominant and far-root carriers

Proposition 5.3. *For a fixed $\Pi_{\max} < \infty$, the bounded-π negative-gamma carrier has at most two displacement zeros, and the bounded-π positive-gamma carrier has at most three. The positive far-root carrier of* (G.8) *has at most two. All counts hold on the entire captured physical interval.*

Proof. Propositions G.1 and G.2 give the two-zero counts on their original connected carriers, with multiplicity, using the covers (G.7) and (G.17). For positive bounded π, Corollary G.7 gives at most two zeros of the same physical F', including the ghost raw tail. Its paired/high overlap, in $\eta = gY_0$, is

$$gK_H(1 + \Pi_{\max}^2 g) < \eta_{\max}/2, \qquad (gk)^2\eta_{\max} < H_{\mathrm{cap}}/2. \tag{5.2}$$

The remaining root-free fibres are contractions by Lemma H.15. Lemma 3.1 then gives the three-zero count on the whole original carrier. □

5.5 The compact cubic block: a common curvature orientation

Proposition 5.4 (The complete compact cubic carrier)**.** *Fix finite U_0, G_0. On the coefficient block*

$$a = -1, \quad p = tU, \quad \gamma = t^2 G, \quad s = t^3, \qquad 0 \le U \le U_0, \quad |G| \le G_0, \tag{5.3}$$

the same displacement has at most two critical points and at most three isolated zeros, counted with multiplicity, on its whole physical carrier under the common positive unscaled raw cap. All captured target cycles lie in this carrier after the common coefficient and marked Hausdorff-collar shrink.

Proof. Put $\tau = t^2$. Proposition 4.1 supplies the actual clipped carrier and its allocation (4.28). Its bounded-action and independent root-collar estimates give the same curvature orientation. The growing range has positive curvature by (G.25), and is identified with the bounded range by the actual resection

$$p_\tau(u) = u + O_{C^3}(\tau(1 + |\log\tau|)), \quad p'_\tau > 0, \qquad 2M + \beta \le u \le 3M - \beta. \tag{5.4}$$

These are estimates for the completed physical map, so

$$\Pi''(x) = 0 \quad \Longrightarrow \quad \Pi'''(x) < 0 \tag{5.5}$$

on the connected lower receiver. Real and fold fibres retain their one-sided through endpoints.

The original ghost tail has the derivative bound (4.26); the core has (4.27). The construction places their comparison and high intervals in strict overlap:

$$\eta_{\mathrm{comp}} < \tau K_H(1 + \tau U^2) < \eta_{\max}/2, \qquad \tau^2\eta_{\max} < h_c/2. \tag{5.6}$$

One final multiplier bound makes $F' > 1/2$ on this complement. The cover in Proposition 4.1 includes every input, retaining the original clips and equality dividers. Its whole-carrier count, equivalently Lemma 3.2, gives the stated multiplicity bounds. Capture is applied after these analytic choices, as in Lemma 2.5. □

5.6 Degree loss: a common orientation of critical points

The compact graph (H.20) and reciprocal graph (H.47) share the physical carrier of Proposition H.10. The reciprocal face $j = 0$ concerns only the one-sided closure of the estimates.

Proposition 5.5 (The complete degree-loss carrier)**.** *Fix the compact positive-unit ranges in Proposition H.4, or the ranges of d, A_d and $0 < j \le j_0$ in Proposition H.5. There are independent active coefficient bounds, one fixed raw-height cap and a final finite-closing multiplier bound for which the complete displacement has at most one critical point on its connected interval of actual through inputs in the original nested windows. Every such point is a simple upward crossing of the same physical F'. The displacement therefore has at most two isolated zeros, counted with multiplicity, and no identity interval. All estimates hold at every retained input, including nonclosed through orbits and positive divider equalities. They are uniform for physical $j > 0$ as $j \downarrow 0$.*

Proof. Write the return in the affine input of Lemma 2.2 as

$$F_\theta(x) = x - f_\theta\big(m\mathfrak{T}_\theta(a_\theta + x)\big) = x - m\Pi_{\theta,m}(x). \tag{5.7}$$

Here $\mathfrak{T}_\theta = d_\theta \circ T_\theta^\infty$ and $f_\theta = C_\theta - C_\theta(0)$. Proposition H.10 gives its actual clipped carrier. Proposition H.14 retains both gains, output translation and ordinary C^3 closing, and proves

$$F'_\theta(x) = 0 \implies F''_\theta(x) > 0, \qquad \#\{x : F'_\theta(x) = 0\} \le 1. \tag{5.8}$$

Its whole-carrier count gives the remaining conclusions. □

5.7 The determinant passage

Proposition 5.6. *Fix $P_D < \infty$, the passive compact coefficient block and original window reserves. For independent small $r_D > 0$, $|\gamma|$, with $\alpha_0 = r_D^2$, $|\alpha_1/r_D| \le P_D$, the displacement on* (I.3) *has at most one critical point and at most two isolated zeros, counted with total multiplicity, after a common coefficient and final multiplier shrink. The raw cap is fixed, positive and unscaled. The assertion holds for every original window, speed or height clip and includes saddle ratio one.*

Proof. Use the single affine $F(x) = x - m\Pi(x)$ of Lemma 2.2. Lemma I.2 supplies the full stopped composition, both exact absolute gains and ordinary C^3 divided closing. On the strict and high sets in (I.18), choose m last so that their complete derivative bounds give $F' > 1/2$. Thus every critical point lies in the actual stopped range.

There (I.17) retains every factor of the same physical derivative:

$$\Pi'(x_b + \sigma_+\delta) = \mathcal{A}_\theta \delta^q e^{R_\theta(\delta)}, \quad q = \wp - 1, \quad |R_\theta| + |\delta R'_\theta| \le C\delta^\alpha(1 + |\log\delta|).$$

The quotient of the two gains is a unit, so $\mathcal{A}_\theta$ is bounded above and below; neither absolute gain needs a lower bound. Let C_0 bound $|\log \mathcal{A}_\theta + R_\theta|$. At a critical point $m\Pi' = 1$, set $t = -\log\delta > 0$. Then

$$-qt \ge L := \log(1/m) - C_0.$$

For $L > 0$, this requires $q < 0$ and $|q| \ge L/t$; in particular $\wp = 1$ produces no critical point. Moreover

$$\frac{|\delta R'_\theta|}{|q|} \le \frac{C}{L} t e^{-\alpha t}(1 + t) \le C_1/L.$$

Choose $L > 2C_1$. The Euler derivative of the logarithm of the full physical gain satisfies $\delta\partial_\delta \log\Pi' < q/2 < 0$ at every critical point. Since $dx/d\delta = \sigma_+ > 0$,

$$F'(x) = 0 \quad \Longrightarrow \quad F''(x) = -m\Pi''(x) > 0. \tag{5.9}$$

The actual stopped/strict and low/high cover (I.18) retains every clipped and intervening input. Lemma 3.1 now gives at most one critical point and two displacement zeros. □

5.8 The signed trace passage

Proposition 5.7. *Fix $0 < p_0 < p_1 < \infty$, the passive compact block and original window reserves. For $\alpha_0 = r^2 a$, $\alpha_1 = rp$, $|a| \le a_*$, $p_0 \le |p| \le p_1$, independent small $r > 0$, $|\gamma|$, and a final common multiplier shrink, the displacement on the original receiver* (I.19) *has at most two isolated zeros for $p > 0$, and at most one for $p < 0$, counted with total multiplicity. Its positive unscaled raw cap and all window, speed and height clips are retained. Both opening signs, zero opening and closed trace endpoints are included.*

Proof. Proposition I.3 gives $F' > 1/2$ for $p < 0$ and $F' = 0 \Rightarrow F'' > 0$ for $p > 0$ on the whole original clipped receiver. Lemma 3.1 gives the stated multiplicity bounds. □

Corollary 5.8. *The nonnegative-opening sector $\alpha_0 \geq 0$ has at most two captured displacement zeros. There is a fixed $P_* < \infty$ such that the negative-opening sector $\alpha_0 = -s^2 < 0$, $p_{\mathrm{src}} = \alpha_1/s \geq P_*$ also has at most two. Both assertions concern the whole physical interval and count multiplicities.*

Proof. First choose the signed trace collar of Proposition 5.7, for example with $1/2 \leq |p| \leq 2$. Choose $P_D > 2/\sqrt{a_*}$. If $\alpha_0 > 0$ and $|\alpha_1| \leq P_D\sqrt{\alpha_0}$, use Proposition 5.6 with $r_D = \sqrt{\alpha_0}$. On the complement put $r = |\alpha_1|$. Then

$$p = \operatorname{sgn}\alpha_1, \qquad a = \alpha_0/\alpha_1^2 < P_D^{-2} < a_*.$$

This is the exact trace scaling of Lemma I.1; the full physical domains are retained in Lemma I.4. The same choice treats $\alpha_0 = 0$, $\alpha_1 \neq 0$, with $a = 0$. Each alternative bounds the displacement on its whole coefficient fibre.

The double-zero coefficient face is a contraction by Corollary 5.1 at every positive height, without introducing a determinant scale.

Finally suppose $\alpha_0 = -s^2 < 0$ and $p_{\mathrm{src}} = \alpha_1/s > 0$. With $r = \alpha_1$, the trace coefficients are exactly

$$p = 1, \qquad a = -p_{\mathrm{src}}^{-2}.$$

Choose $P_* > 2/\sqrt{a_*}$ before the subsequent compact positive-source interval is fixed. Then $p_{\mathrm{src}} \geq P_*$ belongs to Proposition 5.7. Take the maximum of the whole-fibre bounds across these coefficient alternatives. □

5.9 Coefficient-face decomposition and passage domains

We use the variables of Section 4.1 to cover the coefficient space independently of the phase input. The exact coordinate dictionaries place each whole fibre in a regional physical receiver, where the criteria of Section 3 apply. All compact blocks and auxiliary reserves are fixed before the common raw cap and final restrictions in Lemma 5.14.

Lemma 5.9 (Small-source coefficient decomposition)**.** *There are $p_*, s_*, \gamma_* > 0$ such that, for*

$$\alpha_0 = -s^2 < 0, \qquad 0 < s \leq s_*, \qquad |\gamma| \leq \gamma_*, \qquad |p| \leq p_*,$$

the actual passage domains have the coefficient cover in Table 1 and Appendix K.1. The gamma, cubic, degree-loss, and remote-root coordinates are related by the exact dictionaries (K.4)*,* (K.5)*, and* (K.7)*. Their coefficient ranges overlap strictly and all thresholds are fixed before the final raw-neighborhood and multiplier shrink. The face $p \leq 0$ is covered directly by Lemma H.15.*

The proof is in Appendix K.1.

Lemma 5.10 (Central height versus signed-trace gap)**.** *Fix $K_0, L_c > 0$. Suppose $\alpha_0 = -s^2 < 0$, $\alpha_1 > 0$, and put*

$$p = \frac{\alpha_1}{s}, \qquad r_L = \alpha_1, \qquad v - \gamma = r_L X, \qquad z = r_L^2 Y_L.$$

For $p \geq 1$, every through graph satisfying $Y_0 = z(\gamma)/s^2 \leq K_0(1 + p^2)$ obeys $Y_L(L_c) \leq B_c$, with B_c independent of p, s, γ. If the lower boundary of the through family has nonnegative one-sided height d_L on $X = L_c$, then

$$0 < \frac{\mathfrak{g}_L}{r_L^2} = Y_L(L_c) - d_L \leq B_c. \tag{5.10}$$

The proof is in Appendix K.2.

Lemma 5.11 (Large-source trace domains)**.** *There exists $P_* < \infty$ such that, for*

$$\alpha_0 = -s^2 < 0, \qquad p = \frac{\alpha_1}{s} \geq P_*,$$

the part below the high-height threshold belongs to a fixed signed trace block with bounded actual gap. Its terminal and compact strict intervals have the overlap (I.34)*; the remaining regular pieces have bounded complete derivative. The construction is uniform as $p \to \infty$, without a bound on γ/s.*

The proof is in Appendix K.3.

5.10 A coefficient cover and its raw-height budgets

The coefficient alternatives must be fixed before choosing a common height cap. Their section constructions then give a raw-height cap independently of the remaining coefficient radii. In all phase derivatives below, the coefficients and actual finite multiplier are held fixed.

Lemma 5.12 (Coefficient-only cover)**.** *Fix the original nested sections, the passive compact coefficient block, and the finite orders required by the regional preparations. There are fixed thresholds $P_D, p_*, P_*, k_\flat, \Pi_{\mathrm{far}}, \Pi_{\max}, \mathfrak{q}_\flat, U_d, U_{\mathrm{cub}}, G_{\max}$, and fixed overlapping coefficient rectangles, with the following properties.*

For $\alpha_0 \geq 0$, the determinant and signed-trace substitutions (I.1)–(I.2) *cover every nonzero coefficient pair. The double-zero pair is retained separately. For $\alpha_0 = -s^2 < 0$, put $p = \alpha_1/s$. The ranges*

$$p \leq 0, \qquad 0 < p \leq p_*, \qquad p_*/2 \leq p \leq 2P_*, \qquad p \geq P_* \tag{5.11}$$

cover every coefficient fibre. The small-positive range is covered by the overlapping gamma, compact-cubic, degree-loss and remote-root alternatives in Table 1. Here the variables are exactly (4.4) *and* (4.5)*. The degree and remote-root substitutions are* (K.4)*,* (K.5)*, and* (K.7)*. Their full independent rectangles lie in the prescribed coefficient ranges of their preparations. All thresholds and reserves are chosen before a final raw coefficient restriction. Neither a phase input, a return zero, nor a height cap is used to choose a coefficient alternative.*

Proof. Fix a signed compact strict source block about $a = p = 0, Y(0) = 1$. Choose its high-height constant K_{hi}, large enough also for the signed trace block $1/2 \leq |p_L| \leq 3/2$, and put $K_{\mathrm{lo}} = 2K_{\mathrm{hi}}$. The trace preparation on an initial $|a_L| \leq a_0 \leq 1$ uses the fixed central-height range $K_{\mathrm{lo}}(1 + (3/2)^2)$. Its graph barrier fixes the bounded actual-gap range; its terminal construction then fixes Q, R_0, a_*, ξ_0. Choose $P_D > 2/\sqrt{a_*}$. When $\alpha_0 > 0$, use $r_D = \sqrt{\alpha_0}$ if $|\alpha_1| \leq P_D\sqrt{\alpha_0}$. Otherwise put

$$r_L = |\alpha_1|, \qquad p_L = \operatorname{sgn}\alpha_1, \qquad a_L = \alpha_0/\alpha_1^2 < a_*/4.$$

The two coefficient descriptions overlap on $P_D/2 < |\alpha_1|/\sqrt{\alpha_0} < P_D$. For zero opening and nonzero trace, the second description has $a_L = 0$.

For negative opening, first choose the bounded positive-quadratic root-free and ghost reserves, and the independent positive far-root bounds q_0, λ_0, τ_0 in (G.8). Choose the reciprocal preparation bounds $\mathfrak{q}_-, \Pi_-, \tau_-$, with its positive-unit lower bound $\omega_{\min}$, as in (K.6). Put

$$\begin{gathered}\mathfrak{q}_\flat = \tfrac{1}{2}\min\{q_0, \mathfrak{q}_-\}, \qquad \Pi_{\mathrm{far}} = \max\{\lambda_0^{-1/2}, \Pi_-, 2\sqrt{12}\}, \\ \Pi_{\max} = 2\Pi_{\mathrm{far}}.\end{gathered} \tag{5.12}$$

On $0 < \pi \leq \Pi_{\max}$, fix the two bounded-gamma preparations, including the positive ghost raw-tail reserve. Choose $k_\flat > 0$ so that $2k_\flat$ lies below their fixed k-bounds and below $k_d(\mathfrak{q}_\flat)$.

The latter is the bound placing the gamma-degree dictionary in its prescribed compact positive-unit block for $\mathfrak{q} \geq \mathfrak{q}_\flat$. Set $G_{\max} = k_\flat^{-2/3}$. Fix U_d for the large-U cubic-degree dictionary on this G-block and take $U_{\mathrm{cub}} > U_d + 1$. Fix the compact cubic preparation on $[0, U_{\mathrm{cub}}] \times [-G_{\max}, G_{\max}]$.

These substitutions are algebraic. The gamma polynomial $t^3 + 3\epsilon t^2 - \mathfrak{q}t + k^2$ and Vieta's relations give (K.4); the positive-unit entries have compact closure for $\mathfrak{q} \geq \mathfrak{q}_\flat$. On the negative far face, the same relations give $\omega = 1 - a + ja$, $\tau = p^2/\omega^2$, and the full rectangle (K.6). The large-U polynomial $y^3 + 3Gy^2 - Uy + 1$ gives (K.7); its positive units tend uniformly to one for bounded G. Let $\tau_d > 0$ be the resulting compact degree bound. Take both degree radial bounds below $1/2$, including their bounded-source, enlarged positive-energy, raw-image and relative curvature reserves. Include the fixed core/high overlap reserve in τ_0 and choose once

$$0 < p_*^2 < \tfrac{1}{2} \min\{\tau_0, \tau_d, \tau_- \omega_{\min}^2\}. \tag{5.13}$$

Thus $\tau = p^2$ and $\tau = p^2/\omega^2$ already lie in their respective independent rectangles.

If $\gamma \neq 0$ and $k \leq 2k_\flat$, the overlapping π-ranges are $\pi \leq \Pi_{\max}$ and $\pi \geq \Pi_{\mathrm{far}}$. On the latter range split at $\mathfrak{q}_\flat$. The large-$\mathfrak{q}$ dictionary is in the compact degree block, and the small-$\mathfrak{q}$ dictionaries are in the positive far or negative reciprocal rectangle according to the sign of γ. If $\gamma = 0$ or $k \geq k_\flat$, then $|G|^{3/2} \leq k_\flat^{-1}$, and $[0, U_{\mathrm{cub}}] \cup [U_d, \infty)$ covers $U \geq 0$. The k, π, $\mathfrak{q}$, and U descriptions have the strict overlaps just chosen, with their equality faces retained.

For the large positive-source range use the exact trace dictionary (K.11). Its signed opening is $a_L = -p_L^2/p^2$. Choose $P_* > \max\{2, p_*, 3/\sqrt{a_*}\}$, increasing it if necessary for the fixed bounded-gap trace block of Lemma 5.10. Finally fix the compact positive interval $[p_*/2, 2P_*]$ and its endpoint, middle and high reserves. The remaining independent radii r_D, r_L, g, t, s can subsequently be restricted in raw coordinates. This proves the coefficient cover. □

The small-positive-source subcover. The outer overlaps are $P_D/2 < |\alpha_1|/\sqrt{\alpha_0} < P_D$, $p_*/2 < p < p_*$, and $P_* < p < 2P_*$. With $H = z(\gamma)$, the low and high estimates overlap on $K_{\mathrm{hi}}(|\alpha_0| + \alpha_1^2) < H < K_{\mathrm{lo}}(|\alpha_0| + \alpha_1^2)$, where $K_{\mathrm{lo}} = 2K_{\mathrm{hi}}$. Every phase description is restricted to the common receiver of Lemma 5.14.

In the first three rows assume $\gamma \neq 0$, $k \leq 2k_\flat$, using (4.4). The next two use (4.5) when $\gamma = 0$ or $k \geq k_\flat$, with $|G| \leq G_{\max}$.

Table 1: The small-positive-source coefficient subcover. Each regional estimate counts the whole physical fibre.

Range	Regional estimate	Overlap or boundary
$0 < \pi \leq \Pi_{\max}$	Proposition 5.3: positive gamma 3, negative gamma 2	$\Pi_{\mathrm{far}} < \pi < \Pi_{\max}$; real, fold, ghost and root-free faces
$\pi > \Pi_{\mathrm{far}}$, $\mathfrak{q} \geq \mathfrak{q}_\flat$	(K.4); Proposition H.14: 2	Far receivers extend past $\mathfrak{q}_\flat$
$\pi > \Pi_{\mathrm{far}}$, $\mathfrak{q} < \mathfrak{q}_\flat$	Positive gamma: (G.14) and Proposition 5.3; negative gamma: (K.5) and Proposition H.14; both 2	$\mathfrak{q}_\flat < \mathfrak{q}_-$, $\Pi_{\mathrm{far}} \geq \Pi_-$
$0 \leq U \leq U_{\mathrm{cub}}$	Proposition 5.4: 3, including the original raw ghost tail	$k_\flat < k < 2k_\flat$, $U_d < U < U_{\mathrm{cub}}$
$U \geq U_d$	(K.7); Proposition H.14: 2	$U_d < U < U_{\mathrm{cub}}$

The degree-loss phase subcover and its internal dividers are those of Proposition H.10. Each row selects a whole-fibre bound; Proposition 5.16 will take their maximum after the common raw restrictions.

Proposition 5.13 (Regional raw-height budgets)**.** *Fix the coefficient rectangles and all section and overlap reserves of Lemma 5.12. Let $\mathcal{A}_j$ be their finitely many regional alternatives and write*

$$\mathcal{V}_\delta = \{\max(|A-1|, |C-1|, |\gamma|, |\alpha_0|, |\alpha_1|) < \delta\}.$$

There is $b > 0$ such that, for every $0 < h \le b$ and each j, there are independent raw bounds $\delta_j(h) > 0$ and $\bar{m}_j(h) > 0$ for which the actual regional transition identities and complete derivative and curvature estimates hold on their stated phase domains for every $\theta \in \mathcal{A}_j \cap \mathcal{V}_{\delta_j(h)}$ and every $0 \le m \le \bar{m}_j(h)$. These domains together cover every input of

$$\{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < h_\theta(a_\theta + x) < h\}. \tag{5.14}$$

The bounds restrict $(A, C, \gamma, \alpha_0, \alpha_1)$ in the independent raw coordinates and $0 \le m \le \bar{m}_j(h)$. These additional restrictions impose no relation between active radii or root gaps. Each estimate retains its original input and inverse output windows, positive-time passage, speed and stop clips, admitted equality dividers, and intervening nonroot inputs. Its raw observers and divided closing are composed on their full admitted images, with ordinary phase C^3 bounds and all nonlinear curvature terms retained.

Proof. We first construct a fixed signed high-height cap on the original raw sections. Fix the enlarged signed strict source block used in the coefficient lemma. Choose the core cut L and raw port radius $0 < d < \delta$, with $d \le 1/100$, as in Lemma F.9. Let a_H, p_H bound its normalized $|a|, |p|$, and increase L until $p_H/L \le 1/40$, $a_H/L^2 \le 1/4000$. The fixed-core continuity bounds $s_{\mathrm{core}}, \rho_{\mathrm{core}} > 0$ put both entrance ratios in $[1/4, 3/4]$ for independent $s \le s_{\mathrm{core}}$ and $|\gamma|, |A-1|, |C-1| \le \rho_{\mathrm{core}}$.

Choose a fixed transverse tube from these ports to the original sections, and let $\mu_z > 0$ be smaller than its port tolerance and both raw-window margins. With $T = \log(d/(sL))$, the entrance-memory estimate of Lemma F.9, with its fixed $\lambda = 7/5$, gives

$$|w_\sigma(T) - w_\Gamma(d)| \le C\{s + |\gamma| + d|A-1| + d^2|C-1|\}. \tag{5.15}$$

Indeed its initial term is $O(e^{-\lambda T}) = O(s)$, and integration of the two source terms has orders e^{-T} and $e^{-\lambda T}$; the passive terms have the three orders displayed. The constant depends on the fixed ports and strict block, not on the flight length. Choose $s_H > 0$ and independent passive coefficient bounds so that the right side times d^2, plus $2d|\gamma|$, is less than μ_z. The invariant ratio strip from the same lemma keeps both complete tails in the chart. Their raw port errors are therefore below μ_z, so both full images lie in the original windows. The regular observers and divided closing can now be composed on their larger fixed buffers.

At any signed high input freeze $r_0 = \sqrt{H}$. The transformed parameters satisfy $|a| \le K_{\mathrm{hi}}^{-1}$, $|p| \le K_{\mathrm{hi}}^{-1/2}$, and $Y(0) = 1$. If $H \le s_H^2$, this is the same admitted strict block. The two absolute tail variations in (C.24) have the common $r_0^{2-\lambda_\gamma}$ factor; its cancellation, followed by the actual bridges and divided closing, gives a fixed C_H. The dilation is frozen during differentiation. This proves the signed high estimate without a restriction on $\gamma \log r_0$.

Let $z_{\mathrm{geo}} > 0$ be a fixed central reserve below the upper chart boundary constructed before Lemma 2.3, and bound $0 < \ell_\theta(\gamma) \le L_{\mathrm{raw}}$ on the preliminary raw block. Choose

$$b = \tfrac{1}{4} \min\{s_H^2, z_{\mathrm{geo}}, d_{\mathrm{out}}/(4L_{\mathrm{raw}})\} > 0. \tag{5.16}$$

Every signed high input below b consequently has its original raw-domain description. With $|\alpha_0| < d_{\mathrm{out}}/4$, its speed satisfies $\sigma_\theta \le |\alpha_0| + L_{\mathrm{raw}} H < d_{\mathrm{out}}/2$.

Now fix any $0 < h \le b$. The lower and strict preparations have already fixed their compact blocks, chords and positive raw-window reserves. We place their comparison and upper shots

below h by the following independent raw restrictions. In the determinant and trace charts it is enough to require

$$r_D^2 < \frac{h}{8K_{\rm hi}(1+P_D^2)}, \qquad r_L^2 < \frac{h}{4K_{\rm lo}}. \tag{5.17}$$

Their bounded normalized-height complements have the finite strict domains of (I.18) and (I.30); their larger heights are in the signed high block. On the fixed compact positive-ratio region the enlarged strict box $[u_L/2, 3u_H]$ is placed below $h/4$ by $s^2 < h/(12u_H)$. The endpoint admission and its actual matching are those of Lemma F.1.

For a bounded-gamma preparation let k_B, Π_B be its fixed coefficient bounds and $E_B > 0$ its fixed upper core height. Retain the action comparison bound and impose

$$g < \min\left\{1, \frac{E_B}{4K_{\rm hi}(1+\Pi_B^2)}, \frac{1}{2k_B}\sqrt{\frac{h}{E_B}}\right\}. \tag{5.18}$$

Then the normalized high threshold is below $E_B/2$, while $(gk)^2 E_B < h/4$. The actual comparison and upper shots and every intervening graph retain the outward admission used in (G.7) and in the positive paired construction. The positive ghost raw tail is part of the same original domain, including $gku \geq d_0$; its raw derivative estimate is (G.50).

For the compact cubic block let $E_c > 0$ be its fixed upper core height. Retain the exponentially small comparison bound and require

$$\tau_{\rm cub} < \min\left\{1, \frac{E_c}{4K_{\rm hi}(1+U_{\rm cub}^2)}, \tfrac{1}{2}\sqrt{h/E_c}\right\}. \tag{5.19}$$

Here $\tau_{\rm cub} = (-\alpha_0)^{1/3}$. Thus the normalized threshold is below $E_c/2$ and $\tau_{\rm cub}^2 E_c < h/4$. The comparison remains in the actual lip domain, with the bounded/growing resection and the original raw ghost continuation retained in (4.28).

For the positive far-root rectangle, the fixed τ_0 already ensures $\tau K_{\rm hi}(1+\tau) < E_f/2$ and the action comparison, where $E_f > 0$ is the selected upper shot. Since $a = gq\lambda$, $q \leq q_0$, $\lambda \leq \lambda_0$, the raw restriction

$$g < \frac{1}{2q_0\lambda_0}\sqrt{h/E_f} \tag{5.20}$$

gives $a^2 E_f < h/4$. The full lower output and upper core images are the admitted images used in (G.17). No ratio threshold is changed after h is fixed.

For the degree preparations, the exact source dictionary (H.21) gives

$$\alpha_1 = \tau^2\kappa^2\mathcal{R}\omega, \qquad s_s^2 = \tau^3\kappa^4\mathcal{R}^2 B. \tag{5.21}$$

The actual $h_L = 0$ source graph and its real enlarged chord have $Y_0 \leq C_Y(1 - \log(2\tau))$ by (H.5) and the enlarged positive-energy construction in Lemma H.11. Use the degree bounds $\kappa \leq \kappa_*$, $\tau \leq \tau_* < 1/2$ already fixed in the coefficient cover, and put

$$D_* = \tau_*[1 - \log(2\tau_*)], \qquad C_d = C_Y\kappa_*^2\frac{\mathcal{R}_{\max}B_{\max}}{\omega_{\min}}D_*. \tag{5.22}$$

The function $t[1 - \log(2t)]$ increases on $(0, 1/2)$. Consequently the actual trace h^0 satisfies

$$H_\theta(h^0) \leq C_Y\alpha_1\tau\kappa^2\frac{\mathcal{R}B}{\omega}[1 - \log(2\tau)] \leq C_d\alpha_1. \tag{5.23}$$

On the reciprocal graph (H.47), write $a_{{\rm rec},*} = \kappa_* A_{d,\max} < 1/2$ for its fixed opening bound. There

$$\frac{\mathcal{R}B}{\omega} = \frac{dA_d j}{1 - a + ja} \leq \frac{d_{\max}A_{d,\max}j_0}{1 - a_{{\rm rec},*}}.$$

Thus (5.23) holds with

$$C_r = C_Y \kappa_*^2 \frac{d_{\max} A_{d,\max} j_0}{1 - a_{\mathrm{rec},*}} D_* \tag{5.24}$$

in place of C_d, uniformly as $j \downarrow 0$. The independent raw restriction

$$0 < \alpha_1 < \frac{h}{4 \max\{C_d, C_r\}} \tag{5.25}$$

places the upper source shot and its chosen enlarged chord below $h/4$. The fixed positive-label threshold also gives $H_\theta(h^0) > K_{\mathrm{hi}}(|\alpha_0| + \alpha_1^2)$, so that this chord overlaps the same signed high continuation. The fixed p_* and degree ratios are unchanged.

Raw output admission in these degree charts is independent of a reciprocal lower bound on a gain. Indeed $r_s^2 = (\mathcal{R}/\omega)\alpha_1 \le C\alpha_1$; for a fixed $|\lambda_\gamma| < \nu < 1$, the retained absolute factor obeys

$$\Omega = r_s^{2-\lambda_\gamma} \le C\alpha_1^{1-\nu/2} \longrightarrow 0.$$

The stopped raw trace has a fixed window margin, so the full enlarged output chord is admitted before its observer is composed. The exact input buffer, actual matching and every-point allocation are those of Proposition H.10. Actual reciprocal maps use $j > 0$; $j = 0$ retains its one-sided normalized meaning. The raw-Jacobian estimate already holds on the full original raw-window and inverse-output-window domain and can be restricted to any such cap.

All these descriptions are restrictions of the original physical map $\Pi = \mathcal{C}_{\theta,m} \circ T_\theta(a_\theta + \cdot)$. For local nonlinear observers write $\Pi = \psi_- \circ P \circ \iota$, where $\iota = \psi_+^{-1}$ on the actual incoming image. In particular the physical curvature contains all three terms

$$\Pi'' = \psi_-''(P)(P')^2(\iota')^2 + \psi_-'(P)P''(\iota')^2 + \psi_-'(P)P'\iota''. \tag{5.26}$$

Here the derivatives of P are evaluated at $\iota(x)$. The regional estimates include these ordinary observer derivatives, through order three, before any positive affine normalization (3.2). Restriction leaves those estimates unchanged. The divided closing has the ordinary C^3 rows (3.5), including $m = 0$.

Take the finite minimum of the original raw-domain and error bounds and the independent restrictions above. Express $\tau_{\mathrm{cub}} < T$ as $|\alpha_0| < T^3$, $r_D < R$ as $\alpha_0 < R^2$, and $r_L < R$ as $|\alpha_1| < R$. Half this minimum defines $\delta_j(h) > 0$ in the independent raw coordinates. All other normalized coordinates remain in the rectangles already fixed by Lemma 5.12.

Lastly let C_j bound the complete derivative on the regional bounded pieces and let M_j bound the full translated output on the original closing image. Intersect the ordinary closing-domain bound, the regional critical-logarithm bound, and the inequalities $mC_j < 1/2$, $mM_j < a_{\mathrm{cl}}/2$. Their finite positive minimum gives $\bar{m}_j(h) > 0$. The enlarged domain and equality buffers were chosen earlier; their estimates hold at nonclosed inputs as well. This proves the asserted raw budgets. □

5.11 The common raw receiver

We now use the coefficient cover and regional raw budgets to define one interval on which all regional estimates hold for the complete return.

Lemma 5.14 (The common raw receiver)**.** *Fix the coefficient cover and reserves of Lemma 5.12, and let $b > 0$ be the raw budget of Proposition 5.13. For every fixed $0 < h_c \le b$, there are a raw coefficient neighborhood and a fixed $\bar{m} > 0$ such that all covering regional constructions and their every-point estimates hold on the same actual interval*

$$I_\theta(h_c) = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < h_\theta(a_\theta + x) < h_c\}. \tag{5.27}$$

This interval retains the original input and inverse output windows, positive-time passage, speed and stop clips, admitted equality dividers, and all intervening nonroot inputs. The normalized regional descriptions use their exact centres and positive absolute gains and the same ordinary C^3 divided closing. The coefficients and actual m are fixed in every phase derivative. The neighborhood and multiplier bound may depend on h_c, but the coefficient cover and its reserves do not.

Proof. Fix any $0 < h_c \le b$. Proposition 5.13 gives the finitely many independent bounds $\delta_j(h_c), \bar{m}_j(h_c)$ on the full actual regional domains. Intersect their raw neighborhoods, including $|\alpha_0| < d_{\mathrm{out}}/4$. Let $m_{\mathrm{ad}} > 0$ be the ordinary closing-domain bound, and choose

$$\bar{m} = \tfrac{1}{2} \min\{m_{\mathrm{ad}}, \bar{m}_1(h_c), \dots, \bar{m}_N(h_c), (2C_H)^{-1}\} > 0. \tag{5.28}$$

The complete displacement (2.13) is therefore defined on every admitted regional image, with its original divided observer (I.13). All three nonlinear curvature terms in (5.26) remain in its estimates.

The original raw admission set is exactly

$$J_\theta^{\mathrm{out}} = \mathcal{T}_\theta \cap \operatorname{int} K_+^\sharp \cap T_\theta^{-1}(\operatorname{int} K_-^\sharp) \cap \{\sigma_\theta < d_{\mathrm{out}}\}.$$

Scalar uniqueness and first variation order the actual through graphs. Thus $\mathcal{T}_\theta$, both window preimages, the speed clip, and the central-height clip are intervals; central height is strictly increasing by (2.23). Their intersection (5.27) is one interval, possibly empty.

The regional lower, core and high sets cover each admitted through input before clipping. Intersect them with this same interval. Their actual matching identities, inverse images and every-point estimates survive. In particular (I.18), (I.30), (G.7), (G.17), (4.28), and Proposition H.10 retain their admitted positive equality dividers. The compact positive-ratio sets remain the monotone height preimages of Lemma F.1. If a clip removes an overlap, the retained connected interval is still covered by the adjoining enlarged estimates. The natural stopped endpoints are excluded. The original compact-cubic and bounded-positive ghost raw tails retain (4.26) and (G.50), respectively.

The fixed cap is independent of the normalized overlap heights. Proposition 5.13 places all comparison shots and upper core endpoints below $h_c/4$ by later raw restrictions and covers the high complement up to h_c. For degree loss it uses (5.23)–(5.25) on both the compact and reciprocal rectangles. Hence no ratio reserve or coefficient threshold is selected again when h_c is reduced. Set $h_{\mathrm{out}} = h_c$ in (2.24).

Finally (2.12), with $\varepsilon_f = h(\theta)K$ and positive h, gives the actual $m(\theta) \le \bar{m}$ on $K > 0$ after a further raw coefficient shrink. This restriction is independent of $s, |\gamma|$ and of all root gaps. For $K \le 0$ the selected finite channel has no positive-time exit. Thus the receiver and all its regional estimates are fixed before any zero count or geometric capture. □

One admissible fixed choice for the subsequent argument is

$$h_c = b/2 > 0. \tag{5.29}$$

5.12 Pairing the admitted raw tails

On a ghost fibre, a very small central height can come from a first lip whose raw abscissa lies beyond the growing curvature range. The original through interval retains these inputs. We pair their actual inward heights at the original centre and compare the complete input and output gains. The cubic endpoint power and the quadratic nonpositive long-dwell correction control this ratio as the negative lip approaches its root.

Proposition 5.15 (Paired raw-tail contraction)**.** *Fix the original nested raw-section windows, their invariant upper boundary, the passive compact coefficient neighbourhood and the ordinary C^3 regular maps and divided closing of Lemma 2.2. Use one positive unscaled cap h_c in the common receiver* (5.27)*, chosen before the final coefficient and multiplier restrictions. In each of the following regions choose the fixed raw divider d_0 and buffer $a_b = d_0/2$ in the original chart.*

For the compact cubic coefficients (4.10)–(4.11)*, $0 \le U \le U_0$, $|G| \le G_0$, retain the strict ghost fibres $\Delta < 0$ and put $\tau = \tau_{\mathrm{cub}}$. For the bounded positive quadratic coefficients* (G.31)*, retain $\pi_\flat \le \pi \le \Pi$, $-\Delta_0 \le \Delta(k,\pi) < 0$ and independently small $g, k > 0$; choose $d_0 < a_0$, where a_0 is the original positive outward reserve of Corollary J.2.*

Let x_{comp} be the native admitted comparison divider of the corresponding regional construction. Every original through input $x \le x_{\mathrm{comp}}$ has a unique first positive lip. For every such input with respectively $a = \tau u \ge a_b$ or $a = gku \ge a_b$, its actual inward height at the original centre has a unique negative match, with respectively $r_0 \le r < c$ or $r \ge r_c$, $0 < R = kr < \mathfrak{c}$. The matched graph is its original through trajectory, including its original output and closing admissions. The complete physical derivative satisfies

$$0 < \Pi'_{\theta,m}(x) \le \begin{cases} C(d_0)\tau^3, & \textit{compact cubic,} \\ C(d_0)g^3, & \textit{bounded positive quadratic.} \end{cases} \tag{5.30}$$

The constants are uniform in the independent coefficient ranges, the ghost gap, the negative root distance and $0 \le m \le m_0$. All phase derivatives fix the coefficients and m. In particular the estimates retain every original admitted $a \ge d_0$, all its window, speed and height clips, and the inputs between return roots. For the actual finite channel $K_{\mathrm{fin}} > 0$, one last common multiplier restriction gives $F' > 1/2$ on both tails.

Proof. Use the original-domain constructions in Proposition 4.1 for the cubic field and Proposition G.6 for the quadratic field. We identify their matches with the common physical receiver and recall the complete-gain estimates needed for the contraction.

The cubic construction first admits a root-free comparison and a reference pair in a buffered band about $[2M, 3M]$. Its bounded and high-action descriptions overlap through (4.17) and the actual resection (4.21). This construction does not require the cubic action to diverge as the positive lip label grows. The quadratic comparison is the admitted pair of Lemma J.9. On its upper comparison interval, both original raw-section derivatives in (J.60) contain the common factor $(gk)^{2-\lambda_g}$, where $\lambda_g = 12g/(1+2g)$. This factor cancels exactly in their quotient, and (J.59) bounds the remaining core ratio. On the original ghost tail, the negative-root factor is instead absorbed by the actual centre variation, as computed below. The first-lip argument in each native proof applies to every original through input below its comparison: the fixed window floor and scalar order give the hit, and the strict transverse sign at a noncentral hit makes it unique. In the affine physical input its label satisfies $a_x < 0$.

For inward matching, use the actual original-centre maps in (4.23)–(4.24) and (G.54)–(G.56). Their full-range monotonicity and absolute endpoint constructions give

$$\eta_0^-([r_0, c)) = (0, \eta_0^-(r_0)] \quad \text{and} \quad \eta_0^-([r_c, \mathfrak{c}/k)) = (0, \eta_0^-(r_c)]$$

in the cubic and quadratic cases, respectively. The negative memory retains its initial polynomial and the simple-root action diverges at the excluded endpoint. Decreasing positive centre height from the reference to a_b, then using its full-range monotonicity, puts every $a \ge a_b$ in this negative image. The construction and its strict overlap with the growing range are proved in (4.25) and (G.57).

For an input in the common receiver, this matched graph and the original through graph have the same positive height at the original centre. Scalar uniqueness identifies them. Thus

the matched output and closing image are the original admitted ones; the auxiliary match at a_b is made before observation. The raw outward passages and window buffers are those of (4.22), (G.51), Lemma G.4 and Corollary J.3.

For the cubic gain calculation, use (4.26). Its proof keeps the factor $X^{\beta-}$ in the negative-root output and uses the actual central variation to compare it with the positive input gain. For the quadratic calculation, (G.61) absorbs the potentially singular negative-root gain into differentiation with respect to the actual centre height; the positive estimate in the same proof then gives (G.50). In both cases these are complete gain quotients, including the ordinary observers and divided closing. The preceding identification therefore gives (5.30) on the common receiver, without estimating either gain ratio anew.

The regional estimates use the original independent coefficient blocks and root distances. Their intrinsic collar derivatives precede the ordinary C^3 section chain of Lemma 3.3. Restricting to the common receiver preserves the original windows, inverse output, speed, stop and height clips. The quadratic tail is assigned $a \geq d_0$, whereas Proposition G.6 proves its estimate already for $a \geq a_b = d_0/2$. It therefore strictly overlaps the curvature domain $a < d_0$ on $a_b \leq a < d_0$, with $a = d_0$ retained in the tail. Finally choose $m_* \leq m_0$ with $m_* C(d_0) \max\{\tau_0^3, g_0^3\} < 1/2$. Then $F' = 1 - m\Pi' > 1/2$ on both original admitted tails. □

5.13 Whole-carrier coefficient synthesis

Proposition 5.16. *With the coefficient thresholds of Lemma 5.12 and the common receiver $I_\theta(h_c)$ of Lemma 5.14, the following whole-carrier bounds hold after one raw coefficient and multiplier restriction. Isolated zeros are counted with multiplicity.*

coefficient alternative	*number of roots*
$\alpha_0 \geq 0$	2
$\alpha_0 < 0,\ p \leq 0$	1
$\alpha_0 < 0,\ 0 < p \leq p_*$	3
$\alpha_0 < 0,\ p_*/2 \leq p \leq 2P_*$	2
$\alpha_0 < 0,\ p \geq P_*$	2

Here $p = \alpha_1/\sqrt{-\alpha_0}$ when $\alpha_0 < 0$. The alternatives cover every sufficiently small raw coefficient and each controls its whole phase fibre. Their maximum is three.

Proof. Use the fixed coefficient cover of Lemma 5.12 and the raw budgets of Proposition 5.13. Lemma 5.14 has already placed every regional estimate on the same connected interval $I_\theta(h_c)$, with the final coefficient and multiplier restrictions in force.

If $\alpha_0 > 0$, the determinant chart applies when $|\alpha_1| \leq P_D\sqrt{\alpha_0}$, and signed trace applies otherwise. The exact trace dictionary gives $0 < a_L < a_*/4$. Propositions 5.6 and 5.7 give two zeros on determinant or positive trace, and one on negative trace. If $\alpha_0 = 0$ and $\alpha_1 \neq 0$, the trace opening is zero. At $\alpha_0 = \alpha_1 = 0$, every positive height is in the signed high region of Corollary 5.1; the final multiplier bound makes the return contractive. Zero height is stopped.

For $\alpha_0 < 0$ and $p \leq 0$, Lemma H.15 gives at most one zero. For $0 < p \leq p_*$, the coefficient alternatives place each whole fibre in a bounded-gamma, far-root, compact-cubic or degree-loss receiver. Propositions 5.3 and 5.4 give at most three. The remaining original cubic and bounded-positive quadratic ghost tails have $F' > 1/2$ by Proposition 5.15, with both full gains and their original admissions retained. The positive far-root budget (G.14), physical cover (G.17), and the degree-loss carrier and count of Propositions H.10 and H.14 give two in their alternatives. Their independent rectangles and overlap reserves are those fixed by Lemma 5.12, before h_c was selected. The two three-root arguments bound the zeros of the actual lower F' by two and have $F' > 0$ on the overlapping upper continuation. Lemmas 3.2 and 3.1 therefore count the same connected lower/core/high interval. Both nonlinear section maps and all three terms of the physical curvature identity remain in these estimates.

For $p \geq P_*$, the exact large-source dictionary (K.11) has $p_L = 1$ and $a_L = -p^{-2}$; Proposition 5.7 gives two uniformly on this whole sector. On $[p_*/2, 2P_*]$, Lemma F.1 and Proposition 5.2 give two on the whole receiver. The three positive ranges overlap strictly. At every coefficient one covering argument counts all of $I_\theta(h_c)$, including its intervening nonroot inputs and admitted equality dividers. The maximum of the five whole-fibre bounds is three. Proposition 5.17 then applies geometric capture to this already counted receiver. □

5.14 Geometric capture and the upper bound

Proposition 5.17 (Central three-cycle upper bound)**.** *There are a coefficient neighborhood $\mathcal{V}_5$ and $\varepsilon_5 > 0$ such that every field in the five-parameter family with coefficient in $\mathcal{V}_5$ has at most three isolated periodic orbits C with $d_{\mathrm{H}}(\iota_0(C), \Gamma) < \varepsilon_5$.*

Proof. Fix the windows, coefficient cover, reserves, h_c, and multiplier bound of Lemma 5.14. Choose

$$0 < h_{\mathrm{in}} < \min\{h_c/2, d_{\mathrm{in}}/(4L_{\mathrm{raw}})\}, \qquad |\alpha_0| < d_{\mathrm{in}}/4.$$

Every through orbit of central height $H < h_{\mathrm{in}}$ satisfies

$$\sigma_\theta \leq -D_\theta(\gamma, H) = -\alpha_0 - Q_\theta(\gamma)H \leq |\alpha_0| + L_{\mathrm{raw}}H < d_{\mathrm{in}}/2. \tag{5.31}$$

Intersect the analytic neighborhood with $|\alpha_0| < d_{\mathrm{in}}/4$ and call it $\mathcal{V}_{\mathrm{an}}$. Only now apply Lemma 2.5 with these original inner windows, speed bounds and fixed h_{in}. It supplies $\mathcal{V}_0, \varepsilon_0$; set

$$\mathcal{V}_5 = \mathcal{V}_{\mathrm{an}} \cap \mathcal{V}_0, \qquad \varepsilon_5 = \varepsilon_0.$$

Every collar cycle is admitted to the inner windows and height cap, follows the complete marked positive-time circuit with $K > 0$, and gives one zero of (2.13) in $I_\theta(h_c)$ at its unique prime return-section hit.

Each covering alternative in Proposition 5.16 controls all of $I_\theta(h_c)$, including intervening nonroot inputs, so

$$N_{\mathrm{cen}} \leq \max\{2, 1, 3, 2, 2\} = 3. \tag{5.32}$$

The regional arguments control the connected lower/core/high union. On coefficient overlaps, either applicable whole-interval bound is used; local phase counts are not added. Distinct cycles give distinct zeros and transverse return preserves isolation, proving the assertion on this one neighborhood and collar. □

5.15 Concentration of three-cycle parameters

The two-cycle estimate on a compact positive source interval also locates the coefficient sequences that can carry three cycles. We use the canonical slice throughout: for a field F near F_*, put $\theta_F = \theta(\mathscr{C}_0(F))$, with $\mathscr{C}_0$ from (1.7) and θ from Lemma 2.1. Define $\gamma(F), \alpha_0(F), \alpha_1(F)$ by (2.10) at θ_F, and, when $\alpha_0(F) < 0$, set

$$s(F) = \sqrt{-\alpha_0(F)}, \qquad p(F) = \frac{\alpha_1(F)}{s(F)}.$$

Thus the ratio below refers to this fixed analytic slice.

Proposition 5.18 (Uniform two-cycle exclusion on compact positive ratios)**.** *For every fixed $0 < \eta \leq P < \infty$, there are an open coefficient neighborhood $\mathcal{V}_{\eta,P}$ of F_* in the full twelve-coefficient quadratic germ and $\varepsilon_{\eta,P} > 0$ such that every $F \in \mathcal{V}_{\eta,P}$ with $\alpha_0(F) < 0$ and $\eta \leq p(F) \leq P$ has at most two distinct isolated periodic orbits C satisfying*

$$d_{\mathrm{H}}(\iota_*(C), \Gamma_{I4}) < \varepsilon_{\eta,P}.$$

The return estimate holds on one complete physical receiver, retaining the original windows and clips and all inputs between its roots.

Proof. Apply Lemma F.1 and Proposition 5.2 on the enlarged fixed ratio interval $[\eta/2, 2P]$. They give the two-zero multiplicity bound on one complete physical receiver, including its endpoint, middle and high ranges. Choose its cap so that

$$0 < b_{\eta,P} \le \min\{h_c, z_*\}, \qquad L_{\mathrm{raw}} b_{\eta,P} < d_{\mathrm{out}}/4, \qquad |\alpha_0| < d_{\mathrm{out}}/4.$$

All constants may depend on η, P. The ordinary coefficient restriction in (2.12) makes the actual finite multiplier satisfy the final bound used by that proposition.

Now choose $0 < h_{\mathrm{in}} < \min\{b_{\eta,P}/2, d_{\mathrm{in}}/(4L_{\mathrm{raw}})\}$ and $|\alpha_0| < d_{\mathrm{in}}/4$. The height estimate (5.31) and Lemma 2.5 admit every sufficiently close whole cycle to this counted receiver, at its unique prime section hit. Lemma 2.6, with $\mathcal{B} = \{\alpha_0 < 0, \eta \le p \le P\}$ and $N_{\mathrm{cyc}} = 2$, gives the full-germ neighborhood and collar. □

Corollary 5.19 (Three-cycle concentration)**.** *Let $F_n \to F_*$ in the full quadratic coefficient germ. Suppose each F_n has three distinct isolated periodic orbits C_n^1, C_n^2, C_n^3, all of this same field, and*

$$\max_{1 \le j \le 3} d_{\mathrm{H}}(\iota_*(C_n^j), \Gamma_{I4}) \longrightarrow 0.$$

Then $\alpha_0(F_n) < 0$ and $\alpha_1(F_n) > 0$ for all sufficiently large n, and

$$p(F_n) = \frac{\alpha_1(F_n)}{\sqrt{-\alpha_0(F_n)}} \longrightarrow 0.$$

Proof. The whole-coefficient bounds of Proposition 5.16, on the common receiver followed by geometric capture, give at most two cycles when $\alpha_0 \ge 0$, at most one when $\alpha_0 < 0, \alpha_1 \le 0$, and at most two when $\alpha_0 < 0, p \ge P_*$, for one fixed $P_* < \infty$. In the last range the exact trace scaling is $r_L = \alpha_1$, $p_L = 1$, $a_L = -p^{-2}$; its independent small $r_L, |\gamma|$ bounds and positive unscaled cap cover the entire large-ratio sector. The same-field transport used above puts these exclusions on one full coefficient neighborhood and collar. The sequence and all three cycle sequences eventually enter them. Hence eventually $\alpha_0(F_n) < 0$, $\alpha_1(F_n) > 0$, and $0 < p(F_n) < P_*$.

Fix $0 < \eta < P_*$. Apply Proposition 5.18 to $[\eta, 2P_*]$. Its neighborhood and collar are fixed for this η, so the sequence and its three cycles eventually enter them. If $p(F_n) \ge \eta$ there, all three cycles belong to one two-cycle sector, a contradiction. Thus $p(F_n) < \eta$ eventually. This proves the limit. The argument uses one fixed η at a time and requires no common collar as $\eta \downarrow 0$. □

Corollary 5.20 (A consequence in a positive-gamma subregion)**.** *Along any subsequence in Corollary 5.19 for which $\gamma(F_n) > 0$, if*

$$\frac{s(F_n)}{\gamma(F_n)^{3/2}} \longrightarrow 0,$$

then

$$\liminf \frac{p(F_n)}{\sqrt{\gamma(F_n)}} \ge \sqrt{12}.$$

Proof. Fix $0 < \pi_0 < \sqrt{12}$. The second alternative of Lemma H.15 supplies a fixed $k_0 > 0$ and a uniformly contracting complete return when

$$\frac{s}{\gamma^{3/2}} \le k_0, \qquad \frac{p}{\sqrt{\gamma}} \le \pi_0.$$

Its fixed-window endpoint quotient bounds the derivative on the whole physical interval, including intervening inputs. After its coefficient and multiplier restriction, geometric capture gives one neighborhood and collar. The subsequence eventually satisfies the first inequality and enters both; three isolated cycles exclude the second. Thus its lower limit is at least every $\pi_0 < \sqrt{12}$, which proves the assertion. □

6 Three hyperbolic cycles approaching the marked graphic

The complete organizing gain tends to zero as $c \downarrow 1$. We first construct three hyperbolic cycles at fixed $c > 1$ by balancing this gain against the central transition slope, then pass diagonally to $c = 1$ and transfer the cycles to the original marked graphic.

Theorem 6.1. *For every neighborhood $\mathcal{V}$ of F_* in the full space $\mathcal{Q}_2$ of real quadratic vector fields and every $\delta > 0$, there is $F \in \mathcal{V}$ with three geometrically distinct hyperbolic periodic orbits $\gamma_1, \gamma_2, \gamma_3$ satisfying*

$$d_{\mathrm{H}}\big(\iota_*(\gamma_j), \Gamma_{I4}\big) < \delta, \qquad j = 1, 2, 3.$$

The cycles follow both marked branches in the positive-time order $q_* \to p_* \to q_*$. In particular, $\mathrm{Cycl}_{\mathcal{Q}_2}(\Gamma_{I4}, F_*) \geq 3$.

Use H_c, Γ_c from (1.11); $H_1 = H_0$ and $\Gamma_1 = \Gamma$. For $c > 1$, this organizing family appears in [8, (1.1)–(1.3), p. 2], with its variables (x, y, C_0) written here as (X, Y, c).

All coefficient norms below are taken in a fixed norm on the twelve quadratic coefficients, and d_{H} uses the fixed smooth metric of this paper.

6.1 A construction at a fixed limiting parameter

Use (2.1), the organizing family (1.11), and its positive speed f_c in (1.12). Both infinite ends have endpoint $q = (0, 1, 0)$.

Theorem 6.2. *There is $K_0 > 0$ such that, for each fixed $c \in (1, 1 + K_0)$, there is a continuous family $F_{c,\rho}$ of fields (2.1), defined for all sufficiently small $\rho > 0$, with $F_{c,\rho} \to H_c$ in quadratic coefficients. Each $F_{c,\rho}$ has three geometrically distinct hyperbolic periodic orbits $\gamma_{j,c,\rho}$, $1 \leq j \leq 3$, and*

$$\max_{1 \leq j \leq 3} d_{\mathrm{H}}(\iota_0(\gamma_{j,c,\rho}), \Gamma_c) \longrightarrow 0 \qquad (\rho \downarrow 0).$$

The coefficient $C_{c,\rho}$ converges to c.

All estimates below are uniform on the indicated compact phase and coefficient sets. After c is fixed, their constants may depend on c. Every phase derivative keeps the coefficients fixed.

6.1.1 A cubic action map with three transverse roots

Set $k = 3/2$ and

$$P_\varepsilon(\xi) = (\xi + 1)(\xi - \varepsilon)(\xi - k\varepsilon).$$

For $0 < r < 1$ let $u_\varepsilon(r) \in (0, \varepsilon)$ be determined by

$$\int_0^{u_\varepsilon(r)} \frac{\xi}{P_\varepsilon(\xi)}\, d\xi = \int_0^r \frac{s}{P_\varepsilon(-s)}\, ds. \tag{6.1}$$

Both integrals increase strictly and the left diverges at $u = \varepsilon$, giving a unique smooth u_ε with

$$u_\varepsilon'(r) = \frac{r P_\varepsilon(u_\varepsilon(r))}{u_\varepsilon(r) P_\varepsilon(-r)} > 0. \tag{6.2}$$

Define S_ε by

$$b_\varepsilon(r) = -u_\varepsilon(r)^2/2, \qquad S_\varepsilon(b_\varepsilon(r)) = -r^2/2.$$

On compact subintervals of $0 < r < 1$ this defines a smooth increasing map in the phase b.

Lemma 6.3. *There are a compact interval $J \Subset (0,1)$, three disjoint closed subintervals $U_j \subset \operatorname{int} J$, and $\varepsilon_* > 0$ with the following properties. Put*

$$m_\varepsilon = 2\varepsilon^{5/2}, \qquad c_\varepsilon = \frac{\varepsilon^2}{2} - \frac{101}{100}\varepsilon^{5/2}.$$

For $0 < \varepsilon < \varepsilon_*$ the function

$$Q_\varepsilon(r) = \varepsilon^{-5/2}\{c_\varepsilon + b_\varepsilon(r) - m_\varepsilon S_\varepsilon(b_\varepsilon(r))\}$$

has one simple zero in each U_j. Its endpoint signs and derivative lower bounds on these intervals can be chosen independently of ε. Moreover,

$$\|Q_\varepsilon - \Phi\|_{C^1(J)} \le C_J(\sqrt{\varepsilon} + \varepsilon|\log\varepsilon|), \qquad \Phi(r) = \sqrt{\frac{1-r}{8r}} + r^2 - \frac{101}{100}. \tag{6.3}$$

Proof of Lemma 6.3. The partial fractions of $\xi/P_\varepsilon(\xi)$ have residues

$$a_N = -\frac{1}{(1+\varepsilon)(1+k\varepsilon)}, \quad a_P = -\frac{2}{1+\varepsilon}, \quad a_B = \frac{3}{1+k\varepsilon}, \quad a_N + a_P + a_B = 0.$$

The action difference in (6.1) is therefore

$$a_N \log\frac{1+u}{1-r} + a_P \log\frac{\varepsilon - u}{r+\varepsilon} + a_B \log\frac{k\varepsilon - u}{r+k\varepsilon}. \tag{6.4}$$

Put $\varepsilon = h^2$, $u = h^2 - dh^3$ in (6.4). The coefficient of $\log h$ is exactly $3a_P + 2a_B = -3h^2/((1+h^2)(1+kh^2))$; all remaining terms extend smoothly to $h = 0$, with limit $\log((1-r)/(8r)) - 2\log d$. On positive compact (d, r) rectangles the full $C^1_{d,r}$ error is $O(h + h^2|\log h|)$. The limiting d derivative $-2/d$ and the implicit function theorem give

$$d_\varepsilon(r) = \frac{\varepsilon - u_\varepsilon(r)}{\varepsilon^{3/2}}, \qquad \|d_\varepsilon - d_0\|_{C^1(J)} \le C_J(\sqrt{\varepsilon} + \varepsilon|\log\varepsilon|), \quad d_0(r) = \sqrt{\frac{1-r}{8r}}. \tag{6.5}$$

Uniqueness in (6.1) identifies this local solution with u_ε.

The center terms cancel exactly:

$$Q_\varepsilon(r) = d_\varepsilon(r) - \frac{\sqrt{\varepsilon}}{2} d_\varepsilon(r)^2 + r^2 - \frac{101}{100}. \tag{6.6}$$

This and (6.5) prove (6.3).

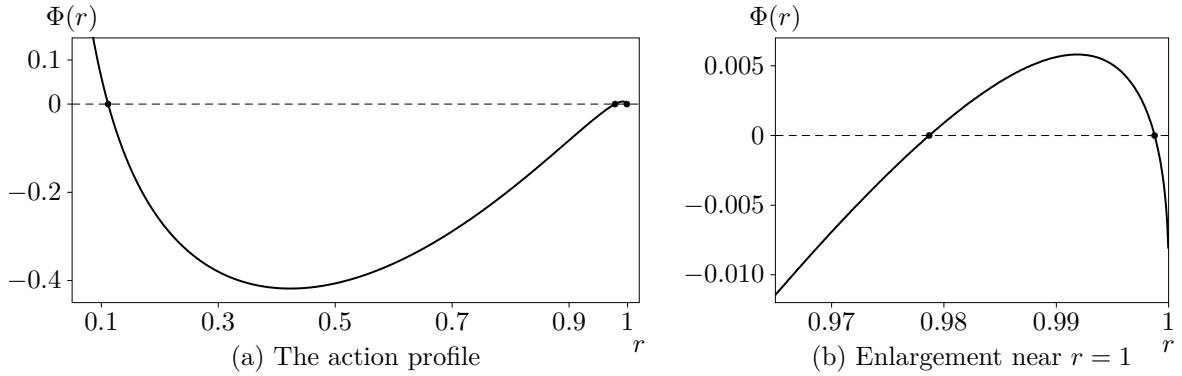
The values at $r = 1/20, 1/2, 49/50, 999/1000$ have signs $+, -, +, -$, as follows by evaluating

$$\frac{1-r}{8r} - \left(\frac{101}{100} - r^2\right)^2 = -\frac{10000r^5 - 20200r^3 + 11451r - 1250}{10000r}.$$

Furthermore

$$\Phi'(r) = 2r - \frac{1}{4\sqrt{2}\, r^{3/2}\sqrt{1-r}}.$$

Its zeros are exactly the solutions of $r^5(1-r) = 1/128$. The left side increases on $(0, 5/6)$ and decreases on $(5/6, 1)$; its maximum is $3125/46656 > 1/128$. Thus Φ' has exactly two zeros. The alternating signs and Rolle's theorem give exactly three roots of Φ, all simple: a nonsimple root and the two intervening Rolle points would give a third root of Φ'. Choose J containing the four test points and choose disjoint neighborhoods U_j of the three roots, indexed in increasing r order, with opposite endpoint signs and $|\Phi'| \ge 2\kappa > 0$ on their union. Reduce ε_* so that (6.3) preserves these signs and gives $|Q'_\varepsilon| \ge \kappa$ there. □



(a) The action profile (b) Enlargement near $r = 1$

Figure 3: Computed graph of $\Phi(r) = \sqrt{(1-r)/(8r)} + r^2 - 101/100$ on $0 < r < 1$. Panel (a) clips the divergent left-end portion as $r \downarrow 0$. The dots mark numerically computed zeros; panel (b) resolves the two near $r = 1$. The three simple zeros are established analytically in Lemma 6.3.

Figure 3 shows the model profile and resolves its two roots near $r = 1$.

The lower construction balances the vanishing outer gain $m_\varepsilon = 2\varepsilon^{5/2}$ with the central slope $S'_\varepsilon(b_\varepsilon(r)) \asymp \varepsilon^{-5/2}$ on the chosen phase interval, as follows from (6.5). Their product remains finite and positive; the calibration below fixes this gain and the return center in one field.

6.1.2 The actual passage through infinity

Choose $K_0 > 0$ small enough that, for $1 < c < 1 + K_0$,

$$\varepsilon = \varepsilon(c) = \left(\frac{1}{2} e^{-4\pi/\sqrt{c-1}}\right)^{2/5} < \varepsilon_*. \tag{6.7}$$

Fix such a c and this ε for the remainder of the construction. Set

$$\begin{gathered} \beta = 1 - (1+k)\varepsilon, \qquad \gamma = \beta\rho/3, \\ \alpha_0 = -k\varepsilon^2\rho^3, \qquad \alpha_1 = ((1+k)\varepsilon - k\varepsilon^2)\rho^2, \\ \mu_3 = -\gamma - \gamma^2, \quad \mu_2 = \alpha_1 + 3\gamma^2, \quad \mu_1 = \alpha_0 - \alpha_1\gamma - 2\gamma^3. \end{gathered} \tag{6.8}$$

The coefficients μ are fixed by c, ρ; A, C will be chosen near $(1, c)$. In the physical field (2.4), substitute $v = \gamma + \rho\xi$, $z = \rho^2\eta$ and the positive time $t_{\text{new}} = \rho t_{\text{raw}}$, giving exactly

$$\begin{aligned} \xi' &= -\rho P_\varepsilon(\xi) - \ell_\rho(\xi)\eta, \\ \eta' &= -\eta\{\xi B_\rho(\xi) + \zeta_\rho(\xi)\eta\}, \\ \ell_\rho &= 1 - A(\gamma + \rho\xi) + C(\gamma + \rho\xi)^2, \\ B_\rho &= 1 + 2\gamma + \rho\xi, \qquad \zeta_\rho = C\rho(\gamma + \rho\xi). \end{aligned} \tag{6.9}$$

Enlarge $J \Subset (0, 1)$; its entry and exit feet remain separated from 0 and the three zeros of P_ε. Choose compact phase intervals I_+, I_- containing their images $-u^2/2, -r^2/2$, with $b_\varepsilon(J) \Subset \operatorname{int} I_+$ and $S_\varepsilon(I_+) \Subset \operatorname{int} I_-$. Choose L large enough that

$$3/8 < 1/2 + b/L^2 < 5/8 \tag{6.10}$$

on both phase intervals and on intervals connecting them to $b = 0$. On $\xi = \pm L$ use the same literal coordinate $b = \eta - L^2/2$. The tail reference uses $b = 0$.

Use the same lip height $h_\rho = (k\varepsilon^2\rho)^{12}$ on both sides. Between the lips use $Q = \rho \log(\eta/h_\rho)$.

Lemma 6.4. *On the enlarged input interval I_+, the positive-time passage of* (6.9) *from $\xi = L$ to $\xi = -L$ exists for small $\rho > 0$. In the literal phase b its map satisfies*

$$\|S_{\varepsilon,\rho;A,C} - S_\varepsilon\|_{C_b^1} \le C_\varepsilon \rho(1 + |\log \rho|) \longrightarrow 0, \tag{6.11}$$

uniformly for A, C in a sufficiently small fixed rectangle about $(1, c)$, and the entire passage has $\xi' < 0$.

Proof of Lemma 6.4. Use the same lip height $h_\rho = (k\varepsilon^2\rho)^{12}$ on both sides. On each outer part put $H = \eta - \xi^2/2$ and use $\log \eta$ as a clock. Direct differentiation gives

$$\frac{dH}{d\log\eta} = -\frac{\rho\xi P_\varepsilon + \xi\eta(\ell_\rho - B_\rho) - \zeta_\rho\eta^2}{\xi B_\rho + \zeta_\rho\eta}. \tag{6.12}$$

For the selected negative H windows, $\xi = \pm\sqrt{2(\eta - H)}$ stays a positive distance from zero. On their fixed enlarged neighborhoods the denominator in (6.12) has the appropriate nonzero sign, while its right side and the mixed derivatives needed for a first phase variation are $O_\varepsilon(\rho)$. Its clock length is $O_\varepsilon(1 + |\log\rho|)$. Integration and the scalar variational equation therefore give

$$|H_{\text{out}} - H_{\text{in}}| + |\partial_{H_{\text{in}}} H_{\text{out}} - 1| \le C_\varepsilon \rho(1 + |\log\rho|). \tag{6.13}$$

Applying this estimate up to a first exit from the enlarged windows excludes such an exit for small ρ.

Starting at a positive lip foot u and integrating outward in reverse time gives at $\xi = L$ $b = -u^2/2 + O_\varepsilon(\rho(1 + |\log\rho|))$ in C_u^1. Its derivative is $-u + o(1)$, bounded away from zero. Thus it can be inverted on the entire enlarged entry interval. The negative lip gives in positive time $b = -r^2/2 + O_\varepsilon(\rho(1 + |\log\rho|))$ in C_r^1. Near each foot, $P_\varepsilon > 0$. Away from its small foot neighborhood, the limiting height $(\xi^2 - u^2)/2$ or $(\xi^2 - r^2)/2$ has a positive lower bound. This lower bound dominates the bounded term ρP_ε, so $\rho P_\varepsilon + \ell_\rho\eta > 0$ throughout both fast parts. The hits of $\xi = \pm L$ are unique and transverse.

Between the lips use $Q = \rho\log(\eta/h_\rho)$. In the region $Q \le 0$ the exact graph equation is

$$Q_\xi = \frac{\xi B_\rho + \zeta_\rho\eta}{P_\varepsilon + \ell_\rho\eta/\rho}, \qquad \eta = h_\rho e^{Q/\rho}. \tag{6.14}$$

The compact traversed ξ interval, with its exit buffer, has $P_\varepsilon \ge p_\varepsilon > 0$. Replace $e^{Q/\rho}$ for $Q > 0$ by a bounded smooth function of Q/ρ equal to the exponential for $Q \le 0$. This defines an open scalar ODE domain and changes no solution before its first return to the lip. For total derivative order at most three in Q, ξ, A, C, the right side differs from ξ/P_ε by at most

$$C_\varepsilon\{\rho + \rho^{11-j}\}, \qquad 0 \le j \le 3, \tag{6.15}$$

where j is the number of Q derivatives. Indeed, $h_\rho/\rho = O_\varepsilon(\rho^{11})$, and each Q derivative costs one power of ρ. The scalar integral equation and its first variational equation converge on the fixed ξ interval.

The limiting solution starting at u is $Q_0(\xi) = \int_u^\xi s/P_\varepsilon(s)\,ds$. It is strictly negative before its negative exit $-r$, except at the entrance, and $Q_0'(-r) = -r/P_\varepsilon(-r) < 0$. Uniform signs of Q_ξ near entrance and exit, and a strictly negative upper bound for Q_0 between them, exclude an early first return and persist under (6.15). Between two buffered negative exit cuts with opposite limiting signs, the implicit function theorem gives a unique first lip hit. The thin field and first variations have $O_\varepsilon(\rho)$ error and the exit has a fixed inverse bound. Composition with (6.13) proves (6.11); the denominator signs and $P_\varepsilon > 0$ on the thin part give $\xi' < 0$ everywhere. □

6.1.3 The two parabolic tails

Choose a small raw half-width $d > 0$. Write $q = |v - \gamma|$, $v = \gamma + \sigma q$ with $\sigma = \pm 1$, and $\widetilde{P}_\rho(x) = \rho^3 P_\varepsilon(x/\rho)$. The exact radial graph equation is

$$z_q = F_{\sigma,\rho}(q,z) = \frac{z\{q(1+2\gamma+\sigma q) + \sigma C(\gamma+\sigma q)z\}}{\widetilde{P}_\rho(\sigma q) + [1 - A(\gamma+\sigma q) + C(\gamma+\sigma q)^2]z}. \tag{6.16}$$

Its initial condition at $q = \rho L$ is $z = \rho^2(L^2/2 + b)$. The negative tail is traversed outward in positive time; the positive tail is constructed outward in reverse time.

Lemma 6.5. *For the enlarged phase windows the tails reach the raw ports $v = \gamma \pm d$ uniquely. Their maps $T_{\sigma,\rho}(b)$ have the decomposition*

$$T_{\sigma,\rho}(b) = a_{\sigma,\rho} + c_{\sigma,\rho}\phi_{\sigma,\rho}(b), \qquad a_{\sigma,\rho} = T_{\sigma,\rho}(0), \quad c_{\sigma,\rho} = \partial_b T_{\sigma,\rho}(0) > 0. \tag{6.17}$$

Uniformly on a fixed small A, C rectangle,

$$\begin{aligned} a_{\sigma,\rho} \to a_{\sigma,0}, \qquad c_{\sigma,\rho}/\rho^2 \to \ell_\sigma > 0 \quad \text{in } C^1_{A,C}, \\ \phi_{\sigma,\rho} \to \mathrm{id} \quad \text{in } C^1_b. \end{aligned} \tag{6.18}$$

On every fixed q interval bounded away from zero, the entire tail family converges to its limiting parabolic graph.

Proof of Lemma 6.5. Since $\gamma = O_\varepsilon(\rho)$ and $q \geq \rho L$,

$$\widetilde{P}_\rho(\sigma q) = O_\varepsilon(q^3 + \rho q^2 + \rho^2 q + \rho^3), \qquad |\widetilde{P}_\rho|/q^2 \leq C_\varepsilon q.$$

For $w = z/q^2$ in $[1/4, 3/4]$ the denominator in (6.16) is at least a positive constant times q^2 after d is reduced. Its logarithmic-radius equation is $w_{\log q} = 1 - 2w + O_\varepsilon(q)$, with the same bound for its first w derivative. The boundary vectors point inward. Equation (6.10) starts all phases strictly inside this band. The band is invariant, the radial speed does not vanish, and the raw port is reached once.

Writing the right side as $z(\beta + \zeta z)/(\widetilde{P} + \ell z)$, where $\widetilde{P} = \widetilde{P}_\rho(\sigma q)$, direct differentiation gives

$$F_{zz} = \frac{2\widetilde{P}(\zeta\widetilde{P} - \beta\ell)}{(\widetilde{P} + \ell z)^3}.$$

Together with the band bounds this yields

$$\begin{aligned} |F_z| &\leq C_\varepsilon(1 + \rho/q + \rho^2/q^2 + \rho^3/q^3), \\ |F_{zz}| &\leq C_\varepsilon q^{-2}, \\ |F_A| + |F_C| \leq C_\varepsilon q^2, &\qquad |F_{zA}| + |F_{zC}| \leq C_\varepsilon. \end{aligned} \tag{6.19}$$

The first majorant is uniformly bounded since $\rho/q \leq 1/L$. For $J = \partial_b z$, $J(\rho L) = \rho^2$ and $J_q = F_z J$. Consequently

$$c'_\varepsilon \rho^2 \leq J \leq C'_\varepsilon \rho^2, \qquad |\partial_b \log J| \leq \int_{\rho L}^{d} |F_{zz}| J \, dq \leq C'_\varepsilon \rho. \tag{6.20}$$

This proves the last assertion of (6.18), since $\phi(0) = 0$ and $\phi' = J(q,b)/J(q,0)$.

At $\rho = 0$ equation (6.16) has a unique graph in the band with $z_0(q) = q^2/2 + O(q^3)$. It is obtained by starting $z(a) = a^2/2$ and letting $a \downarrow 0$. The inward-band comparison gives $|z/q^2 - 1/2| \leq Cq$ for these initial data. Any two bounded-band limiting graphs have distance

at most Ca^2 at $q = a$; the bound for F_z propagates this distance to $q = d$, proving uniqueness. For the actual reference $b = 0$, coefficient convergence on compact $q > 0$ intervals, $z \le Cq^2$, and propagation from a small fixed a give convergence to this graph.

The parameter-independent initial height gives $z_{A,q} = (\partial_z F)z_A + F_A$ and the analogous equation for C. The majorants (6.19) give $|z_A| + |z_C| \le Cq^3$; their integral equations converge to the unique zero-initial-value equations at $q = 0$, proving $C^1_{A,C}$ convergence of $a_{\sigma,\rho}$. Finally,

$$\log(c_{\sigma,\rho}/\rho^2) = \int_{\rho L}^{d} F_z(q, z(q, 0))\, dq.$$

Its A, C derivatives have integrands $F_{zA} + F_{zz}z_A$ and $F_{zC} + F_{zz}z_C$, bounded by $C(1 + q)$. Pointwise convergence for $q > 0$ and these integrable majorants give the first-slope limits in (6.18). The same bounds show ℓ_σ and their reciprocals are bounded on the coefficient rectangle. □

6.1.4 The finite connection and two coefficient directions

At $\rho = 0, A = 1, C = c$ the two limiting tail graphs are the ends of the organizing parabola, whose raw height is

$$z_c(v) = \frac{v^2}{1 + \sqrt{1 + cv^2}}. \tag{6.21}$$

The finite arc between $v = -d$ and $v = d$ is a compact affine arc. It includes the two crossings $Y = 0,\ X = \pm\sqrt{c}$. At these points

$$\dot{X} = c \pm \sqrt{c} > 0, \qquad \dot{Y} = c \pm c^{3/2} \ne 0.$$

Along the entire arc $f_c \ge (c-1)/2$. Choose a fixed narrow affine tube and buffered raw section windows. After the coefficient rectangle and ρ are reduced, the finite-time flow stays in this tube and hits the positive port transversely. It defines an increasing map E_ρ from negative to positive raw height. Its mixed derivatives through order two converge uniformly to those of E_0. The same transverse finite-time estimates give a uniform C^3 norm and slope bounded away from zero. Lemma 6.5 admits the port heights and all phase images to these buffered domains before composition.

Define

$$\Delta_\rho = a_{+,\rho} - E_\rho(a_{-,\rho}), \qquad \Lambda_\rho = E'_\rho(a_{-,\rho})c_{-,\rho}/c_{+,\rho}. \tag{6.22}$$

These functions converge in $C^1_{A,C}$ by the finite-time estimates and the normalized slope limits in (6.18). The limiting return center varies nontrivially with A, while C changes the complete gain along the connected family $A = 1$. The triangular Jacobian below allows both conditions to be imposed on the same field.

Lemma 6.6. *At $(A, C) = (1, c)$,*

$$\Delta_0 = 0, \quad \partial_C\Delta_0 = 0, \quad \partial_A\Delta_0 > 0, \qquad \Lambda_0 = e^{-4\pi/\sqrt{c-1}}, \quad \partial_C\Lambda_0 = 2\pi(c-1)^{-3/2}\Lambda_0 > 0. \tag{6.23}$$

Proof of Lemma 6.6. For $A = 1$ the whole $C > 1$ family has the invariant parabola, so $\Delta_0(1, C) = 0$. Write $K = c - 1$ and

$$\mathcal{E}_c(X) = ((X + 1)^2 + K)^2 \exp\left[-\frac{4}{\sqrt{K}} \arctan\frac{X + 1}{\sqrt{K}}\right].$$

For $W = Y - (X^2 - c)/2$, the exact affine equation at $\mu = 0, C = c$ is

$$\dot{W} = 2XW + (1 - A)X^2, \qquad \mathcal{E}'_c/\mathcal{E}_c = 2X/f_c.$$

The graph variation in the A direction therefore satisfies

$$W_{A,X} = (2X/f_c)W_A - X^2/f_c.$$

The tail bound $z_A = O(q^3)$ gives $W_A = O(X)$ at infinity. Since $\mathcal{E}_c \asymp X^4$, the free homogeneous coefficient vanishes, and the selected variations are

$$W_A^-(X) = -\mathcal{E}_c(X)\int_{-\infty}^{X} \frac{t^2}{f_c(t)\mathcal{E}_c(t)}\,dt, \qquad W_A^+(X) = \mathcal{E}_c(X)\int_{X}^{\infty} \frac{t^2}{f_c(t)\mathcal{E}_c(t)}\,dt.$$

The positive integrand is $O(|t|^{-4})$ at both ends, so $W_A^+ - W_A^-$ is positive at the positive port. At fixed $v = d$,

$$W_z = \frac{d^2 - z_c(d)}{z_c(d)^3} > 0,$$

conversion to raw height gives $\partial_A \Delta_0 > 0$.

Truncate the limiting tails at $v = \pm a$. The outer derivative is E_0' times the negative-tail first variation divided by the positive one, tending to $\Lambda_0 = E_0'\ell_-/\ell_+$ as $a \downarrow 0$. The parabola endpoints are $X = \pm X_a$, $X_a = a/z_c(a) \to \infty$, with equal fixed-$v$ conversion factors W_z. Thus the derivative is $\mathcal{E}_c(X_a)/\mathcal{E}_c(-X_a)$: its polynomial ratio tends to one and its arctangent difference to π, giving $\Lambda_0(1,c) = e^{-4\pi/\sqrt{K}}$. For every C in the rectangle, $\Lambda_0(1,C) = e^{-4\pi/\sqrt{C-1}}$, so differentiation gives $\partial_C \Lambda_0(1,c) = 2\pi K^{-3/2} e^{-4\pi/\sqrt{K}}$. □

6.1.5 A single calibration and three simple actual roots

Equation (6.7) gives $\Lambda_0(1,c) = m_\varepsilon$ exactly. Consider the two equations

$$G_\rho(A,C) = \big(\Delta_\rho - c_{+,\rho}c_\varepsilon,\ \Lambda_\rho - m_\varepsilon\big) = 0. \tag{6.24}$$

By (6.18), $c_{+,\rho} \to 0$ together with its parameter derivatives. Thus $G_\rho \to G_0$ in $C^1_{A,C}$ and $G_0(1,c) = 0$. Lemma 6.6 gives

$$\det DG_0(1,c) = (\partial_A \Delta_0)(\partial_C \Lambda_0) > 0.$$

On a small fixed ball in the previous buffered domains with $C > 1$, C^1 convergence makes $(A,C) \mapsto (A,C) - DG_0(1,c)^{-1}G_\rho(A,C)$ a contraction into itself for small ρ. The resulting coefficient pair satisfies

$$(A_\rho, C_\rho) \longrightarrow (1,c). \tag{6.25}$$

The calibrated pair is continuous in $\rho > 0$, since all actual hits are transverse. No estimate uniform in $c \downarrow 1$ is needed for this inverse.

Evaluate all maps at this one coefficient pair. Normalize $E_\rho \circ T_{-,\rho}$ by its exact center and slope:

$$E_\rho(T_{-,\rho}(b)) = E_\rho(a_{-,\rho}) + E_\rho'(a_{-,\rho})c_{-,\rho}\widehat{\phi}_{-,\rho}(b).$$

Here $\widehat{\phi}_{-,\rho} = N_{c_{-,\rho}} \circ \phi_{-,\rho}$, with $\psi = E_\rho$ and $z_0 = a_{-,\rho}$ in (3.6). The buffered C^3 bounds, $c_{-,\rho} = O(\rho^2)$, and (6.18) give $\widehat{\phi}_{-,\rho}(0) = 0$ and C^1_b convergence to the identity.

The return equation is $T_{+,\rho}(b) = E_\rho(T_{-,\rho}(S_{\varepsilon,\rho}(b)))$. Dividing its difference by the positive $c_{+,\rho}$ and using (6.24) gives exactly

$$D_\rho(b) = c_\varepsilon + \phi_{+,\rho}(b) - m_\varepsilon \widehat{\phi}_{-,\rho}(S_{\varepsilon,\rho}(b)). \tag{6.26}$$

For $D_\varepsilon = c_\varepsilon + b - m_\varepsilon S_\varepsilon(b)$, the core bound, (6.20), and the buffered finite connector give, on the common enlarged interval and uniformly in the coefficient ball,

$$\|D_\rho - D_\varepsilon\|_{C^1_b} \le C_\varepsilon \rho(1 + |\log \rho|). \tag{6.27}$$

This remains true at (6.25). Keep $\varepsilon = \varepsilon(c)$ fixed and put $\mathcal{Q}_\rho(r) = \varepsilon^{-5/2} D_\rho(b_\varepsilon(r))$. The chain rule gives the explicit estimate

$$\|\mathcal{Q}_\rho - Q_\varepsilon\|_{C^1(J)} \le \varepsilon^{-5/2} \max(1, \|b'_\varepsilon\|_{C^0(J)}) \, \|D_\rho - D_\varepsilon\|_{C^1_b} \longrightarrow 0. \tag{6.28}$$

All coefficients, including A_ρ, C_ρ, are fixed in this r derivative. For the fixed ε, choose the same ρ small enough to preserve the three endpoint sign pairs and the derivative lower bound of Lemma 6.3. There is exactly one root $r_{j,\rho}$ in each U_j, and $|\mathcal{Q}'_\rho(r_{j,\rho})| \ge \kappa/2$.

Each root closes one actual circuit. The central passage and both tails have strictly decreasing v in positive time. For each $\rho > 0$ all flight times are finite: on the thin part $-\xi' \ge \rho p_\varepsilon$, the fast parts are compact with nonzero speed, and the outward tail speed is at least a positive constant times q^2 with $q \ge \rho L > 0$. The finite connection lies in a compact affine tube whose spherical distance from q is positive. The inner section $\xi = L$ tends to q, so for small ρ it has no further hit during the circuit. Thus the three roots give three different prime periodic orbits.

They are hyperbolic in the actual raw input coordinate. Indeed $h(r) = T_{+,\rho}(b_\varepsilon(r))$ is positive and

$$h_r = c_{+,\rho} \phi'_{+,\rho} b'_\varepsilon(r) < 0, \qquad h_r \ne 0,$$

by (6.2). If R_ρ is the first return and $F_\rho(h) = h - R_\rho(h)$, then the exact identity

$$F_\rho(h(r)) = c_{+,\rho} \varepsilon^{5/2} \mathcal{Q}_\rho(r)$$

gives, at each root,

$$F_{\rho,h} = \frac{\varepsilon^{5/2} \mathcal{Q}'_\rho}{\phi'_{+,\rho} b'_\varepsilon} \ne 0. \tag{6.29}$$

The common gain $c_{+,\rho}$ cancels; positivity of the first-return derivative and $R'_\rho \ne 1$ prove hyperbolicity.

6.1.6 Whole-orbit convergence

The three roots remain in the same compact phase intervals. On the central passage $|\xi| \le L$ and $0 < \eta \le C_{\varepsilon,L}$. At $\xi = 0$ the thin orbit has $Q \le 0$, hence all three satisfy

$$0 < z(\gamma) = \rho^2 \eta(0) \le \rho^2 h_\rho = (k\varepsilon^2)^{12} \rho^{14} \longrightarrow 0. \tag{6.30}$$

Fix $0 < d_0 < d$. On $d_0 \le |v - \gamma| \le d$, both tails converge uniformly to their organizing ends, by Lemma 6.5 and (6.25). The finite connection converges uniformly, in affine coordinates and finite time, to the corresponding organizing arc. Each of these arcs is parameterized over a full common interval, with endpoints converging to the corresponding organizing endpoints. Consequently both directed Hausdorff distances tend to zero on these pieces.

The remaining tail pieces have $z \le C|v - \gamma|^2$ and $|v| \le d_0 + O(\rho)$. The central piece has $|v - \gamma| \le \rho L$ and $z \le C\rho^2$. For these pieces $Y > 0$, and the true spherical embedding is

$$\iota_0(X, Y) = \frac{(v, 1, z)}{\sqrt{1 + v^2 + z^2}}.$$

Their distance to q is at most $C(d_0 + \rho)$. The omitted organizing ends have distance at most Cd_0 to q. Conversely, (6.30) provides an actual point tending to q on each of the three cycles, so every omitted organizing point is within $Cd_0 + o(1)$ of that cycle. First let $\rho \downarrow 0$, then $d_0 \downarrow 0$. This proves simultaneous whole-orbit Hausdorff convergence and Theorem 6.2. The finite part containing $Y = 0$ uses the affine embedding throughout.

Corollary 6.7 (Stability order)**.** *For each fixed $c \in (1, 1+K_0)$ and all sufficiently small $\rho > 0$, order the three constructed cycles of $F_{c,\rho}$ by their roots $r_{1,\rho} < r_{2,\rho} < r_{3,\rho}$. The first and third cycles are attracting, and the second is repelling.*

Proof. The roots of Φ, in increasing r order, cross downward, upward and downward. The two C^1 estimates (6.3) and (6.28) preserve these signs for $\mathcal{Q}'_\rho$. Since $b'_\varepsilon < 0$ and $\phi'_{+,\rho} > 0$, identity (6.29) gives signs $+,-,+$ for $1 - R'_\rho$. Positivity of the actual first-return derivative proves the stated stability order. □

Corollary 6.8 (Limits of the three return multipliers)**.** *Let $r_1^0 < r_2^0 < r_3^0$ be the three zeros of Φ and let $M_j(c,\rho) > 0$ be the derivative of the positive first return at the cycle $\gamma_{j,c,\rho}$, ordered by increasing r. For $\varepsilon = \varepsilon(c)$, let $r_{j,\varepsilon}$ be the zero of Q_ε in U_j and put $\mu_j(c) = m_\varepsilon S'_\varepsilon(b_\varepsilon(r_{j,\varepsilon})) > 0$. Then*

$$\begin{gathered} \lim_{\rho\downarrow 0} M_j(c,\rho) = \mu_j(c), \qquad \lim_{c\downarrow 1} \mu_j(c) = L(r_j^0), \\ L(r) = 8\sqrt{2}\, r^{5/2}\sqrt{1-r}. \end{gathered} \tag{6.31}$$

The limiting multipliers satisfy

$$0 < L(r_1^0), L(r_3^0) < \tfrac{1}{2}, \qquad \tfrac{3}{2} < L(r_2^0) < 2. \tag{6.32}$$

Proof. For fixed c and $\varepsilon = \varepsilon(c)$, (6.28) gives $r_{j,\rho} \to r_{j,\varepsilon}$. Evaluate all maps at the calibrated coefficients (6.25). In the actual raw input coordinate $h = T_{+,\rho}(b)$, the first return is

$$R_\rho = E_\rho \circ T_{-,\rho} \circ S_{\varepsilon,\rho} \circ T_{+,\rho}^{-1}.$$

At $b = b_\varepsilon(r_{j,\rho})$, its derivative is exactly

$$\begin{aligned} M_j(c,\rho) &= \frac{E'_\rho(T_{-,\rho}(S_{\varepsilon,\rho}(b)))T'_{-,\rho}(S_{\varepsilon,\rho}(b))S'_{\varepsilon,\rho}(b)}{T'_{+,\rho}(b)} \\ &= m_\varepsilon \frac{\widehat{\phi}'_{-,\rho}(S_{\varepsilon,\rho}(b))S'_{\varepsilon,\rho}(b)}{\phi'_{+,\rho}(b)} > 0. \end{aligned} \tag{6.33}$$

The second equality uses the exact calibration $\Lambda_\rho = m_\varepsilon$. The core and tail C^1 limits, including $\widehat{\phi}_{-,\rho} \to \mathrm{id}$, give the first limit in (6.31).

All derivatives keep the coefficients fixed. From $u_\varepsilon = \varepsilon - \varepsilon^{3/2} d_\varepsilon$ and $S_\varepsilon(b_\varepsilon(r)) = -r^2/2$ we obtain

$$\begin{gathered} b'_\varepsilon(r) = \varepsilon^{5/2}(1 - \sqrt{\varepsilon}\, d_\varepsilon(r))d'_\varepsilon(r), \\ m_\varepsilon S'_\varepsilon(b_\varepsilon(r)) = \frac{-2r}{(1-\sqrt{\varepsilon}\, d_\varepsilon(r))d'_\varepsilon(r)} \longrightarrow L(r) \quad \text{uniformly on } J. \end{gathered} \tag{6.34}$$

Indeed (6.5) gives $d_\varepsilon \to d_0$ in $C^1(J)$ and $d'_0(r) = -1/(4\sqrt{2}\, r^{3/2}\sqrt{1-r})$ is bounded away from zero there. By (6.3), $r_{j,\varepsilon} \to r_j^0$. Together with $\varepsilon(c) \to 0$, this proves (6.31).

The bounds needed for (6.32) follow from exact rational brackets. The sign of $\Phi(r)$ on $(0,1)$ is the sign of

$$\frac{1-r}{8r} - \left(\frac{101}{100} - r^2\right)^2 = -\frac{10000r^5 - 20200r^3 + 11451r - 1250}{10000r}.$$

Direct substitution gives the endpoint signs in the following table. Since $L(r)^2 = 128r^5(1-r)$ increases up to $r = 5/6$ and decreases afterwards, rational endpoint evaluation also gives its indicated bounds throughout each bracket. The endpoints have common denominator 10^5:

j	$(10^5 a_j, 10^5 b_j)$	$(\operatorname{sgn}\Phi(a_j), \operatorname{sgn}\Phi(b_j))$	$L(r)^2$ $(a_j \le r \le b_j)$
1	$(11159, 11160)$	$(+,-)$	$(1/1000, 1/400)$
2	$(97868, 97869)$	$(-,+)$	$(12/5, 5/2)$
3	$(99875, 99876)$	$(+,-)$	$(3/20, 1/5)$

Each bracket contains r_j^0, since Φ has exactly three roots. Taking positive square roots gives (6.32). □

6.2 Endpoint attainment

Lemma 6.9 (Endpoint attainment)**.** *There are fields F_n in (2.1), with $F_n \to H_0$ in quadratic coefficients, and three geometrically distinct hyperbolic periodic orbits $\gamma_{j,n}$ of each F_n, $j = 1, 2, 3$, such that*

$$\max_{1\le j\le 3} d_{\mathrm{H}}(\iota_0(\gamma_{j,n}), \Gamma) \longrightarrow 0.$$

For every fixed choice of admissible windows in Lemma 2.5, all three orbits eventually follow the same marked positive-time circuit $q \to p \to q$, meeting its return section once in their prime periods. Their positive first-return multipliers $M_{j,n}$ satisfy

$$M_{j,n} \longrightarrow L(r_j^0), \qquad j = 1, 2, 3,$$

with the notation of Corollary 6.8.

Proof. Set $c_n = 1 + K_0/(2n)$, with K_0 from Theorem 6.2. For each n, first fix c_n, then choose $0 < \rho_n \le 1/n$ so that the theorem applies and the same field $F_n := F_{c_n,\rho_n}$ and its three hyperbolic orbits $\gamma_{j,n} := \gamma_{j,c_n,\rho_n}$ satisfy

$$\begin{gathered} \|F_n - H_{c_n}\| < \frac{1}{n}, \qquad \max_{1\le j\le 3} d_{\mathrm{H}}\big(\iota_0(\gamma_{j,n}), \Gamma_{c_n}\big) < \frac{1}{n}, \\ \max_{1\le j\le 3} |M_j(c_n, \rho_n) - \mu_j(c_n)| < \frac{1}{n}. \end{gathered} \tag{6.35}$$

This choice may depend on c_n and controls the actual coefficient $C_{c_n,\rho_n} - c_n$. The last condition is available by Corollary 6.8 after c_n is fixed; its second limit gives the asserted convergence of $M_{j,n} = M_j(c_n, \rho_n)$ on this same diagonal. Continuity of H_c gives

$$\|F_n - H_0\| \le \frac{1}{n} + \|H_{c_n} - H_1\| \longrightarrow 0. \tag{6.36}$$

Pair the organizing parabolas at the same X; their heights differ by $(c_n - 1)/2$. Bounded $D\iota_0$ and comparability of the fixed metric to the spherical metric give one constant M_Γ with

$$d_{\mathrm{H}}(\Gamma_{c_n}, \Gamma) \le M_\Gamma(c_n - 1) \longrightarrow 0. \tag{6.37}$$

Both closures have the same endpoint $q = (0, 1, 0)$ and no equatorial arc. The triangle inequality gives

$$\max_{1\le j\le 3} d_{\mathrm{H}}(\iota_0(\gamma_{j,n}), \Gamma) \le \frac{1}{n} + M_\Gamma(c_n - 1) \longrightarrow 0. \tag{6.38}$$

The same pairing gives convergence of the marked parts $X \le -1$ and $X \ge -1$, whose cut is $p_{c_n} = (-1, (1 - c_n)/2) \to (-1, 0)$. The positive speed (1.12) orders them as $q \to p_{c_n} \to q$. For any fixed admissible windows, speed bounds, and inner height, (6.36) and (6.38) admit all three cycles to Lemma 2.5 for all sufficiently large n. It gives their marked circuit, transverse hits, and prime return. □

Proof of Theorem 6.1. The inverse coefficient isomorphism $\mathscr{C}_0^{-1}$ from (1.7) transports the fields of Lemma 6.9 to fields converging to F_*, hence eventually into $\mathcal{V}$. The projective inverse S^{-1} in (1.8) is Lipschitz on the compact hemisphere, with constant L_-, and sends Γ to Γ_{I4}. The transported cycles therefore have distances at most

$$L_- \max_{1\le j\le 3} d_{\mathrm{H}}\big(\iota_0(\gamma_{j,n}), \Gamma\big),$$

Choose n so that this is less than δ, the field lies in $\mathcal{V}$, and capture holds. The fixed coordinate change and positive time factor preserve hyperbolicity, distinctness, and the marked order $q_* \to p_* \to q_*$. □

The strengthened diagonal in (6.35) realizes Theorem 6.1 with first-return multipliers converging to $L(r_1^0), L(r_2^0), L(r_3^0)$. For all sufficiently large n, these multipliers lie in $(0, 1/2), (3/2, 2), (0, 1/2)$, respectively. The fixed coordinate change and positive time factor preserve the return multipliers, giving attracting, repelling and attracting cycles in order of increasing r.

6.3 The coefficient face of the same-field triples

The construction also identifies how the three cycles enter the small positive ratio sector. Its calibration varies A, C and preserves the three coefficients prescribed in (6.8). We compute the ratio directly from those coefficients, independently of the upper estimates.

Proposition 6.10. *The fields in Lemma 6.9 can be chosen with*

$$\alpha_0 < 0 < \alpha_1, \qquad p \longrightarrow 0^+, \qquad \gamma/s \longrightarrow +\infty.$$

In their positive-gamma coordinates $g = \gamma$, $k = s/g^{3/2}$, $\pi = p/\sqrt{g}$ and $\mathfrak{q} = k\pi$, they satisfy

$$k \longrightarrow 0, \qquad \pi \longrightarrow 5/\sqrt{2}, \qquad \mathfrak{q} \longrightarrow 0.$$

Their two positive axis roots in the coordinate $(v - \gamma)/(gk)$ have squared half-gap tending to $1/72$. In particular they remain on the bounded positive-gamma real-root side, separated from the fold. The same field and its three complete hyperbolic cycles meet every prescribed positive upper cutoff on p.

Proof. For each fixed $c > 1$, write

$$\epsilon = \left(\tfrac{1}{2} e^{-4\pi/\sqrt{c-1}}\right)^{2/5}, \qquad \beta = 1 - \tfrac{5}{2}\epsilon.$$

Restrict the fixed c-range so that $0 < \epsilon < 1/10$. Equation (6.8), with its fixed root ratio $3/2$, gives

$$\gamma = \frac{\beta\rho}{3}, \qquad \alpha_0 = -\frac{3}{2}\epsilon^2\rho^3, \qquad \alpha_1 = \left(\frac{5}{2}\epsilon - \frac{3}{2}\epsilon^2\right)\rho^2. \tag{6.39}$$

These are the actual coefficients after calibration: the equations (6.24) change only A, C. Moreover a field H_θ already in the slice has, by local uniqueness in Lemma 2.1, slice parameter $\theta(H_\theta) = \theta$ and affine/time parameters $(I, 0, 1)$. Consequently after the fixed inverse transport $\mathscr{C}_0^{-1}$ these same coefficients define (1.15) for the original field.

Taking the square root and dividing in (6.39) gives

$$s = \sqrt{3/2}\,\epsilon\rho^{3/2}, \qquad p = \frac{5 - 3\epsilon}{\sqrt{6}}\sqrt{\rho}, \qquad \frac{\gamma}{s} = \frac{\beta}{3\sqrt{3/2}\,\epsilon\sqrt{\rho}}. \tag{6.40}$$

For $c_n \downarrow 1$, the endpoint construction first fixes c_n, then chooses one $\rho_n \le 1/n$ for the actual coefficient difference and all three whole-cycle distances. Thus $0 < p_n \le 5/\sqrt{6n} \to 0$, while $\gamma_n/s_n \to +\infty$. The actual coefficient C_{c_n,ρ_n} tends to c_n with the chosen accuracy; it need not equal c_n. No estimate for the inverse calibration uniform as $c \downarrow 1$ is used.

The gamma coordinates computed from the same path are

$$k = \frac{9\epsilon}{\sqrt{2}\,\beta^{3/2}}, \qquad \pi = \frac{5 - 3\epsilon}{\sqrt{2\beta}}, \qquad \mathfrak{q} = \frac{9\epsilon(5 - 3\epsilon)}{2\beta^2}. \tag{6.41}$$

The translated roots in (6.8) are $-\rho, \epsilon\rho, 3\epsilon\rho/2$. Dividing by $gk = 3\epsilon\rho/\sqrt{2\beta}$ gives

$$-r = -\frac{\sqrt{2\beta}}{3\epsilon}, \qquad a = \frac{\sqrt{2\beta}}{3}, \qquad b = \sqrt{\beta/2}.$$

Hence

$$\frac{(b-a)^2}{4} = \frac{\beta}{72} \longrightarrow \frac{1}{72}, \qquad kr = \frac{3}{\beta} \longrightarrow 3. \tag{6.42}$$

In particular $\pi \to 5/\sqrt{2} > \sqrt{12}$, on the separated real-root side, with an escaping negative root at finite further radius. These formulas describe this constructed family; they impose no gamma-chart limit on other three-cycle sequences.

For any neighborhood, collar and positive ratio cutoff, take a sufficiently large n in this same sequence. Coefficient convergence, the simultaneous Hausdorff limits and $p_n \to 0^+$ meet all three requirements at once. The fixed inverse coordinate change and positive time preserve the three hyperbolic cycles of that single field. □

The one-cycle estimate at $p = 0$ and these strictly positive-ratio triples concern different coefficient limits. The realization passes through the unbounded γ/s branch. A bounded-γ/s restriction would remove this sharpness family.

7 The sharp small positive ratio sector

The estimate down to $p = 0$ uses the coefficient cover and original physical passages directly. It is not obtained by sending the lower endpoint of a compact-positive-ratio estimate to zero. The preceding lower construction supplies its sharpness independently.

Proof of Theorem 1.2. Use Proposition 5.16 on the common receiver of Lemma 5.14, with $h_c = b/2$. Its fixed reserve (5.13) gives $0 < p_{\mathrm{s}} \le p_*$ for which every $0 < p \le p_{\mathrm{s}}$ is covered before the independent raw radii and final multiplier are restricted. The proposition counts the full connected interval with multiplicity, including both original ghost tails; no further regional count is needed.

At $p = 0$, $\alpha_1 = 0$. Lemma H.15 bounds the actual transition Jacobian on the full original domain for $\alpha_0 \le 0, \alpha_1 \le 0$. With the divided closing and one further multiplier shrink, $F' > 1/2$, so there is at most one isolated zero and it is simple. These are fixed multiplier bounds; (2.12) makes the actual m satisfy them by an ordinary coefficient shrink, independently of s, γ and all root distances.

Apply geometric capture with these fixed choices, as in Proposition 5.17. This gives one slice neighborhood and one collar for both statements. Use Lemma 2.6 with $\mathcal{B} = \{\alpha_0 < 0,\ 0 \le p \le p_{\mathrm{s}}\}$, and intersect with its one-cycle version on $p = 0$ if necessary. The resulting full-germ neighborhood and collar are common to the whole small-ratio sector. The stronger return-multiplicity assertion is the slice count above, preserved at the corresponding physical sections.

Finally Proposition 6.10 supplies same-field hyperbolic triples with coefficient and whole-orbit Hausdorff convergence and $p_n \to 0^+$. Given $\mathcal{W}, \delta, \eta$, choose one sufficiently large n so that its field lies in $\mathcal{W}$, all three distances are below δ, and $0 < p_n < \min\{p_{\mathrm{s}}, \eta\}$. These are the three cycles calibrated by one coefficient pair in Theorem 6.2, proving the sharpness clause. □

7.1 Transfer to the full quadratic germ

Proof of Theorem 1.1. Proposition 5.17 supplies one slice neighborhood and one complete-graphic collar with at most three isolated cycles. Apply Lemma 2.6 with the whole slice parameter space and $N_{\mathrm{cyc}} = 3$. This gives the stated upper bound on one ordinary full

twelve-coefficient neighborhood and collar. Proposition 6.10, independently of that count, gives one field with three hyperbolic complete cycles in every smaller neighborhood and collar. Together these prove exact cyclicity three with the asserted marking and quantifiers. □

The counting argument uses one ordered physical interval, a common orientation of curvature zeros or displacement critical points, and a bounded complete derivative on its complement. The family-specific inputs are the cubic cancellation (D.22), relative division (D.12), and cancellation of the two physical tail gains in (C.24). The root and degree-loss estimates establish these inputs where the ordinary through-domain margins vanish. The finite coefficient cover then gives one neighborhood after fixing the raw cap and overlap reserves.

A Scalar transitions and physical section calculus

The regional passages require estimates on normalized scalar maps, on derivatives with moving initial sections, and on local Dulac maps. We develop those tools here. Regular adapters and divided closing maps are handled by the identities of Section 3.

A.1 Normalized scalar transitions

Subtracting a reference orbit and dividing by its first variation removes the affine part of a scalar transition. The remaining equation vanishes to second order in the normalized phase variable. An integrable bound on that equation gives uniform endpoint maps even when the unnormalized first variation has no uniform bound.

Lemma A.1 (Normalized scalar transitions)**.** *Let $k \geq 1$. Consider equations*

$$\eta_r = N_\vartheta(r,\eta), \qquad L \leq r \leq R_\vartheta, \qquad N_\vartheta(r,0) = \partial_\eta N_\vartheta(r,0) = 0,$$

on $|\eta| \leq a$, where L and $a > 0$ are fixed. Suppose that, uniformly in the parameters,

$$\|N_\vartheta(r,\cdot)\|_{C^{k+1}([-a,a])} \leq g(r), \qquad g \in L^1([L,\infty)).$$

Then there is a fixed neighborhood of zero on which all endpoint maps have bounded C^{k+1} norms and positive first derivatives bounded away from zero. Their inverses have the same bounds on a fixed smaller neighborhood of zero.

Suppose also that $R_\vartheta \to \infty$ and the kernels converge in $C^0([L,M];C^{k+1}_\eta)$ for every fixed $M > L$. Then the endpoint maps and their inverses converge in C^{k+1} on smaller neighborhoods. For a chosen parameter derivative ∂_ℓ, convergence in C^k holds for the differentiated maps if the differentiated kernels converge in $C^0([L,M];C^k_\eta)$, have a common integrable majorant, and

$$\|N_\vartheta(R_\vartheta,\cdot)\partial_\ell R_\vartheta\|_{C^k} \longrightarrow 0.$$

The convergence is joint in an additional compact parameter family if these local convergences, $R_\vartheta \to \infty$, and the vanishing endpoint term are uniform in that family.

Proof. Taylor's formula gives $|N_\vartheta(r,\eta)| \leq g(r)|\eta|^2/2$. Comparison with $y' = g(r)y^2/2$ keeps all solutions from a common enlarged initial interval inside $|\eta| < a$. The first variation is

$$\exp\left(\int_L^r \partial_\eta N_\vartheta(t,\eta(t))\,dt\right) > 0.$$

It and its reciprocal are bounded. Induction in the triangular variational equations and Gronwall's inequality bound all remaining rows through order $k+1$. The slope lower bound gives a common image interval, and the inverse identity gives the inverse bounds.

Subtracting the integral equations and their variations gives convergence on $[L, M]$; each remaining tail costs at most $C \int_M^\infty g$. Let the parameters tend to their limit and then $M \to \infty$. For the parameter derivative, the variation satisfies

$$Z_r = \partial_\eta N_\vartheta \, Z + \partial_\ell N_\vartheta, \qquad Z(L) = 0.$$

The endpoint derivative adds $N_\vartheta(R_\vartheta, \eta(R_\vartheta))\partial_\ell R_\vartheta$. The same argument proves C^k convergence under the endpoint assumption: differentiating a coefficient $N_{\eta^h}(r, \eta)$ adds $N_{\eta^{h+1}}(r, \eta)Z$, which requires the extra phase order. Finally, C^{k+1} convergence of the endpoint maps A_ϑ and

$$\partial_\ell A_\vartheta^{-1} = -\left(\frac{\partial_\ell A_\vartheta}{A'_\vartheta}\right) \circ A_\vartheta^{-1}$$

give C^k convergence of the differentiated inverses. Each estimate is uniform in the additional compact family under the stated assumptions. □

For a graph equation $Y_r = F(r, Y)$, a reference solution Y_* and its first variation $j > 0$, the substitution $\eta = (Y - Y_*)/j$ gives the exact kernel

$$N(r, \eta) = j\eta^2 \int_0^1 (1 - t) F_{YY}(r, Y_* + tj\eta)\, dt. \tag{A.1}$$

For $h \geq 2$, $\partial_\eta^h N = j^{h-1} F_{Y^h}(r, Y_* + j\eta)$. At fixed r, η, a parameter derivative is therefore

$$\begin{aligned} \partial_\ell \partial_\eta^h N = j^{h-1}\{&(h-1)(\partial_\ell \log j) F_{Y^h} + \partial_\ell^{\exp} F_{Y^h} \\ &+ F_{Y^{h+1}}(Y_{*,\ell} + j_\ell \eta)\}. \end{aligned} \tag{A.2}$$

Here exp holds the graph coordinates fixed. The orders zero and one follow by integrating in η, since $N(r, 0) = N_\eta(r, 0) = 0$. Thus C^3-valued parameter derivatives require F_{Y^4} and N_{η^4}. Integrating the indicated parameter derivative gives the sequential parameter secants used below. This requires neither a mixed parameter derivative nor convergence of j.

A quantitative form controls how close the normalized map is to the identity when only an upper bound on the first variation is available. For an actual graph family write $J_n = \partial_b^n Y$, $Q_n = J_n/J_1$, and $a_n = F_{Y^n} J_1^{n-1}$. Direct differentiation gives

$$Q_2' = a_2, \qquad Q_3' = 3a_2 Q_2 + a_3, \qquad Q_4' = a_2(4Q_3 + 3Q_2^2) + 6a_3 Q_2 + a_4. \tag{A.3}$$

If, uniformly on an enlarged label interval, $Q_n(L) = O(\varepsilon)$ and $\int |a_n|\, dr = O(\varepsilon)$, $2 \leq n \leq 4$, these identities give $Q_n = O(\varepsilon)$. Since $\partial_b \log J_1 = Q_2$, the endpoint map in $b - b_*$, normalized by its exact value and slope at b_*, and its inverse on a smaller common interval, are $\mathrm{id} + O_{C^3}(\varepsilon)$ for sufficiently small ε. This conclusion is independent of the size of $J_1(b_*) > 0$.

A.2 Moving-section variations and inward continuation

Moving a section changes both the initial height and the time of departure. The following identities retain both effects, including the endpoint terms that appear after using height as the independent variable.

Lemma A.2 (Scalar differentiation on moving sections)**.** *Fix the coefficients of a scalar graph $z_x = f(x, z)$, with initial curve $(x_l(a), z_l(a))$, and let D be a phase vector field, commuting with differentiation at fixed x. Write $V_j = D^j z$. Through order three its equations and moving initial rows are*

$$\begin{gathered} (V_1)_x = f_z V_1, \qquad (V_2)_x = f_z V_2 + f_{zz} V_1^2, \\ (V_3)_x = f_z V_3 + 3 f_{zz} V_1 V_2 + f_{zzz} V_1^3, \\ B_1 = Dz_l - f_l Dx_l, \quad B_2 = DB_1 - f_{z,l} B_1 Dx_l, \\ B_3 = DB_2 - (f_{z,l} B_2 + f_{zz,l} B_1^2) Dx_l, \qquad V_j(x_l; a) = B_j. \end{gathered} \tag{A.4}$$

For $f = z(n+\zeta z)/(M+Qz)$ on a positive graph with positive denominator, put $\mathscr{D} = M + Qz$. The complete fixed-port first derivative, including an observer χ, is

$$\begin{aligned} f_z &= \frac{d}{dx}\log\frac{z}{\mathscr{D}} + \frac{M' + (Q'+\zeta)z}{\mathscr{D}}, \\ D\chi(z_T) &= \chi'(z_T)B_1\frac{(z/\mathscr{D})_T}{(z/\mathscr{D})_l}\exp\int_{x_l}^{x_T}\frac{M' + (Q'+\zeta)z}{\mathscr{D}}\,dx. \end{aligned} \tag{A.5}$$

For $z_l = \ell_0 M(a)$, $n = xB$, and $b = -a^2/(2\sigma^2)$, this becomes

$$\begin{aligned} \Gamma_b' &= \sigma^2\chi'(z_T)\frac{z_T}{\mathscr{D}_T}B(a)K(a)e^{I(a)}, \quad I = \int_a^{x_T}\frac{M' + (Q'+\zeta)z}{\mathscr{D}}\,dx, \\ K &= 1 - \frac{M'(a)}{aB(a)}(1+\ell_0 Q(a)) + \frac{\ell_0\zeta(a)M(a)}{aB(a)}. \end{aligned} \tag{A.6}$$

If instead $r_\eta = V(r,\eta)$, with moving data (η_l, r_l), write $P_j = D^j r|_\eta$, $l_j = D^j\eta_l$, $t_j = D^j\eta_T$, where r_T is fixed. Then

$$\begin{aligned} P_{1,l} &= Dr_l - V_l l_1, \quad P_{2,l} = DP_{1,l} - V_{r,l}P_{1,l}l_1, \\ (P_1)_\eta &= V_r P_1, \quad (P_2)_\eta = V_r P_2 + V_{rr}P_1^2, \\ t_1 &= -P_{1,T}/V_T, \quad t_2 = -\{P_{2,T} + 2V_{r,T}P_{1,T}t_1 + (V_{r,T}V_T + V_{\eta,T})t_1^2\}/V_T. \end{aligned} \tag{A.7}$$

For $I = \int_{\eta_l}^{\eta_T} J(r(a,\eta),\eta)\,d\eta$, its full rows are

$$\begin{aligned} DI &= J_T t_1 - J_l l_1 + \int J_r P_1\,d\eta, \\ D^2 I &= \int (J_{rr}P_1^2 + J_r P_2)\,d\eta + (J_r P_1)_T t_1 - (J_r P_1)_l l_1 \\ &\quad + J_T t_2 + J_{\eta,T}t_1^2 - J_l l_2 - (J_{r,l}Dr_l + J_{\eta,l}l_1)l_1. \end{aligned} \tag{A.8}$$

The observer and divided-closing rows use Lemma 3.3, specifically (3.3) *and* (3.5)*. Coefficient-only gains and affine calibrations are frozen under D.*

Proof. Differentiate the graph equation at fixed x; differentiating $V_j(x_l(a),a) = B_j(a)$ gives the initial recursion. Rational differentiation proves (A.5). For the inverse graph differentiate twice at fixed η, and then twice the identity $r(a,\eta_T(a)) = r_T$. Leibniz differentiation gives (A.8); at the lower endpoint the total radial derivative is Dr_l, whereas at the upper endpoint it is zero. The observer and closing claims follow from the cited ordinary C^3 identities. □

In the inward passages, the ratio of height to the axis polynomial decays on the action clock. The next lemma estimates the resulting integral without discarding the dependence on its terminal clock time.

Lemma A.3 (Stable ratio integrals with moving endpoints)**.** *At fixed action clock q, let a phase field D commute with ∂_q. For $\kappa > 0$, consider*

$$W_q = \mathcal{F} = W\{-(1+\omega W)/(1+QW) + \kappa H\}, \qquad W(0) = \ell_0, \qquad L = (Q-\omega)W/(1+QW).$$

Suppose $Q, \kappa H, \omega$ and their required two clock rows are uniformly bounded, $1+QW$ is a positive unit, and both $\mathcal{F}/W$ and $\mathcal{F}_W$ are at most $-c_0 < 0$ on $0 \le W \le \ell_0$. For some $0 \le \rho \le \rho_0$, assume the first three phase rows of $\kappa H, Q, \omega$, and mixed rows containing a phase derivative of total order at most three, are bounded by $C\rho e^{\alpha q}$, with $\alpha < c_0/12$. There are an integer N and $0 < c < c_0/2$, depending only on the stated bounds, such that

$$0 < W \le \ell_0 e^{-c_0 q}, \qquad |D^j W| + |D^j L| \le C\rho(1+q)^N e^{-cq}, \quad 1 \le j \le 3. \tag{A.9}$$

The needed mixed rows obey these bounds; pure clock rows have the same exponential bound without ρ. For a moving T, put $T_j = D^j T$ and $I_T = \int_0^T L\,dq$. All endpoint contributions are

$$\begin{aligned} DI_T &= \int_0^T DL\,dq + L_T T_1, \\ D^2 I_T &= \int_0^T D^2 L\,dq + 2(DL)_T T_1 + (L_q)_T T_1^2 + L_T T_2, \\ D^3 I_T &= \int_0^T D^3 L\,dq + 3(D^2 L)_T T_1 + 3(DL_q)_T T_1^2 + (L_{qq})_T T_1^3 \\ &\quad + 3(DL)_T T_2 + 3(L_q)_T T_1 T_2 + L_T T_3. \end{aligned} \tag{A.10}$$

If $v_q = -\kappa P(v)/(vB(v))$, $H = P'/(vB)$, and $W = Y/P(v)$, its exact polynomial memory is

$$W(q) = \ell_0 \frac{P(a)}{P(v(q))} e^{-q+I(q)}, \qquad Y_T = \ell_0 P(a) e^{-T+I_T} = P(v_T) W(T). \tag{A.11}$$

The corresponding rows of $W(T)$ are the ordinary chain rule $DW = W_D + W_q T_1$, $D^2 W = W_{DD} + 2W_{Dq} T_1 + W_{qq} T_1^2 + W_q T_2$, and its third row with coefficients $1, 3, 3, 1, 3, 3, 1$. Thus the interior integral contributions are $O(\rho)$; the full moving-limit rows also retain their terminal contributions, which are at most $C(1+T)^N e^{-cT}(1 + \sum_{j=1}^3 |T_j|)^3$. The constants depend only on the stated coefficient bounds, c_0, ℓ_0, ρ_0, independently of the flight length.

Proof. The negative logarithmic rate gives the first-exit bound for W. Differentiate its equation three times. Each row has diagonal $\mathcal{F}_W$; the explicit coefficient forcing is $O(\rho e^{\alpha q} W)$, and other terms are products of lower rows. Successive variation of constants absorbs at most three coefficient growth factors into the fixed exponential decay and proves (A.9). Differentiating these equations in q gives the mixed rows. Leibniz differentiation and the ordinary chain rule give the terminal formulas. Since $\partial_q \log P(v) = -\kappa H$, integrating the ratio equation gives (A.11). □

When the radial route is fixed, the same decay controls ordinary derivatives with respect to the initial height, even for arbitrarily long flights.

Lemma A.4 (Initial-height rows on a frozen ratio route)**.** *Consider the ratio equation and density $\mathcal{F}, L$ of Lemma A.3, with coefficients and radial endpoints fixed. On its full prescribed buffer $0 \le W \le \ell_0$, assume the stated positive-unit denominator, the two negative margins $\mathcal{F}/W, \mathcal{F}_W \le -c_0 < 0$, and bounded $Q, \omega, \kappa H$. Let $W(0; w) = w$, and, for a fixed finite flight $T \ge 0$, put*

$$\mathcal{N}_T(w) = \int_0^T L(q, W(q; w))\,dq, \qquad e_w = w\partial_w.$$

Uniformly in T and $0 \le w \le \ell_0$, the ordinary extension at $w = 0$ satisfies

$$\mathcal{N}_T(0) = 0, \qquad |e_w^j \mathcal{N}_T(w)| \le Cw \quad (0 \le j \le 3), \qquad \sum_{j=1}^3 |\partial_w^j \mathcal{N}_T(w)| \le C.$$

The normalized nonlinear unit $e^{\mathcal{N}_T}$ and its reciprocal have bounded ordinary C^3 norms. In the polynomial-memory setting, $Y_T = Y_0 e^{-T+\mathcal{N}_T(w)}$, with $Y_0 = P(v_0)w$.

Proof. The buffer and negative logarithmic margin give $0 \le W(q; w) \le we^{-c_0 q}$. Rational height differentiation bounds $\mathcal{F}_{WW}, \mathcal{F}_{WWW}$ and the first three height derivatives of L. Apply the triangular equations (A.4) to $V_j = \partial_w^j W$, with $V_1(0) = 1$, $V_2(0) = V_3(0) = 0$. Successive variation of constants gives $|V_j(q)| \le Ce^{-c_0 q}$, $1 \le j \le 3$. Every term in $\partial_w^j L(q, W(q; w))$ contains a decaying variation, so its integral is uniformly bounded. Also $|L| \le CW$ gives $|\mathcal{N}_T| \le Cw$. The ordinary extension at zero and the expansion of $(w\partial_w)^j$ give the weighted rows. Exponentiation and (A.11) prove the final claims. □

After the long inward flight, a short central segment remains. Factoring out its linear propagator gives a nonlinear error controlled by the incoming height, uniformly as that propagator tends to zero.

Lemma A.5 (Compact inward height continuation)**.** *For a frozen coefficient field, suppose the actual equation on a fixed segment, oriented from $b > 0$ to zero, is $Y_v = A(v)Y + K(v,Y)Y^2$, with linear propagator $m(b) = 1$, $0 < m(v) \le 1$. On its prescribed height buffer $0 \le Y \le h_0$, assume the ordinary height derivatives of K through order three are bounded by C/κ_c, for a local parameter $\kappa_c > 0$. For $0 < y \le h_0/2$ and y/κ_c sufficiently small, its actual continuation satisfies*

$$Y(0;y) = m(0)ye^{\mathcal{N}(y)}, \qquad |(y\partial_y)^j \mathcal{N}(y)| \le Cy/\kappa_c, \quad 0 \le j \le 3. \tag{A.12}$$

In composition with $y = y(a)$, with the field, segment and κ_c fixed, put $\Lambda_j = D^j \log y$ and $e = y\partial_y$. The three rows of $\mathcal{N}(y(a))$ are $e\mathcal{N}\,\Lambda_1$, $e^2\mathcal{N}\,\Lambda_1^2 + e\mathcal{N}\,\Lambda_2$, and $e^3\mathcal{N}\,\Lambda_1^3 + 3e^2\mathcal{N}\,\Lambda_1\Lambda_2 + e\mathcal{N}\,\Lambda_3$. No reciprocal bound on $m(0)$ is used. Constants depend only on the stated height bounds and fixed segment, independently of the extracted multiplier.

Proof. Write $Y = m(v)yu(v;y)$. The integral equation for u, with $u(b) = 1$, has coefficients and its first three Euler forcings bounded by Cy/κ_c on $1/2 \le u \le 2$. First exit and three triangular integral estimates give $u^{\pm 1} \le C$ and $|e^j \log u(0;y)| \le Cy/\kappa_c$. The displayed composition rows follow by the chain rule. □

A.3 An exterior spectral estimate

The outward cubic, quadratic and degree-loss passages share a scalar graph equation. The quadratic coefficient determines a power in its first variation, while the remaining terms control the error in its logarithmic derivatives. We use logarithmic height as the independent variable to estimate these derivatives directly, including the moving endpoints.

Lemma A.6 (Exterior spectral estimate)**.** *Fix a terminal cut $d > 0$, a small lip constant $\ell_0 > 0$, compact bounds on A, C, and positive constants c_0, C_0. Put*

$$M = tx^3 + \mu_2 x^2 + \mu_1 x + \mu_0, \qquad B = 1 + b + tx,$$
$$Q = 1 - A(v + tx) + C(v + tx)^2, \quad \zeta = tC(v + tx), \quad |t| \le 1.$$

All coefficients are fixed under $E = a\partial_a$. On an enlarged open interval of $0 < a < d/2$, suppose, pointwise in a, that some $0 < \rho \le \rho_0$ satisfies

$$\begin{aligned} &c_0\rho \le m(x) := M(x)/x^2 \le C_0\rho \quad (a \le x \le 2a), \\ &|b| + |v| + |\mu_2| + |t|a + |\mu_1|/a + |\mu_0|/a^2 \le C_0\rho. \end{aligned} \tag{A.13}$$

For sufficiently small $|t|d, \rho_0$, depending only on these fixed bounds, the graph

$$z_x = \frac{z(xB + \zeta z)}{M + Qz}, \qquad z(a) = \ell_0 M(a) \tag{A.14}$$

reaches $x = d$ with positive height and denominator. An ordinary C^3 observer χ, with $\chi' > 0$, is appended on the open preimage of its prescribed buffered terminal image. Set

$$\nu = \frac{2\mu_2}{1+b}, \quad \lambda = 2\nu, \quad e = 2 - \lambda > 3/2, \quad \Theta = |t|a + |\mu_1|/a + |\mu_0|/a^2, \quad L_\rho = 1 + |\log\rho|,$$

$$H_\chi = \sup\{|z(\log\chi')'| + |z(\log\chi')' + z^2(\log\chi')''|\}, \qquad \mathcal{T} = |t|d + |\mu_2| + |\mu_1|/d + |\mu_0|/d^2 + H_\chi,$$

where the supremum is on that buffer and is finite. For any coefficient-only $\sigma > 0$, use $\beta = -a^2/(2\sigma^2)$ and $\Gamma(\beta) = \chi(z_d)$. Then $\Gamma' > 0$, and

$$|E \log \Gamma' + \lambda| + |E^2 \log \Gamma'| \le C\{\Theta L_\rho + \mathcal{T}(a/d)^e\}. \tag{A.15}$$

The same rows hold after a positive coefficient-only affine calibration. For intrinsic terminal height take $\chi = \mathrm{id}$, so $H_\chi = 0$. No reciprocal bound on σ or bound on $\lambda \log a$ is used.

Proof. Set $W = z/x^2$, $\tau = \log(x/a)$, $N = B + \zeta x W$, and $D = m + QW$. Then $W_\tau = F = W(N/D - 2)$. On $[a, 2a]$, N, Q are positive units and $D \asymp \rho + W$ for bounded W. Below a small fixed w_0, $F \ge cW/(\rho + W)$. Starting at $W_l = \ell_0 m(a)$, the time to reach w_0 is at most

$$C\rho \log \frac{w_0}{\ell_0 c_0 \rho} + C w_0 < \log 2,$$

by choosing w_0 first and then ρ_0. At the boundaries $w_0 < 1/2$, $w_1 = 3/4$, the formula $F(x, w) = 1 - 2w + O_w(\rho)$ makes the strip invariant up to $2a$. Beyond it, $|m(x)| \le |t|d + C\rho \ll w_0^2$ preserves the strip, gives positive-unit D and $F_W \le -c < 0$. Thus the graph reaches d, with $c\rho \le W \le w_1$, even if m changes sign.

The bounded strong graph with $b = v = \mu_j = 0$, keeping t, A, C, is $W_* = 1/2 + O(|t|x)$, by the integral from zero for $xU' = -2U + O(|t|x)$, $U = W - 1/2$. Strip contraction gives

$$|W(d) - W_*(d)| \le C\rho + C(a/d)^c. \tag{A.16}$$

Where this comparison applies, the original buffer margin and the permitted cutoff choices put the terminal height in the observer's domain. Elsewhere we retain the original terminal admission condition.

Put $r = \log x$, $\eta = \log z$. At fixed height, $W_r = -2W$ and $r_\eta = V = D/N$. The raw cocycle (A.5) has the exact density split

$$\begin{aligned} \frac{2m + xm' + (xQ' + \zeta x)W}{N} &= \nu + J, \\ J = \frac{(3 - \nu)tx + \mu_1/x + tx[-A + (3 - \nu)C(v + tx)]W}{N}&. \end{aligned} \tag{A.17}$$

Here $2m + xm' = 3tx + 2\mu_2 + \mu_1/x$; μ_0 cancels only in the density. Let $I = \int_{\eta_l}^{\eta_T} J\, d\eta$, with $\eta_l = \log(\ell_0 M(a))$. The near height length is $O(L_\rho)$; on the strip $d\eta \le C\, d\log x$. Thus every partial integral is bounded by $C\{\Theta L_\rho + |t|d + |\mu_1|/a\}$.

Use (A.7) with $P_j = E^j r|_\eta$, $l_j = E^j \eta_l$, $t_j = E^j \eta_T$, $E r_l = 1$. The polynomial gives bounded l_1, l_2, $V_l, V_{r,l} = O(\rho)$, and $P_{1,l} = 1 + O(\rho) > 0$. For $\mathscr{B} = EW_l - F_l$, the exact first cocycle gives

$$\begin{aligned} P_1 &= \frac{-\mathscr{B}(1 + \ell_0 Q(a))}{\ell_0 N}(a/x)^e \left(\frac{W}{\ell_0 m(a)}\right)^\nu e^I, \\ 0 < P_1 &\le C(a/x)^e, \qquad |P_2| \le C P_1. \end{aligned} \tag{A.18}$$

The initial factor equals $B(a)K(a) > 0$, with K from (A.6). Since $|\nu| \le C\rho$ and $c\rho \le W \le w_1$, its height ratio to power ν is a unit for either sign, proving the first bound without bounding $a^{-\lambda}$. Near the lip, fixed-height differentiation gives $|V_{rr}| \le C(\rho + W)$, whereas $d\eta = N/D\, d\log x$. On the strip all these factors are bounded. Hence

$$|V_{rr}|\, d\eta \le C\, d\log x, \qquad (P_2/P_1)_\eta = V_{rr} P_1.$$

Its integral is bounded by $C \int_a^d (a/x)^e\, d\log x$, proving the second bound. The two fixed-terminal rows then give $|t_1| + |t_2| \le C(a/d)^e$.

Rational differentiation at fixed height gives $|J_r| + |J_{rr}| \le C(|t|x + |\mu_1|/x)$. In the two full Leibniz rows (A.8), the lower terms cost $C\Theta$, the near integrals cost $C\Theta L_\rho$, and

$$\int_{2a}^{d} (|t|x + |\mu_1|/x)(a/x)^e \, d\log x \le C(|t|a + |\mu_1|/a).$$

Every upper term retains $(a/d)^e$. Therefore

$$|EI| + |E^2 I| \le C\{\Theta L_\rho + (|t|d + |\mu_1|/d)(a/d)^e\}. \tag{A.19}$$

The exact polynomial-lip formula becomes

$$\Gamma' = \sigma^2 a^{-\lambda} B(a) K(a) \ell_0^{-\nu} m(a)^{-\nu} \frac{W_T}{m(d) + Q(d) W_T} z_T^\nu \chi'(z_T) e^I. \tag{A.20}$$

Both E rows of $\log(BK)$ are $O(\Theta)$. Also $Em, E^2 m = O(\Theta)$, $m(a) \asymp \rho$, and $|\nu|, \Theta \le C\rho$, so those of $-\nu \log m(a)$ cost $C\Theta$. At fixed d, the first two height-logarithmic rows of $\log[W_T/(m(d) + Q(d)W_T)]$ cost $C|m(d)|$, and those of $\nu \log z_T$ cost $C|\nu|$. The observer rows are precisely those defining H_χ. Their terminal chain rules and the integral bounds prove (A.15), since $a^{-\lambda}$ contributes $-\lambda$ and zero to the first and second Euler rows. □

A.4 Exact identities on moving polynomial sections

For the cubic, quadratic and degree-drop scalings consider

$$\dot\xi = -\epsilon\{P(\xi) + Q(\xi)Y\}, \qquad \dot Y = -Y\{\xi B(\xi) + \zeta(\xi)Y\}, \tag{A.21}$$

where $\epsilon > 0$ and B, Q are positive units, with transverse sections

$$Y = \ell_0 P(\xi), \qquad 0 < \ell_0 \ll 1, \tag{A.22}$$

Put

$$A(\xi) = \int^{\xi} \frac{xB(x)}{P(x)}\, dx, \qquad \mathcal{D} = \frac{P(\xi)}{\xi B(\xi)} \partial_\xi.$$

Lemma A.7 (Exact moving-section identities)**.** *On a positive graph of* (A.21) *with $P > 0$ and $P + QY > 0$, put $W = Y/P$. Then*

$$\frac{d}{d\xi}\{\epsilon \log Y - A(\xi)\} = \frac{W}{1 + QW}\left\{\zeta - Q\frac{\xi B}{P}\right\}. \tag{A.23}$$

If this graph joins the two lips at u and $-r$, normalize $A(0) = 0$ and write $A_+(u) = A(u)$, $A_-(r) = A(-r)$. With $P_+(u) = P(u)$, $P_-(r) = P(-r)$, one has

$$A_+(u) - A_-(r) = \epsilon\{\log P_+(u) - \log P_-(r) + R(u, r)\}. \tag{A.24}$$

where ϵR is the integral of the right-hand side of (A.23) *from u to $-r$ along this graph. The two endpoint polynomials and the whole integral are retained.*

For $f = dY/d\xi$, its nonlinear derivatives and the inward action density $dq = |dA|/\epsilon$, away from $\xi = 0$, are

$$\partial_Y^n f = (-1)^{n-2} n! \frac{PQ^{n-2}(\zeta P - \xi B Q)}{\epsilon(P + QY)^{n+1}}, \quad n \ge 2, \qquad |d\xi| = \frac{\epsilon P}{|\xi| B}\, dq. \tag{A.25}$$

For a moving initial curve $(\xi_l(x), Y_l(x))$, let $Y_T(x)$ be its graph value at a fixed terminal abscissa and let φ_T be a fixed regular terminal adapter. Write J for the scalar first variation. Then

$$\frac{d}{dx}\varphi_T(Y_T(x)) = \varphi_T'(Y_T) J(\xi_T, \xi_l)\{Y_l'(x) - f(\xi_l, Y_l)\xi_l'(x)\}. \tag{A.26}$$

In the label $b = -x^2/2$, this derivative is multiplied by $-1/x$. Coefficient-only affine calibrations leave logarithmic slope derivatives unchanged. The quantitative bounds for the actual polynomial passages are established separately below and in Appendix J.

Proof. Division of the graph equation and integration give the first two identities; rational differentiation gives (A.25). The initial variation at fixed abscissa is $Y_l' - f_l\xi_l'$; its scalar variational equation gives (A.26). The remaining statements are the ordinary chain rule. □

A.5 A compact-spectrum local Dulac estimate

The following local Dulac estimate is used in the compact positive-ratio passage and in the stopped determinant chart. Its logarithmic remainder bounds remain uniform when the characteristic ratio crosses one.

Lemma A.8 (Uniform local positive Dulac estimate)**.** *Consider a compact analytic family of hyperbolic saddles with fixed compatible entrance and exit sections. Suppose the positive tangential and normal rates L, ν are bounded above and away from zero and $\rho = \nu/L \in [\rho_0, \rho_1] \Subset (0, \infty)$. Writing θ for the family parameter, in a positive local coordinate H which vanishes at the incoming separatrix, the local Dulac map has the form*

$$\mathcal{D}(H) = c_\theta H^\rho e^{\mathcal{R}_\theta(H)}, \qquad c_\theta > 0.$$

For every $0 < \epsilon < \min(1, \rho_0)$, there is a uniform constant $C < \infty$ such that

$$\left|(H\partial_H)^j \mathcal{R}_\theta(H)\right| \le C H^\epsilon (1 + |\log H|), \qquad 0 \le j \le 2.$$

The estimate includes $\rho = 1$, and all local coordinate changes preserve the positive-time orientation.

Proof. Straighten the invariant axes. A positive time factor gives $u' = Lu$. Linearize the one-dimensional stable-axis equation and then replace v by $ve^{h(u)}$, where $h'(u) = (b(u,0) - \nu)/(Lu)$. The quotient is removable at zero. The remaining normal-coefficient defect vanishes on both axes, so on one common small rectangle the equation is

$$u' = Lu, \qquad v' = -\nu v(1 + uvG_\theta(u, v)), \qquad \rho = \nu/L \in [\rho_0, \rho_1].$$

These changes use no division by $\rho - 1$, preserve positive time, and have common finite derivative bounds on the compact family. Take the entrance (H, v_0) and the exit $u = u_0$. All phase derivatives below hold the coefficients and these cuts fixed. Put $\mathsf{B} = -\rho v G_\theta$. The graph equation is

$$\partial_u \log v = -\rho/u + \mathsf{B}(u, v).$$

After shrinking the rectangle, v decreases. Its integral equation first gives $v(u; H) \le C(H/u)^\rho$, and then

$$I_\rho(H) := H^\rho \int_H^{u_0} u^{-\rho}\, du \le C H^{\min(1,\rho)}(1 + |\log H|).$$

The bound is uniform through $\rho = 1$; no separate bound for $(1 - \rho)^{-1}$ is taken.

Let $\mathcal{E} = H\partial_H$, $P = \mathcal{E}\log v$, and $Z = \mathcal{E}^2 \log v$, at fixed u. Differentiating the moving initial condition $v(H; H) = v_0$, as in (A.4), gives

$$\begin{aligned} P(H) &= \rho - H\mathsf{B}(H, v_0), \quad Z(H) = O(H), \\ P_u &= \mathsf{B}_{\log v} P, \qquad\qquad Z_u = \mathsf{B}_{\log v} Z + \mathsf{B}_{\log v \log v} P^2. \end{aligned}$$

The two logarithmic height derivatives of $\mathsf{B} = -\rho v G$ are $O(v)$. Variation of constants therefore bounds $P > 0$ and gives $|P(u_0) - \rho| + |Z(u_0)| \le C(H + I_\rho(H))$. For

$$\mathcal{R}(H) = \log \frac{v(u_0; H)}{v_0 (H/u_0)^\rho},$$

one has $\mathcal{E}\mathcal{R} = P(u_0) - \rho$ and $\mathcal{E}^2\mathcal{R} = Z(u_0)$; its value is bounded by the integral of B. These prove the stated estimate for every fixed $0 < \epsilon < \min(1, \rho_0)$. The endpoint $H = 0$ is a separatrix endpoint, not a through orbit. □

B Strong projection and physical section heights

The outgoing passages are parametrized by their projection to the equatorial axis. We construct this projection for a common normal-time field, with bounds independent of the length of the logarithmic corridor. An integrating coordinate then converts the axis action into the height of the actual orbit on a fixed section.

Lemma B.1. *Fix $0 < w_0 < 1/4$. Let h, c, k be C^4 functions on $\mathbb{R}$, with*

$$\|h\|_{C^4} + \|c-1\|_{C^4} + \|k\|_{C^4} \le \epsilon,$$

and let V denote the field

$$u' = h(u) + c(u)w, \qquad w' = w\{1 - 2h(u) + (k(u) - 2c(u))w\}. \tag{B.1}$$

For ϵ sufficiently small, this field has a C^3 projection $\widehat{\pi}$ to $w = 0$, satisfying $V\widehat{\pi} = h(\widehat{\pi})$ and $\widehat{\pi}(u, 0) = u$. Uniformly on $\mathbb{R} \times [0, w_0]$,

$$\Big\| \widehat{\pi} - u - \tfrac{1}{2}\log(1-2w) \Big\|_{C^3} \le C\epsilon.$$

Writing $q = e^u$, $Y = q^2 w$, and $\pi(q, Y) = e^{\widehat{\pi}(u,w)}$, we have

$$c_1 w \le \log q - \log \pi \le C_1 w, \qquad 1 \le q/\pi < 2, \qquad \partial_u \widehat{\pi} \ge \tfrac{1}{2}. \tag{B.2}$$

There is also a uniformly C^3 function J in (u, w), with $J = O(w)$, for which $Z = Ye^{-J}$ satisfies $V \log Z = 1$ when $Y > 0$. The constants depend on w_0 and the indicated coefficient bounds.

Proof. Let ϕ_h^T be the axis flow $u' = h(u)$. Transporting the backward orbit's base coordinate by this flow defines the footpoint:

$$\begin{aligned} \widehat{\pi}(u, w) &= \lim_{T\to\infty} \phi_h^T(u(-T)) \\ &= u - \int_0^\infty D\phi_h^T(u(-T))c(u(-T))w(-T)\, dT, \\ \pi(q, Y) &= \exp(\widehat{\pi}(\log q, Y/q^2)). \end{aligned} \tag{B.3}$$

The integral identity follows by differentiation in T. Write the backward field G, with $n = 1-2h$, $d_n = k - 2c$, as

$$\dot{U} = -h(U) - c(U)W, \qquad \dot{W} = -n(U)W - d_n(U)W^2.$$

The strip $0 \le W \le w_0$ is invariant. After decreasing ϵ, $n + 2d_n W \ge \lambda := \frac{1}{2}(1 - 4w_0) > 0$ and $W(T) \le we^{-\lambda T}$. For initial directions $i, j, k \in \{u, w\}$, put $z_i = (U_i, W_i)$, $M = DG(U, W)$. The variational rows are

$$\begin{aligned} \dot{z}_i &= Mz_i, \\ \dot{z}_{ij} &= Mz_{ij} + D^2G[z_i, z_j], \\ \dot{z}_{ijk} &= Mz_{ijk} + \sum_{\text{cyc}} D^2G[z_{ij}, z_k] + D^3G[z_i, z_j, z_k]. \end{aligned} \tag{B.4}$$

The normal diagonal is at most $-\lambda$, the base diagonal is $O(\epsilon)$, and the product of their off-diagonal couplings is $O(\epsilon W)$. Integrating the normal row first therefore leaves an integrable Volterra coupling in the base row. Higher normal forcings contain W or a lower normal variation, while pure base forcings have $O(\epsilon)$ coefficients. Induction gives

$$|D^j U(T)| \le C(1+T)^N e^{C_3\epsilon T}, \qquad |D^j W(T)| \le C(1+T)^N e^{-(\lambda - C_3\epsilon)T}, \quad 1 \le j \le 3.$$

The scalar axis-flow equations give the same growth bound through order four. Thus the integrand of (B.3), through three ordinary derivatives, is bounded by $C(1+T)^N e^{-(\lambda - C\epsilon)T}$, with an additional w in order zero. This proves uniform C^3 dependence; comparison with $(h,c,k) = (0,1,0)$ gives the stated error. More generally, row j uses axis-flow order $j+1$, giving C^4 under the small C^5 bounds used in the compact-positive passage. Flow translation gives $V\widehat{\pi} = h(\widehat{\pi})$.

Exponentiation gives $\pi(q,0) = q$, $V\pi = b(\pi)$, where $b(q) = qh(\log q)$. Positivity of the projection integral and comparison with its unperturbed value prove (B.2).

Finally set

$$J(q,Y) = \int_0^\infty k(U(T))W(T)\,dT, \qquad Z = Ye^{-J}. \tag{B.5}$$

The same majorant gives three ordinary derivatives in (u,w) and $J = O(w)$. Translation along the flow gives $VJ = kw$. Since (B.1) gives $V\log Y = 1 + kw$, we obtain $V\log Z = 1$. □

On a logarithmic corridor with fixed-length collars, extend the same triple (h,c,k) by smooth retractions with uniformly bounded fourth derivatives. This preserves the coefficient bounds and $V\log Y = 1 + kw$, independently of corridor length. Each application checks that its physical passage stays in the unchanged region.

We now relate the divergence of the axis action at a stopped endpoint to a physical height tending to zero. The next lemma makes this relation quantitative without dividing by the axis rate at that endpoint.

Lemma B.2 (Physical height from the axis action)**.** *Use the preceding projection for a field unchanged on $Q/2 \le q \le 2R$, and write $h(q) = h(\log q)$ in physical coordinates. Suppose $b(q) = qh(q) > 0$ on $[Q/2, R)$, $b \ge b_0 > 0$ on $[Q/2, Q]$, and $\mathcal{A}(s) = \int_Q^s b(q)^{-1}\,dq \to \infty$ at R. For $2Q < R_0 < R < \infty$ and every $R_0 \le s < R$, the strong section $Y/q^2 = w_0$ with footpoint s has a unique backward crossing of $q = Q$, of height $y(s) > 0$. With $\mathcal{D} = b(s)\partial_s$,*

$$\log y = -\mathcal{A}(s) + 2\log s + \log U(s) + E_Q(y), \qquad E_Q(y) = J(Q,y) + \mathcal{A}(\pi(Q,y)), \tag{B.6}$$

where U, U^{-1} and three logarithmic derivatives of U are bounded, $E_Q(0) = 0$, and $\|E_Q\|_{C^3} \le C_Q$. If also $b \le P_*$ on $[Q, R_0]$, then

$$0 < y(s) \le e_0 := C_Q R_0^2 e^{-(R_0 - Q)/P_*}.$$

Once $C_Q e_0$ is small, y decreases to zero at R, and

$$\mathcal{D}\log y = -K, \qquad |K-1| + |\mathcal{D}K| \le C\epsilon + C_Q e_0, \qquad K \ge \tfrac{1}{2}. \tag{B.7}$$

The constants are independent of R and of the axis rate at R.

Proof. The footpoint bounds give a unique strong-section abscissa $q_s(s) = sU_s(s) \in (s, 2s)$, with $q_s' \asymp 1$ and three bounded logarithmic derivatives of U_s. In backward normal time both w and the footpoint decrease. While $Q/2 \le \pi < R$, $u' = h(\pi) + cw + O(\epsilon w) > 0$. The axis reaches $Q/2$ in finite backward time for each $s < R$; there $q < Q$. Thus the physical orbit crosses $q = Q$ exactly once, with $0 < y < w_0 Q^2$ and $Q/2 < \pi(Q,y) < Q$. The entire segment stays in the unchanged corridor.

The first integral $\log Z - \mathcal{A}(\pi)$ gives (B.6), with the explicit unit

$$U(s) = w_0 (q_s/s)^2 e^{-J(q_s, w_0 q_s^2)}.$$

The fixed-interval lower bound for b bounds E_Q on the whole possible height segment. Since $\mathcal{D}\log(s^2 e^{-\mathcal{A}}) = 2h - 1 \le -1/2$, $y(s) \le C_Q R_0^2 e^{-\mathcal{A}(R_0)} \le e_0$. Differentiating this identity after the small-height bound gives

$$K = \frac{1 - 2h - \mathcal{D}\log U}{1 - yE_Q'(y)}.$$

The Euler bounds imply $\mathcal{D}\log U = O(\epsilon)$ and $\mathcal{D}^2\log U = O(\epsilon^2)$. Using $\mathcal{D}y = -Ky$ and the bounded derivatives of E_Q proves (B.7). Finally the exact height identity and $\mathcal{A}(s) \to \infty$, for each fixed coefficient fibre with finite R, give $y(s) \to 0$. □

C Normalized transitions on the parabolic tails

Above the coefficient height scale, the infinity transition consists of two parabolic tails joined by a bounded connector. Normalization by each tail's exact value and first variation gives convergent nonlinear maps, while the common singular power cancels in the ratio of the two absolute variations. This gives the signed high-height estimate of Corollary 5.1 for the actual return (1.4).

All constants are uniform on the stated compact parameter sets. A positive unit is bounded above and away from zero; O_{C^k} denotes ordinary phase derivatives through order k unless weights are displayed. Choose the finite intrinsic normal-form regularity before constructing the maps. Exterior observers use ordinary C^3 bounds on their actual buffered images.

Put $\alpha_0 = s^2a,\ \alpha_1 = sp,\ v = \gamma + sX,\ z = s^2Y$. For $s > 0$, a positive change of time gives

$$\begin{aligned} X' &= a + pX - Y + A(\gamma + sX)Y - 3\gamma X^2 - sX^3 - C(\gamma + sX)^2Y, \\ Y' &= -Y\{(1 + 2\gamma)X + sX^2 + Cs(\gamma + sX)Y\}. \end{aligned} \tag{C.1}$$

Here s is a general positive scale. For negative opening one may take $s = \sqrt{-\alpha_0}$, whereas normalizing the height of a reference orbit allows a to have either sign.

C.1 The limiting transition

At $s = \gamma = 0$, the equation is

$$X' = a + pX - Y, \qquad Y' = -XY. \tag{C.2}$$

We call a complete orbit *strict* if it is a graph on $\mathbb{R}$ with $Y > 0$ and $D = Y - a - pX > 0$. Positive time then corresponds to decreasing X. A compact strict family means a compact family of such graphs, with uniform positive margins for these two inequalities on each fixed compact X-interval.

Removing the common quadratic, linear and logarithmic terms at each end gives finite labels for these complete graphs.

Lemma C.1. *On a compact strict family the limits*

$$B_\pm = \lim_{X \to \pm\infty} \{Y - X^2/2 - 2pX - (2a - 4p^2)\log|X|\} \tag{C.3}$$

exist. They give increasing coordinates on the family and depend real-analytically on the parameters and the orbit. The corresponding maps and their inverses have bounded derivatives of every prescribed finite order on smaller compact strict families.

Proof. Bound $|a|, |p|$ by $M \geq 1$. On either outward branch the graph equation $Y_X = XY/(Y - a - pX)$, with $r = |X|$, gives $Y_r \geq rY/(Y + M(1+r))$. Below $Y = M(1+r)$, $(\log Y)_r \geq 1/(4M)$ for $r \geq 1$; above $r = 2M$ this line is a lower barrier. Compactness supplies a common entrance; after enlarging the cut, $Y \geq r^2/8$ and $Y_r = r + O(1)$, uniformly. Writing $c = 2a - 4p^2$ and $Y = X^2/2 + 2pX + c\log|X| + \mathcal{B}$, direct expansion in a parabolic corridor gives

$$|\mathcal{B}_X| \leq C\frac{1 + \log|X| + |\mathcal{B}|}{X^2}, \qquad |\partial_{\mathcal{B}}\mathcal{B}_X| \leq CX^{-2}.$$

Gronwall's inequality gives boundedness and convergence. Beyond a fixed large cut, the prescribed-limit integral equation is a contraction on a common complex neighborhood, since its denominator is comparable to X^2. Holomorphic Picard iteration and finite-interval dependence give analyticity. The positive scalar first variation and its reciprocal are bounded, proving the coordinate and inverse assertions. □

Denote the scattering map by

$$S_{a,p}(b) = B_-, \qquad b = B_+. \tag{C.4}$$

Its curvature has a particularly simple sign.

Proposition C.2. *On every strict component,*

$$S_{a,p}(b) = b - p\mathcal{I}(a,p,b), \qquad \mathcal{I}_{bb} > 0. \tag{C.5}$$

In particular, the curvature of $S_{a,p}$ has sign $-\operatorname{sgn} p$ when $p \neq 0$.

Proof. When $p = 0$, reflection $X \mapsto -X$ reverses the flow and gives $S_{a,0} = \mathrm{id}$. Analytic division in p defines $\mathcal{I}$. For $\sigma = \partial_b Y > 0$ and $\tau = \partial_b^2 Y$, the graph equation gives

$$\sigma_X = -\frac{X(a+pX)}{D^2}\sigma, \qquad \tau_X = -\frac{X(a+pX)}{D^2}\tau + \frac{2X(a+pX)}{D^3}\sigma^2.$$

Integrating the equation for τ/σ from the incoming to the outgoing end yields

$$\frac{S''}{S'} = -2\int_{\mathbb{R}} \frac{X(a+pX)\sigma}{D^3}\, dX.$$

To evaluate this integral, use

$$D_X = X - p + \frac{X(a+pX)}{D}, \qquad (\sigma/Y)_X = -X\sigma/D^2.$$

The integral of $(\sigma/D)_X$, followed by that of $(\sigma/Y)_X$, therefore gives

$$\frac{S''(b)}{S'(b)} = -p\int_{\mathbb{R}} \frac{\sigma(X,b)}{D(X,b)^2}\, dX. \tag{C.6}$$

All boundary terms vanish by (C.3). Thus $\mathcal{I}_{bb} = S' \int_{\mathbb{R}} \sigma D^{-2}\, dX > 0$, also at $p = 0$ by continuity. □

C.2 Normalization at a reference orbit

Fix a strict reference orbit and label nearby orbits by b, with reference value b_0. For positive s, let $h(b)$ be their height on Σ_+, and put

$$k(b) = \mathcal{C}_{\theta,m}(T_\theta(h(b))).$$

Thus $h - a_\theta - mk = 0$ is exactly the periodic-orbit equation. The maps h, k include all transitions to the fixed sections. Their slopes may vanish at different rates, so normalize their values and first derivatives separately:

$$H(b) = b_0 + \frac{h(b) - h(b_0)}{h'(b_0)}, \quad K(b) = b_0 + \frac{k(b) - k(b_0)}{k'(b_0)}, \quad \Phi = K \circ H^{-1}. \tag{C.7}$$

The input $\beta = H(b)$ is affine in the actual incoming height. Reversing the sign of the return equation and dividing by $mk'(b_0) > 0$ gives exactly

$$D(\beta) = c_0 + c_1\beta + \Phi(\beta) = 0. \tag{C.8}$$

The constants retain both centers and both slopes. No bound on their ratio is assumed.

Proposition C.3 (Convergence on the actual sections)**.** *On each compact strict family, Φ extends continuously in phase C^3 to $s=\gamma=0$, with limit the positive affine normalization of $S_{a,p}$ in* (C.7)*. Let Φ^{int} denote the normalized transition before the exterior observers. Its maps and inverses have every prescribed finite phase order; near $p=0$, when the division intervals remain strict, its s-derivative for $s\geq 0$ and p-derivative at $s=0$ extend in each prescribed C^k, using one extra intrinsic phase order. Moreover, on common enlarged domains,*

$$\|\Phi-\Phi^{\mathrm{int}}\|_{C^3}=O(s^{2-\nu}),\qquad 0<\nu<1.$$

Consequently the actual s-secant extends continuously in C^3 at zero. At $p=s=0$, for small γ, $\Phi=\mathrm{id}$. All assertions are uniform in passive coefficients and $0\leq m\leq m_0$.

Proof. Ordinary ODE dependence controls the compact core. On each outward tail, set $X=\sigma r$, $\sigma=\pm 1$; then

$$Y_r=F_\sigma(r,Y)=\frac{(1+2\gamma+\sigma sr)rY+\sigma Cs(\gamma+\sigma sr)Y^2}{\ell_\sigma Y+3\gamma r^2+\sigma sr^3-\sigma pr-a},\quad \ell_\sigma=1-A(\gamma+\sigma sr)+C(\gamma+\sigma sr)^2. \tag{C.9}$$

Choose a small raw radius δ, then a large core radius L, then the bounds on s,γ. The tail ends at $R_s=\delta/s$, where uniformly regular maps join $v=\gamma\pm\delta$ to $\Sigma_\pm$.

Write

$$L_\gamma=1-A\gamma+C\gamma^2,\quad W_\gamma=\frac{1-4\gamma}{2L_\gamma},\quad \lambda_\gamma=\frac{12\gamma}{1+2\gamma}.$$

At $s=p=a=0$, $W=Y/r^2$ has equilibrium W_γ and radial exponent $-2+\lambda_\gamma$. Fix $0<\nu<1$, with $|\lambda_\gamma|<\nu$. A small band about W_γ is invariant; its logarithmic equation has perturbation $O(sr+|p|/r+|a|/r^2)$. Variation of constants gives

$$W-W_\gamma=O(sr+|p|/r+r^{-2+\nu}),\qquad L\leq r\leq R_s. \tag{C.10}$$

This includes the logarithmic term for bounded a and makes the denominator comparable to r^2, uniformly on the compact strict family after choosing L.

Let $Y_*(r)=Y(r,b_0)$ and $j(r)=Y_b(r,b_0)>0$. For the rational expression $F=(\alpha Y^2+\beta Y)/(\ell Y+m)$, polynomial division gives the exact identity

$$F_{Y^h}=(-1)^h h!\,\frac{m(\alpha m-\beta\ell)\ell^{h-2}}{(\ell Y+m)^{h+1}},\qquad h\geq 2. \tag{C.11}$$

Using the coefficients in (C.9) and the corridor gives

$$F_Y-\lambda_\gamma/r=O(s+|p|r^{-2}+r^{-3+\nu}), \tag{C.12}$$

$$|F_{Y^j}|\leq C_j\{sr^{-2j+2}+|\gamma|r^{-2j+1}+|p|r^{-2j}+|a|r^{-2j-1}\},\quad j\geq 2. \tag{C.13}$$

At fixed r,Y, their explicit coefficient derivatives satisfy

$$|\partial_s^{\exp}F_{Y^h}|\leq C_h r^{-2h+2},\qquad |\partial_p^{\exp}F_{Y^h}|_{s=0}\leq C_h r^{-2h+1},\quad h\geq 2. \tag{C.14}$$

Integrating (C.12), using $sR_s=\delta$, gives

$$cr^{\lambda_\gamma}\leq j(r)\leq Cr^{\lambda_\gamma}. \tag{C.15}$$

The parameter variations, at fixed r, satisfy

$$|Y_{*,s}|\leq Cr^3,\quad |(\log j)_s|\leq Cr,\qquad |Y_{*,p}|_{s=0}\leq Cr,\quad |(\log j)_p|_{s=0}\leq C. \tag{C.16}$$

For the height rows, apply variation of constants to $F_s = O(r^2)$, $F_p|_{s=0} = O(1)$, with propagator (C.15). For the slope rows, $(\log j)_{\ell r} = F_{Y\ell} + F_{YY} Y_{*,\ell}$ is respectively $O(1)$ and $O(r^{-2})$.

Subtracting the exact affine part gives

$$\eta = (Y - Y_*)/j, \qquad \eta_r = \mathcal{N}(r, \eta), \tag{C.17}$$

with kernel (A.1). On a bounded η-interval this remainder can also be written exactly as

$$\mathcal{N} = \frac{j\eta^2 m(\alpha m - \beta\ell)}{D_*^2(D_* + \ell j\eta)}, \qquad D_* = \ell Y_* + m. \tag{C.18}$$

Both denominators are comparable to r^2. The phase and parameter formula (A.2), together with (C.13), (C.14) and (C.16), gives

$$\|\mathcal{N}\|_{C^k} \le C\{sr^{-2+\nu} + |\gamma| r^{-3+\nu} + |p| r^{-4+\nu} + |a| r^{-5+\nu}\}, \tag{C.19}$$

$$\|\mathcal{N}_s\|_{C^k} \le C r^{-2+\nu}, \qquad \|\mathcal{N}_p\|_{C^k, s=0} \le C r^{-3+\nu}. \tag{C.20}$$

The reference and slope are chosen by finite ODE dependence at the fixed core cut. Thus $Y_{*,\ell}$, j_ℓ include their motion while the normalized initial phase stays fixed. Every further phase row adds $j/r^2 = O(r^{-2+\nu})$, so the bounds include the extra order needed for the C^k-valued parameter variation.

These are the integrable majorants of Lemma A.1. The moving endpoint contributes

$$\|\mathcal{N}(R_s, \cdot) R_s'\|_{C^k} = O(s^{1-\nu}) \longrightarrow 0.$$

Finite-interval ODE dependence supplies local kernel convergence. The lemma gives joint convergence, common domains and inverse bounds.

For each exterior map R, its actual interval is $z = z_* + d_s\eta$, where $d_s = s^2 j(R_s) = O(s^{2-\nu})$. Lemma 3.3 gives on the admitted buffered image

$$\frac{R(z_* + d_s\eta) - R(z_*)}{R'(z_*) d_s} = \eta + O_{C^3}(d_s), \tag{C.21}$$

including inverse and divided closing. With these observers denoted by $E_\pm$, the exact composition is $\Phi = E_- \circ \Phi^{\mathrm{int}} \circ E_+^{-1}$. The intrinsic C^4 bound and Lemma 3.4 give the claimed $O_{C^3}(s^{2-\nu})$ error. Its quotient by s tends to zero, requiring only phase derivatives of the exterior maps.

At $p = s = 0$ a first integral is

$$\begin{gathered} B_{a,\gamma}(X, Y) = -Y^{-\nu_\gamma} \left(\frac{X^2}{2} - \frac{L_\gamma}{1 - 4\gamma} Y \right) - \frac{a}{1 + 2\gamma} \log Y \, \frac{1 - e^{-\nu_\gamma \log Y}}{\nu_\gamma \log Y} - a \log 2, \\ \nu_\gamma = \frac{6\gamma}{1 + 2\gamma}, \end{gathered} \tag{C.22}$$

with the quotient continued at zero. Its Y-derivative is positive on the strict core. Use it as the initial orbit label, shifted to b_0 at the reference orbit. The same label occurs on the two tails. Implicit differentiation gives $Y_{bb}/Y_b = O(r^{-2+\nu})$, and the analogous higher-order estimates. Therefore both normalized tail limits are the identity. Together with (C.21) this proves $\Phi = \mathrm{id}$. At $\gamma = 0$, the limiting tail labels for general p are (C.3), which proves the stated scattering limit. □

During intrinsic parameter integration, hold m independently fixed in $[0, m_0]$. Its divided exterior map extends to both signs of the finite opening; only the actual $m(\theta) > 0$ enters the cycle equation.

C.3 Exact division in the perturbation parameters

On common enlarged strict domains, put

$$\Phi(\beta) = \beta + pU(\beta) + sV(\beta), \qquad U = \int_0^1 \Phi_p^{\mathrm{int}}(a, tp, 0, \gamma; \beta)\, dt,$$
$$V = \int_0^1 \Phi_s^{\mathrm{int}}(a, p, ts, \gamma; \beta)\, dt + \frac{\Phi - \Phi^{\mathrm{int}}}{s}. \tag{C.23}$$

The two integrals divide successively in p and s, retaining the actual exterior remainder. Both secants have continuous bounded C^3 limits: the first uses only the intrinsic p-derivative at $s = 0$; the last quotient is $O_{C^3}(s^{1-\nu})$. The strict margins hold throughout both intrinsic integration intervals, with fixed references and common inverse images.

C.4 Cancellation of the absolute tail derivatives

Use the actual affine incoming coordinate of Lemma 2.2 and write

$$\Pi(x) = \mathcal{C}_{\theta,m}(T_\theta(a_\theta + x)), \qquad \Delta(x) = x - m\Pi(x).$$

The fixed raw ports, their regular bridges, and the divided closing map are all included in this same transition.

Proposition C.4. *On any fixed compact strict source family with its two outward tails, the complete transition satisfies $0 < \Pi'(x) \le C$, uniformly for small positive s, small γ, and $0 \le m \le m_0$. The constant may depend on the compact family. This includes a small signed a-block about $a = p = 0$, with central height near one.*

On the negative-opening source family $a = -1$, there are constants $K_H, z_*, C_H > 0$ such that, for either sign of small γ,

$$Y_0 > K_H(1 + p^2), \qquad 0 < s^2 Y_0 < z_* \quad \Longrightarrow \quad 0 < \Pi'(x) \le C_H.$$

Here C_H is independent of p and of the tested phase point. After one common multiplier shrink, $\Delta' > 1/2$ on this whole high-height interval.

Proof. Fix a compact strict family and a core cut L_t. With $y_c = Y(0)$, ordinary first variation on $[-L_t, L_t]$ gives

$$A_\sigma = \frac{\partial Y(\sigma L_t; y_c)}{\partial y_c} \in [c_A, C_A], \qquad \sigma = \pm.$$

The coefficient scale s is held fixed in this differentiation. Equations (C.12) and (C.15), before affine normalization, give on each outward tail

$$J_\sigma(R) = (R/L_t)^{\lambda_\gamma} U_\sigma(R), \qquad \lambda_\gamma = \frac{12\gamma}{1 + 2\gamma}, \qquad 0 < c \le U_\sigma, U_\sigma^{-1} \le C.$$

At ports $v = \gamma \pm d_H$, $R = d_H/s$, the raw variations are

$$\frac{dz_\sigma}{dy_c} = s^{2-\lambda_\gamma}(d_H/L_t)^{\lambda_\gamma} A_\sigma U_\sigma(R), \qquad T'(z_+) = \frac{A_- U_-(R)}{A_+ U_+(R)}. \tag{C.24}$$

Here T is the passage between the raw ports. The common power cancels; distinct fixed radii add only the bounded factor $(d_-/d_+)^{\lambda_\gamma}$. Regular bridges are derivative units, and $\mathcal{C}'_{\theta,m} = f'_\theta(md_\theta)d'_\theta$ is a uniform positive unit by Lemma 2.2. Thus the derivative of the complete transition in the actual incoming coordinate is bounded on the fixed compact strict family.

For the signed block $|a|, |p| \le 1/8$, $3/4 \le y_c \le 5/4$, the half-graphs satisfy

$$3/4 + r^2/4 \le Y(r) \le 5/4 + r^2, \qquad Y - a - \sigma pr \ge 39/64.$$

These follow by scalar comparison; below $Y/r^2 = 1/4$ the logarithmic vector points upward, and above $3/4$ it points downward beyond a common fixed cut. Thus this is a compact strict family with uniform parabolic entrances.

For the negative-opening high-height range, freeze $q = Y_0^{-1/2}$ at a selected point and rescale the source field to

$$(\bar{a}, \bar{p}, \bar{\gamma}, \bar{s}) = (-Y_0^{-1}, pY_0^{-1/2}, \gamma, sY_0^{1/2}), \qquad y(0) = 1.$$

Large K_H, followed by small z_* and a coefficient shrink, places these parameters in one compact strict source block about $y = 1 + x^2/2$. Apply the preceding bound with scale $\bar{s}$. The same constant C_H then works at every tested point, and $mC_H < 1/2$ gives $\Delta' > 1/2$. The scale is frozen during the phase derivative, with no bound on $\gamma \log \bar{s}$. □

C.5 Extension through the height cap

The derivative bound joins the low-height count to the fixed raw cap. The overlap condition in the next lemma ensures that this continuation introduces no additional critical point.

Lemma C.5 (Extension through the height cap)**.** *On one ordered physical carrier, suppose a connected lower interval has a prescribed critical-point bound and overlaps a bounded-height interval where the complete transition derivative is uniformly bounded. Write its scaled height as $\eta = \tau Y_0$, $\tau > 0$, and suppose these two intervals cover $0 < \eta \le \eta_{\max}$. If Proposition C.4 applies above $Y_0 = K_H(1 + p^2)$ and*

$$\tau K_H(1 + p^2) < \eta_{\max}, \tag{C.25}$$

then a common multiplier shrink adds no critical point on the complement of the lower interval, up to the captured raw-height cap.

Proof. The bounded derivative gives $F' > 0$ on the bounded-height part after shrinking m. Put $Z = z_*/s^2$. If $Z \le K_H(1 + p^2)$, the displayed inequality puts every captured height below $\eta_{\max}/\tau$. Otherwise the intervals $(0, \eta_{\max}/\tau]$ and $(K_H(1 + p^2), Z)$ overlap below the cap and cover it. The high-height estimate gives $F' > 0$ on the added interval. These are restrictions of the same completed map; scalar order retains all inputs between captured roots. □

D Degenerating saddle transitions: action coordinates

We use an action coordinate to describe a large negative source-energy interval. Let β be defined by $Y + \log Y = X^2/2 + \beta$ for the unperturbed negative source. The passage consists of two parabolic tails joined by a thin region near the equator. Its limiting transition is determined by an action integral, but its curvature vanishes on the reversible coefficient set. Absolute convergence alone does not control the curvature near that set. We estimate the nonlinear part of the actual transition relative to the distance to that set. Keeping the section maps and their inverses in physical coordinates allows this estimate to be used in (1.4).

D.1 The equation and the section coordinates

The local scales below are related to the raw coefficients in Section 4.1.

Fix a scale $t > 0$ on each coefficient fibre, keeping it fixed as the orbit varies. Start with (C.1) at $a = -1$, write $p = t\Lambda_1$, $s = t^3\Lambda_2$, $\gamma = t^2 k$, and put $\xi = tX$, $\eta = t^2 Y$. After this

phase scaling and a positive change of time, the same form (C.1) holds in (ξ,η) with the new coefficients

$$(a_{\mathrm{new}},p_{\mathrm{new}},\gamma_{\mathrm{new}},s_{\mathrm{new}})=(-t^2,t^2\Lambda_1,t^2k,t^2\Lambda_2). \tag{D.1}$$

Thus $p_{\mathrm{new}}=tp$, $s_{\mathrm{new}}=s/t$, and the equation is

$$\begin{aligned}\dot\xi&=-\eta-t^2P_\Lambda(\xi)+At^2L(\xi)\eta-Ct^4L(\xi)^2\eta,\\ \dot\eta&=-\eta\{\xi+t^2\xi(2k+\Lambda_2\xi)+Ct^4\Lambda_2L(\xi)\eta\},\\ P_\Lambda(\xi)&=1-\Lambda_1\xi+3k\xi^2+\Lambda_2\xi^3,\qquad L(\xi)=k+\Lambda_2\xi.\end{aligned} \tag{D.2}$$

The passive coefficients A,C range over a fixed compact neighborhood of $(1,1)$. Physical tails have $\Lambda_2>0$; at $\Lambda_2=0$ we will use limits of normalized maps.

For the unperturbed negative source, $Y+\log Y=X^2/2+\beta$. Hence its rescaled energy is

$$\widehat b=t^2\beta+t^2\log(2t^2),\qquad \eta+t^2\log(2\eta)=\xi^2/2+\widehat b. \tag{D.3}$$

Energy-label differences are exactly t^2 times source-label differences; the action integral below keeps these transverse labels.

We specify the intervals on which the estimate will be used. Choose compact intervals

$$J\Subset\operatorname{int}\widehat J_+\Subset(-\infty,0),\qquad \widehat J_-\Subset(-\infty,0),$$

and a compact coefficient set. For $b\in\widehat J_+$, put $x=\sqrt{-2b}$ and define $r>0$ by

$$A_-(r;\Lambda)=A_+(x;\Lambda),\qquad A_+(x;\Lambda)=\int_0^x\frac{u}{P_\Lambda(u)}\,du,\quad A_-(r;\Lambda)=\int_0^r\frac{u}{P_\Lambda(-u)}\,du. \tag{D.4}$$

We require that the entry and exit lie in fixed compact subintervals of $(0,\infty)$, that $-r^2/2$ lies in $\operatorname{int}\widehat J_-$ with a fixed margin, and that

$$P_\Lambda\ge p_0>0 \tag{D.5}$$

on a common neighborhood of every traversed segment $[-r,x]$. The common neighborhood extends to both sides of the limiting exit. Monotonicity of A_- then gives comparison exit cuts with opposite action differences. Enlarged intervals provide the smaller common images needed for composition and inversion.

Choose a fixed L beyond the entry and exit intervals. The positive height pieces between the two sections $\xi=\pm L$ and the lips are required to lie in fixed transverse neighborhoods. Each outward tail lies in a parabolic corridor and reaches the prescribed raw section, followed by its actual regular transitions. The divided finite closing map is included on its side. Its multiplier m is held fixed as an independent passive parameter in $0\le m\le m_0$ during parameter division, as in Section C; the cycle equation uses its actual value $m>0$. These requirements concern the whole enlarged input and output intervals.

There is a useful energy coordinate for all small signed k. Put

$$\begin{aligned}\ell&=1-At^2k+Ct^4k^2,\qquad \nu_k=\frac{6t^2k}{1+2t^2k},\\ \mathcal B_{t,k}(\xi,\eta)&=-\frac12\left[\eta^{-\nu_k}\left(\xi^2+\frac{1}{3k}-\frac{2\ell\eta}{1-4t^2k}\right)-\frac{1}{3k}\right].\end{aligned} \tag{D.6}$$

The apparent pole is removable, with

$$\mathcal B_{t,0}=\eta+t^2\log\eta-\xi^2/2. \tag{D.7}$$

Direct differentiation of (D.2) gives $\dot{\mathcal{B}}_{t,k} = 0$ when $\Lambda_1 = \Lambda_2 = 0$. Shrink the parameters so that $1 + 2t^2k$, $1 - 4t^2k$, and ℓ are positive. At the two fixed cuts use the separate coordinates

$$b_\pm = \mathcal{B}_{t,k}(\pm L, \eta) + t^2 \log 2. \tag{D.8}$$

Here η stays in a compact positive interval and $\partial_\eta \mathcal{B}_{t,k}$ is bounded above and away from zero. These coordinates and their inverses are smooth in $t^2, k, b_\pm$. They agree with (D.3) at $k = 0$; away from that set, their relation is part of the actual section maps.

Let $S_{t,\lambda,k}$, $\lambda = (\Lambda_1, \Lambda_2)$, be the passage from b_+ to b_-, and let $\Gamma_\pm$ be the maps from these separate coordinates to the actual input and divided output heights. For references $b^*_\pm$ in the enlarged intervals, set

$$\begin{aligned} c_\pm = \Gamma'_\pm(b^*_\pm) > 0, \qquad a_\pm = \Gamma_\pm(b^*_\pm) - c_\pm b^*_\pm, \\ \phi_\pm(b) = b^*_\pm + \frac{\Gamma_\pm(b) - \Gamma_\pm(b^*_\pm)}{c_\pm}, \qquad \Gamma_\pm = a_\pm + c_\pm \phi_\pm. \end{aligned} \tag{D.9}$$

On the respective raw images $\mathcal{I}_\pm = \Gamma_\pm(\operatorname{int} \widehat{J}_\pm)$ and normalized images $\mathcal{U}_\pm = \phi_\pm(\operatorname{int} \widehat{J}_\pm)$, put

$$\Pi_{t,\Lambda} = \phi_- \circ S_{t,\lambda,k} \circ \phi_+^{-1} : \mathcal{U}_+ \longrightarrow \mathcal{U}_-. \tag{D.10}$$

If necessary the domain is restricted to the smaller input interval whose image has the prescribed output margin. The estimates below ensure that $J \subset \mathcal{U}_+$ for small parameters.

To see the role of both affine factors, write the actual cycle equation as $h - a_{\mathrm{f}} - m h_{\mathrm{out}} = 0$, where $a_{\mathrm{f}} = a_\theta = C_\theta(0)$ and $h_{\mathrm{out}} = \mathcal{C}_{\theta,m}(T_\theta(h))$ is the divided output height from Lemma 2.2. Substitution of (D.9) gives, without a change of zeros,

$$D(u) = a_+ - a_{\mathrm{f}} - m a_- + c_+ u - m c_- \Pi_{t,\Lambda}(u) = 0. \tag{D.11}$$

Thus u is affine in the actual incoming height. The two positive first derivatives may tend to zero at different rates; neither their equality nor a bounded ratio is required.

The zero count therefore requires curvature control of the completed map (D.10). Lemma 3.5 expresses this curvature with both section contributions; the relative comparison below estimates them separately.

D.2 A relative estimate near the reversible coefficients

For a fixed reference $b_* \in J$, write $\mathfrak{T} f = f - f(b_*) - (b - b_*) f'(b_*)$. On a fixed compact interval the quotient norm satisfies

$$\|[f]\|_{C^3/\,\mathrm{Aff}} \asymp \|\mathfrak{T} f\|_{C^3} \asymp \|f''\|_{C^1}.$$

This quotient isolates the derivatives needed for the curvature count. The affine coefficients remain in the actual displacement (D.11).

Theorem D.1 (Relative convergence on the actual sections)**.** *Fix the enlarged intervals, the entry and exit margins, the tail and regular-transition neighborhoods described above. There exist $t_0, \lambda_0, k_0, C > 0$, depending only on these data and the compact passive coefficient set, such that the following holds. For*

$$0 < t \le t_0, \qquad 0 < \rho_{\mathrm{id}} := (\Lambda_1^2 + \Lambda_2^2)^{1/2} \le \lambda_0, \qquad \Lambda_2 \ge 0, \qquad |k| \le k_0,$$

put $\theta_i = \Lambda_i / \rho_{\mathrm{id}}$. On the fixed smaller interval J, the map (D.10) *satisfies*

$$\left\| \frac{[\Pi_{t,\Lambda} - \mathrm{id}]}{\rho_{\mathrm{id}}} - \theta_1[U_0] - \theta_2[V_0] \right\|_{C^3/\,\mathrm{Aff}} \le C\{t^2(1 + |\log t|) + \rho_{\mathrm{id}} + |k|\}, \tag{D.12}$$

where

$$U_0(b) = -\frac{2}{3}(-2b)^{3/2}, \qquad V_0(b) = \frac{2}{5}(-2b)^{5/2}. \tag{D.13}$$

The coefficient k is independently signed. The constants are uniform in the two separate reference values, held fixed during parameter integration, and in $0 \le m \le m_0$. No positive lower bound on θ_2 is needed. At $\Lambda_2 = 0$ the assertion concerns the one-sided normalized limits.

Proof. On each outward tail use (C.9) with dictionary (D.1) and the separate initial label (D.8). With L fixed before shrink, the enlarged windows have $\eta \asymp r^2$ and denominator $\asymp r^2$ throughout division. Thus the tail proof of Proposition C.3 applies with its exact $j \asymp r^{\lambda_\gamma}$, kernel (C.17), and bounds (C.19)–(C.20) through one extra phase order. It gives bounded C^3-valued intrinsic s-rows and p-rows at $s = 0$, including the moving-end error $O_{C^3}(s^{1-\nu})$. These differentiations concern only the tails.

At $p = s = 0$, (D.6) is a first integral; implicit differentiation gives identity normalized intrinsic limits. As in (C.23), integrate the intrinsic tails first in p at $s = 0$, then in s at fixed p, holding the two reference values separately fixed. Since $\partial_{\Lambda_1} = t^2\partial_p$ and $\partial_{\Lambda_2} = t^2\partial_s$, the intrinsic maps and inverses differ from the identity by $Ct^2(|\Lambda_1| + \Lambda_2)$ in C^3.

At the tested coefficients, each raw gain for $\Lambda_2 > 0$ satisfies

$$d \le C(t^2\Lambda_2)^{2-\nu} \le Ct^2\rho_{\mathrm{id}}, \qquad 0 < \nu < 1.$$

The ordinary observers, including divided closing, therefore add $O_{C^3}(d)$ by Lemma 3.3. Thus

$$\|\phi_\pm - \mathrm{id}\|_{C^3} + \|\phi_\pm^{-1} - \mathrm{id}\|_{C^3} \le Ct^2(|\Lambda_1| + \Lambda_2). \tag{D.14}$$

At $\Lambda_2 = 0$ use the one-sided normalized limit. Inverses are on fixed smaller common images; no exterior coefficient derivative is needed.

For the core, work in a small independent (λ, k)-block with strict core margin and construct $S_{t,\lambda,k}$ and its first λ-derivatives in C^3_b, keeping $t > 0, k$ fixed. Choose an integer $M > 20$, and cut each lip at $y = \eta/t^2 = t^M$. On a lip use $H = \eta - \xi^2/2$ and $\ell_\eta = \log\eta$. Equation (D.2) gives exactly

$$\frac{dH}{d\ell_\eta} = t^2\Phi_\pm, \qquad \Phi_\pm = \frac{\xi\eta(2k + \Lambda_2\xi + AL - Ct^2L^2) - \xi P_\Lambda + Ct^2\Lambda_2 L\eta^2}{\xi\{1 + t^2(2k + \Lambda_2\xi)\} + Ct^4\Lambda_2 L\eta}. \tag{D.15}$$

The signed root $\xi^2 = 2(\eta - H)$ is bounded away from zero on each lip, as is the denominator. Thus the required derivatives of $\Phi_\pm$ in H, λ, k are bounded. The logarithmic clock has length $O(1 + |\log t|)$, giving a change $O(t^2(1 + |\log t|))$ through total order five; the positive-height pieces to $\pm L$ add $O(t^2)$. Equation (D.6) therefore gives

$$b = H + O_{C^5}(t^2(1 + |\log t|)), \tag{D.16}$$

At $k = 0$, the error is $t^2\log(2\eta)$, largest at $\eta = t^{M+2}$; the removable continuation in k has the same mixed bound through order five. The positive entry is $\sqrt{-2b} + O_{C^5}(t^2(1 + |\log t|))$; the negative exit has the corresponding estimate about $-\sqrt{-2b}$.

In the thin core put $Q = t^2\log y$. Its exact graph equation is

$$Q_\xi = \frac{\xi(1 + 2t^2k + t^2\Lambda_2\xi) + Ct^6\Lambda_2(k + \Lambda_2\xi)y}{P_\Lambda(\xi) + \ell_\Lambda(\xi)y} = \frac{\xi}{P_\Lambda(\xi)} + R, \quad \ell_\Lambda = 1 - At^2L + Ct^4L^2. \tag{D.17}$$

Until the first return to $y = t^M$, one has $0 < y \le t^M$. The root-free margin and direct differentiation give

$$|\partial_Q^j\partial_\lambda^\alpha R| \le C\{t^2 + t^{M-2j}\}, \qquad j + |\alpha| \le 5, \tag{D.18}$$

with the same bounds for the required ξ-derivatives. The negative powers come only from differentiating $y = e^{Q/t^2}$. For dependence on an open domain, replace this exponential above the lip by $t^M\chi((Q - Q_{\mathrm{lip}})/t^2)$, where $Q_{\mathrm{lip}} = Mt^2 \log t$, $\chi(z) = e^z$ for $z \le 1$, and χ is smooth, positive and constant for $z \ge 2$. This changes no first-exit orbit, preserves (D.18), and keeps the denominator positive.

The two comparison exit cuts from (D.4) have opposite action differences. Starting at the positive lip, Q decreases, stays a fixed amount below the lip near $\xi = 0$, and increases strictly on the negative side away from zero. The comparison cuts consequently give a unique actual first exit. At that exit, $Q_\xi = -r/P_\Lambda(-r) + o(1)$ is bounded away from zero. The differentiated integral equation for (D.17), followed by this implicit exit equation, therefore has uniform derivatives through total order five. With the two lip maps it gives

$$\begin{aligned} &\|S_{t,\lambda,k} - S_{0,\lambda,k}\|_{C_b^3} + \|\partial_\lambda S_{t,\lambda,k} - \partial_\lambda S_{0,\lambda,k}\|_{C_b^3} \\ &\qquad \le C\{t^2(1 + |\log t|) + t^{M-10}\}, \qquad S_{0,\lambda,k}(b) = -r^2/2. \end{aligned} \tag{D.19}$$

No derivative with respect to t at zero is asserted.

The exact energy and even $P_{0,k}$ give $S_{t,0,k} = S_{0,0,k} = \mathrm{id}$. At $\lambda = k = 0$, (D.4) gives

$$\partial_{\Lambda_1} S_{0,0,0} = -2x^3/3 = U_0(b), \qquad \partial_{\Lambda_2} S_{0,0,0} = 2x^5/5 = V_0(b).$$

Its derivatives are smooth on the compact root-free block. The identity on the coefficient ideal permits the exact division

$$S_{t,\lambda,k} - \mathrm{id} = \int_0^1 \partial_\lambda S_{t,v\lambda,k}\, \lambda\, dv.$$

Using (D.19) in this integral, and then the smoothness of the limiting action, gives

$$\|S_{t,\lambda,k} - \mathrm{id} - \Lambda_1 U_0 - \Lambda_2 V_0\|_{C^3} \le C\rho_{\mathrm{id}}\{t^2(1 + |\log t|) + \rho_{\mathrm{id}} + |k|\}. \tag{D.20}$$

Thus division by ρ_{id} precedes the limit; the even coefficient k creates no additional row.

The same construction bounds S in C^4. Apply Lemma 3.4 with (D.14) to obtain

$$\|\Pi_{t,\Lambda} - S_{t,\lambda,k}\|_{C^3} \le Ct^2\rho_{\mathrm{id}}. \tag{D.21}$$

Only the core needs the extra phase derivative for evaluation at ϕ_+^{-1}. Combining with (D.20) proves (D.12). □

The outgoing tail must first be estimated in its own intrinsic label. If an outgoing graph is written as $\eta(\xi, S(b))$, then

$$\partial_b^2 \eta(\xi, S(b)) = \eta_{BB}(\xi, S(b))(S')^2 + \eta_B(\xi, S(b))S''.$$

The second term need not decay on that tail. Estimating the map in B first and then composing retains this contribution.

D.3 Zeros of the completed displacement

The action transition has a curvature formula useful beyond the small coefficient neighborhood. Differentiating (D.4), with $x = \sqrt{-2b}$, gives

$$S'_\Lambda = \frac{P_\Lambda(-r)}{P_\Lambda(x)} > 0, \qquad \frac{S''_\Lambda}{S'_\Lambda} = \frac{x + r}{P_\Lambda(x)}\left(3\Lambda_2 - \frac{\Lambda_1}{xr}\right), \quad S_\Lambda = -r^2/2. \tag{D.22}$$

The terms containing k cancel. Moreover

$$\frac{d(xr)}{db} = -\frac{r}{x} - \frac{xP_\Lambda(-r)}{rP_\Lambda(x)} < 0. \tag{D.23}$$

Thus, away from $\lambda = 0$, every possible curvature zero is simple and has the same crossing orientation.

Corollary D.2. *On a compact family satisfying the section, tail, and root-free margins above, assume $\Lambda_2 \geq 0$ and that the physical coefficients satisfy $k \to 0$ whenever $\rho_{\mathrm{id}} \to 0$, uniformly on the family. For example, the condition $|k|^3 \leq C\Lambda_2^2$ suffices. There is a common $t_0 > 0$ such that every actual completed displacement* (D.11)*, for $0 < t \leq t_0$, has at most three isolated zeros on its input interval, counted with multiplicity.*

The same conclusion holds for connected coefficient-dependent intervals J_Λ if their coefficient–label graph has compact closure and the enlarged input and output intervals, both exit margins, and all actual transition neighborhoods have the same uniform margins.

Proof. At $\rho_{\mathrm{id}} = 0$, the exact first integral and the separate normalized tail limits give $\Pi = \mathrm{id}$. The displacement is therefore affine: it has at most one zero unless it vanishes identically, in which case it has no isolated zero. This coefficient boundary is understood as a normalized limit; actual raw passages have $\Lambda_2 > 0$.

Near the coefficient ideal, put $x = \sqrt{-2b}$. The limiting rows satisfy

$$U_0'' = -2/x, \qquad V_0'' = 6x, \qquad \frac{(\theta_1 U_0 + \theta_2 V_0)''}{2/x} = -\theta_1 + 3\theta_2 x^2.$$

If the last expression vanishes, compactness of the positive x-interval and $\theta_1^2 + \theta_2^2 = 1$ give a positive lower bound on θ_2. Its derivative in b is $-6\theta_2$. When θ_2 is smaller than a fixed positive constant, the expression is instead bounded away from zero. This proves a uniform alternative of no curvature zero or one simple downward crossing. The relative C^3 estimate preserves that alternative after a common shrink of t, ρ_{id}, k.

On the complementary compact coefficient annulus, the tail estimate still applies as $p = t^2\Lambda_1$, $s = t^2\Lambda_2$, $\gamma = t^2 k$ tend uniformly to zero. The strict core margin permits the same lip and first-exit argument, giving absolute C^3 convergence to S_Λ without division through the ideal. Hence $\|[\Pi - S_\Lambda]\|_{C^3/\,\mathrm{Aff}} \to 0$. Equations (D.22) and (D.23) give the same uniform crossing alternative on this compact annulus. Finitely overlapping the two coefficient regions gives one bound on t.

Lemma 3.2 gives at most one curvature zero and three displacement zeros since $D'' = -mc_-\Pi''$. The affine relation (D.11) preserves multiplicity. Compactness of the coefficient–label graph supplies the same constants fibrewise on J_Λ. □

E Small characteristic ratios, finite middle passages, and fixed exit

At degree loss, the two incident saddle exponents vanish at different rates. We compare their small-power maps using an action coordinate, exact local invariants and backward shooting from a fixed exit. The degree-loss section joins these descriptions on their actual raw receivers and includes the divided closing.

E.1 Two-saddle comparison and action matching

On this coefficient face the two incident saddles lie on different scales, and both characteristic ratios tend to zero. Their local maps must be compared after the intervening nonlinear flight. We first derive its scalar action equation and the actual local section coordinates. Section H.2 then constructs their physical receiver and follows the passage over the full action range.

E.2 The two saddle scales

Let $0 < \kappa, \tau \ll 1$, $\delta = \kappa\tau$. In coordinates which put the negative saddle at $u = -1$, the other two equatorial roots are $u = a$ and $u = c$. Write

$$p_{\mathrm{d}} = -\mathcal{R}(u+1)(u-a)(u-c), \tag{E.1}$$

where

$$a = \kappa A_{\mathrm{d}}, \qquad a + c = 1 - 3\nu. \tag{E.2}$$

The quantities $A_{\mathrm{d}}, \mathcal{R}, c, c-a, 1-3\nu$ and their reciprocals belong to fixed positive compact sets; A, C vary in fixed compact sets. Thus a tends to zero while -1 and c remain separated. The vector field has the form

$$\dot{u} = \delta p_{\mathrm{d}} - Kw, \qquad \dot{w} = -w(L + Ew),$$

with

$$\begin{aligned} K &= 1 - \delta\mathcal{R}A(u+\nu) + C\delta^2\mathcal{R}^2(u+\nu)^2, \\ L &= u\{1 + \delta\mathcal{R}(u+2\nu)\}, \qquad E = C\delta^2\mathcal{R}^2(u+\nu). \end{aligned} \tag{E.3}$$

On the intervening axis segment put $h = -\delta p_{\mathrm{d}} > 0$. Positive time runs from the cut $u_+ = a(1-\eta_+)$ to $u_- = -1 + \eta_-$, with fixed $0 < \eta_\pm < 1$. The orbit equation is

$$w_u = \frac{w\{L(u) + E(u)w\}}{h(u) + K(u)w}. \tag{E.4}$$

The tangential eigenvalues at a and -1 are $\delta\ell_+$ and $-\delta\ell_-$, whereas their normal eigenvalues are $-b_+$ and b_-, where

$$\begin{aligned} \ell_+ &= \mathcal{R}(1+a)(c-a), \quad b_+ = a\{1 + \delta\mathcal{R}(a+2\nu)\}, \\ \ell_- &= \mathcal{R}(1+a)(1+c), \quad b_- = 1 - \delta\mathcal{R}(1-2\nu). \end{aligned}$$

Define the small characteristic ratios used in the local maps by

$$\chi = \delta\ell_+/b_+, \qquad \beta = \delta\ell_-/b_-, \qquad q = \beta/\chi. \tag{E.5}$$

Thus χ is the exponent for the inverse positive Dulac map, whereas β is the exponent for the forward negative Dulac map. The normal rate at the positive saddle is of order κ, while that at the negative saddle is bounded away from zero. Hence χ is of order τ, β is of order $\kappa\tau$, and $q = \kappa Q_{\mathrm{d}}$ with Q_{d} a positive unit. In particular,

$$\beta|\log a| + q|\log a| \longrightarrow 0. \tag{E.6}$$

The inequality $\beta \ll \chi$ will determine the sign in the comparison of the two Dulac maps. To transfer that sign to the return, we also need the nonlinear transition between their local sections.

E.3 A transition in accumulated action

Writing $A_m = L/h$ separates the linear normal equation in (E.4) from its nonlinear part. The following calculation also applies to general coefficients for which $h > 0$:

$$w_u = A_m(u)w + \frac{E(u) - A_m(u)K(u)}{h(u) + K(u)w}\, w^2, \qquad h > 0. \tag{E.7}$$

Let $J' = A_m J,\ J(u_+) = 1$, and write $h_- = h(u_-), J_- = J(u_-)$. For a fixed $\lambda > 0$, put

$$\rho = \frac{h_- J}{J_- h}, \qquad w = \lambda h \rho V.$$

The factor J/J_- removes the linear amplification between the cuts, and $\rho(u_-) = 1$. This normalization isolates the remaining nonlinear change of height. In the saddle configuration above the amplification is exponentially large in $1/\beta$; its exact value is given by the partial fractions in (E.16). Direct substitution retains the denominator in the equation for V:

$$dV = -\frac{\lambda K V^2}{1 + \lambda K \rho V}\, d\rho + \frac{\lambda \rho (E - K h'/h) V^2}{1 + \lambda K \rho V}\, du. \tag{E.8}$$

For bounded $V = V(u_+)$, an early negative cut gives the actual section coordinates before the nonlinear amplification layer. Larger values of V require a different description of the same passage. Section E.5 uses the coordinate $\Theta = \beta \log V$ on a bounded nonnegative interval. For a positive limiting Θ in that interval, V is exponentially large in $1/\beta$, and the orbit leaves the small-height region before the negative saddle. Its height on a fixed exit section then supplies a regular coordinate. These are two overlapping parts of the orbit interval for the same coefficients.

E.4 Actual section coordinates at both stopped endpoints

For bounded amplified height, the middle passage is close to its linear normalization up to an early negative cut. We stop there and use exact saddle invariants to continue to both actual sections. This retains the nonlinear section maps needed to compare the two small powers.

Lemma E.1 (Uniform actual-section matching)**.** *Consider* (E.1)–(E.3)*, with the compact coefficient conditions stated there. Fix $V_1 < \infty$ and let V be the raw middle input in* (E.8)*. Choose the local cuts sufficiently small in the order in the proof, before decreasing κ, τ. On $0 \le V \le V_1 + 1$ there is an increasing change $\widehat{V} = V + O_{C^3}(\epsilon_\theta)$, fixing zero, where $\epsilon_\theta/\chi \to 0$. The two actual section coordinates, including the finite closing map on the negative side, have the form*

$$x_{\mathrm{in}} = x_b + P_* \Psi(\widehat{V}^\chi), \qquad f_\theta(m t_{\mathrm{out}}) = b_* + a_* D(\beta^{-\beta} C(\widehat{V})^\beta), \qquad P_*, a_* > 0, \tag{E.9}$$

for $m > 0$. Put $e_C(v) = vC'(v)/C(v) > 0$ for $v > 0$. Uniformly for $0 < V \le V_1 + 1$,

$$\begin{gathered} C(0) = 0, \quad C'(0) = 1, \quad \|C - \mathrm{id}\|_{C^3} \le \epsilon_\theta, \qquad \left| \frac{V e_C'(V)}{e_C(V)} \right| = o(\chi), \\ D' > c_D > 0, \qquad |D''/D'| \le C_D, \qquad \sup_{0 \le t \le (V_1+2)^\chi} |t\Psi''(t)/\Psi'(t)| \longrightarrow 0, \qquad P_* \to 0. \end{gathered} \tag{E.10}$$

The maps $\widehat{V}, C, D, \Psi$ extend in C^3 to their zero endpoints; this does not assert ordinary C^3 regularity of the compositions with the small powers of $\widehat{V}$. The normalized estimates have a uniform one-sided extension as $j = 1/c \downarrow 0$ at the remote-root face. Actual raw section maps and their positive affine scales are asserted only for $j > 0$.

Proof. We first construct local coordinates in which the saddle invariant is exact and the derivative bounds remain uniform as its exponent vanishes. After dividing by the positive normal rate, and reversing time at the positive saddle, both fields on a fixed small box take the form

$$\dot{x} = -\rho x p(x) + c(x) y, \qquad \dot{y} = y\{1 + k(x) y\}, \qquad \rho \in \{\beta, \chi\}. \tag{E.11}$$

Here $p(0) = 1$, $p > c_p > 0$, and p, c, k have uniform bounds through any prescribed finite order. At the negative saddle $x = u + 1, y = w$, $c = K/L < 0$; at the positive saddle $x = (u - a)/a, y = w/a^2$, $c > 0$. Both $|c|$ and its reciprocal are bounded. These assertions follow by substituting the displayed scalings in (E.3); in particular the positive normal rate is divided out only after scaling by a.

Put $h = -\rho x p(x)$. Construct the footpoint π by backward axis transport as in Lemma B.1, and put $J = \int_0^\infty k(X(-T))Y(-T)\,dT$, using bounded extensions. Here $Y(-T) \le ye^{-T/2}$, $h = O_{C^4}(\rho)$, and the couplings $c_x Y, k_x Y^2$ have integrable mass. The same triangular argument bounds tangential rows by $C(1+T)^N e^{C\rho T}$ and normal rows by $C(1+T)^N e^{-(1/2-C\rho)T}$. For a normalized coefficient parameter ϑ, its new forcing is $-(h_\vartheta + c_\vartheta Y, k_\vartheta Y^2)$: the tangential term adds polynomial factors, while normal terms keep a decaying height factor. Thus the backward integrals and their mixed rows through five phase derivatives and one parameter derivative have an integrable majorant $C(1+T)^N e^{-(1/2-C\rho)T}$, after fixing the finite order and then decreasing ρ. These mixed rows use axis-flow derivatives through order seven, supplied by the prescribed finite coefficient bounds.

Consequently $\dot\pi = h(\pi)$, and $Z = ye^{-J}$ satisfies $\dot Z = Z$. The axis coordinate

$$\ell(x) = x \exp \int_0^x \frac{p(s)^{-1} - 1}{s}\, ds$$

has uniform finite-order bounds and positive derivative. Hence $\Xi = \ell(\pi)$ gives the exact invariant

$$\dot\Xi = -\rho\Xi, \qquad I = \Xi Z^\rho, \qquad \dot I = 0. \tag{E.12}$$

This is a uniformly regular local coordinate change. Since $\pi_y(x, y) = -c(x) + o_\rho(1) + O(y)$ and $|c|$ is bounded below, choose the height strip first and then ρ to preserve its sign. All these bounds concern intrinsic coordinates, before extracting raw section scales.

Fix a permissible small upper height $y_{\max}$ in the negative box. Choose $x_- > 0$ small compared with this height, and then $\eta_0 > 0$ sufficiently small compared with both, as quantified in the output construction below. Finally fix $0 < \eta_- < \eta_0$ and put $u_0 = -1 + \eta_0$. This cut lies in the negative box, and strictly before the cut u_- in positive time. If $J_m(u) = \exp \int_{u_+}^u L/h$, put

$$\varepsilon_+ = \frac{\lambda h_-}{a^2 J_m(u_-)}, \qquad \varepsilon_0 = \frac{\lambda h_- J_m(u_0)}{J_m(u_-)}.$$

The raw heights at u_+ and u_0 are respectively $a^2\varepsilon_+ V$ and $\varepsilon_0 M_0(V)$. Partial fractions (E.16) imply, for some fixed $c_0 > 0, N < \infty$,

$$\varepsilon_+ + \varepsilon_0 \le C\beta^{-N} e^{-c_0/\beta}, \qquad c \le \varepsilon_0^\beta \le C, \qquad \varepsilon_+^\chi \le C e^{-c_0/(2q)}. \tag{E.13}$$

To check uniformity, $\beta \le Ca$, so every inverse power of a is bounded by a power of β^{-1}. The logarithmic terms in the partial fractions are $q \log a$, $\beta \log a$, and $\beta \log \beta$, all tending to zero. The action between u_0 and u_- has a fixed positive limit, while the full action to u_- has the positive limit in (E.19). These facts give all three estimates, with the last obtained by multiplying the logarithm of the first by $\chi = \beta/q$.

Before u_0, the middle map satisfies

$$M_0(0) = 0, \quad M_0'(0) = 1, \qquad \|M_0 - \mathrm{id}\|_{C^3([0, V_1 + 1])} \le C\beta^{-N} e^{-c_0/(2\beta)}. \tag{E.14}$$

Indeed $\rho'/\rho = (L - h')/h$ has at most one zero, a minimum; the endpoint actions give $\sup_{[u_0, u_+]} \rho \le C\beta^{-N} e^{-c_0/\beta}$. The ordinary equation is

$$V_u = \mathscr{F} = \frac{\lambda\rho(E - KL/h)V^2}{1 + \lambda K \rho V}.$$

On $[0, V_1 + 2]$ its denominator is at least one, and the integral of its C^3_V norm is at most $C\beta^{-N}(1 + |\log \beta|)e^{-c_0/\beta}$. The first-exit and triangular-variation argument of Lemma A.1 proves the estimate; $\mathscr{F} = O(V^2)$ gives $M_0'(0) = 1$. No ordinary u-derivative at the shrinking positive cut is needed.

On the positive middle cut write the coordinate restrictions as $\Xi_{+,m}(y), Z_{+,m}(y)$, and let $z_+ = Z'_{+,m}(0) > 0$. Define

$$\widehat{V} = \frac{Z_{+,m}(\varepsilon_+ V)}{z_+ \varepsilon_+} \left(\frac{-\Xi_{+,m}(\varepsilon_+ V)}{-\Xi_{+,m}(0)} \right)^{1/\chi}.$$

The weak coordinate is negative on this cut. Taylor's formula and (E.13) give $\widehat{V} = V + O_{C^3}(\varepsilon_+/\chi)$, with $\widehat{V}'(0) = 1$. Its inverse has the same bounds. Choose a fixed incoming local cut $x = x_+ > 0$ in the positive box. The stopped orbit $\Xi = 0$ intersects it at a height $y_+ > c > 0$, since $\Xi_y < 0$. On a fixed chord above that height the function

$$L_+(y) = -\Xi_{+,\mathrm{out}}(y) Z_{+,\mathrm{out}}(y)^\chi$$

is increasing, vanishes at y_+, and has a uniformly bounded C^3 inverse. The exact invariant gives

$$L_+(y) = \Sigma_+ \widehat{V}^\chi, \qquad \Sigma_+ = -\Xi_{+,m}(0)(z_+\varepsilon_+)^\chi \le Ce^{-c_0/(2q)}.$$

Continue this chord to the actual positive section. In $(X, Y) = (u/a, w/a^2)$ the limiting stopped graph is $Y = (X^2 - 1)/2$, $X \ge 1 + x_+$. A smaller chord has positive height and denominator margins to a fixed large cut and then enters a common parabolic band. Apply the outgoing-tail normalization (C.12)–(C.19) and the fixed-port admission argument of Lemma F.9. Both use only this half-family and its common band. If d_+ is the exact raw derivative at y_+, then $p = (x_{\mathrm{in}} - x_b)/d_+$ gives $L_+(y) = \Phi_+(p)$, $\Phi_+(0) = 0$, with positive derivative units and uniform C^3 inverse bounds. Retain the absolute gain $d_+ = \Omega a^{2-\lambda_\gamma} U_+ \le C$, where U_+ is a positive unit. Set

$$\sigma_+ = \Phi_+^{-1}(\Sigma_+), \qquad \Psi(t) = \frac{\Phi_+^{-1}(\Sigma_+ t)}{\sigma_+}, \qquad P_* = d_+\sigma_+.$$

Then $\sup |t\Psi''/\Psi'| \le C\Sigma_+$, giving the positive assertions without any bound on unscaled raw derivatives.

At the negative cut u_0, denote the coordinate restrictions by $\Xi_{-,m}(w), Z_{-,m}(w)$, and put $\xi_0 = \Xi_{-,m}(0) > 0$, $z_0 = Z'_{-,m}(0) > 0$. For v in a fixed bounded interval set

$$\mathcal{G}(v) = \frac{Z_{-,m}(\varepsilon_0 v)}{z_0 \varepsilon_0} \left(\frac{\Xi_{-,m}(\varepsilon_0 v)}{\xi_0} \right)^{1/\beta}, \qquad C = \mathcal{G} \circ M_0 \circ (V \mapsto \widehat{V})^{-1}.$$

This is the actual strong-coordinate correction on the early cut. Its C^3 difference from the identity is $O(\varepsilon_0/\beta)$; the cut η_0 is fixed. Together with (E.14) this yields $C'(0) = 1$ and the claimed estimate for $C - \mathrm{id}$, with $\epsilon_\theta = C\beta^{-N}e^{-c_0/(3\beta)}$. Hadamard division gives $C(v) = vc(v)$, $\|c - 1\|_{C^2} \le C\epsilon_\theta$, so $|ve'_C/e_C| \le C\epsilon_\theta = o(\chi)$.

On the fixed outer negative cut $u = -1 - x_-$, $\Xi_w > 0$, so the stopped branch $\Xi = 0$ meets it at a height $w_- > cx_- > 0$. On a slightly larger positive-height chord its invariant

$$I_-(w) = \Xi_{-,\mathrm{out}}(w) Z_{-,\mathrm{out}}(w)^\beta$$

has $I'_- > c > 0$ and uniformly bounded C^3 norm. Writing $r = \beta^{-\beta} C(\widehat{V})^\beta$, equality of the two invariants is precisely

$$I_-(w) = \alpha_\theta r, \qquad \alpha_\theta = \xi_0(z_0\varepsilon_0\beta)^\beta.$$

The constant α_θ is bounded above and away from zero by (E.13). Also $\alpha_\theta \le C\eta_0$, while $r \le 1 + o(1)$ on the fixed amplified-height interval. The choices above can be made with $\pi(-x_-, y_{\max}) \ge c y_{\max} > 0$, and then with

$$(1+\epsilon)\sup_\theta \alpha_\theta < \tfrac{1}{2}\inf_\theta I_-(y_{\max}).$$

The whole invariant interval consequently lies strictly inside the output chord, with a uniform buffer. The inverse function theorem gives $w = D_0(r)$ on $[0, 1+\epsilon]$, with uniform C^3 and inverse bounds. At $r = 0$ its output is $w_- > 0$. The extension at the excluded endpoint $\widehat{V} = 0$ is thus obtained on a positive-height chord.

From this cut the limiting stopped graph is $w = (u^2 - 1)/2$. Its buffered chord has the same finite-cut margins and negative-tail band. The preceding normalization and admission argument, now on the negative half-tail, gives $T(w) = t_b + \Omega U_- B_-(w)$, with $B'_- > c$ and uniform C^3 map and inverse bounds. In particular $|T''/T'| \le C$, although its raw slope may vanish. For the divided finite closing map, the chain rule gives

$$\frac{(f_\theta(mT)/m)''}{(f_\theta(mT)/m)'} = \frac{T''}{T'} + m\frac{f''_\theta(mT)}{f'_\theta(mT)}T'.$$

The last term is bounded on the fixed output window since $0 < \Omega \le C$. Normalizing its exact value and positive slope at a reference point, and composing with D_0, gives D and the constants b_*, a_* in (E.9). This proves the negative assertions.

Finally, in the remote-root chart the coefficients on both local boxes are regular functions of j. The source-tail scales are exactly Ω and $\Omega a^{2-\lambda_\gamma}$; $|\lambda_\gamma| < 1$ keeps both bounded above. No reciprocal bound on these raw scales was used. The fixed-cut action has no $\beta \log j$ term, so every estimate above is uniform as $j \downarrow 0$, as a one-sided family of physical passages. □

E.5 The fixed-exit continuation

For nonzero limiting action $\Theta = \beta \log V$, orbits of (E.1)–(E.4) leave the small-height region before the negative saddle. At the fixed exit $u_c \in (-1, u_-)$ the height $W = w(u_c)$ provides a regular coordinate. Shooting backwards from this section constructs the scalar inverse phase and its C^2 limit without using the positive return-section map.

E.6 The action interval and the exit height

For $J(u) = \exp\int_{u_+}^{u} L/h$, put

$$\rho = \frac{h(u_-)J(u)}{J(u_-)h(u)}, \qquad \Gamma_\beta = -\beta\log\rho, \qquad H_\beta = h/\beta. \tag{E.15}$$

The exact partial fractions are

$$\frac{L}{h} = -\frac{1}{\beta(u+1)} + \frac{1}{\chi(a-u)} - \frac{1 + 1/\beta + 1/\chi}{c-u}. \tag{E.16}$$

Thus

$$\Gamma'_\beta = -L/H_\beta + \beta H'_\beta/H_\beta, \qquad H_\beta = \frac{b_-(u+1)(a-u)(c-u)}{(1+a)(1+c)}. \tag{E.17}$$

On negative compact intervals this tends in C^3 to

$$H_{0,c} = \frac{(-u)(1+u)(c-u)}{1+c}, \qquad \Gamma_c(u) = \log\frac{(1+u)(c-u_-)}{(1+u_-)(c-u)}. \tag{E.18}$$

Indeed, $\Gamma_c' = -u/H_{0,c}$, and both actions vanish at u_-. The limit action labels the point at which an orbit leaves the small-height region. From that point to the fixed exit, the limiting graph equation is $w_u = u$. These two limits will parametrize the inverse transition.

At the shrinking positive cut, integration of (E.16) gives

$$\beta \log J(u_-) = \log \frac{(1+u_+)(c-u_-)}{(1+u_-)(c-u_+)} + q \log \frac{(a-u_+)(c-u_-)}{(a-u_-)(c-u_+)} + \beta \log \frac{c-u_-}{c-u_+}.$$

By (E.6),

$$\Gamma_\beta(u_+) \longrightarrow \Gamma_{-,c} := \Gamma_c(0) = \log \frac{c-u_-}{c(1+u_-)} > 0. \tag{E.19}$$

Also $(L-h')' = 1 + o(1) > 0$; hence $\Gamma_\beta' = -\beta(L-h')/h$ has at most one zero, a maximum.

Choose $0 < \Delta < \frac{1}{4} \inf_c \Gamma_{-,c}$, and set

$$r_1 = \Gamma_c^{-1}(\Gamma_{-,c} - \Delta), \qquad r_1^\sharp = \Gamma_c^{-1}(\Gamma_{-,c} - 3\Delta/4).$$

Choose fixed cuts u_c, u_R and $\epsilon > 0$ with

$$-1 < u_c < u_- - 2\epsilon, \qquad r_1^\sharp(c) < u_R < 0, \qquad \Gamma_c(u_R) > \Gamma_{-,c} - \Delta/2.$$

These choices are uniform in the compact c-family. Define

$$\mathcal{W}(r) = \frac{u_c^2 - r^2}{2}, \quad \mathcal{J}_c = \mathcal{W}([u_-, r_1]), \quad \mathcal{J}_c^\sharp = \mathcal{W}([u_- - \epsilon, r_1^\sharp]). \tag{E.20}$$

They are positive compact intervals with a uniform strict inclusion.

E.7 Convergence of the inverse transition

We first work only with the scalar equation (E.4), without assuming any positive return-section map. Let $\mathcal{K}_c$ be a compact interval strictly inside $(0, u_c^2/2)$, with uniform endpoint margins, and prescribe $w(u_c; W) = W$ for $W \in \mathcal{K}_c$. Write

$$r(W) = -\sqrt{u_c^2 - 2W}, \qquad d_{\mathcal{K}} = \inf_{c, W \in \mathcal{K}_c} \{\Gamma_{-,c} - \Gamma_c(r(W))\} > 0.$$

Increase $u_R < 0$ towards zero if necessary, so that it lies beyond all $r(W)$ and $\Gamma_c(u_R) > \Gamma_{-,c} - d_{\mathcal{K}}/4$. We follow the backward shot through its positive-height part, the thin transition layer, and the remaining small-height part. The next three first-exit arguments continue it to u_+ and estimate its phase. They apply, in particular, to $\mathcal{J}_c^\sharp$ and to any fixed larger chord still inside $(0, u_c^2/2)$. Define

$$\phi_\beta(u; W) = \beta \log \frac{w(u; W)}{h(u)} + \Gamma_\beta(u). \tag{E.21}$$

Its value at u_+ gives the exact inverse phase

$$\Psi_\theta(W) = \phi_\beta(u_+; W) - \beta \log \lambda = \beta \log V =: \Theta. \tag{E.22}$$

The differential equation for this phase is

$$\phi_\beta' = \frac{r(Eh - KL)}{H_\beta(1 + Kr)}, \qquad r = w/h. \tag{E.23}$$

The phase changes by only a small amount after the orbit has entered the small-height part. This will allow the leaving point of the limiting graph to determine the inverse phase.

Lemma E.2. *For fixed $\omega_0 > 0$, the graph meets $w = \beta\omega_0$ at a unique point $\sigma_\beta(W) < u_R$, and, uniformly on $\mathcal{K}_c$,*

$$\|\sigma_\beta - \mathcal{W}^{-1}\|_{C^2} \le C(\omega_{\mathrm{d}} + \beta|\log\beta|) \longrightarrow 0.$$

Here ω_{d} bounds in C^3 the differences $H_\beta - H_{0,c}, K-1, L-u, E$ on the fixed negative rectangle.

Proof. On that rectangle $L + Ew \le -\ell < 0$. The inverse graph, $U_\beta(W;W) = u_c$, satisfies

$$\partial_w U_\beta = \frac{K}{L+Ew} + \frac{\beta H_\beta}{w(L+Ew)}.$$

Between $\beta\omega_0$ and the fixed upper height, its right side differs from $1/u$ in L^1_w by at most $C\omega_{\mathrm{d}} + C\int_{\beta\omega_0}^{W_{\max}} \beta\, dw/w$. Gronwall's inequality and the first two triangular variational equations give the asserted C^2 bound. To justify existence, apply this estimate only up to a first exit from a fixed rectangle about the limiting graphs. Their endpoints $r(W) < u_R$ stay strictly inside, so an exit is impossible. Negative nonzero speed at the selected height gives uniqueness and transversality. □

Lemma E.3. *Uniformly on $\mathcal{K}_c$,*

$$\|\phi_\beta(u_R;W) - \Gamma_\beta(\sigma_\beta(W))\|_{C^2} \le C\beta.$$

Proof. On this fixed negative rectangle, $g_\beta = -(L-h')/H_\beta \ge g_0 > 0$, and the exact normal equation is

$$\beta r' = \frac{r\{-g_\beta + \beta r(E - KH'_\beta/H_\beta)\}}{1+Kr}.$$

Set $\zeta = (u-\sigma)/\beta$, so $r_\zeta = f_\beta$, $r(0;\sigma) = \omega_0/H_\beta(\sigma)$. On the fixed bounded r-interval, $f_r \le -a_0 < 0$, $|f_u| + |f_{uu}| \le Cr$, and $|f_{ur}| + |f_{rr}| \le C$. The triangular variation argument of Lemma A.1, with these decaying parameter forcings, gives

$$|\partial^j_\sigma r| \le C_j(1+\zeta^j)e^{-c\zeta}, \qquad j = 0,1,2.$$

A first-exit argument retains the rectangle up to u_R. For $G(\sigma,\zeta) = F_\beta(\sigma+\beta\zeta, r)$, where F_β is the right side of (E.23), put $T = (u_R-\sigma)/\beta$, $J = \beta\int_0^T G\, d\zeta$. The moving endpoint must be retained:

$$J_\sigma = \beta\int_0^T G_\sigma\, d\zeta - G(\sigma,T), \qquad J_{\sigma\sigma} = \beta\int_0^T G_{\sigma\sigma}\, d\zeta - 2G_\sigma(\sigma,T) + \beta^{-1}G_\zeta(\sigma,T).$$

The integral terms are $O(\beta)$; all boundary terms are exponentially small because $u_R - \sigma \ge d_0 > 0$. Thus $J = O_{C^2_\sigma}(\beta)$. At the crossing, $\phi_\beta - \Gamma_\beta = \beta\log(\omega_0/H_\beta(\sigma_\beta)) = O_{C^2_W}(\beta)$. Compose with the preceding C^2 crossing map. □

Lemma E.4. *For some fixed $d > 0$ and finite integer n,*

$$\|\phi_\beta(u_+;W) - \phi_\beta(u_R;W)\|_{C^2} \le C\beta^{-n}e^{-d/\beta}.$$

Proof. The strict cuts and preceding estimates give the fixed phase gap

$$\phi_\beta(u_R;W) \le \Gamma_{-,c} - 3d_{\mathcal{K}}/4, \qquad \inf_{[u_R,u_+]}\Gamma_\beta \ge \Gamma_{-,c} - d_{\mathcal{K}}/2.$$

The second bound follows from endpoint convergence and the fact that Γ_β has at most one critical point, a maximum. Equation (E.16) gives bounded L/H_β: at the shrinking cut its only potentially large term satisfies $q/(a-u) \le q/(a\eta_+) \le C$. Thus F_β and its first two r-derivatives are bounded; ordinary u-derivatives at that cut are not needed.

Since $h, K > 0$, the shot continues while it stays in the rectangle. The identity $r = e^{(\phi_\beta - \Gamma_\beta)/\beta}$ and the phase gap give $r \le Ce^{-d/\beta}$: up to a first loss, the phase change is at most $C\int r\,du$, too small to close half the gap. For its first two phase variations, $r_\phi = r/\beta$, $r_{\phi\phi} = r/\beta^2$. The scalar triangular variation argument therefore has exponentially small integrated coefficients times finite powers of β^{-1}. It preserves the bounded W-variations at u_R and gives the claimed C^2 error. □

Proposition E.5. *The scalar backward-shooting solutions are defined up to u_+, and, jointly in the compact c-family, on every chord $\mathcal{K}_c$ above,*

$$\Psi_\theta \longrightarrow \Psi_{0,c} = \Gamma_c \circ \mathcal{W}^{-1} \quad \text{in } C^2. \tag{E.24}$$

More precisely, for the coefficient error ω_{d} in Lemma E.2,

$$\|\Psi_\theta - \Psi_{0,c}\|_{C^2(\mathcal{K}_c)} \le C\{\omega_{\mathrm{d}} + \beta(1 + |\log\beta|)\}. \tag{E.25}$$

Using $\mathcal{K}_c = \mathcal{J}_c^\sharp$, its inverse $\mathcal{H}_\theta$ is defined on

$$0 \le \Theta \le \Gamma_{-,c} - \Delta \tag{E.26}$$

and satisfies $0 < c_H \le \mathcal{H}'_\theta \le C_H$ and $|\mathcal{H}''_\theta/\mathcal{H}'_\theta| \le C_H$.

Proof. The exact derivative (E.17), together with $\Gamma_\beta(u_-) = \Gamma_c(u_-) = 0$, gives $\Gamma_\beta - \Gamma_c = O_{C^3}(\omega_{\mathrm{d}} + \beta)$ on the fixed negative interval. Combining the three estimates in (E.22) and composing with the uniformly C^2 crossing $\sigma_\beta(W)$ gives (E.25); the exponentially small remainder is absorbed into $O(\beta)$. All derivatives here are in W with the coefficients fixed. For $r = \mathcal{W}^{-1}(W)$, $\Psi'_{0,c}(W) = 1/H_{0,c}(r) > 0$ with uniform upper and lower bounds. The larger intervals in (E.20) give a fixed image margin around (E.26). The uniform inverse function theorem applies. Explicitly,

$$u_{*,c}(\Theta) = \frac{c(1+u_-)e^\Theta - (c-u_-)}{(1+u_-)e^\Theta + (c-u_-)}, \qquad \mathcal{H}_{0,c}(\Theta) = \frac{u_c^2 - u_{*,c}(\Theta)^2}{2}.$$

In particular, $\Theta = 0$ has a positive interior height on the fixed exit section. □

The strict gap between the fixed-exit action endpoint and a negative axis cut is

$$\Gamma_{-,c} - \Gamma_c(-\sigma) = \log\frac{c+\sigma}{c(1-\sigma)} > 0, \qquad 0 < \sigma < 1. \tag{E.27}$$

Proposition H.13 uses this gap to place the fixed-exit construction and the bounded-height construction on overlapping actual receivers. It then proves the completed-return curvature estimate on their union, which is the physical interval used by the coefficient cover.

F The compact positive-ratio passage

We next treat the coefficient face $\alpha_0 = -s^2 < 0$ with $p = \alpha_1/s$ in a fixed interval $[p_0, P] \subset (0, \infty)$. The low-height orbits pass between a positive equatorial saddle and a negative saddle escaping to $X = -\infty$, whose characteristic number tends to zero.

The escaping saddle prevents a comparison on fixed compact source sections alone. We parametrize its passage by the point where the strong leaf meets the equator. An axis integral measures the normal expansion all the way to that saddle. Matching it with the positive Dulac map gives a curvature estimate in the input of the same return, uniformly as the negative characteristic number tends to zero. We then join this endpoint estimate to the compact middle and the high interval, where the derivative is bounded.

Lemma F.1 (Domains of the compact positive-ratio passage)**.** *For $p_0 \le p \le P$, sufficiently small $s > 0, |\gamma|$, and one fixed sufficiently small bound on $z(\gamma)$, the actual through interval is covered by the negative-saddle endpoint, a compact family of strict source passages, and the high-height interval. Consecutive pieces overlap in actual trajectories. The completed transition has negative curvature on the endpoint and compact middle, and bounded first derivative on the high-height interval. All cuts and estimates are uniform on the fixed p-interval.*

F.1 The equatorial roots and the strong foliation

With $a = -1$, write (C.1) as

$$X' = -(M + LY), \qquad Y' = -Y(XB + \zeta Y),$$

where

$$\begin{aligned} M &= 1 - pX + 3\gamma X^2 + sX^3, & B &= 1 + 2\gamma + sX, \\ L &= 1 - A(\gamma + sX) + C(\gamma + sX)^2, & \zeta &= Cs(\gamma + sX). \end{aligned} \tag{F.1}$$

Lemma F.2. *The roots of M are simple and have the order $-R < x_+ < x_f$, where $R \to \infty$, $x_+ \to p^{-1}$. For $M_-(q) = M(-q)$, $B_-(q) = B(-q)$, put $\kappa = -M'_-(R)$ and $\rho_- = \kappa/(RB_-(R))$. Then*

$$sR = 3\gamma + p/R + R^{-2}, \tag{F.2}$$

and

$$\kappa \ge cp_0, \qquad 0 < \rho_- \le C(R^{-1} + |\gamma| + sR) \longrightarrow 0. \tag{F.3}$$

At x_+, the rates $L_+ = -M'(x_+)$, $\nu_+ = x_+B(x_+)$ give $0 < \rho_0 \le \rho_+ = \nu_+/L_+ \le \rho_1 < \infty$.

Proof. On compact sets $M \to 1 - pX$, yielding x_+ and the negative sign immediately to its right. Since $M(0) > 0$, the cubic has a second positive root and a negative root, both escaping uniformly. Substitution at $-R$ proves (F.2). The factorization

$$M = s(X + R)(X - x_+)(X - x_f), \qquad sRx_+x_f = 1$$

gives $\kappa = x_+^{-1}(1 + x_+/R)(1 + R/x_f) \ge cp_0$. Direct differentiation also gives $\rho_- = (2p/R + 3\gamma + 3/R^2)/B_-(R)$. The positive rates converge on the compact p-interval. □

Fix $Q \gg P$, then $R_0 > 2Q$, and shrink the coefficients so that

$$d_Q = Q^{-2} + P/Q + |\gamma| + sR \ll 1. \tag{F.4}$$

In $q = -X$ the negative tail is

$$q' = M_-(q) + L_-(q)Y, \qquad Y' = Y\{qB_-(q) + \zeta_-(q)Y\}, \tag{F.5}$$

with

$$\begin{aligned} M_- &= 1 + pq + 3\gamma q^2 - sq^3, & B_- &= 1 + 2\gamma - sq, \\ L_- &= 1 - A(\gamma - sq) + C(\gamma - sq)^2, & \zeta_- &= Cs(sq - \gamma). \end{aligned}$$

Let $\mathcal{V}$ denote (F.5). Use the positive normal time $d\tau_n = qB_-(q)\,dt$, and write $V = \mathcal{V}/(qB_-(q))$. In $u = \log q$, $w = Y/q^2$, put $c = L_-/B_-$ and $k = \zeta_- q/B_-$. The axis speed is

$$h_-(q) = \frac{M_-(q)}{q^2B_-(q)} = \frac{q^{-2} + p/q + 3\gamma - sq}{B_-(q)} = O(d_Q). \tag{F.6}$$

The exact normal-time equations are

$$u' = h_- + cw, \qquad w' = w\{1 - 2h_- + (k - 2c)w\}.$$

The functions h_-, $c-1$, k and their first five Euler derivatives $D_q^j = (q\partial_q)^j$ are $O(d_Q)$ on $Q/2 \le q \le 2R$ and fixed logarithmic collars. Extend this triple consistently as in Lemma B.1. Its hypotheses hold with $\epsilon = Cd_Q$; choosing Q first and then the small coefficient bounds makes this quantity sufficiently small. The projection is therefore constructed directly from the negative-tail field, independently of either return-map estimate.

It gives a strong projection $\pi(q, Y)$ for V with uniform scaled bounds on the whole corridor, including the root R. The induction through one further order, using the fifth coefficient derivatives, gives a bounded C^4 norm for $\log \pi - \log q$ in $(\log q, Y/q^2)$. Throughout this section, the footpoint refers to this normal-time projection. In particular, its invariance equation is

$$\pi(q,0) = q, \qquad \pi_* V = b_-(r)\partial_r, \qquad b_-(r) = \frac{M_-(r)}{rB_-(r)}, \qquad r = \pi(q,Y). \tag{F.7}$$

Use the section

$$Y/q^2 = w_0, \qquad 0 < w_0 < 1/4. \tag{F.8}$$

At the limiting corner, the leaf over r satisfies $Y - q^2/2 = -r^2/2$, and meets the section transversely at $q = r(1-2w_0)^{-1/2}$. The uniform projection estimates give a unique intersection over every $R_0 \le r < R$, of the form

$$q_s(r) = rU_s(r), \qquad U_s \asymp 1, \qquad q_s'(r) \asymp 1, \qquad |(r\partial_r)^j \log U_s| \le C \quad (1 \le j \le 3).$$

We use this footpoint r to parametrize the strong section.

Figure 4 shows the compact core, long tails and root configuration.

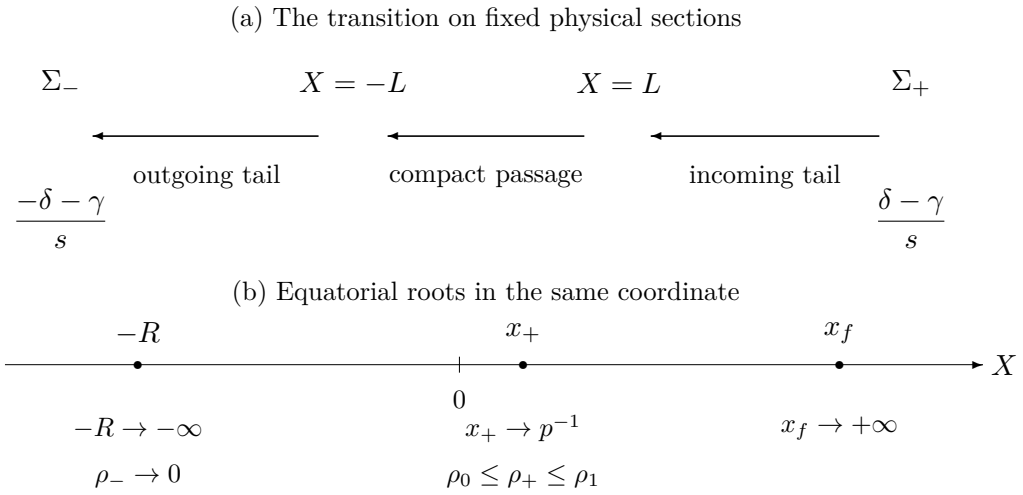


Figure 4: Two consequences of the rescaling $v = \gamma + sX$, $z = s^2 Y$, for $\alpha_0 = -s^2$ and $p \in [p_0, P]$. The positions and distances are schematic. In (a), a fixed compact core does not include the tails leading to $\Sigma_\pm$; the arrows denote the three factors of the positive-time transition. In (b), the ordered zeros of M are shown as $s, \gamma \to 0$. The negative incident saddle escapes and its characteristic number vanishes, although the positive incident saddle remains on a compact scale. The axis arrow gives the orientation of X, not the flow.

F.2 An integrating coordinate up to the negative saddle

The linear normal equation along the negative axis is $d\log Y/dq = qB_-(q)/M_-(q)$. Its integral suggests the coordinates

$$\mathcal{A}_-(r) = \int_Q^r \frac{qB_-(q)}{M_-(q)}\,dq, \qquad \mathcal{D}_- = \frac{M_-(r)}{rB_-(r)}\partial_r. \tag{F.9}$$

Then $\mathcal{D}_-\mathcal{A}_- = 1$, and $\mathcal{A}_- \to \infty$ at R. Thus $\mathcal{A}_-$ covers the entire interval up to the saddle, although the upper bound R for the footpoint tends to infinity. The next lemma relates this linear integral to the height of the actual orbit on the fixed cut $q = Q$.

Lemma F.3. *The height $Y_Q(r)$ at the fixed cut $q = Q$ of the orbit through the strong section over r satisfies*

$$\log Y_Q = c_\theta - \mathcal{A}_-(r) + \mathcal{R}_-(r), \qquad R_0 \le r < R, \tag{F.10}$$

where, for $j = 1, 2$,

$$|\mathcal{D}_-^j \mathcal{R}_-| \le C(d_Q + e_{R_0}), \qquad e_{R_0} = C_Q R_0^2 e^{-c(R_0-Q)/P} \longrightarrow 0. \tag{F.11}$$

The value of $\mathcal{R}_-$ need not be small.

Proof. Apply Lemma B.2 to the triple just constructed. The polynomial in (F.5) gives $M_-(q)/q \ge p_0/2$ on $[Q/2, Q]$ after shrinking the coefficients, and $b_- > 0$ up to its simple root R. After R_0 is fixed, the same polynomial gives $b_- \le CP$ on $[Q, R_0]$. Thus the lemma constructs the whole actual height interval within $Q/2 < q < 2R$, and yields

$$\log Y_Q = -\mathcal{A}_-(r) + 2\log r + \log \mathcal{U}(r) + E_Q(Y_Q), \tag{F.12}$$

with the unit and fixed-cut bounds stated there. In particular,

$$0 < Y_Q(r) \le e_{R_0} = C_Q R_0^2 e^{-c(R_0-Q)/P}. \tag{F.13}$$

Choose R_0 large and then shrink the coefficient neighborhood. Taking $c_\theta = 0$ and $\mathcal{R}_- = 2\log r + \log \mathcal{U} + E_Q(Y_Q)$, (B.7) gives $\mathcal{D}_-\mathcal{R}_- = 1 - K$ and $\mathcal{D}_-^2\mathcal{R}_- = -\mathcal{D}_- K$, proving the claimed bounds. The lemma also gives $Y_Q \downarrow 0$ at R. □

F.3 Derivatives on the return sections

We now continue from the strong section to the actual return section, retaining the footpoint power and the divided closing. In the following proof, w_0 keeps the small-lip restriction $0 < w_0 < 1/4$ of (F.8); only w_1 is chosen in $(1/2, 1)$.

Lemma F.4. *Let $\Gamma_-(r)$ denote the resulting coordinate on the negative return section. Then*

$$-\Gamma_-'(r) = \Omega_-(\theta) r^{1-\lambda_\gamma} U_-(r) > 0, \qquad \Omega_- > 0, \tag{F.14}$$

where U_- and its reciprocal are bounded and

$$|\mathcal{D}_-^j \log U_-| \le C(d_Q + e_{R_0}), \qquad j = 1, 2. \tag{F.15}$$

Proof. Fix w_0, $w_1 \in (1/2, 1)$, $\mu = 3/2$, and the small physical width δ before its nested output windows. All phase derivatives fix coefficients and multiplier. On the strong section the inverse of $\upsilon = \widehat{\pi}(u, w_0)$, $\upsilon = \log r$, is $u = a(\upsilon) = \log q_s(r)$. The projection bounds give

$$a_1 \asymp 1, \qquad |a_2| + |a_3| \le C, \qquad 1 < q_s/r < 2, \qquad a_j = \partial_\upsilon^j a. \tag{F.16}$$

Also, for $h = M_-/(r^2 B_-)$,

$$\mathcal{D}_- = h\partial_\upsilon, \qquad |h| + |h_\upsilon| \le C d_Q. \tag{F.17}$$

The scalar graph and its logarithmic-radius equation are

$$Y_q = f = \frac{Y(qB_- + \zeta_- Y)}{M_- + L_- Y}, \qquad W_u = F(u, W) = W\left\{\frac{B_- + ZW}{D + L_- W} - 2\right\}, \tag{F.18}$$

where $W = Y/q^2$, $D = q^{-2} + p/q + 3\gamma - sq$, and $Z = q\zeta_-$. On $w_0 \le W \le w_1$, after choosing Q and shrinking the coefficients,

$$D + L_- W \ge c, \qquad F(u, w_0) > c, \qquad F(u, w_1) < -c, \qquad F_W \le -\mu. \tag{F.19}$$

The needed coefficient and state derivatives are bounded; explicitly,

$$qf_Y = F_W + 2, \qquad F_{WW} = \frac{2D(ZD - B_-L_-)}{(D + L_-W)^3}, \qquad F_{WWW} = -\frac{6DL_-(ZD - B_-L_-)}{(D + L_-W)^4}. \tag{F.20}$$

Thus the strip is invariant and $q' > 0$. Since $\sup_{r<R} q_s(r) < q_H = (\delta+\gamma)/s$, every orbit reaches this raw exit once, even when $q_s > R$.

To extract the long-tail power, compare with

$$F^0_\gamma(W) = W\left\{\frac{1 + 2\gamma}{3\gamma + L_\gamma W} - 2\right\}, \quad L_\gamma = 1 - A\gamma + C\gamma^2, \quad w_\gamma = \frac{1 - 4\gamma}{2L_\gamma}.$$

Here $F^0_\gamma(w_\gamma) = 0$, $(F^0_\gamma)'(w_\gamma) = -2 + \lambda_\gamma$, $(F^0_\gamma)'' = O(|\gamma|)$, and $F - F^0_\gamma = O(sq + p/q + q^{-2})$ in C^1_W. Scalar contraction, with $q_0 = q_s(r)$, consequently gives

$$\left| f_Y - \frac{\lambda_\gamma}{q} \right| \le C\left(s + pq^{-2} + q^{-3} + \frac{|\gamma|}{q}(q_0/q)^\mu\right). \tag{F.21}$$

The last term retains entrance memory and has integral at most $|\gamma|/\mu$, independently of the tail length. Hence

$$I(r) = \int_{q_0}^{q_H} (f_Y - \lambda_\gamma/q)\, dq, \qquad |I(r)| \le C(\delta + |\gamma| + p/q_0 + q_0^{-2}). \tag{F.22}$$

For the fixed-entry first variation,

$$(J_Y)_q = f_Y J_Y, \tag{F.23}$$

so the exact cocycle is

$$\frac{J_Y(q)}{J_Y(q_0)} = (q/q_0)^{\lambda_\gamma} \exp \int_{q_0}^{q} \{f_Y(\widetilde{q}, Y(\widetilde{q})) - \lambda_\gamma/\widetilde{q}\}\, d\widetilde{q}. \tag{F.24}$$

The moving initial value $W(a(v); v) = w_0$ gives

$$J := \partial_v W = -F(a, w_0)a_1 \exp \int_a^u F_W\, dv < 0, \qquad |J| \le Ce^{-\mu(u-a)}. \tag{F.25}$$

Differentiate its logarithm using the triangular variation argument of Lemma A.1. The first integral forcing is $F_{WW}J$; in the second row it is $F_{WWW}J^2 + F_{WW}J\partial_v \log(-J)$. The moving lower-limit terms involve only a_1, a_2, a_3 and the bounded entrance derivatives of F. Integrating the decaying forcings in order yields

$$|\partial_v \log(-J)| \le C, \tag{F.26}$$

$$|\partial_v^2 \log(-J)| \le C. \tag{F.27}$$

No reciprocal of J is estimated separately.

For raw admission choose $0 < \eta \ll \delta$, then require $2sR < \eta$. Every orbit crosses $sq = \eta$ in the strip. Contraction on the fixed remaining raw interval, compared with $w_\Gamma(x) = (\sqrt{1 + x^2}+1)^{-1}$, gives

$$|W_H - w_\Gamma(\delta + \gamma)| \le C(\eta/(\delta + \gamma))^\mu + C_{\eta,\delta}(|A - 1| + |C - 1| + |\gamma| + sp + s^2).$$

The raw height is $z_H = (\delta+\gamma)^2 W_H$; its organizing height shifts by $O(\delta|\gamma|)$. Choose η/δ from the fixed window reserve, then shrink the coefficients. On this whole admitted image the divided adapter $\chi = \mathcal{C}_{\theta,m}$ has the ordinary C^3 bounds of Lemma 3.3. Set $\Gamma_-(r) = \chi(z_H(r))$. Its observer identities and the preceding two rows give

$$|\partial_\upsilon \log(-\Gamma'_-)| + |\partial_\upsilon^2 \log(-\Gamma'_-)| \le C. \tag{F.28}$$

Finally separate $F_W = -2 + \lambda_\gamma + (qf_Y - \lambda_\gamma)$ in the moving cocycle:

$$-z'_H = \Omega_\theta r^{1-\lambda_\gamma} \widehat{U}_-, \qquad \Omega_\theta = s^{2-\lambda_\gamma}(\delta+\gamma)^{\lambda_\gamma}, \tag{F.29}$$

$$\widehat{U}_- = F(a, w_0) a_1 (q_s/r)^{2-\lambda_\gamma} e^{I(r)}, \qquad U_- = \chi'(z_H)\widehat{U}_-. \tag{F.30}$$

The factors $\widehat{U}_-, U_-$ are positive units, with Ω_θ kept separate. Subtracting $(1-\lambda_\gamma)\upsilon + \log\Omega_\theta$ in (F.28) bounds two ordinary derivatives of $\log U_-$. Equation (F.17) then gives $\mathcal{D}_- \log U_- = O(d_Q)$ and $\mathcal{D}_-^2 \log U_- = O(d_Q^2)$, proving the assertion. The same bounds before the adapter give $\mathcal{D}_-^j z_H = O((sr)^{2-\lambda_\gamma} d_Q^j)$, $j = 1, 2$. □

At the other saddle the characteristic number remains compact. Its Dulac estimate must still include the actual incoming coordinate, because the curvature is taken in the physical affine input.

Lemma F.5 (The fixed-cut height and the actual positive half-gate)**.** *For fixed $p \in [p_0, P] \Subset (0,\infty)$, physical windows, and then fixed Q, there is a common positive radius H_* such that the map from the local positive-saddle exit to $q = Q$ is $\Psi_\theta(\upsilon)$, with $\Psi_\theta(0) = 0$, bounded C^3 norm, and derivative bounded above and away from zero. The actual incoming half-gate $x = \Gamma_+(H)$, on a slightly larger interval than $[0, H_*]$, satisfies*

$$\Gamma'_+ > 0, \qquad \left|\frac{\Gamma''_+}{\Gamma'_+}\right| + \left|\frac{\Gamma'''_+}{\Gamma'_+}\right| \le C_Q.$$

Its exact value-and-slope normalization and its inverse on a smaller normalized image have uniform C^3 bounds. Its full raw image is admitted by the preassigned incoming window. For every fixed $0 < \overline{H} \le H_*$, $y_+(\overline{H}) = Y_Q^+(\overline{H})$ has a uniform positive lower bound. No uniform lower bound for the raw Γ'_+ or bound for the raw inverse is asserted.

Proof. The fixed rescaled axis segment from the outgoing saddle cut to $q = Q$ has a positive speed margin. Its finite hitting map Ψ therefore has $\Psi(0) = 0$, positive derivative units and uniform C^3 bounds; no raw-tail scale enters this map.

For the incoming half-map, follow the stable separatrix backwards. In the limiting field its denominator $D = Y + 1 - pX$ cannot first vanish, since $D_X = XY/D - p > 0$ near a proposed zero; a quadratic upper barrier excludes finite blow-up. The compact half-family consequently enters a common parabolic band at a fixed $X = L_0$. A buffered chord has a finite hitting map $A_\theta(H) = Y(L_0)/L_0^2$ with positive derivative units and uniform C^3 bounds, including $H = 0$.

On the outward tail $F_w \le -\lambda < 0$, and the state derivatives through order three are bounded. The relative-jet argument (A.3), with $0 < J_1 \le e^{-\lambda t}$, bounds $J_2/J_1, J_3/J_1$. Exact value-and-slope normalization thus gives the asserted map and inverse bounds. The fixed-port argument of Lemma F.9 admits its whole raw chord: that argument needs only this common parabolic band, retains the entrance memory and $O(d|\gamma|)$ port shift, and reaches the original window through its fixed raw tube. Compose the ordinary bridge on this admitted image.

Finally Lemma A.8 and Ψ give $y_+(H) = c_+ H^{\rho_+} e^{\mathcal{R}_+(H)}$, with c_+ a positive unit. Thus for fixed $0 < \overline{H} < 1$, $y_+(\overline{H}) \ge c_Q \overline{H}^{\rho_1} e^{-M_Q} > 0$. This height bound involves no lower bound on the raw incoming slope. □

Lemma F.6. *There is a fixed incoming coordinate $H > 0$ at x_+ for which the height on $q = Q$ is*

$$Y_Q^+(H) = c_+ H^{\rho_+} e^{\mathcal{R}_+(H)}. \tag{F.31}$$

For $0 < \epsilon < \min(1, \rho_0)$, the Euler derivatives of $\mathcal{R}_+$ through order two are bounded by $CH^\epsilon(1 + |\log H|)$. The continuation to the input x in (2.13) has the form $x = \Gamma_+(H)$, with

$$\Gamma'_+(H) = \Omega_+(\theta) U_+(H) > 0, \tag{F.32}$$

where U_+ and its reciprocal are bounded and its first two Euler logarithmic derivatives have the same vanishing bound.

Proof. Apply Lemma A.8 to the rates in Lemma F.2, then compose with the fixed-cut map Ψ above. Hadamard division gives $\Psi(v) = \Psi'(0) v e^{j(v)}$, $j(0) = 0$, with bounded C^2 norm. Its first two Euler rows along the Dulac map are $O(H^{\rho_+})$, so this composition preserves the asserted remainder. For the incoming half-gate set $\Omega_+ = \Gamma'_+(0)$, $U_+ = \Gamma'_+(H)/\Gamma'_+(0)$. The preceding relative C^3 bounds give $\mathcal{E} \log U_+, \mathcal{E}^2 \log U_+ = O(H)$, $\mathcal{E} = H\partial_H$. The final input differs from raw height only by the translation a_θ. □

F.4 Endpoint curvature and the total count

Lemma F.7 (Exhaustion of the low central heights)**.** *There are fixed $R_a < R_b$, $R_a > 2Q$, and $u_L > 0$, chosen before the final coefficient shrink, such that every actual through orbit with $0 < u := Y(0) \le 2u_L$ is represented uniquely by the same negative strong footpoint $R_b < r < R$ and by the positive Dulac coordinate $0 < H < \overline{H}$. The identification on the fixed cut is*

$$N_\theta(u) = y_\theta(r) = Y_Q^+(H), \qquad N_\theta(u) := Y(Q; Y(0) = u).$$

All actual input and output windows are those already fixed; zero height is not included as a through orbit.

Proof. Start at (Q, y), $0 < y < w_0 Q^2$. Until the strong-section hit, the footpoint $p_f = \pi(q, Y)$ increases below R, and the projection estimates give

$$(\log q)' \ge h(\log p_f) + (1 - C\epsilon) w > 0, \qquad q < 2p_f < 2R, \qquad (\log w)' \ge \nu_0 > 0.$$

Thus there is one transverse hit in finite normal time, inside the unchanged corridor; it may have $q_s > R$. Scalar first variation and $\partial_q(Y - w_0 q^2) = q_s F(\log q_s, w_0) > 0$ give $dq_s/dy < 0$, hence $dr/dy < 0$. The reverse construction in Lemma B.2 has $y_r = -Ky/b < 0$, $K \ge 1/2$, and $y \to 0$ at R. It is therefore a bijection $[R_a, R) \to (0, y(R_a)]$, identical to the forward construction by uniqueness.

On $0 \le q \le Q$, choose a small signed height strip with $M_- + L_- Y > c > 0$. Its finite map satisfies $N(0) = 0$, $c_Q \le N' \le C_Q$, with common finite-order bounds. Choose $\overline{H}$ for the positive Euler errors, then R_a so that $e_{R_a} < \inf y_+(\overline{H})$ and the fixed-cut errors are small; take $R_0 = R_a$. On every fixed footpoint interval the exact action-height formula bounds the unit and action, giving $\inf y(R_a) > 0$. Choose $R_b > R_a$ with $e_{R_b} < \inf y(R_a)$, then require $R > 2R_b$. The same fixed-interval formula gives

$$y(R_b) = R_b^2 \mathcal{U}(R_b) e^{-\mathcal{A}(R_b) + E_Q(y(R_b))} \ge \eta_b > 0.$$

Choose $2u_L$ below the domain of N and η_b/C_Q. Then $N(u) < y(R_b)$ selects a unique $r > R_b$; it is also below $y_+(\overline{H})$ and selects a unique positive Dulac preimage. Equality at $q = Q$ identifies the orbit, so both previously admitted gates and the divided closing apply throughout this interval. □

The exhaustion lemma identifies the same orbit in the two endpoint coordinates. Equating their descriptions of Y_Q gives

$$\rho_+ \log H + \mathcal{R}_+(H) = -\mathcal{A}_-(r) + \mathcal{R}_-(r) + c_\theta. \tag{F.33}$$

Set $\mathcal{D} = H\partial_H$ and $K_- = 1 - \mathcal{D}_-\mathcal{R}_- \geq 1/2$. Then

$$Hr_H = -\frac{\rho_+ + \mathcal{D}\mathcal{R}_+}{K_-}\frac{M_-(r)}{rB_-(r)} < 0. \tag{F.34}$$

By (F.13) and (F.31), throughout this endpoint

$$H \leq C_Q R_0^{2/\rho_0} e^{-c(R_0-Q)/(P\rho_1)}.$$

Thus the same choice of R_0 makes all positive Dulac remainders small on the entire interval $R_0 \leq r < R$.

Proposition F.8. *The complete map $x = \Gamma_+(H) \mapsto \Pi_\theta(x) = \Gamma_-(r(H))$ has $\Pi''_\theta(x) < 0$ on this endpoint. Its displacement has at most two zeros counted with multiplicity.*

Proof. Write $y(H) = Y_Q^+(H)$ and $\mathcal{O}(y(r)) = \Gamma_-(r)$. Put $b = M_-/(rB_-)$, $\mathcal{D}_- = b\partial_r$, and $K = -\mathcal{D}_- \log y \geq 1/2$. The action-height jets give $K = 1 + O(d_Q + e_{R_a})$ and $\mathcal{D}_- K = O(d_Q + e_{R_a})$. The coefficient quotients are

$$M'_-/(rB_-) = (p/r + 6\gamma - 3sr)/B_- = O(d_Q), \tag{F.35}$$

$$M_-/(r^2B_-) = (r^{-2} + p/r + 3\gamma - sr)/B_- = O(d_Q), \tag{F.36}$$

$$sM_-/(rB_-^2) = O(d_Q). \tag{F.37}$$

Thus $b' = O(d_Q)$. The identity $\mathcal{O}' = (-\Gamma'_-)b/(yK)$, together with Lemma F.4, gives

$$e_{\mathcal{O}} := 1 + y\frac{\mathcal{O}''}{\mathcal{O}'} = -\frac{\mathcal{D}_-\log(-\Gamma'_-) + b' - \mathcal{D}_-\log K}{K}, \qquad |e_{\mathcal{O}}| \leq Cd_Q + C_Q e_{R_a}.$$

No reciprocal negative saddle rate occurs. Set $\mathsf{A} = \rho_+ + H\mathcal{R}'_+$. In the actual input $x = \Gamma_+(H)$, the full chain rule is

$$H\partial_H \log \Pi'_x = -1 + \mathsf{A}e_{\mathcal{O}} + H\partial_H \log \mathsf{A} - H\partial_H \log \Gamma'_+. \tag{F.38}$$

Choose Q first. For this fixed cut choose $\overline{H}$ so that $|\mathsf{A} - \rho_+| \leq \min(1, \rho_0/2)$ and the last two terms each have absolute value at most $1/16$. Choose R_a using the positive fixed-cut image margin, as in Lemma F.7, and then shrink the coefficients so that $|e_{\mathcal{O}}| \leq 1/[16(\rho_1 + 1)]$. The right side is at most $-13/16$. Since $\Pi'_x > 0$, $H\Gamma'_+ > 0$,

$$\frac{\Pi_{xx}}{\Pi'_x} = \frac{H\partial_H \log \Pi'_x}{H\Gamma'_+} < 0.$$

For the physical multiplier $m > 0$, $\Delta'' = -m\Pi'' > 0$; Rolle's theorem with multiplicities gives the endpoint count. □

Lemma F.9 (Fixed parabolic tails and the physical sections)**.** *Fix the physical sections $\Sigma_\pm = \{v = \pm\delta\}$, their nested windows, and $0 < d < \delta \leq 1/32$, with $d \leq 1/100$. On an enlarged compact strict source family, one can choose a fixed L, and then sufficiently small bounds on the raw coefficients and on $s > 0$, so that the two outward graphs reach $v = \gamma \pm d$. Their entire images are admitted by uniformly regular increasing maps*

$$\mathcal{B}_{\sigma,\theta} : z(\gamma + \sigma d) \longmapsto z(\sigma\delta), \qquad \sigma = \pm 1.$$

These maps and their inverses have uniform phase C^3 bounds on slightly larger fixed windows. The divided closing map is defined on the full negative image for $0 \leq m \leq m_{\mathrm{ad}}$, where $m_{\mathrm{ad}} > 0$ is fixed before any zero-counting multiplier bound.

Proof. Choose $|\gamma| < \min\{d/2, (\delta - d)/2, 1/500\}$. For $r = Le^t$, $\xi = sr$, write $w = Y/r^2$ on either tail. Its exact equation is

$$\dot{w} = F_\sigma(t, w) = w\left(\frac{n_{0,\sigma} + n_{1,\sigma} w}{\ell_\sigma w + d_\sigma} - 2\right),$$

where

$$v_\sigma = \gamma + \sigma\xi, \qquad \ell_\sigma = 1 - Av_\sigma + Cv_\sigma^2,$$
$$d_\sigma = 3\gamma + \sigma\xi - \sigma p/r - a/r^2, \quad n_{0,\sigma} = 1 + 2\gamma + \sigma\xi, \qquad n_{1,\sigma} = \sigma C\xi v_\sigma.$$

The entrance L is chosen on the enlarged limiting family first. Continuous dependence on the fixed core then puts both entrance values strictly inside $[1/4, 3/4]$. Increase L, if necessary, so that $|p|/L \le 1/40$ and $|a|/L^2 \le 1/4000$, and require $A, C \in [9/10, 11/10]$ and $sL < d$. The fixed core is retained when these choices are made. On

$$1/4 \le w \le 3/4, \quad \xi \le 1/100, \quad |\gamma| \le 1/500, \quad |p|/r \le 1/40, \quad |a|/r^2 \le 1/4000$$

the denominator is bounded below and the vector field points strictly inward at both edges. Differentiating this rational quotient, as in (F.20), gives

$$F_w \le -\lambda, \qquad \lambda = 7/5, \qquad |F_{ww}| \le 10, \qquad |F_{ww}| \le 240|d_\sigma|. \tag{F.39}$$

Thus the band remains invariant up to $T = \log(d/(sL))$, independently of its length.

The invariant band controls the graph throughout the long tail. To compose with the regular transitions, we must also place its whole image inside the fixed physical windows. At the organizing coefficients the exact radial graph is

$$w_\Gamma(\xi) = \frac{1}{\sqrt{1 + \xi^2} + 1}, \qquad \xi\partial_\xi w_\Gamma = F_\sigma^0(\xi, w_\Gamma).$$

On the preceding coefficient box, the mean-value formula gives

$$|F_\sigma - F_\sigma^0| \le 100(|\gamma| + |A - 1|\xi + |C - 1|\xi^2 + |p|/r + |a|/r^2).$$

For $T = \log(d/(sL))$, contraction gives the uniform error

$$\mathcal{E}_\theta(L, T) = e^{-\lambda T}|w_\sigma(0) - w_\Gamma(sL)| + 100\int_0^T e^{-\lambda(T-t)}\left(|\gamma| + |A - 1|sLe^t + |C - 1|(sLe^t)^2 + \frac{|p|}{L}e^{-t} + \frac{|a|}{L^2}e^{-2t}\right)dt.$$

In particular $|w_\sigma(T) - w_\Gamma(d)| \le \mathcal{E}_\theta$. This retains the entrance memory; $1 < \lambda < 2$ bounds both decaying-power convolutions. For fixed L and bounded a, p, $\mathcal{E}_\theta \to 0$ uniformly as $s, \gamma, A - 1, C - 1 \to 0$. With $g(v) = \sqrt{1 + v^2} - 1$, the actual port heights satisfy

$$|z_\sigma - g(\gamma + \sigma d)| \le d^2\mathcal{E}_\theta(L, T) + 2d|\gamma|.$$

On the fixed annulus $d/2 \le |v| \le 2\delta$, the organizing graph has $-D_0(v, g(v)) \ge v^2/4 \ge d^2/16$. Choose a fixed tube about this graph, and then shrink the raw coefficients so that $-D_\theta \ge d^2/32$ there. The actual graph equation

$$z_v = \frac{z(R_\theta(v) + Cvz)}{[1 - Av + Cv^2]z - P_\theta(v)}$$

has bounded state derivatives through order three in the tube. Gronwall's inequality, with the port error just obtained, gives admission to that tube by a first-exit argument. Its finite physical length then gives the asserted regular maps and inverse bounds. First place their entire images strictly inside the prescribed inner windows. Next choose m_{ad} so that $md_\theta(K_-^\sharp)$ lies strictly inside the fixed domain of f_θ. Only after these domain assertions do we compose the divided closing map. □

Lemma F.10 (Actual curvature on a compact positive source block)**.** *Fix $0 < p_0 \le p \le P$ and a buffered positive central-height interval $0 < u_- \le u \le u_+$. For $a = -1$, after a common coefficient shrink the complete transition has $\Pi_{xx} < 0$ on this whole block in the actual affine input. The assertion is uniform for $0 \le m \le m_{\mathrm{ad}}$.*

Proof. We first construct the limiting strict block. On the two half-graphs $X = \sigma r$,

$$Y_r = \frac{rY}{Y + 1 - \sigma p r}, \qquad Y(0) = u.$$

The height increases and is bounded above by $u_+ + 1 + r^2 + 2Pr$. The negative denominator is automatically positive. On the positive half, at a sufficiently small positive denominator its derivative $rY/D - p$ is positive, which excludes a first zero. More quantitatively,

$$D \ge \min\{1/2, u_-/(4P^2)\} > 0.$$

These inequalities exclude finite blow-up and give both infinite half-graphs. A common parabolic entrance is fixed as follows. For $Q = \max\{4, 16P\}$, $w = Y/r^2$, and $t = \log r$, the source equation satisfies

$$0 < w \le 1/4 \implies (\log w)_t \ge 2/3, \qquad w \ge 3/4 \implies (\log w)_t \le -2/5.$$

Its entrance bounds at Q are uniform. After a fixed additional logarithmic time both half-graphs enter the strict parabolic band.

The resulting enlarged strict family is admitted on the actual sections by Lemma F.9. On this compact family, Proposition C.2 gives $S'' = -p\mathcal{I}_{bb}$ with a uniform negative margin since $p \ge p_0$. The positive affine normalization of S has the same margin up to bounded positive factors. Proposition C.3 with phase order two therefore gives $\Phi'' < 0$ after a common coefficient shrink.

Retain the exact centres and gains of the actual maps h, k in (C.7), including both bridges and divided closing. With $c_h = h'(b_0) > 0$, $c_k = k'(b_0) > 0$, one has

$$\Pi_{xx}(x) = \frac{c_k}{c_h^2}\Phi''\left(b_0 + \frac{x + a_\theta - h(b_0)}{c_h}\right) < 0. \tag{F.40}$$

The curvature sign is preserved by the positive prefactor, even when both absolute gains tend to zero. □

Lemma F.11 (Paired first variations at high central height)**.** *For the fixed compact positive-ratio region, there are fixed $K_H, z_*, C_H > 0$ such that the complete actual transition has $0 < \Pi'_x \le C_H$ on*

$$u = Y(0) > K_H(1 + p^2), \qquad 0 < s^2 u < z_*.$$

The constant is independent of the tested phase point. The actual multiplier restriction $mC_H < 1/2$ gives $\Delta' > 1/2$.

Proof. The derivative assertion is Proposition C.4 with $a = -1$. Its paired absolute variations (C.24) cancel the common $s^{2-\lambda_\gamma}$ power before the actual bridges and ordinary divided closing are composed. For the high range freeze $H = s^2 u$; the signed normalization in Corollary 5.1 gives the same C_H at every phase point.

For the domain assertion, apply Lemma F.9 to that signed compact source block, fixing its core, tail entrance, ports and raw-bridge tubes before shrinking the coefficients and z_*. This admits both full raw images, including the positive-height equality and double-zero faces. Shrink z_* further so that

$$\sigma_\theta \le -D_\theta(\gamma, H) = H(\ell_\gamma - \bar{a}) \le 2H < d_{\mathrm{out}}.$$

Restrict the receiver to $H < \min\{h_{\mathrm{out}}, z_*\}$, and choose $m < \min\{m_{\mathrm{ad}}, 1/(2C_H)\}$ last. □

Proof of Lemma F.1. Fix the coefficient interval, finite smoothness, physical windows and closing-domain cap m_{ad}. Lemma F.7 chooses $Q, \overline{H}, R_a = R_0, R_b, u_L$ in that order and matches all $0 < u \le 2u_L$ through the fixed-cut equality $N(u) = y(r) = Y_Q^+(H)$. Its full raw images are admitted. Independently fix the core, tail entrance and raw tubes for Lemma F.11, obtaining K_H, C_H, z_*. Choose $u_H > \max\{2u_L, K_H(1+P^2)\}$ and apply Lemma F.10 on $[u_L/2, 3u_H]$. Intersect these coefficient bounds, including $R > 2R_b$, without changing Q.

Choose the unscaled cap and coefficient bounds so that

$$0 < h_{\mathrm{cap}} \le z_*, \qquad L_{\mathrm{raw}} h_{\mathrm{cap}} < d_{\mathrm{out}}/4, \qquad |\alpha_0| < d_{\mathrm{out}}/4, \quad 0 < \ell_\theta(\gamma) \le L_{\mathrm{raw}}.$$

Then every through orbit with $H = z(\gamma) < h_{\mathrm{cap}}$ satisfies $\sigma_\theta \le -D_\theta(\gamma, H) = -\alpha_0 + \ell_\theta(\gamma)H < d_{\mathrm{out}}/2$. Let J be the original buffered receiver clipped by $H < \min\{h_{\mathrm{cap}}, h_{\mathrm{out}}\}$, in the affine coordinate $x = z_{\mathrm{in}} - a_\theta$. The final global choice takes $h_{\mathrm{out}} \le h_{\mathrm{cap}}$, as in Section 5.11.

The strictly increasing central height $u(x) = z_\theta(\gamma; a_\theta + x)/s^2$ gives the relative open intervals

$$I_E = J \cap \{0 < u < 2u_L\}, \qquad I_M = J \cap \{u_L < u < 2u_H\}, \qquad I_H = J \cap \{u > u_H\}.$$

They cover J, including all four internal equality levels, and have physical overlaps $u_L < u < 2u_L$ and $u_H < u < 2u_H$. The endpoint estimate holds on the larger $r > R_a$ domain, the middle estimate on $[u_L/2, 3u_H]$, and the high estimate below z_*. Original window and speed clips preserve this cover; the fixed-cut match has its own image margin. Raw overlap lengths need no positive lower bound.

Uniqueness identifies these as restrictions of the same completed map (2.13). Proposition F.8 and Lemma F.10 give $\Pi_{xx} < 0$ on $I_E \cup I_M$, while Lemma F.11 gives $0 < \Pi_x' \le C_H$ on I_H. All cuts, caps and estimates precede the final multiplier restriction and geometric capture. □

G Gamma-dominant and compact cubic passages

When axis roots coalesce, the incoming and outgoing root distances can tend to zero independently. We estimate the complete return through these limits in the gamma-dominant and compact cubic charts. The discriminant separates real, multiple and ghost roots, and the far-root ranges are retained. In each case the estimates apply to the same physical input used for zero counting.

G.1 Gamma-dominant and root-coalescence faces

In the chart (4.4), negative γ leaves two separated incident roots as $k \to 0$. For positive γ, the positive roots may coalesce and the negative root escapes. These two geometries require different comparisons: a physical endpoint power law on the negative face, and weighted curvature on the positive face. We first treat the negative face and the positive far-root face, then the bounded positive face. In each case the displacement is the original $F(x) = x - m\Pi_{\theta,m}(x)$ of (2.13), with $x = h - a_\theta$. Fix the original passive compact set, nested windows and their reserves before the regional cuts. Every carrier below is

$$I_\theta = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < H(x) := z_\theta(\gamma; a_\theta + x) < H_{\mathrm{cap}}\},$$

where $0 < H_{\mathrm{cap}} < z_*$ is a positive unscaled cap chosen before the coefficient and multiplier bounds. Thus every set retains the original input window, inverse output window, stop and speed clips, and all intervening nonroot inputs. Phase derivatives fix every coefficient and m; a stopped finite channel contributes no cycle.

Proposition G.1 (Negative quadratic face)**.** *Fix $\Pi_{\max} < \infty$ and a sufficiently small routing bound $k_\flat$ from Lemma 5.9. For $\gamma = -g < 0$, $0 < k \le 2k_\flat$, and $0 \le \pi \le \Pi_{\max}$, independently*

small g and m give the physical endpoint law (G.6). *The endpoint, core and high intervals cover the original I_θ, with strict estimate overlaps and positive dividing heights retained. Every critical point of the same F is simple and has $F'' > 0$. Consequently the whole coefficient fibre has at most one critical point and at most two isolated displacement zeros, counted with multiplicity.*

Proof. We first match the two root passages on fixed normal sections, retaining the middle multiplier in their logarithmic actions. The exterior maps then turn this match into the physical power law. A comparison orbit joins its endpoint interval to the bounded core and high-height interval, so the final critical-point argument applies to one connected carrier.

Put $\epsilon = gk$. The exact dictionary is $v - \gamma = \epsilon\xi$, $z = \epsilon^2\eta = s^2 Y_0$, $\eta = gY_0$, $s = g^{3/2}k$. After positive time scaling,

$$\dot\xi = -gP - Q\eta, \qquad \dot\eta = -\eta(H_0 + H_1\eta),$$
$$P = 1 - \pi\xi - 3\xi^2 + k\xi^3, \quad Q = 1 + Ag(1 - k\xi) + Cg^2(1 - k\xi)^2,$$
$$H_0 = (1 - 2g)\xi + kg\xi^2, \quad H_1 = -Ckg^2(1 - k\xi).$$

Its incident roots $a > 0, -b < 0$, their distances from zero, $|P'(a)|, P'(-b)$, and normal rates are positive units on the closed (π, k) block. The third root lies beyond a fixed large cut L. Choose independent fixed normal sections and inner weak cuts at the two incident roots. The intrinsic construction (E.11)–(E.12) has kernels $C(1+T)^N e^{-(c-Cg)T}$; four differentiated integral equations give uniformly bounded strong projections and normal coordinates. With inward footpoint distances $t, r > 0$, put $A_+ = -H_0(a)/P'(a)$, $A_- = -H_0(-b)/P'(-b)$, $\nu = A_+/A_-$. These are positive units. The anchored normal-time actions satisfy

$$J_+(t) = -A_+ \log t + j_+(t), \quad J_-(r) = -A_- \log r + j_-(r),$$
$$|(\delta\partial_\delta)^j (j_\pm(\delta) - j_\pm(0))| \le C\delta, \quad j \le 3. \tag{G.1}$$

Indeed, in the intrinsic equations $\dot\Xi = -\rho\Xi$, $\dot Z = Z$, $\rho = g/A_\pm$, $|\Xi| Z^\rho$ is constant and $J = -(g/\rho)\log(|\Xi_n|/|\Xi_m(0)|)$; its axis derivative is H_0/P. Pole subtraction gives the displayed smooth remainders. On each actual cut write $|\Xi_n| = \delta a_n(\delta)$, $Z_n = b_n(\delta)$, $Z_m = z_0 w U_m(w)$. The positive units have uniform C^3 bounds; Taylor division uses the intrinsic fourth derivative. The exact passage is

$$w = w_0 e^{\Psi_g(w)}, \qquad w_0 = (b_n/z_0)e^{-J/g}, \qquad \Psi_g = \rho^{-1}\log(|\Xi_m(0)|/|\Xi_m(w)|) - \log U_m(w).$$

Here $|\Psi_g| \le Cw/g$, $|\partial_w^j \Psi_g| \le C/g$, $j \le 3$. Choose each fixed collar before g, so that $w_0 \le \delta^{A_{\min}/(2g)}$. Then $1 - w\Psi_g'(w) \ge 1/2$ and $w \asymp w_0$. Implicit Euler differentiation gives $|(\delta\partial_\delta)^j \log(w/w_0)| \le Cg^{-(j+1)}w$, $j \le 3$, hence ordinary C^3 extension to zero. Thus, independently on both sections, $w_\pm = u_\pm e^{-J_\pm/g}$, where $u_\pm$ and their inverses are uniform C^3 units.

Take w to be the raw height η on each fixed inner weak cut, where the normal restriction is $Z_m = z_0 w U_m(w)$. On the core $P \ge p_c > 0$, the scalar equation is $f = \eta(H_0 + H_1\eta)/(gP + Q\eta)$. Choose an actual input chord $0 \le w \le e^{-K_0/g}$, with K_0 fixed large enough that a first-exit comparison keeps its whole passage and output inside the two weak-cut buffers. Here $|f_{\eta^h}| \le C_h g^{-h}$. The initial-height rows $V_j = \partial_w^j \eta$, through order five, satisfy

$$V_j' = f_\eta V_j + \sum_{h=2}^{j} f_{\eta^h} \mathcal{B}_{j,h}(V_1, \dots, V_{j-h+1}),$$

where $\mathcal{B}_{j,h}$ are the partial Bell polynomials. The linear kernel is only $O(g^{-1})$; successive variation of constants puts the higher inverse powers in the forcing and gives $\|M_g\|_{C^5} + \mu_g^{-1} \le e^{K/g}$,

after enlarging K. For the two fixed physical cuts the linear multiplier is exactly

$$g\log\mu_g=\int_{\xi_{i,+}}^{\xi_{i,-}}\frac{H_0}{P}\,d\xi. \tag{G.2}$$

Taylor's integral division $N_g(w)=\mu_g^{-1}\int_0^1(1-t)M_g''(tw)\,dt$ therefore gives the actual middle map

$$M_g(w)=\mu_g w(1+wN_g(w)),\qquad \mu_g^{\pm1}+\|N_g\|_{C^3}\le e^{K/g}. \tag{G.3}$$

Choose t_0 before g, with its action exceeding all these fixed exponents. The weak input then lies on the actual positive core chord, and $g(t\partial_t)^j\log(1+wN_g(w))=O(t^N)$, $j\le 3$, for any fixed N. Matching the actual heights, while retaining $g\log\mu_g$, gives

$$\begin{gathered}A_-\log r-A_+\log t=C_\theta+e_+(t)+e_-(r),\qquad |C_\theta|\le C,\\ |(t\partial_t)^j e_+|\le C(t+t^N),\quad |(r\partial_r)^j e_-|\le Cr,\quad j\le 3.\end{gathered} \tag{G.4}$$

The negative height is strictly monotone from zero, and $A_- - r\partial_r e_- > 0$. Reducing t_0 admits a unique actual output in its independent fixed negative collar. Consequently

$$\begin{gathered}r=A_\theta t^\nu e^{E_\theta(t)},\qquad c\le A_\theta, A_\theta^{-1}\le C,\\ |(t\partial_t)^j E_\theta|\le Ct^\beta,\quad j\le 3,\qquad 0<\beta<\min(1,\inf\nu).\end{gathered} \tag{G.5}$$

Continue both sections to $\pm L$. Their fixed-normal fast graphs have limit $\eta=(\xi^2-u^2)/2$; at the cuts their heights, denominators and gap derivatives are units. At a fixed small raw cut d, differences of the bounded L-labels are $O(\epsilon^{2-\lambda_\gamma})$, while the reference graph has $z/d^2=1/2+O(d)+o(1)$. The raw corridor first variation is $C(d_1/d)^{\lambda_\gamma}$ times a unit. Choosing d first and then g therefore propagates the reference to the original organizing-arc windows uniformly on the retained passive compact set. This admits the whole outward chord, including its limiting zero end, before applying its ordinary observers; it imposes no condition on $g\log k$. The exact tail calculation (C.12)–(C.19) yields

$$d_\pm=\epsilon^{2-\lambda_\gamma}U_\pm,\qquad \lambda_\gamma=-12g/(1-2g),\qquad c\le U_\pm, U_\pm^{-1}\le C.$$

Its first remainder is integrable with majorant $C(\epsilon+g\pi R^{-2}+R^{-3+\zeta})$, $0<\zeta<1$; the higher normalized kernels give ordinary C^3 embeddings. Only d_- includes the divided closing, whose first derivative is $f'(md)d'$. With exact centers and slopes retained separately,

$$x=x_{\rm st}+d_+\phi_+(t),\qquad \Pi(x)=y_{\rm st}+d_-\phi_-(r(t)),\qquad \phi_\pm(0)=0,\quad \phi_\pm'(0)=1.$$

These maps and their inverses are uniformly C^3. Differentiating this identity, and putting $q=(x-x_{\rm st})/d_+=\phi_+(t)$, gives

$$\begin{gathered}\Pi'(x)=B_{\theta,m}q^{\nu-1}e^{R(q)},\qquad B_{\theta,m}=(d_-/d_+)A_\theta\nu,\\ c\le B_{\theta,m}, B_{\theta,m}^{-1}\le C,\qquad |(q\partial_q)^j R|\le Cq^\beta,\quad j\le 2.\end{gathered} \tag{G.6}$$

In fact $R(\phi_+(t))=E+\log(1+tE'/\nu)+\log\phi_-'(r)-\log\phi_+'(t)+(\nu-1)\log(t/\phi_+(t))$. The third Euler row in (G.5) controls the two Euler derivatives of this expression. The absolute amplitude $B_{\theta,m}d_+^{1-\nu}$ need not be a unit.

Choose the comparison $t_c=t_0/2$; its matched r_c lies in a fixed positive interval. On the two outer pieces beyond $a-t_c/2$ and $-b+r_{\min}/2$, its fast height has a fixed positive margin. The intervening piece has $P\ge p_c$. Choose a fixed upper central shot $\eta_{\rm core}>0$; its limiting graph is $\eta_{\rm core}+\xi^2/2$. Both bounding shots have buffered outward admission as above. Scalar order and $Q>0$ admit every intervening shot $\eta_c\le\eta_0\le\eta_{\rm core}$, including nonroots, and their raw central

speed is at most $\epsilon^2(g+Q(0)\eta_{\rm core}) < d_{\rm out}$. For $D = gP + Q\eta$ and $f = \eta(H_0 + H_1\eta)/D$, use (A.5), with residual $[gP' + (Q' + H_1)\eta]/D$. On the middle piece the last fraction is bounded by $C + Cgk$; on the outer pieces the comparison height bounds D below. The endpoint quotients are units, so integration bounds the core first variation and its inverse. The two outward derivatives are $\epsilon^{2-\lambda_\gamma}$ times units on this entire chord. Their ratio, regular bridges and closing give $0 < \Pi' \le C_c$. The limiting $t = 0$ orbit stops, whereas every $t > 0$ passes; Lemma 2.3 identifies $x_{\rm st}$ with the excluded mother endpoint. The endpoint extends through t_0, strictly beyond the comparison.

The comparison action gives $\eta_c \le Cg^{-N}e^{-c/g}$. The signed high threshold and the ordered final shrink are

$$H_H = \epsilon^2 g K_H(1+g\pi^2), \qquad \eta_c < \eta_H := gK_H(1+g\pi^2) < \eta_{\rm core}/2, \qquad \epsilon^2\eta_{\rm core} < H_{\rm cap}/2. \tag{G.7}$$

Hence the endpoint, closed core and signed high packages strictly overlap and cover I_θ on the original height axis. The equality $H = H_H$ belongs to the core; original clips are imposed on all three. Choose m last so $m\max(C_c, C_H) < 1/2$. It remains to control critical points in the endpoint interval, including parameters for which the exponent ratio approaches one. On the endpoint, $F' = 1 - e^L$, $L = \log(mB) + (\nu-1)\log q + R(q)$. If $\nu \ge 1$, then $L < 0$. Otherwise, for $z = 1 - \nu > 0$, a critical point satisfies $z\log(1/q) \ge M := \log(1/m) - C$, so

$$qL_q = -z + O(q^\beta) \le -z + Ce^{-\beta M/z} < -z/2.$$

This choice is uniform as $z \downarrow 0$. Since $q_x = 1/d_+ > 0$, every actual critical point has $F'' > 0$. The core and high add none; Lemma 3.1 gives the whole-carrier count. □

Proposition G.2 (The positive far-root face)**.** *For $\gamma = g > 0$, there are independent bounds*

$$0 < \lambda = \pi^{-2} \le \lambda_0, \qquad 0 < q = k\pi \le q_0, \qquad 0 < \tau = g\pi^2 = p^2 \le \tau_0 \tag{G.8}$$

such that the actual lower interval has strictly negative completed-map curvature. Its bounded core and signed high complement strictly overlap and have $F' > 1/2$ after one multiplier shrink. They cover the original raw-capped I_θ, including all nonroot inputs and original clips. The same physical F has at most one critical point and at most two isolated zeros, counted with multiplicity, on every fibre.

Proof. The negative incident root now escapes in the scaled coordinate. We estimate its outward map first below one third of the root radius and then in a root collar rescaled by that radius. The exact corridor exponent keeps both estimates uniform as the root escapes; their action weights make the resulting section correction small in physical curvature.

Put $\chi = q\lambda^2$, $a = s/p = \tau\chi$, $g = \tau\lambda$, $s = \tau^{3/2}\chi$, $v = g + aU$, $z = a^2Z$; at the center $Z = \tau Y_0$. After positive time scaling the exact field is

$$U' = -\tau P_+(U) - Q_+(U)Z, \qquad Z' = -Z\{UB_+(U) + Ca(g+aU)Z\},$$

$$\begin{aligned} P_+ &= 1 - U + 3\lambda U^2 + \chi U^3, \qquad P_- = 1 + T + 3\lambda T^2 - \chi T^3, \\ B_+(U) &= 1 + 2g + aU, \quad B_-(T) = 1 + 2g - aT, \\ Q_+(U) &= \ell(g+aU), \quad Q_-(T) = \ell(g-aT), \quad \ell(v) = 1 - Av + Cv^2. \end{aligned} \tag{G.9}$$

The moving lips are $Z = \tau\ell_0 P_\pm$, with labels $b_+ = -U^2/2, b_- = -T^2/2$. The source quartet is $(-\tau, \tau, \tau\lambda, \tau\chi)$; reflection at a fixed negative base T_b gives $(-\tau/T_b^2, -\tau/T_b, \tau\lambda, -\tau\chi T_b)$ without changing the positive-time orbit order. The first positive root is $A_+ = 1 + O(\lambda+\chi)$. Writing $w_* = q\lambda T_*$, the unique negative-side root satisfies

$$w_* = 3 + q/w_* + q^2\lambda/w_*^2, \quad 3 < w_* < 4, \qquad T_* \asymp (q\lambda)^{-1}, \quad \chi T_* \le 4\lambda. \tag{G.10}$$

Its slope is $c_- = 2 + 3\lambda T_* + 3/T_*$. The action residues $r_+ = A_+B_+(A_+)/[-P'_+(A_+)] \asymp 1$ and $r_- = T_*B_-(T_*)/c_- \asymp \lambda^{-1}$ are retained exactly. Define

$$J_+(U) = \int_0^U \frac{xB_+}{P_+}\,dx, \qquad A_-(T) = \int_0^T \frac{xB_-}{P_-}\,dx,$$
$$\mathcal{D}_\pm = \frac{P_\pm}{xB_\pm}\partial_x, \quad H_\pm = \frac{P'_\pm}{xB_\pm}, \quad h_\pm = \frac{P_\pm}{x^2B_\pm}.$$

With $R = T^{-1}, d = \chi T, E = T\partial_T$, $H_- = (R + 6\lambda - 3d)/B_-$, $h_- = (R^2 + R + 3\lambda - d)/B_-$, $\mathcal{D}_- = h_-E$. Repeated differentiation gives

$$\frac{|\mathcal{D}_-^j H_-|}{h_-} \le C_j(R+\lambda)^{j-1}(R+d), \qquad \int_{T_b}^{T_*} |\mathcal{D}_-^j H_-|\,dA_- \le C_j(T_b^{-1} + \chi T_*), \quad j = 1,2,3. \tag{G.11}$$

Here $dA_- = d\log T/h_-$; the retained factor $R + d$ avoids any $\lambda \log T_*$ loss. Also $\tau H_- \le 1/4$. The reflected coefficient $\varpi_- = -C\tau a(g - aT)P_-/(TB_-)$ and its first four radial rows are $O(\tau^3\lambda^3)$, while those of $Q_- - 1$ are $O(g)$. Choose ℓ_0 with $\ell_0 \max(|Q_-|, |\varpi_-|) \le 1/4$. The positive root collar has the same oriented margins on its fixed bounded cuts.

Apply Lemma A.3 to $\vartheta = Z/(\tau P)$, $q_a = (A(x_l) - A(x))/\tau$, with $D = \mathcal{D}_l$ on either entrance, $\kappa = \tau$, $\omega = \varpi$, $\rho = \tau$, and $\alpha = 0$. At fixed clock, endpoint transport is $\mathcal{D}$ at the running point; the coefficient rows follow from (G.11) and the bounded positive collar. The margins above give the two negative diagonals, and $\partial_{q_a} = -\tau\mathcal{D}$ supplies the clock rows. Thus $\vartheta \le \ell_0 e^{-q_a/4}$, and the exact memory is

$$\vartheta = \ell_0 \frac{P(x_l)}{P(x)} e^{-q_a + V}, \qquad V = \int_0^{q_a} \frac{(Q - \varpi)\vartheta}{1 + Q\vartheta}\,dq,$$

with three bounded action rows. In particular the initial polynomial factor is retained through the long negative flight. Fix inner cuts $U_i > 0$, $0 < T_i < T_b$ on a positive core, with $A_-(T_b) - A_-(T_i) \ge c_i > 0$, and put $J_i = J_+(U_i)$, $A_i = A_-(T_i)$. At either fixed inner cut the terminal clock is $q_i = (A(x_l) - A(x_i))/\tau$, with $\mathcal{D}_l q_i = 1/\tau$ and $\mathcal{D}_l^2 q_i = \mathcal{D}_l^3 q_i = 0$. All three total rows in (A.10) are bounded: their terminal densities decay as $e^{-c/\tau}$, absorbing the factors through τ^{-3}. Writing $V_\pm$ for the complete terminal integrals, the actual raw weak heights are

$$w_+(U) = \tau\ell_0 P_+(U) e^{-[J_+(U) - J_i]/\tau + V_+(U)},$$
$$w_-(T) = \tau\ell_0 P_-(T) e^{-[A_-(T) - A_i]/\tau + V_-(T)}.$$

The same five-row scalar core construction, with g replaced by τ, gives $M(w) = \mu_\theta w(1 + wN_\theta(w))$ on its actual weak chord, with $\mu_\theta^{\pm 1} + \|N_\theta\|_{C^3} \le e^{K/\tau}$. Its exact first multiplier has $\log \mu_\theta = (A_i - J_i)/\tau$. Fix the cuts, large radial cut and normal height first; enlarge K to include the fixed action-bracket reserves, then choose $X_0 > 0$ with $J_+(A_+ - X_0) - J_i > 4K + 1$, and $0 < X_1 < X_0$. The actual weak memory and three action rows are $O(\tau^{-N} e^{-(3K+1)/\tau})$ on $A_+ - X_0 \le U < A_+$. The exact equality $M(w_+(U)) = w_-(T)$ now gives

$$J_+(U) - A_-(T) = \tau\{\log P_+(U) - \log P_-(T) + R_0(U,T)\}, \quad |R_0| + \sum_{1 \le i+j \le 3} |\mathcal{D}_+^i \mathcal{D}_-^j R_0| \le C. \tag{G.12}$$

Here the remainder has the separated extension

$$R_0(U,T) = V_+(U) - V_-(T) + \log\{1 + w_+(U)N_\theta(w_+(U))\}. \tag{G.13}$$

It is defined on the product of the actual positive weak chord and negative inward domain. Its mixed rows with $i, j > 0$ are exactly zero; the pure rows through order three are those just

estimated. Only then is it restricted to the actual center-image intersection. Scalar order gives its unique matched $T_b \le T < T_*$; at both comparison labels $A_+ - X_j$, fixed action brackets and $A_-(T) \to T - \log(1+T)$ put T_j in fixed positive compact intervals. Differentiation yields

$$T_U = \frac{UB_+P_-}{TB_-P_+}(1+\varrho), \qquad 1+\varrho = \frac{1-\tau H_+ - \tau \mathcal{D}_+ R_0}{1-\tau H_- + \tau \mathcal{D}_- R_0}, \qquad |\mathcal{D}_+^i \mathcal{D}_-^j \varrho| \le C\tau \quad (i+j \le 2).$$

Let $\Gamma_\pm(b_\pm) = d_\pm + c_\pm \phi_\pm(b_\pm)$ be the actual raw embeddings, including the regular bridges and output closing; the positive center is measured in $x = h - a_\theta$. With $S(b_+) = -T^2/2$, $\widehat{\Pi} = \phi_- \circ S \circ \phi_+^{-1}$ and $\Pi(d_+ + c_+ w) = d_- + c_- \widehat{\Pi}(w)$. Put $L_\pm = \mathcal{D}_\pm \log \Gamma'_\pm$, where the primes refer to their own $b_\pm$. Apply (3.8) with $u = U, v = T, W = (U+T)/(UT)$, retaining both section terms. In this weight the polynomial part is $3\chi UT - 1$, apart from $O(\tau\lambda)$. The relative correction is $O(\tau)$; the section term has bounded weight $UB_+T/(U+T)$. We now estimate its actual log-slopes.

At the positive root, pole subtraction and the intrinsic normal clock retain the embedding power $(A_+ - U)^{1+\alpha_+}$, $\alpha_+ = \tau/r_+$. The normal distance kernel and its first three weighted rows have majorant $C(W + \tau u)$, whose normal-clock integral is $O(\sigma + \tau(1 + |\log \tau|))$. Its fixed normal-to-raw ordinary C^3 observer therefore gives

$$|L_+| \le C_0\sigma + C_\sigma X_0 + C_\sigma \tau (1 + |\log \tau|)^3.$$

Choose σ, then X_0, to make the fixed terms small; reducing X_0 strengthens the earlier action gap.

For the negative outward graph put $D = \tau P_- + Q_- Z$, $n = RB_-$, $\zeta = -Ca(g - aR)$, $f = Z(n + \zeta Z)/D$. It starts at the true moving value $Z_l(T) = \tau \ell_0 P_-(T)$ and ends at the fixed raw port $R_{\mathrm{out}} = \delta_-/a$. The original observer ψ is composed only after raw admission. Use (A.6) with $M = \tau P_-$, $x_l = T$, $\sigma = 1$, and $\chi(Z) = \psi(a^2 Z)$. It gives

$$\Gamma'_-(b_-) = a^2 \psi'(a^2 Z_T) \frac{Z_T}{D_T} B_-(T)[1 + \ell_0 \varpi_- - \tau H_-(1 + \ell_0 Q_-)] e^{I(T)},$$

where $I = \int_T^{R_{\mathrm{out}}} [\tau P'_- + (Q'_- + \zeta)Z]/D \, dR$. This retains the exact moving initial factor and first gain. The two logarithmic-slope rows use only the ordinary third observer derivative, by (3.3).

For $T_b \le T \le T_*/3$, apply Lemma A.6 in the raw coordinate $x = aR$, with starting abscissa aT, terminal cut δ_-, and

$$t = -1, \quad (\mu_2, \mu_1, \mu_0) = (3g, \tau a, \tau a^2), \quad b = 2g, \quad v = g, \quad \sigma = a.$$

Its input label is exactly $-T^2/2$. With the comparison mass $\kappa = \tau/T^2 + \tau/T + g + aT$, the root bound $aT_* < 4g$ gives $m(x) \asymp \kappa$ on $[aT, 2aT]$, and all coefficient bounds in (A.13) hold. The same margin holds for $T < 7T_*/20$, providing a strict overlap with the root chart beginning at $T_*/3$. Here $\Theta \le \kappa \le C\tau$, the terminal coefficient is bounded, and $(aT/\delta_-)^{2-\nu} \le C\tau$, where $\nu = 12g/(1+2g)$ is the lemma's exact spectral exponent. Thus, for $E = T\partial_T$, its two rows give

$$|E \log \Gamma'_-| + |E^2 \log \Gamma'_-| \le C\tau(1 + |\log \tau|)^3.$$

The observer is used on its original admitted image; neither a nor the comparison mass is differentiated as a coefficient.

For the rescaled root range below, the corridor from $R_c = LT_*$ retains $y = Z/R^2$, $Q_g = \ell(g)$, $y_g = (1-4g)/(2Q_g)$, $\nu = 12g/(1+2g)$ exactly. For $m_R = \tau/R^2 + \tau/R + 3g - aR$, $B = 1 + 2g - aR$, $Q = \ell(g - aR)$, $\zeta_R = -CaR(g - aR)$, $D_y = m_R + Qy$, the exact equations are

$$Ry_R = y\left(\frac{B + \zeta_R y}{D_y} - 2\right), \qquad R(y - y_g)_R = -\frac{2Q_g y}{D_y}(y - y_g) + S,$$

$$S = \frac{y}{D_y}\{(B - 1 - 2g) - 2(m_R - 3g) + [\zeta_R - 2(Q - Q_g)]y\}.$$

In a fixed band about y_g, the damping is $\geq k_0 > 0$, $|S| \leq C\xi(R)$, $\xi = aR + \tau/R + \tau/R^2$. Choose the raw radius first, then L and the coefficient bounds. The actual shots enter the inner band, and the two boundary signs give its strict continuation to δ_-/a. Variation of constants yields

$$\int_{R_c}^{R_{\rm out}} |y - y_g| \frac{dR}{R} \leq C\left(|y(R_c) - y_g| + \delta_- + \frac{\tau}{R_c} + \frac{\tau}{2R_c^2}\right).$$

The exact first density is $Rf_Z = (Bm_R + 2\zeta_R m_R y + \zeta_R Q y^2)/D_y^2$. At $(3g, 1 + 2g, Q_g, 0, y_g)$ it equals ν, and its derivative in y is $O(g)$. Hence

$$|Rf_Z - \nu| \leq C(g|y - y_g| + \xi), \qquad j(R) = (R/R_c)^\nu U(R), \qquad c \leq U, U^{-1} \leq C.$$

Here j is the linear propagator. Extracting its exact exponent makes the residual density integrable uniformly along the long corridor, without a $g^2 \log q$ loss. Rational division also gives

$$|f_{ZZ}| \leq C(aR^{-2} + gR^{-3} + \tau R^{-4} + \tau R^{-5}), \quad |f_{ZZZ}| \leq C(aR^{-4} + gR^{-5} + \tau R^{-6} + \tau R^{-7}).$$

For $1/3 \leq \theta = T/T_* < 1$, use the coefficient-only radius T_*, $r = R/T_*$, $W = Z/T_*^2$. The exact field is

$$r' = g\widehat{P}(r) + Q(r)W, \quad W' = Wn(r, W), \quad \widehat{P} = (1 - r)[a_0 + (a_0 + a_1)r + w_* r^2],$$

where $a_0 = q^2\lambda/w_*^2$, $a_1 = q/w_*$, $Q = \ell(g - gw_* r)$, $n = r(1 + 2g - gw_* r) - Cg^2 w_*(1 - w_* r)W$. On the actual compact strip $1/3 \leq r \leq 1 + C\sigma$, n is a positive unit. The selected strong graph $r = \phi(W)$ at $(1, 0)$ has coefficient denominators $jB_* + g\widehat{c} \geq cj$, $B_* = 1 + 2g - gw_*$, $\widehat{c} = w_* + a_1 + 2a_0$. Its power-series majorant gives a fixed normal interval and uniform C^4 bounds. Put $x_- = 1 - \theta$, $W_0 = g\ell_0 x_- \widehat{C}(\theta)$, and $u = \phi(W) - r$. Then $u_0 = x_-(1 + O(g))$, with three weighted rows, and exactly

$$(\log u)_{\log W} = -\alpha_- - K(W, u), \quad \alpha_- = g\widehat{c}/B_* = \tau/r_-,$$
$$|(W\partial_W)^i(u\partial_u)^j K| \leq C(W + gu), \quad i + j \leq 3.$$

Taylor division of the rational normal field also gives $\alpha_- + K \geq c(g + W)$, so the actual orbit reaches $W = \sigma$ once. Its integrated densities are $\int W\, d\log W \leq \sigma$, $\int gu\, d\log W \leq Cg(1 + |\log g|)$. Use (A.4) with $s = \log W$, $z = \log u$, $f = -\alpha_- - K(W, e^z)$, and $D = E_* := \theta(1 - \theta)\partial_\theta$ at fixed s. Its initial recursion retains the moving clock $s_0 = \log W_0$. The two integrated densities above give

$$u_\sigma = g^{\alpha_-} x_-^{1+\alpha_-} e^{A(\theta)}, \qquad |A| + \sum_{j=1}^{3} |E_*^j A| \leq C[\sigma + g(1 + |\log g|)],$$

and $r'_\sigma = g^{\alpha_-} x_-^{\alpha_-} e^A(1 + \alpha_- - x_- A_\theta) > 0$. The fixed normal continuation to $r = L$ has height $H_g(r_0) = \sigma + (L^2 - r_0^2)/2 + O_{C^3}(g)$, with $-H'_g$ a unit. At the coefficient-only cut $R_c = LT_*$, its full initial Bell rows are $T_*^2 H_g^{(j)}$. The same corridor and rational densities give $\int |f_{ZZ} U_1| \leq Cg$ and $\int (|f_{ZZ} U_2| + |f_{ZZZ} U_1^2|) \leq Cg$. Thus the two Bell ratios are bounded, including the third observer row. For $\Psi_*(r_0) = \psi(a^2 Z_{\rm out}(r_0))$, the exact gain is

$$-\Psi'_* = a^2\psi'(a^2 Z_{\rm out}) T_*^2[-H'_g](\delta_-/(aLT_*))^\nu U(R_{\rm out}), \qquad \Omega_* \asymp (aT_*)^{2-\nu}\delta_-^\nu L^{-\nu}.$$

After removing this exact gain, $G = (\Psi_* - d_*)/\Omega_*$ and its inverse have ordinary C^3 bounds, with $-G'$ a unit. Hence

$$\Gamma'_-(b_-) = \frac{\Omega_*}{T_*^2}\frac{-G'(r_\sigma)}{\theta} g^{\alpha_-} x_-^{\alpha_-} e^A (1 + \alpha_- - x_- A_\theta), \qquad |E_* \log \Gamma'_-| + |E_*^2 \log \Gamma'_-| \le C.$$

Neither Ω_* nor $x_-^{\alpha_-}$ has been replaced by a unit. Factor $P_- = (1 - T/T_*)C_*(T)$, where $C_* = 1 + (1 + T_*^{-1})T + \chi T_* T^2$. On this range $\mathcal{D}_- = [C_*/(T^2 B_-)]E_*$, with $C_*/(T^2 B_-) \le C\lambda$. Combining both ranges proves the actual consumed estimate

$$|L_-| \le C\{\tau(1 + |\log \tau|)^3 + \lambda\}, \qquad T_b \le T < T_*. \tag{G.14}$$

In the root range, the residual logarithmic derivatives are bounded, and the factor $C_*/(T^2 B_-)$ makes their action derivatives small. Together with the positive-section and relative-variation estimates, this gives

$$|\varrho| + \max_{i+j\le 2} |\mathcal{D}_+^i \mathcal{D}_-^j \varrho| + |L_-| \le C\{\tau(1+|\log\tau|)^3 + \lambda\}, \qquad |L_+| \le C_0\sigma + C_\sigma[X_0 + \tau(1+|\log\tau|)^3]. \tag{G.15}$$

Therefore the full physical identity is

$$\frac{UTP_+\phi'_+}{U+T}\frac{\widehat{\Pi}''}{\widehat{\Pi}'} = 3\chi UT - 1 + E_f, \quad |E_f| \le \epsilon_0 + C_\sigma[\tau(1 + |\log\tau|)^3 + \lambda], \quad \epsilon_0 < 1/8. \tag{G.16}$$

Independent q_0, λ_0, τ_0 make this strictly negative. The exact endpoint powers also remain: $x_- = T_* - T$ obeys $x_- \sim C_\theta (A_+ - U)^{(r_+ + \tau)/(r_- + \tau)}$, while the two physical embeddings carry powers $1 + \tau/r_\pm$. The physical exponent is $r_+/r_- \asymp 3\lambda$; only weighted endpoint rows and pointwise C^3 on the open passage are used.

It remains to establish raw admission and the whole upper complement. The two comparison shots $U = A_+ - X_j$ have positive-unit heights at fixed cuts $L_+ > A_+ + 1$, $L_- > \max T_j + 1$, since their fast invariants give $Z_+(L_+) = \frac{1}{2}[L_+^2 - (A_+ - X_j)^2] + O(\tau(1+|\log\tau|))$ and $Z_-(L_-) = \frac{1}{2}(L_-^2 - T_j^2) + O(\tau(1+|\log\tau|))$. Choose a fixed upper fast shot and $\eta_{\max} > 0$; scalar order bounds every intervening central shot through that height at both cuts. Before applying an observer, compare their ratios to the passive parabolic references

$$t(Y_{\mathrm{par},\pm})_t = Y_{\mathrm{par},\pm}\left\{\frac{1 \pm t + Ct^2 Y_{\mathrm{par},\pm}}{\pm t + \ell(\pm t) Y_{\mathrm{par},\pm}} - 2\right\}, \qquad Y_{\mathrm{par},\pm}(0) = 1/2.$$

Their analytic recursion has denominator $j + 2$, and the original prepared passive compact set retains their strict raw-window margins. The actual/reference ratio fields have negative y-derivative and difference $O(g + s^2/t^2 + sp/t)$. Logarithmic variation of constants at each fixed raw radius yields

$$|y(\delta_\pm) - Y_{\mathrm{par},\pm}(\delta_\pm)| \le C[(aR_c/\delta_\pm)^{c_0} + g + \tau/R_c + \tau/R_c^2].$$

Thus all comparison and upper core shots are admitted uniformly in the independent rectangle. For the entire lower output family, the raw-height spectral slope (A.20), with identity observer, gives $|h_T| \le Ca^{2-\nu}\delta_-^\nu T^{1-\nu}$, hence total variation $\le C(aT_*)^{2-\nu}\delta_-^\nu$. The raw version of the root-range gain gives the same bound. Since $aT_* \le 4g$, the total is $O(g^{3/2})$, smaller than the fixed reference buffer. The bounded positive normal family has gain $O(a^{2-\nu})$ and the same input admission. Ordinary observers are composed only on these admitted images; the original smaller windows, speed and stop clips are then intersected.

On the fixed core, $D = \tau P + QZ$ and $f_Z = d\log(Z/D)/dU + [\tau P' + (Q' + \zeta)Z]/D$. The inner positive piece has $P \ge p_c$, the outer positive piece has the comparison's $Z \ge z_c$, and the negative

fixed slab has $P_- \geq p_c$. The quotient is uniformly integrable and endpoint Z/D are units, giving bounded core variation and inverse. The positive tail has $m_+ = \tau/R^2 - \tau/R + 3g + aR$, $B_+ = 1 + 2g + aR$, $Q_+ = \ell(g + aR)$, $\zeta_{R,+} = CaR(g + aR)$; the same exact y_g, ν corridor applies. Both absolute outward gains are $a^{2-\nu}\delta_\pm^\nu$ times units. Their common factor cancels in the actual Π', so $0 < \Pi' \leq C_c$ through $\eta_{\max}$. All ordinary observer and divided closing rows are those of Lemma 2.2; their third chain rule uses only f''', d''', retaining each small gain.

The action gap gives $0 < \eta_{c,1} < \eta_{c,0} \leq C\tau^{-N}e^{-c/\tau}$. Fixing the shots and caps first, the independent shrink gives

$$\begin{aligned} \eta_{c,1} < \eta_{c,0} < \eta_H &:= \tau K_H(1+\tau) < \eta_{\max}/2, \\ H_H = a^2\eta_H = K_H(s^2 + (sp)^2), &\qquad a^2\eta_{\max} < H_{\mathrm{cap}}/2. \end{aligned} \tag{G.17}$$

On the original height axis, lower through the X_0 shot, core starting at the X_1 shot, and high above H_H cover I_θ with strict overlaps. The positive equality $H = H_H$ belongs to the core; scalar order retains all clipped nonroot inputs. Choose m last so $m\max(C_c, C_H) < 1/2$. The upper two packages have $F' > 1/2$; on the lower one $\Pi'' < 0$, so every critical point of the same F has $F'' > 0$. Lemma 3.1 proves the whole-fibre bound, excluding its natural stopped endpoint. □

G.2 Cubic and quadratic source compactifications

The chart (4.5) covers the compact cubic limits of the small-p region. Proposition 4.1 proves their complete return and whole-fibre count. The estimates below supply its growing tails, where a compact phase comparison no longer applies. We then return to the bounded positive quadratic face of (4.4).

On a growing positive tail, the lip label is unbounded while its raw abscissa remains small. The next estimate controls the logarithmic derivatives of the actual exterior map in that regime. Keeping its absolute first gain will later allow composition with the independent negative section map.

Lemma G.3 (Complete section derivatives on a growing cubic tail)**.** *Fix a sufficiently small terminal cut $\delta > 0$. Let*

$$\begin{aligned} M(x) = x^3 + \mu_2 x^2 + \mu_1 x + \mu_0, &\qquad B_0(x) = 1 + b + x, \\ Q_0(x) = 1 - A(v+x) + C(v+x)^2, &\qquad \zeta_0(x) = C(v+x). \end{aligned}$$

with bounded A, C. Hold all coefficients fixed as the starting point a varies. On an enlarged interval $I \subset (0, d_0]$, with $d_0 < \delta/2$, suppose uniformly for $a \in I$

$$|b| + |v| \leq C_0 a, \qquad |\mu_j| \leq C_0 a^{3-j}\ (j = 0,1,2), \qquad M(x) \geq c_0 x^3 \quad (a \leq x \leq \delta).$$

Use the actual positive graph

$$z_x = \frac{z\{xB_0(x) + \zeta_0(x)z\}}{D_0(x,z)}, \qquad D_0 = M + Q_0 z, \qquad z(a) = \ell_0 M(a), \tag{G.18}$$

and an increasing regular terminal adapter χ, with uniformly bounded C^3 norm and inverse on its enlarged image. For fixed positive σ, label the input by $b_+ = -a^2/(2\sigma^2)$, and write $\Gamma_+(b_+) = \chi(z_\delta)$. After choosing ℓ_0, d_0 small, this continuation and every separate affine normalization ϕ_+ satisfy

$$|(a\partial_a)^j \log \phi_+'| \leq Ca(1 + |\log a|)^4, \qquad j = 1, 2. \tag{G.19}$$

The exact positive factor σ^2 in the first derivative is retained; no bound on its reciprocal is required.

Proof. Set $E = a\partial_a$ and $m(x) = M(x)/x^2$. On $a \le x \le 2a$, the coefficient bounds and the polynomial margin give

$$c_0 a \le m(x) \le C_1 a, \qquad |b| + |v| + |\mu_2| + a + |\mu_1|/a + |\mu_0|/a^2 \le C_1 a.$$

Thus Lemma A.6 applies with $t = 1$ and comparison mass $\rho = a$, including its near-lip strip entry. This mass is not differentiated; all polynomial coefficients and σ remain fixed. The adapter is evaluated on its stated enlargement of the generated terminal image. In the actual raw-section applications, (A.16), with $\rho = a$, and the original fixed terminal-image margin admit the whole family after choosing d_0 within its original cutoff reserve. The stated ordinary C^3 adapter and inverse bounds give a uniform bound for its two logarithmic-height rows on the enlarged terminal image.

The common spectral estimate has

$$\lambda = \frac{4\mu_2}{1+b} = O(a), \qquad \Theta = a + |\mu_1|/a + |\mu_0|/a^2 \le C_1 a.$$

Its terminal coefficient $\mathcal{T}$ is bounded on the stated buffer, and $\mathcal{T}(a/\delta)^{2-\lambda} \le Ca$, after the permitted d_0 shrink. Consequently

$$|E \log \Gamma'_+ + \lambda| + |E^2 \log \Gamma'_+| \le Ca(1 + |\log a|).$$

The physical gain is given exactly by (A.6),

$$\Gamma'_+ = \sigma^2 \chi'(z_\delta) \frac{z_\delta}{D_0(\delta, z_\delta)} B_0(a) J_0(a) e^{I(a)}, \qquad J_0 = 1 + O(a), \tag{G.20}$$

where $J_0 = K$ and I are the exact factors in that formula. This retains the factor σ^2 without a reciprocal bound. Since $\lambda = O(a)$, both uncorrected logarithmic rows satisfy the stronger first-logarithmic-power bound. Frozen affine calibration leaves them unchanged, proving (G.19). □

Lemma G.4 (Weighted cubic tail on the actual sections)**.** *Fix a compact set of coefficients $0 \le U \le U_0$, $|G| \le G_0$, and use the field* (4.10)–(4.11)*. On its root-free positive fibres choose M sufficiently large, independently of τ_{cub}, and retain*

$$u \ge 2M, \qquad a = \tau_{\mathrm{cub}} u \le d_0, \qquad r_0 \le r < c.$$

Here c is the simple negative root and the two moving sections are $Y = \ell_0 P$, with $\eta = \tau_{\mathrm{cub}} Y$. The exterior maps are the actual continuations to the fixed raw sections, followed by the regular input adapter and the divided finite-side output adapter, respectively. These continuations use the nested domains constructed by (4.16)*. There are constants, uniform on these domains, such that, with*

$$E = u\partial_u, \quad \mathcal{D}_+ = \frac{P(u)}{uB(u)}\partial_u, \quad \mathcal{D}_- = \frac{P(-r)}{rB(-r)}\partial_r, \quad \epsilon_u = \tau_{\mathrm{cub}}(1+u), \quad L_u = 1 + |\log \epsilon_u|,$$

the exact action matching has the form

$$A_+(u) - A_-(r) = \tau_{\mathrm{cub}}\{\log P(u) - \log P(-r) + R_+(u) - R_-(r)\}, \tag{G.21}$$

where

$$|R_+| + |R_-| \le C, \qquad |E^j R_+| \le C\epsilon_u, \qquad |\mathcal{D}_-^j R_-| \le C\tau_{\mathrm{cub}} \quad (1 \le j \le 3). \tag{G.22}$$

The relative variation in (G.30) *obeys*

$$|\varrho| + \frac{|\mathcal{D}_+\varrho| + |\mathcal{D}_-\varrho|}{u+r} \le C\epsilon_u L_u^4. \tag{G.23}$$

If $b_+ = -u^2/2$, $b_- = -r^2/2$, *and* $\phi_\pm$ *are the separately affine-normalized actual section embeddings, then*

$$|E\log\phi_+'| + |E^2\log\phi_+'| \le C\epsilon_u L_u^4, \qquad |\mathcal{D}_-^j\log\phi_-'| \le C\tau_{\mathrm{cub}}(1+|\log\tau_{\mathrm{cub}}|)^4 \quad (j=1,2). \tag{G.24}$$

Every prime on $\phi_\pm$ *is taken in its own label* $b_\pm$. *The full composition* $\Pi = \phi_- \circ S \circ \phi_+^{-1}$, *with* $S(-u^2/2) = -r^2/2$, *therefore satisfies*

$$\phi_+' \frac{P(u)}{u+r}\frac{\Pi''}{\Pi'} = 3 - \frac{U}{ur} + E_u, \qquad |E_u| \le C\epsilon_u L_u^4. \tag{G.25}$$

The inward action estimates and the two exterior estimates are obtained separately on their actual domains.

Proof. We first continue the two inward graphs to the original centre and differentiate their exact height match. We then estimate the two exterior embeddings separately. The negative embedding requires a root-collar calculation because its first variation vanishes there. Substitution in the completed-curvature identity gives the final estimate.

Choose a small fixed $\delta_c > 0$ such that P is a positive unit on $[-2\delta_c, 2\delta_c]$, and so that $2\delta_c < r_0$. On the root-free positive fibre, put

$$q = \frac{A_+(u) - A_+(v)}{\tau_{\mathrm{cub}}}, \quad W = \frac{Y}{P(v)}, \quad D(v) = \frac{P(v)}{vB(v)},$$
$$H(v) = \frac{P'(v)}{vB(v)}, \qquad \varpi(v) = \frac{C\tau_{\mathrm{cub}}^3(G+v)P(v)}{vB(v)}.$$

For $\delta_c \le v \le u$, graph division gives the exact equations

$$v_q = -\tau_{\mathrm{cub}}D(v), \qquad W_q = W\left\{-\frac{1+\varpi W}{1+QW} + \tau_{\mathrm{cub}}H\right\}, \qquad W(0) = \ell_0. \tag{G.26}$$

In particular, $\tau_{\mathrm{cub}}H = O(\epsilon_u)$, $Q-1 = O(\epsilon_u)$, and $\varpi = O(\epsilon_u^3)$. Fix ℓ_0, then d_0, small enough that the braces and the derivative of the right-hand side with respect to W, for $0 \le W \le \ell_0$, are both less than $-c_1 < 0$. A first-exit argument gives

$$0 < W(q) \le \ell_0 e^{-c_1 q}. \tag{G.27}$$

For unbounded u, Euler differentiation keeps the moving entrance under control. At fixed action clock q,

$$Ev = \kappa(u)D(v), \qquad \kappa(u) = \frac{u^2B(u)}{P(u)} \asymp u^{-1}. \tag{G.28}$$

The first three Euler derivatives of κ are $O(u^{-1})$. On $[M,u]$, $D(v) = O(v^2)$, and each further application of $D(v)\partial_v$ increases polynomial degree by at most one. On $[\delta_c, M]$, this vector field and its first four derivatives are uniformly bounded, including when the positive factor approaches a double root: they contain P, rather than P^{-1}. Differentiating (G.28) inductively therefore gives

$$|E^j\log v| \le C_j \quad (1 \le j \le 4), \qquad |E^j(\tau_{\mathrm{cub}}H)| + |E^j(Q-1)| + |E^j\varpi| \le C_j\epsilon_u \quad (0 \le j \le 3).$$

Because the transported field contains P, these bounds remain uniform when the root-free positive minimum tends to zero. Apply Lemma A.3 with $D = E, \kappa = \tau_{\mathrm{cub}}, \rho = \epsilon_u, \alpha = 0$ on

$\delta_c \le v \le u$. The needed clock rows follow from $\partial_q = -\tau_{\mathrm{cub}} D(v)\partial_v$: on the growing part its degree growth is controlled by $\tau_{\mathrm{cub}} v \le \epsilon_u$, and on the compact part its coefficients are bounded. The strict diagonal chosen above gives

$$|E^j W(q)| \le C\epsilon_u (1+q)^N e^{-c_2 q}, \qquad 1 \le j \le 3. \tag{G.29}$$

The nonlinear logarithmic correction on this tail is exactly

$$V_+(u) = \int_0^{q_c(u)} \frac{(Q-\varpi)W}{1+QW}\, dq, \qquad q_c(u) = \frac{A_+(u) - A_+(\delta_c)}{\tau_{\mathrm{cub}}}.$$

The fixed-clock and terminal formulas of the same lemma give $|V_+| \le C$ and $|E^j V_+| \le C\epsilon_u$, $1 \le j \le 3$: indeed, $|E^j q_c| \le C_j/\epsilon_u$ and $q_c \ge [A_+(2M) - A_+(M)]/\tau_{\mathrm{cub}} \ge a_M/\tau_{\mathrm{cub}}$. Every boundary term is therefore at most $C\tau_{\mathrm{cub}}^{-N} e^{-c_2 a_M/\tau_{\mathrm{cub}}}$.

For the negative inward tail replace (P, B, Q, ϖ) by $(P(-v), B(-v), Q(-v), -C\tau_{\mathrm{cub}}^3 (G - v)P(-v)/(vB(-v)))$. At fixed q, $\mathcal{D}_-$ transports to $P(-v)[vB(-v)]^{-1}\partial_v$; its coefficient rows in the braces are $O(\tau_{\mathrm{cub}})$ on $[\delta_c, c]$, including the root factor. The same lemma with $\rho = \tau_{\mathrm{cub}}$ gives $|V_-| \le C$, $|\mathcal{D}_-^j V_-| \le C\tau_{\mathrm{cub}}$. Since $r \ge r_0 > 2\delta_c$, its complete terminal rows are exponentially small.

Continue both tails to $\xi = 0$ using Lemma A.5, with radial coordinate $v = |\xi|$ oriented from δ_c to zero and $\kappa_c = \tau_{\mathrm{cub}}$. Here $P \ge p_c > 0$, the linear inward propagator is at most one, and the nonlinear coefficient has ordinary height derivatives through order three bounded by C/τ_{cub}. For the incoming heights $y_\pm = Y(\pm\delta_c)$, the exact memory (A.11) gives $y_\pm \le C\tau_{\mathrm{cub}}^{-N} e^{-c/\tau_{\mathrm{cub}}}$; its three logarithmic phase rows grow at most polynomially in τ_{cub}^{-1}. The cited continuation and composition rows therefore make the remaining nonlinear correction exponentially small, with those phase derivatives. The exact cocycle (A.23) gives the actual heights on the common section:

$$\begin{aligned} \log Y_0^+(u) &= \log(\ell_0 P(u)) - A_+(u)/\tau_{\mathrm{cub}} + R_+(u), \\ \log Y_0^-(r) &= \log(\ell_0 P(-r)) - A_-(r)/\tau_{\mathrm{cub}} + R_-(r). \end{aligned}$$

with (G.22). Equality of these heights is exactly the physical matching condition. This proves (G.21); it also supplies a separated extension $R_0(u, r) = R_+(u) - R_-(r)$, whose genuinely mixed derivatives vanish.

The separated height formulas also determine the relative first variation and its weighted derivatives. Write $P_+ = P(u)$, $P_- = P(-r)$, $B_+ = B(u)$, $B_- = B(-r)$, and use radial derivatives in $H_+ = P_+'/(uB_+)$, $H_- = P_-'/(rB_-)$. Differentiating (G.21) gives

$$r_u = \frac{uB_+P_-}{rB_-P_+}(1+\varrho), \qquad 1+\varrho = \frac{1 - \tau_{\mathrm{cub}} H_+ - \tau_{\mathrm{cub}}\mathcal{D}_+ R_+}{1 - \tau_{\mathrm{cub}} H_- - \tau_{\mathrm{cub}}\mathcal{D}_- R_-}. \tag{G.30}$$

Since $\mathcal{D}_+ = (P_+/(u^2 B_+))E$, the numerator differs from one by $O(\epsilon_u)$, with the same Euler bound. The denominator differs from one by $O(\tau_{\mathrm{cub}})$, with two bounded action variations of that order. This proves (G.23) (even with L_u^4 omitted). Both denominators are positive after shrinking d_0 and then τ_{cub}.

For the positive exterior map apply Lemma G.3 with

$$\sigma = \tau_{\mathrm{cub}}, \quad a = \tau_{\mathrm{cub}} u, \quad (\mu_2, \mu_1, \mu_0) = (3\tau_{\mathrm{cub}} G, -\tau_{\mathrm{cub}}^2 U, \tau_{\mathrm{cub}}^3), \quad b = 2\tau_{\mathrm{cub}} G, \quad v = \tau_{\mathrm{cub}} G.$$

For $u \ge 2M$, the coefficient bounds hold and the leading cubic term gives $M(x) \ge cx^3$ on $[a, \delta]$, uniformly on the fixed compact block. In the translated coordinate $x = v - \gamma$, the original positive raw port is $x_T = \delta - \tau_{\mathrm{cub}} G$. The proof of the raw lemma is uniform for terminal cuts in a fixed neighborhood of δ; x_T is coefficient-only and is held fixed under E. Thus it applies

at this actual port. The corridor supplies its enlarged terminal image. Since $a \asymp \epsilon_u$, (G.19) is the positive assertion of (G.24).

Put $\lambda_G = 12\tau_{\mathrm{cub}}G/(1+2\tau_{\mathrm{cub}}G)$. For the negative exterior map, first restrict to $r_0 \le r \le c - x_0$, with $x_0 > 0$ fixed and small. Here $P_-(r) = P(-r)$ has a fixed positive margin. Put $v = -\xi$, $B_-(v) = B(-v)$, $Q_-(v) = Q(-v)$, and use the increasing height clock $s = \log\eta$. The fast invariant $H = \eta - v^2/2$ satisfies the exact equation

$$H_s = \tau_{\mathrm{cub}}\mathscr{F}(H, e^s), \qquad \mathscr{F} = \frac{v\eta(B_- - Q_-)/\tau_{\mathrm{cub}} - C\tau_{\mathrm{cub}}(G-v)\eta^2 - vP_-(v)}{vB_-(v) - C\tau_{\mathrm{cub}}^2(G-v)\eta}.$$

On a fixed box with $v \ge r_0/2$, the denominator is a positive unit, $(B_- - Q_-)/\tau_{\mathrm{cub}}$ is smooth and bounded, and $v = \sqrt{2(e^s - H)}$ gives $|\partial_H^i\partial_s^j\mathscr{F}(H, e^s)| \le C$ for $i + j \le 3$. The initial rows are

$$s_l = \log\{\tau_{\mathrm{cub}}\ell_0 P_-(r)\}, \qquad H_l = -r^2/2 + \tau_{\mathrm{cub}}\ell_0 P_-(r).$$

The ordinary r-derivatives of $s_l - \log\tau_{\mathrm{cub}}$ are bounded. The moving initial recursion of (A.4) therefore gives the three initial variations of $-r^2/2$, with errors $O(\tau_{\mathrm{cub}})$. Choose a fixed small $\eta_* > 0$. First exit and the three triangular equations on $s_l \le s \le \log\eta_*$, whose length is $O(1 + |\log\tau_{\mathrm{cub}}|)$, give

$$H(\log\eta_*; r) = -r^2/2 + O_{C^3_r}(\tau_{\mathrm{cub}}(1 + |\log\tau_{\mathrm{cub}}|)).$$

This is an actual outward passage: initially $D = \tau_{\mathrm{cub}}P_- + Q_-\eta > 0$, and at a possible first zero $D_s = Q_-\eta > 0$. On the fixed positive-height continuation to a sufficiently large fixed cut $v = L > c + 1$, the graph equation is an $O_{C^3}(\tau_{\mathrm{cub}})$ perturbation of $\eta_v = v$. Hence

$$\eta_L(r) = \tfrac{1}{2}(L^2 - r^2) + O_{C^3_r}(\tau_{\mathrm{cub}}(1 + |\log\tau_{\mathrm{cub}}|)).$$

The cut and its buffered chord lie in the original corridor (4.16). From this cut to the raw port use the negative tail with source quartet $(a, p, \gamma, s) = (-\tau_{\mathrm{cub}}, \tau_{\mathrm{cub}}U, \tau_{\mathrm{cub}}G, \tau_{\mathrm{cub}})$. For a fixed $0 < \nu < 1$, (C.19) gives

$$\int_L^{\delta/\tau_{\mathrm{cub}}+G} \|\mathcal{N}\|_{C^3}\, dv \le C\tau_{\mathrm{cub}} \int_L^{\infty} (v^{-2+\nu} + v^{-3+\nu} + v^{-4+\nu} + v^{-5+\nu})\, dv \le C\tau_{\mathrm{cub}}.$$

After extraction of its exact first slope, the tail and its inverse are $\mathrm{id} + O_{C^3}(\tau_{\mathrm{cub}})$. Its physical slope is $d_{\mathrm{c}} = \tau_{\mathrm{cub}}^{2-\lambda_G}U_{\mathrm{c}}$, with U_{c} a positive unit. The original ordinary C^3 observer and divided closing add $O_{C^3}(d_{\mathrm{c}})$ by (C.21). Calibrating value and slope at a frozen reference in this compact interval gives $\widetilde{\phi}_- = \mathrm{id} + O_{C^3}(\tau_{\mathrm{cub}}(1 + |\log\tau_{\mathrm{cub}}|))$ in $b_- = -r^2/2$. Since $\mathcal{D}_-$ has bounded coefficients and derivatives here, the two logarithmic slope rows have the same bound. They are unchanged by a frozen positive affine calibration, so the original ϕ_- satisfies (G.24) on this interval. Near the root, the first variation vanishes and the preceding compact normalization no longer gives a bounded inverse. We extract its endpoint power before estimating logarithmic derivatives.

The root equation gives the exact factorization

$$P_-(r) = (c - r)L_-(r), \qquad L_-(r) = r^2 + (U/c + c^{-2})r + c^{-1}, \qquad L_-(c) = c^2 + U + 2/c > 0.$$

Thus c and L_-, with their required finite coefficient derivatives, are uniform units on a fixed negative collar, independently of the positive bottleneck. Put $x = c - r$, $y = \eta$. Dividing by the positive normal rate $rB(-r)$ gives

$$\dot{x} = -\beta x p(x) - k_1(x)y, \qquad \dot{y} = y\{1 + k_2(x)y\}, \qquad \beta = \frac{\tau_{\mathrm{cub}}L_-(c)}{cB(-c)} = O(\tau_{\mathrm{cub}}),$$

Here the coefficients are exactly

$$p(x) = \frac{cB(-c)L_-(c-x)}{L_-(c)(c-x)B(-c+x)},$$
$$k_1(x) = \frac{Q(-c+x)}{(c-x)B(-c+x)}, \qquad k_2(x) = -\frac{C\tau_{\mathrm{cub}}^2(G-c+x)}{(c-x)B(-c+x)}.$$

In particular $p(0) = 1$, p, k_1 are positive units and $k_2 = O_{C^7}(\tau_{\mathrm{cub}}^2)$ on a fixed box. Use the normal-time construction (E.11)–(E.12) in the proof of Lemma E.1, with $\rho = \beta$, $h = -\beta xp$, $c = -k_1$, $k = k_2$, and selected coefficient parameter τ_{cub} at fixed U, G, A, C. The explicit coefficients have a bounded joint order-seven budget: $h = O_{C^7_x}(\beta)$, k_1 is a positive unit, and their selected τ_{cub}-rows are bounded. The normal coefficient and its parameter rows retain their height factors. The projection at phase order five uses axis-flow order six and the selected parameter row. For a fixed normal decay rate $0 < \lambda < 1$, its integrands have the majorant $C(1+T)^N e^{-(\lambda - C\beta)T}$, integrable after $C\beta < \lambda/2$. Thus the exact coordinates (Ξ, Z) and invariant $I = \Xi Z^\beta$ have five phase derivatives and one intrinsic τ_{cub}-derivative, as required for the divisions below. In the following coefficient-dependent units, a subscript c records dependence on τ_{cub}. On the moving curve $y = \tau_{\mathrm{cub}}\ell_0 P_-$, these coordinates have the form

$$\Xi_m = xu_{\mathrm{c}}(x), \qquad Z_m = \tau_{\mathrm{cub}} x v_{\mathrm{c}}(x), \qquad I(x) = \theta_{\mathrm{c}} x^{1+\beta} A_{\mathrm{c}}(x), \quad \theta_{\mathrm{c}} = \tau_{\mathrm{cub}}^\beta, \quad A_{\mathrm{c}} = u_{\mathrm{c}} v_{\mathrm{c}}^\beta,$$

with positive C^4 units $u_{\mathrm{c}}, v_{\mathrm{c}}, A_{\mathrm{c}}$. To check the coefficient differences before taking the root limit, put $Y_{\mathrm{c}}(x) = \tau_{\mathrm{cub}} x \ell_0 L_-(c-x)$. Then

$$\partial_{\tau_{\mathrm{cub}}} u_{\mathrm{c}}(x) = \frac{(\partial_{\tau_{\mathrm{cub}}}\Xi)(x, Y_{\mathrm{c}}(x))}{x} + \Xi_y(x, Y_{\mathrm{c}}(x))\ell_0 L_-(c-x), \qquad v_{\mathrm{c}} = \ell_0 L_-(c-x) e^{-J(x, Y_{\mathrm{c}}(x))}.$$

The first numerator vanishes at $x = 0$; its parameter-valued fifth phase row gives C^4 after Taylor division. In the second term the moving factor x has already cancelled, so Ξ_y needs only four phase rows. The formula for v_{c} removes the factor $\tau_{\mathrm{cub}} x$ before differentiation. Integrating these intrinsic parameter rows gives $u_{\mathrm{c}} - u_0, v_{\mathrm{c}} - v_0 = O_{C^4}(\tau_{\mathrm{cub}})$, and hence $A_{\mathrm{c}} - A_0 = O_{C^4}(\tau_{\mathrm{cub}})$. Also $\theta_{\mathrm{c}} = 1 + O(\tau_{\mathrm{cub}} |\log \tau_{\mathrm{cub}}|)$.

At a fixed outer cut beyond $-c$, the stopped trajectory has positive height. The invariant is a regular increasing coordinate there, including $I = 0$, as in (E.12). Choose its buffered invariant chord first, then the smaller lip collar so $I(x)$ lies strictly inside it. Choose the upper horizontal boundary with invariant values above that chord throughout the local strip. This is possible at $\tau_{\mathrm{cub}} = 0$, where $\pi = c - \sqrt{r^2 - 2\eta}$, and the strict margin persists for small τ_{cub}. Thus no orbit of the lip collar exits through that boundary. The outward speed starts positive. At a possible first zero of $\mathscr{D} = \tau_{\mathrm{cub}} P_- + Q_-\eta$, one has $\dot r = 0$ and $\dot{\mathscr{D}} = Q_-\dot\eta > 0$, excluding loss of that speed. Increasing height and the invariant chord then give the actual first hit of the outer cut at positive height, uniformly for $0 < x \le x_0$. Continue this fixed chord to the negative raw section, before either the regular output adapter or the finite closing map. After extracting the intrinsic exact first slope, write

$$T(x) = t_{\mathrm{st}} + d_{\mathrm{c}} F_{\mathrm{c}}(I(x)), \qquad d_{\mathrm{c}} = \tau_{\mathrm{cub}}^{2-\lambda_G} U_{\mathrm{c}} > 0,$$

where U_{c} is a positive unit and $F_{\mathrm{c}}, F_{\mathrm{c}}^{-1}$ have uniform C^4 bounds on a fixed invariant chord. Here the finite positive-height passage has $O(\tau_{\mathrm{cub}})$ coefficient changes. On its subsequent tail, (C.19), integrated with the frozen negative quartet, bounds the nonlinear normalized kernel by $O(\tau_{\mathrm{cub}})$ through four phase orders. Thus the intrinsic, separately normalized maps satisfy $F_{\mathrm{c}} = F_0 + O_{C^3}(\tau_{\mathrm{cub}})$; no convergence rate for the extracted slope d_{c} is used.

Since $b_- = -r^2/2$, one has $db_-/dx = r = c - x > 0$. Consequently the exact first derivative is

$$T'_{b_-} = d_{\mathrm{c}}\theta_{\mathrm{c}} x^\beta K_{\mathrm{c}}(x, x^\beta), \qquad K_{\mathrm{c}}(x,z) = \frac{F'_{\mathrm{c}}(\theta_{\mathrm{c}} xz A_{\mathrm{c}}(x))\{(1+\beta)A_{\mathrm{c}}(x) + xA'_{\mathrm{c}}(x)\}}{c-x}.$$

The function K_{c} is a positive C^3 unit on a fixed rectangle $0 \le x \le x_0$, $0 \le z \le 1$, and $K_{\mathrm{c}}(0,z)$ is independent of z. At $\tau_{\mathrm{cub}} = 0$ the outer graph is $\eta(\xi) = (\xi^2 - r^2)/2$. Its intrinsic, separately normalized section map is affine in b_-. The regular factors therefore cancel exactly at $z = 1$, giving a positive constant k_0, independent of x, for which

$$\|K_{\mathrm{c}}(\cdot, 1) - k_0\|_{C^2} \le C\tau_{\mathrm{cub}}(1 + |\log \tau_{\mathrm{cub}}|).$$

The affine limiting map is what makes the logarithmic derivative small; boundedness of the regular factors alone would not suffice.

To differentiate along $z = x^\beta$, use $E_x = x\partial_x + \beta z\partial_z$. For $0 < x \le x_0 < 1$,

$$x|1 - x^\beta| \le C\beta.$$

The uniform C^3 bounds for K_{c}, its constancy in z at $x = 0$, and the preceding cancellation imply

$$|E_x \log K_{\mathrm{c}}| + |E_x^2 \log K_{\mathrm{c}}| \le C\tau_{\mathrm{cub}}(1 + |\log \tau_{\mathrm{cub}}|) \quad \text{on } z = x^\beta.$$

Indeed the differences of the x-derivatives between z and 1 have factors $1 - z$, accompanied by x or x^2; all remaining terms contain β. Since

$$\mathcal{D}_- = -\frac{L_-(c - x)}{(c - x)B(-c + x)}E_x, \qquad E_x \log x^\beta = \beta,$$

we obtain the first two action-logarithmic bounds for T'_{b_-}, including their one-sided limits as $x \downarrow 0$. The derivative itself may vanish like x^β.

Finally append the original increasing ordinary C^3 output adapter ψ and the divided closing map. Put

$$\Psi_{\theta,m} = \mathfrak{f}_{\theta,m} \circ \psi, \qquad \mathfrak{f}_{\theta,m}(q) = f_\theta(mq)/m, \qquad \Gamma_- = \Psi_{\theta,m} \circ T.$$

Admission to the original raw and closing windows precedes these compositions. Apply (3.3) and (3.5) with $D = \mathcal{D}_-$ and primes in b_-. They bound the two logarithmic-slope corrections using only the ordinary C^3 norms and positive first derivative of $\Psi_{\theta,m}$, uniformly for $0 \le m \le m_0$. The invariant formula retains the small gain: $|DT| + |D^2T| \le Cd_{\mathrm{c}}$, and $d_{\mathrm{c}} \le C\tau_{\mathrm{cub}}^2$, since $\lambda_G = O(\tau_{\mathrm{cub}})$. Both additional logarithmic rows are therefore $O(d_{\mathrm{c}})$, using only the ordinary C^3 adapter and closing bounds of Lemma 3.3. Thus the actual complete output satisfies (G.24); affine normalization fixes its logarithmic rows.

We can now combine the inward match with both exterior estimates. Apply Lemma 3.5 with $v = r$, $P_+ = P(u)$, $P_- = P(-r)$, and $W = u + r$. In (3.8), put $R_B = B_+/B_-$. The first term differs from $(P'(u)/u + P'(-r)/r)/(u + r) = 3 - U/(ur)$ by $O(\epsilon_u)$. The second is $O(\epsilon_u)$, since $R_B - 1 = \tau_{\mathrm{cub}}(u + r)/B_-$ and $r_u = O(P_-/(u^2 r))$. The third is controlled by (G.23), and the last by (G.24), since $P_+/(u^2(u + r))$ is bounded. For the negative section term use

$$\frac{\tau_{\mathrm{cub}}(1 + |\log \tau_{\mathrm{cub}}|)^4}{u + r} \le C\epsilon_u(1 + |\log \epsilon_u|)^4.$$

To see this uniformly, write $\epsilon_u = \tau_{\mathrm{cub}}(1 + u)$, use $1 + |\log \tau_{\mathrm{cub}}| \le 1 + |\log \epsilon_u| + \log(1 + u)$, and note that $(1 + \log(1 + u))^4/(1 + u)^2$ is bounded. This proves (G.25) in the actual affine input coordinate. □

Proposition G.5 (Positive quadratic face)**.** *Fix $\Pi < \infty$. For $\gamma = g > 0$, $0 < k < k_0$, and $0 < \pi \le \Pi$, the actual low-height through interval on the real/fold side and its fixed adjacent root-free band has the moving-section descriptions below. Every zero of the complete map's curvature on this interval is simple and is a downward crossing in the same increasing physical*

input. The bounded-height complement has uniformly bounded complete derivative. On the root-free side outside this band, the complete return is a contraction after a coefficient shrink. The domains and estimates persist through the actual discriminant $\Delta = 0$, whose limit at $k = 0$ is $\pi^2 = 12$.

Proof. We use the curvature principal part $3 - \pi/(uR)$, with the homogeneous section powers retained. Its zeros require separate arguments in the joint growing corner and near two stopped endpoints. The latter calculation uses fixed normal sections when b/a is near two. The remaining growing ranges have a strict curvature sign, and a paired derivative bound supplies the upper complement.

Fix a preliminary $0 < \pi_{\mathrm{pre}} < \sqrt{12}$. All comparison estimates are uniform for $\pi_{\mathrm{pre}} \le \pi \le \Pi$. The final $\pi_\flat \in (\pi_{\mathrm{pre}}, \sqrt{12})$ will be chosen after the comparison margin. For $0 < \pi \le \pi_\flat$, Lemma H.15 gives contraction after the final coefficient shrink.

In $x = gk\xi$ the exact polynomial and system are

$$\begin{aligned} P(\xi) &= 1 - \pi\xi + 3\xi^2 + k\xi^3 = (\mathfrak{c} + k\xi)\{(\xi - d)^2 - \Delta\}, \\ \dot\xi &= -g\{P + Q_+ Y\}, \qquad \dot Y = -Y\{\xi B_+ + Ckg^3(1 + k\xi)Y\}, \end{aligned} \tag{G.31}$$

where

$$\mathfrak{c} = 3 + 2kd, \qquad \mathfrak{c}(d^2 - \Delta) = 1, \qquad \pi = 2\mathfrak{c}d - k/\mathfrak{c}, \qquad B_+ = 1 + g(2 + k\xi).$$

Also $Q_+ = 1 - Ag(1 + k\xi) + Cg^2(1 + k\xi)^2$, and the physical coordinates are $x = gk\xi$, $z = g^3k^2Y$. At $k = 0$, $d = \pi/6$ and $\Delta = (\pi^2 - 12)/36$. The curves $Y = \ell_0 P$ are transverse by the analogue of (4.13), and the reciprocal raw equation (4.16) joins them to the fixed sections. On a provisional compact coefficient block about the fold, and the real block up to $\pi = \Pi$, $a, a^{-1}, d, d^{-1}, \mathfrak{c}, \mathfrak{c}^{-1}$ are bounded where the corresponding real root is defined. The original cocycles of Corollaries J.6 and J.4 give a finite zeroth corrected-action bound K_0. Fix $r_c > 0$ and a finite action threshold greater than $K_0 + \sup \widetilde{A}_-(r_c) + 1$. For a fixed small $\delta_c > 0$, the real factorization gives

$$P(v) \le C(a - v), \qquad vB_+(v) \ge c > 0, \quad a - \delta_c \le v \le a - X_{\mathrm{up}}.$$

Thus $J_+(a - X_{\mathrm{up}}) \ge c \log(\delta_c/X_{\mathrm{up}})$, uniformly as $b - a \to 0$, and the fold has stronger inverse-distance divergence. Choose $X_{\mathrm{up}} = X > 0$ to pay this threshold. Choose the retained ghost width $\Delta_0 > 0$ sufficiently small afterwards so that the same margin holds on $-\Delta_0 \le \Delta < 0$. Then choose $\pi_\flat$ with $0 < 12 - \pi_\flat^2 < 18\Delta_0$. Retain $|\Delta| \le \Delta_0$ and the compact real block up to $\pi = \Pi$. Outside this band on the root-free side, $\pi < \pi_\flat$ after shrinking k, so it is already a contraction. In the remainder of the proof,

$$\pi_\flat \le \pi \le \Pi. \tag{G.32}$$

Put $u_{\mathrm{th}} = a - X$, $a = d - \sqrt{\Delta}$, on the real/fold side and $u_{\mathrm{th}} = d - X$ on the ghost side. The comparison $u_* = a - X/2$, respectively $d - X/2$, uses this same X in Lemma J.9. The independent degenerations are $g \to 0$, $k \to 0$, and the positive root gap.

Fix a translated raw half-width $0 < d_q < \delta/2$, small enough for the parabolic corridor estimates below, and shrink the coefficient neighborhood so that $|\gamma| < d_q/2$. The cuts $x = \pm d_q$, equivalently $v = \gamma \pm d_q$, lie on the corresponding sides of $v = \gamma$ and strictly within the strip bounded by the fixed sections $\Sigma_\pm = \{v = \pm\delta\}$. The uniformly regular segments from these cuts to $\Sigma_\pm$, and their inverses, are included in the actual section adapters used below.

For real or double roots the positive lip interval ends at a. For $\Delta < 0$, choose $0 < d_{\mathrm{in}} < d_0 < d_q/2$ and restrict to the component containing u_{th} of the preimage of the enlarged input and output section intervals, with

$$u \ge u_{\mathrm{th}}, \qquad x = gku < d_0.$$

Take smaller nested intervals with $x < d_{\mathrm{in}}$. Indeed $z = \ell_0(x^3 + 3gx^2 - \pi k g^2 x + k^2 g^3)$ on the lip; for $d_{\mathrm{in}} \le x \le d_q$ this is separated from the organizing arc when $\ell_0 d_q$ is small. The localization lemma and the common transverse sign therefore place every actual low-height cycle in the smaller interval. On the positive continuation use the relative outer cut $x = Lgku$, and on the negative continuation $x = -LgR$, with $L > 1$ fixed large. At these cuts $\mathcal{W} = 1/2 - 1/(2L^2) + o(1)$; the corridor equation carries both traces to $x = \pm d_q$, even across an axis root at positive height. This specifies the actual interval used below. Its endpoint on a root-free fibre is a finite section boundary, not $u = \infty$.

Put $R = kr$,

$$\widehat{P}(R) = k^2 P(-R/k) = (\mathfrak{c} - R)\{(R + kd)^2 - k^2\Delta\}.$$

Here $B_-(R) = 1 + g(2 - R)$. Define the two actions on fixed inner cuts by $J'_+ = uB_+/P(u)$ and $A'_- = rB_-/P(-r)$, and put $\mathcal{D}_+ = P(u)(uB_+)^{-1}\partial_u$. In either negative coordinate, $\mathcal{D}_- = \widehat{P}(R)(RB_-)^{-1}\partial_R = P(-r)(rB_-)^{-1}\partial_r$. Retaining both logarithmic terms in the moving-lip matching gives

$$\widetilde{J}_+ = J_+ - g\log P, \qquad \widetilde{A}_- = A_- - g\log P(-r), \qquad |\widetilde{J}_+ - \widetilde{A}_-| \le K_0.$$

Here K_0 is independent of g, k, and $\widetilde{A}_-(r) \ge c\log(1 + r) - C$. The comparison action was chosen above $K_0 + \sup \widetilde{A}_-(r_c)$. Every retained low-height input then has $r \ge r_c$, while the comparison output itself is bounded above independently of k. We define ϱ by the exact relative variation

$$R_u = \frac{uB_+\widehat{P}(R)}{RB_-P(u)}(1 + \varrho) > 0.$$

On the growing negative tail let $\mathcal{D}_- = \widehat{P}(R)(RB_-)^{-1}\partial_R$. Its logarithmic action derivative is

$$H_- = \frac{2\mathfrak{c} - 3R}{1 + g(2 - R)} + O(r^{-1} + k), \qquad |\mathcal{D}_-^j H_-| \le C(R + r^{-1}), \quad j \ge 1. \tag{G.33}$$

In a fixed simple-root collar at $R = \mathfrak{c}$, the rational function H_- and its required derivatives are bounded, while $\mathcal{D}_-$ contains the factor $\mathfrak{c} - R$. Consequently $\mathcal{D}_-^j H_- = O(\mathfrak{c} - R)$ for $j \ge 1$, and $dA_- \asymp dR/(\mathfrak{c} - R)$ there. Together with the growing-tail estimate this gives $\int |\mathcal{D}_-^j H_-|\, dA_- \le C$. In the Bell densities (A.25), the factor P cancels with the action change of variable. The factor $\mathfrak{c} - R$ is retained in the exact logarithmic cocycle and the last segment is treated in the fixed simple-root coordinate. No power $k^{-C/g}$ is generated. More explicitly, on $r_c \le r \le (\mathfrak{c} - \ell_-)/k$, $\mathcal{D}_- = V(kr, r^{-1})(kr\partial_{kr} - r^{-1}\partial_{r^{-1}})$, with V a positive unit. The integral of $kr + r^{-1}$ against $dA_- \asymp dr/r$ is bounded on this interval. The remaining fixed collar $\mathfrak{c} - 2\ell_- < R < \mathfrak{c}$ has normal rate RB_- bounded below and a simple factor $\mathfrak{c} - R$. The two estimates overlap at a fixed R-cut, where the Leibniz boundary terms cancel. No action integral crosses that root.

Only the root-free case can have $u \to \infty$. We first treat the joint quadratic corner, where both radial action speeds are bounded. Put $q = u^{-1}$, $S = ku$, and $t = r^{-1}$. Choose a small $\delta_q > 0$, fix $0 < \delta < \delta_q/2$, and choose M large enough for $M\delta \gg \Pi$ and the action gap below. All these constants precede the choices of g, k. The homogeneous quadratic field has first integral

$$Y^{-\lambda_g/2}\left(\kappa_g Y - \frac{\xi^2}{2}\right), \qquad \lambda_g = \frac{12g}{1 + 2g}, \qquad \kappa_g = \frac{g(1 - Ag + Cg^2)}{1 - 4g}.$$

Its radial exponent is the same at the two fixed raw sections. On $u, r \ge 2M$, $S, R \le \delta_q$, the exact ratio and variational equations, after this power is extracted, give, with

$$h_+ = \frac{P(u)}{u^2 B_+(u)}, \quad h_- = \frac{\widehat{P}(R)}{R^2 B_-(R)}, \quad \theta_+ = u^{-1} + ku, \quad \theta_- = r^{-1} + kr,$$

$$\sum_{i+j\le 2}|\mathcal{D}_+^i\mathcal{D}_-^j\varrho|\le\varepsilon_g(\theta_++\theta_-),$$
$$\mathcal{D}_\pm\log\phi'_\pm=-\lambda_g h_\pm+G_\pm,\qquad \varepsilon_g=Cg(1+|\log g|)^4. \tag{G.34}$$
$$|G_\pm|+|\mathcal{D}_\pm G_\pm|\le\varepsilon_g\theta_\pm,$$

The two $-\lambda_g h_\pm$ terms come from the same homogeneous power and must be combined before estimating the curvature. These bounds hold in the joint corner; the range $S\to\infty$, where $h_+\asymp 1+S$, will instead use radial derivatives. Lemma J.7 gives the first line on the product of the two separate actual domains, including the mixed second row; restriction to their original central-image intersection gives the matched graph. Lemma J.1 gives the section rows with the stronger factor $Cg(1+|\log g|)$, on each admitted raw image. The negative observer includes the divided closing of Lemma 2.2; its ordinary C^3 logarithmic-slope bounds apply on those original buffered images, with its small gain and m fixed under the phase derivative.

We shall also use absolute negative-side estimates on the larger range $r\ge r_c$, $0<R<\mathfrak{c}$. Write the exact logarithmic matching as

$$J_+(u)-A_-(r)=g\{\log P(u)-\log P(-r)+\mathcal{R}_+(u)-\mathcal{R}_-(r)\}.$$

Coefficient-only constants may be included in either $\mathcal{R}_\pm$. The two functions are obtained by continuing each actual lip to the same central section. Thus their separated extension and its derivatives refer to the actual orbit. The complete negative inward action remainder and the negative outward embedding give

$$|\mathcal{D}_-^j\mathcal{R}_-|\le Cg\quad(1\le j\le 3),\qquad |\mathcal{D}_-^j\log\phi'_-|\le C\varepsilon_g\quad(j=1,2). \tag{G.35}$$

Here and below primes on ϕ_- are in the label $-R^2/2$. Changing from $-r^2/2$ only changes its frozen affine slope. The first estimate is Corollary J.4 at the original centre, retaining the initial polynomial and every coefficient-only action constant. In particular,

$$N_-=-g\mathcal{D}_-\log Z_-=1-gH_--g\mathcal{D}_-\mathcal{R}_-,\qquad |\mathcal{D}_-^j(N_--1+gH_-)|\le Cg^2,\quad 0\le j\le 2.$$

Since H_- is bounded, $N_-\ge 1/2$. The second estimate is Corollary J.3, with the stronger $Cg(1+|\log g|)$ bound, including original-buffer admission, p-weighted initial variations, both moving endpoints and the ordinary C^3 divided closing. Neither estimate enlarges the intersection of actual central images.

For $\mathcal{T}(-u^2/2)=-R^2/2$ and $\Pi=\phi_-\circ\mathcal{T}\circ\phi_+^{-1}$, put $B_{\rm rat}=B_+/B_-$ and $\mathcal{V}=\mathcal{D}_++(1+\varrho)\mathcal{D}_-$. Lemma 3.5, with $v=R$, $P_-=\widehat{P}$, and $W=R+S$, gives (3.8) for the weighted curvature

$$\mathscr{K}=\phi'_+\frac{P(u)}{R+S}\frac{\Pi''}{\Pi'}. \tag{G.36}$$

The derivative d/du follows the matched graph. In the joint corner, exact logarithmic differentiation and the preceding estimates give

$$\phi'_+\frac{P(u)}{R+ku}\frac{\Pi''}{\Pi'}=F_\lambda(u,R)+E(u,R), \tag{G.37}$$

where, with $q=u^{-1}$ and $t=r^{-1}$,

$$F_\lambda=(3-\lambda_g)-\frac{N_\lambda}{uR},\qquad N_\lambda=(1-\lambda_g)\pi-\lambda_g(t-q), \tag{G.38}$$

and, in the joint corner $u,r\ge 2M$, $S,R\le\delta_q$,

$$|E|+|(\mathcal{D}_++(1+\varrho)\mathcal{D}_-)E|\le C\varepsilon_g\left(1+\frac{1}{uR}\right). \tag{G.39}$$

To verify (G.38), use

$$\frac{P'(u)/u - \widehat{P}'(R)/R}{R+ku} = 3 - \frac{\pi}{uR}$$

and

$$\frac{\widehat{P}/R^2 - P/u^2}{R+ku} = \frac{ku-R}{u^2R^2} + \frac{\pi}{uR} - 1.$$

The matching rows of Lemma J.7 apply on the product domain to $N_\pm = 1 - gH_\pm - g\mathcal{D}_\pm\mathcal{R}_\pm$, $n = 1+\varrho = N_+/N_-$, with $N_- \geq 1/2$.

To estimate the curvature error and its derivative along the match, put $W = R+S$, $\mathcal{V} = \mathcal{D}_+ + n\mathcal{D}_-$, and $A_\pm = B_\pm(H_\pm - \lambda_g h_\pm)$. Direct polynomial calculation gives

$$\begin{aligned} A_+ &= (6-3\lambda_g) + (\lambda_g - 1)\pi q + (3-\lambda_g)S - \lambda_g q^2, \\ A_- &= (6-3\lambda_g) + (1-\lambda_g)\pi t - (3-\lambda_g)R - \lambda_g t^2. \end{aligned}$$

Since $B_+ - B_- = gW$, the exact error decomposition is

$$\begin{aligned} E_B &= -gA_-/B_- - (B_+/W)\mathcal{V}\log(B_+/B_-), \\ E_n &= -(B_+/W)\{\varrho(H_- - \lambda_g h_-) + \mathcal{V}\log n\}, \\ E_s &= (B_+/W)(G_+ - nG_-), \qquad E = E_B + E_n + E_s. \end{aligned}$$

The first two action rows of $h_\pm, B_\pm, H_\pm$ are bounded. Thus $|E_B| + |\mathcal{V}E_B| \leq Cg$. The total-two matching row gives the required bound on E_n, including $\mathcal{V}^2\log n$. For the section term, retain the exact derivative

$$\mathcal{V}(G_+ - nG_-) = \mathcal{D}_+G_+ - (\mathcal{V}n)G_- - n^2\mathcal{D}_-G_-.$$

Lemma J.1 bounds these terms with the stronger factor $Cg(1+|\log g|)$. Finally $(q+S+t+R)/W = 1 + 1/(uR)$, and $|\mathcal{V}\log W| \leq C$. These identities give (G.39) on the actual matched joint graph. The other coefficient ranges are treated below.

On the joint corner q and t are small. The lower bound (G.32) and a further common shrink give

$$N_\lambda = (1-\lambda_g)\pi - \lambda_g(t-q) \geq \frac{1}{4}\pi_\flat.$$

Put $z = uR$ and $a_0 = 3 - \lambda_g$. At a zero of the actual $\mathscr{K} = F_\lambda + E$, write $|E| \leq \delta_g(1 + 1/z)$, with $\delta_g \to 0$. Then

$$\frac{N_\lambda - \delta_g}{a_0 + \delta_g} \leq z \leq \frac{N_\lambda + \delta_g}{a_0 - \delta_g}.$$

All actual curvature zeros therefore lie in a fixed positive compact z-interval. The exact derivative there is

$$\mathcal{V}F_\lambda = \frac{N_\lambda(h_+ + nh_-) + \lambda_g h_+ q - \lambda_g n h_- t}{z} \geq c > 0,$$

after M is large and g small. The differentiated error is absorbed on this compact interval. If $w = \phi_+(-u^2/2)$ is the actual affine input, then $\mathcal{V}w = -P(u)\phi_+'/B_+ < 0$. Because $\mathscr{K}$ is a positive multiple of Π''/Π', every curvature zero on this matched joint graph has $\Pi'''(w) < 0$. To obtain this orientation on the entire low-height interval, we next treat the mixed and remaining growing ranges, followed by the bounded positive range. There is also a mixed range $u \geq 2M$, $r_c \leq r \leq 2M$. If $S \geq \delta$, the action comparison below already forces $R \geq R_0(\delta) > 0$, which contradicts $R \leq 2Mk$. On $S \leq \delta$, the positive half of the joint-corner calculation still applies: its action speed is bounded. The negative end is now compact in r, and (G.35) supplies its absolute estimates. In (3.8) these give $\mathscr{K} = 3 - \pi/(uR) + E_M$, with $|E_M| \leq C_M\varepsilon_g/(R+S)$.

Relative to $1/(uR)$, it is bounded by $C_M\varepsilon_g\, ur/(u+r) \le 2MC_M\varepsilon_g$. A zero would require $uR \ge c_M > 0$, and hence $ku = uR/r \ge c_M/(2M)$. The action comparison below applies already for $r \ge r_c$; with this last fixed threshold in place of δ, it forces $R \ge R_0 > 0$, contrary to $R = kr \le 2Mk$. Thus the mixed range contains no zero. The remaining corners introduce none. If $u \ge 2M$, $R \ge \delta$, and $S \le \delta$, the same positive estimates and (G.35), including its simple-root collar, give an error $O_\delta(\varepsilon_g)$ in (3.8). The exact polynomial principal part is $3 - \pi/(uR)$; the choice $M\delta \gg \Pi$ leaves a positive margin. For the last range $S \ge \delta$, fix $0 < \overline{R} < \frac{1}{2}\inf \mathfrak{c}$. For $r \ge r_c$ and $R \le \overline{R}$, subtraction of the exact logarithmic cocycles gives

$$\widetilde{J}_+(u) \ge \frac{1-4g}{3}\log\frac{1}{k} - C_\delta, \qquad \widetilde{A}_-(r) \le \frac{1-4g}{3}\left(\log\frac{1}{k} + \log R\right) + C_\delta. \tag{G.40}$$

The upper bound for $\widetilde{A}_-$ is used only on this interval, which stays away from its logarithmic singularity at $R = \mathfrak{c}$. Write $c_g = (1-4g)/3$, and shrink g so that $c_g \ge c_* > 0$. Since $|\widetilde{J}_+ - \widetilde{A}_-| \le K_0$, the displayed inequalities give $c_g \log R \ge -K_0 - 2C_\delta$. Together with the alternative $R > \overline{R}$, this proves the uniform bound

$$R \ge R_0(\delta) := \min\{\overline{R}, \exp[-(K_0 + 2C_\delta)/c_*]\} > 0.$$

Consequently $uR \ge \delta R_0(\delta)/k \to \infty$. We prove the required estimate on this range in the radial derivative $E = u\partial_u$, keeping the conversion to action derivatives explicit. Set

$$a = gS = gku, \qquad L_a = 1 + |\log a|, \qquad h_+ = \frac{3 - \pi/u + u^{-2} + S}{1 + 2g + gS}.$$

For $S \ge \delta$, $a < d_0$, and k small,

$$h_+ \asymp_\delta 1 + S, \qquad |E^j h_+| \le C_\delta h_+ \quad (j = 1, 2), \qquad \frac{h_+}{R+S} \le C_\delta, \qquad \mathcal{D}_+ = h_+ E. \tag{G.41}$$

In particular the first bound in (G.34) is not used here.

Corollary J.5, with $k_0 \le \delta/(2M)$, gives the actual same-centre Euler-three rows

$$|E^j \mathcal{R}_+| \le C_\delta a, \qquad 1 \le j \le 3. \tag{G.42}$$

Its strict ghost domain retains the moving initial polynomial, terminal terms and full compact multiplier; the real/fold through interval is empty here. The action conversion remains $\mathcal{D}_+ = h_+ E$.

Differentiating the separated action matching gives exactly

$$1 + \varrho = \frac{1 - gH_+ - g\mathcal{D}_+\mathcal{R}_+}{1 - gH_- - g\mathcal{D}_-\mathcal{R}_-}, \qquad H_+ = \frac{P'(u)}{uB_+(u)}.$$

Both denominators remain positive. Since $g|E^j H_+| \le C_\delta a$, $gh_+ \le C_\delta a$, (G.35) and (G.42) imply

$$|\varrho| + |E\varrho| \le C_\delta a, \qquad |\mathcal{D}_-\varrho| \le C_\delta g, \qquad |\mathcal{V}\log(1+\varrho)| \le C_\delta(h_+ a + g). \tag{G.43}$$

Thus an action derivative of ϱ need not be small before division by $R + S$.

Choose $d_0 < a_0$ from Corollary J.2. Its actual positive raw graph, with $k_0 \le \delta/(2M)$, original observer and frozen $(gk)^2$ first gain, gives the stronger $C_\delta a L_a$ bound; hence

$$|E^j \log \phi'_+| \le C_\delta a L_a^4, \qquad j = 1, 2. \tag{G.44}$$

We now substitute in (3.8). Its polynomial term has principal part $3 - \pi/(uR)$. Since $B_{\mathrm{rat}} - 1 = g(S+R)/B_-$ and $\widehat{P}_R/R$ is bounded for $R \geq R_0$, the remaining part of that term is $O_\delta(a)$. The second term is exactly

$$-\frac{gSh_+}{R+S} - \frac{gB_+\widehat{P}(1+\varrho)}{RB_-^2(R+S)},$$

and is again $O_\delta(a)$. The third and the positive section term are bounded by (G.41), (G.43), and (G.44). For the complete negative section, including its root collar and finite closing, use (G.35) and

$$\frac{g(1+|\log g|)^4}{R+S} \leq C_\delta a L_a^4.$$

Indeed $a = gS$, $S \geq \delta$, and $(1+\log S)^4/S^2$ is bounded for $S \geq 1$; the interval $\delta \leq S \leq 1$ is compact. We have proved

$$\mathscr{K} = 3 - \frac{\pi}{uR} + E_S, \qquad |E_S| \leq C_\delta a(1+|\log a|)^4. \tag{G.45}$$

First decrease d_0 so that this error is small on $a < d_0$, then decrease g, k. The lower bound on uR gives strict positivity on the entire large-S range.

It remains to justify the bounded positive range and its overlap with the preceding estimates. On $u \leq 3M$, Corollary J.6, prepared on $u < 4M$, supplies $|\mathcal{D}_+^j \mathcal{R}_+| \leq C_M g$, $1 \leq j \leq 3$, and $|\mathcal{D}_+^j \log \phi_+'| \leq C_M g(1+|\log g|)$, $j = 1, 2$. Thus $N_+ = 1 - gH_+ - g\mathcal{D}_+\mathcal{R}_+ \geq 1/2$, with $|\mathcal{D}_+^j(N_+ - 1 + gH_+)| \leq C_M g^2$, $j \leq 2$, on its original real/fold interval or strict ghost preimage.

On bounded moving-lip abscissae the ratio equation, the growing negative estimate (G.33), and the full section maps give

$$\mathscr{K} = 3 - \frac{\pi}{uR} + E_{\mathrm{lip}}, \qquad |E_{\mathrm{lip}}| + |\mathcal{D}_+ E_{\mathrm{lip}}| + |\mathcal{D}_- E_{\mathrm{lip}}| \leq \frac{C\varepsilon_g}{R+k}, \quad \mathscr{K} = \phi_+' \frac{P(u)}{R+ku} \frac{\Pi''}{\Pi'}. \tag{G.46}$$

These action rows consume the bounded positive and absolute negative suppliers on their separate actual images, with both moving initial factors, endpoint rows and physical gains retained.

On a real-root fibre factor

$$P(\xi) = (\mathfrak{c} + k\xi)(\xi - a)(\xi - b), \qquad 0 < a \leq b, \quad \mathfrak{c} = 3 + k(a+b), \quad \mathfrak{c}ab = 1, \quad \pi = \mathfrak{c}(a+b) - kab.$$

Here $a, a^{-1}, b, \mathfrak{c}, \mathfrak{c}^{-1}$ are bounded. At the simultaneous stopped endpoint $(u, R) = (a, \mathfrak{c})$, the limiting principal value at $g = k = 0$ is

$$3 - \frac{\pi}{a\mathfrak{c}} = 2 - \frac{b}{a}.$$

The endpoint value vanishes at $b/a = 2$; this is distinct from the double-root condition $\pi = \sqrt{12}$ at $k = 0$. Outside a fixed neighborhood of $b/a = 2$, this value has a nonzero margin. A possible curvature zero then has $R \geq c_F > 0$ and $P(u) + \widehat{P}(R) \geq c_0 > 0$. Since

$$\frac{d}{du} = \frac{uB_+}{P(u)}\{\mathcal{D}_+ + (1+\varrho)\mathcal{D}_-\}, \qquad \frac{uB_+}{P(u)} \leq C(1+R_u)$$

on this zero neighborhood, the error derivative is at most $C\varepsilon_g(1+R_u)$, whereas $d(3 - \pi/(uR))/du \geq c(1+R_u)$. For $R \to 0$, the principal term instead dominates the error relatively. This proves the crossing assertion except in a fixed joint-root neighborhood of $b/a = 2$. The double root $a = b$ belongs to the separated case. The adjacent root-free block has the same

endpoint margin: the arctangent action gives its continuation, and no passage beyond $a = d$ is added on $\Delta = 0$.

Near $b/a = 2$, the endpoint principal value can vanish while both action speeds tend to zero. We therefore need errors whose ordinary derivatives are small in the two independent root distances. The root gap $b - a$ is a positive unit in this range. Use the actual fixed normal sections constructed in Lemma J.8, in the positive (ξ, gY) and negative (R, gk^2Y) root boxes. Their strong footpoints are $u = a - h$ and $R = \mathfrak{c} - t$, with $h, t > 0$. The retained interval is the intersection of the actual matched graph with $h < h_0$, $t < t_0$, where h_0, t_0 are fixed first. It is one terminal interval because $t'(h) > 0$. The normal rates in these two boxes are positive units, independently of g, k. In this fixed-root calculation, regard A_- as a function of R; the identity $rB_-\, dr/P(-r) = RB_-\, dR/\widehat{P}(R)$ preserves its value. Lemma J.8 supplies the pole subtractions with residues $\mathfrak{a}_\pm$ and uniform C^3 regular parts. All coefficient-only action constants, including possible $\log k$, remain fixed. The normal-footpoint quotient is $\nu_{\mathrm{fp}} = \mathfrak{a}_+/\mathfrak{a}_-$, without an added g in either residue. The exact residue identity is

$$3 - \frac{\pi}{a\mathfrak{c}} - \frac{g\widehat{P}_R(\mathfrak{c})}{\mathfrak{c}B_-(\mathfrak{c})} = \frac{b-a}{a}\left(\frac{\mathfrak{a}_+}{\mathfrak{a}_-} - 1\right). \tag{G.47}$$

This identifies the endpoint value of the actual weighted curvature; it does not assert that every nearby curvature zero lies on the coefficient resonance itself.

On the actual matched intersection, the same lemma gives (J.50), (J.51), and (J.52). Its coefficient memory $C_{g,k}$ and both homogeneous gains are retained without size or reciprocal bounds; the separated matching errors vanish with h, t, and $|\partial_h \varrho| + |\partial_t \varrho| \le C\varepsilon_g$.

Put $E_* = -g\widehat{P}_R(\mathfrak{c})/(\mathfrak{c}B_-(\mathfrak{c}))$. The axis correction in the exact curvature identity is

$$E^{\mathrm{ax}} = -\frac{gkP}{u(R+ku)B_+} - \frac{g\widehat{P}_R}{RB_-} - \frac{gB_+\widehat{P}}{R(R+ku)B_-^2}.$$

It has endpoint value E_*. The remaining matching and section terms, before estimating, are

$$\begin{aligned} E^{\varrho} &= -\frac{B_+\varrho(\widehat{P}_R - \widehat{P}B_{-,R}/B_-)}{RB_-(R+ku)} - \frac{B_+\{\mathcal{D}_+\varrho + (1+\varrho)\mathcal{D}_-\varrho\}}{(R+ku)(1+\varrho)}, \\ E^{\mathrm{sec}} &= \frac{B_+}{R+ku}\{\mathcal{D}_+ \log \phi_+' - (1+\varrho)\mathcal{D}_- \log \phi_-'\}. \end{aligned}$$

Write $E_0 = E^{\mathrm{ax}} - E_* + E^{\varrho} + E^{\mathrm{sec}}$. Here $P \asymp h$, $\widehat{P} \asymp t$, and all displayed denominators are positive units with bounded ordinary endpoint derivatives. The separated quotient calculation in Lemma J.8 bounds the ordinary h, t derivatives of ϱ, $\mathcal{D}_+\varrho$ and $\mathcal{D}_-\varrho$ by $C\varepsilon_g$. For the section terms, division of the second rows in (J.52) by $h\gamma_+$, respectively $t\gamma_-$, gives

$$|\partial_h(\mathcal{D}_+ \log \phi_+')| + |\partial_t(\mathcal{D}_- \log \phi_-')| \le C\varepsilon_g.$$

The axis correction satisfies $E^{\mathrm{ax}} - E_* = O(g(h+t))$, with ordinary endpoint derivatives $O(g)$. Hence the product rule gives $|\partial_h E_0| + |\partial_t E_0| \le C\varepsilon_g$ on the separated product. Restricting to the actual $t = t(h)$ yields

$$\mathscr{K} = 3 - \frac{\pi}{uR} + E_* + E_0, \qquad |E_0| \le C\varepsilon_g(h+t), \qquad \left|\frac{dE_0}{dh}\right| \le C\varepsilon_g(1 + t'(h)), \tag{G.48}$$

Since

$$\frac{d}{dh}\left(3 - \frac{\pi}{uR}\right) = -\frac{\pi}{u^2R} - \frac{\pi t'(h)}{uR^2} \le -c(1+t'(h)),$$

the actual curvature is strictly decreasing at every possible zero, even when both action rates tend to zero independently.

To join this interval to the moving lip, use the actual nonlinear resection of Lemma J.8. Equation (J.53) retains the added g's from the two initial polynomials. Its positive-unit residues give uniform inclusions between suitably nested fixed endpoint neighborhoods; they do not give a linear comparison of the internal root distances. Equation (J.54) identifies the same completed physical map after the two exterior embeddings. Only positive affine changes of its physical input and output remain, so its curvature zeros agree. Both inputs increase with the corresponding positive endpoint distance. Hence every zero has the same downward orientation on the overlap. The bounded construction is used on $u \le 3M$. On $u \ge 2M$, the mixed range covers $r \le 2M$; for $r \ge 2M$, the remaining alternatives are $S, R \le \delta$, $R \ge \delta, S \le \delta$, and $S \ge \delta$. They are covered respectively by the joint corner, the absolute negative-tail estimate, and (G.45). The fixed thresholds are used with the enlarged overlaps specified above, so these constructions exhaust the prescribed interval. Every curvature zero has this same strict downward orientation in the common physical input.

Put $u_* = a - X_{\mathrm{up}}/2$ on the real/fold side and $u_* = d - X_{\mathrm{up}}/2$ on the ghost side. In the same original physical input, Lemma J.9 gives $0 < \Pi'_\theta \le C_*$ on the entire admitted interval $[\eta_{\mathrm{comp}}, \eta_{\max}]$, where $\eta_{\mathrm{comp}} = gZ_+(u_*)$. The bounded comparison cocycle gives $\eta_{\mathrm{comp}} < gZ_+(u_{\mathrm{th}}) < \eta_{\max}$ after a common coefficient shrink. Hence this interval overlaps the lower lip interval strictly and retains every intervening input in the original buffers. The common choice $mC_* < 1/2$ gives $F' > 1/2$ on its bounded-height complement, uniformly through $\Delta = 0$. On $\Delta = 0$, the formal branch $u > d$ approaches the equilibrium in infinite time and is not included. □

G.3 The admitted raw tail on a bounded positive quadratic fibre

The growing curvature calculation uses $gku < d_0$. On a ghost fibre, the original physical interval can also contain first lips beyond this divider. We keep that tail and compare its complete input and output gains through their common central height. The negative gain is controlled by the sign of its density during the long root dwell.

Proposition G.6 (The original bounded-positive ghost raw tail)**.** *Fix the passive compact set, original raw sections, nested windows, ordinary C^3 observers and finite closing of Lemma 2.2. Fix $\pi \in [\pi_\flat, \Pi]$ and the retained strict ghost band $\Delta(k, \pi) < 0$ of* (G.31)*. The parameters $g, k > 0$ approach zero independently. Keep the original master*

$$F(x) = x - m\Pi_{\theta,m}(x), \qquad x = h - a_\theta, \qquad \Pi_{\theta,m} = \mathcal{C}_{\theta,m} \circ T_\theta(a_\theta + \cdot),$$

and one positive unscaled cap $0 < h_c < \min(H_{\mathrm{cap}}, z_*)$, chosen before the final coefficient and multiplier bounds. Its original carrier is

$$I_\theta = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < H(x) = z_\theta(g; a_\theta + x) < h_c\}. \tag{G.49}$$

Choose the fixed auxiliary $d_0 < a_0$ and its buffer $a_\flat = d_0/2$ inside the original chart, where a_0 is the positive outward reserve of Corollary J.2. There are fixed $g_{\mathrm{tail}}, k_{\mathrm{tail}} > 0$; the full-carrier coefficient rectangle uses the minima of these bounds and the native lower, paired and high bounds. Every actual lower input whose positive first lip has raw abscissa $a = gku \ge a_\flat$ satisfies

$$0 < \Pi'_{\theta,m}(x) \le C(d_0)g^3. \tag{G.50}$$

This includes every original admitted $a > d_0$, through all original window, inverse-image, speed, stop and height truncations, and all intervening nonroot inputs. The estimate holds uniformly in the independent coefficient rectangle and $0 \le m \le m_0$. With the actual finite channel $K_{\mathrm{fin}} > 0$, choose m_* last so the whole retained raw tail has $F' > 1/2$. No new receiver or first-lip cutoff is imposed.

Proof. Put $\epsilon = gk$, $\eta = gY$, so $v - g = \epsilon\xi$, $z = \epsilon^2\eta$. Use (G.31) in the original $\eta = gY$ coordinate, with $P = 1-\pi\xi+3\xi^2+k\xi^3$, $B = 1+g(2+k\xi)$, $Q = \ell(g+\epsilon\xi)$, $\ell(v) = 1-Av+Cv^2$, and $H_1 = Ckg^2(1+k\xi)$. On the ghost fibre $P > 0$ on the entire positive segment. In the translated raw positive abscissa $t = v - g$, define

$$M(t) = t^3 + 3gt^2 - \pi kg^2 t + k^2 g^3, \quad B_0(t) = 1 + 2g + t,$$
$$Q_0(t) = \ell(g+t), \quad \zeta_0(t) = C(g+t), \quad D_0 = M + Q_0 z.$$

Thus $M(\epsilon\xi) = \epsilon^2 gP(\xi)$. Let $t_* = \delta - g$ be the actual original positive raw port. The inverse of its incoming passage obeys

$$z_t = f_0(t,z) = \frac{z[tB_0 + \zeta_0 z]}{D_0}, \qquad z(a) = \ell_0 M(a). \tag{G.51}$$

The lip height ℓ_0 is fixed using the original input-window reserve. On $a_b \le a \le t_*$, independently small g, k give $M(t) \asymp t^3$ on $[a, t_*]$. With Q_0 a positive unit, $f_0 \le Ct + Cz$; Gronwall gives $z \le C\delta^2$ inside the fixed upper chart boundary. The numerator is positive, so $z \ge \ell_0 M(a) > 0$. Hence this is an actual regular outward flight. Its original input observer χ_+ is used only on admitted port images. A nonadmitted zero-flight endpoint at $a = t_*$ is an auxiliary graph value and is not assigned an observer value.

Equation (A.6), with $\sigma = \epsilon$, $K_l = K$, $I_+ = I$, and $x_T = t_*$, gives

$$\Gamma'_{+,a} = -a\chi'_+(z_{\rm port})\frac{z_{\rm port}}{D_0(t_*, z_{\rm port})}B_0(a)K_l(a)e^{I_+(a)}. \tag{G.52}$$

The original transverse lip margin makes K_l a positive unit. All admitted remaining factors have bounds and reciprocal bounds depending on d_0. For the original label $b_u = -u^2/2$,

$$-\Gamma'_{+,a} \ge c(d_0)a, \qquad c(d_0)\epsilon^2 \le \Gamma'_{+,b_u} \le C(d_0)\epsilon^2. \tag{G.53}$$

The exact input gain $\epsilon^2 = (gk)^2$ is retained.

We next obtain an absolute first-variation estimate for the positive inward passage on the whole raw tail. Let $\eta_0^+(u) > 0$ be its height at the original center $\xi = 0$. For $f = \eta(\xi B + H_1\eta)/(gP + Q\eta)$, put $W = \eta/(gP)$, $H_+ = P'/(\xi B)$, $\varpi = H_1 gP/(\xi B)$. Moving-initial differentiation is exact:

$$\partial_u \log \eta_0^+ = -\frac{uB_+}{gP_+}\Lambda_+(u), \qquad \Lambda_+ = \frac{K_l(a)}{1+\ell_0 Q_+}\exp\int_u^0 \left(f_\eta - \frac{f}{\eta}\right)d\xi,$$
$$f_\eta - f/\eta = \frac{\xi B}{gP}\frac{W(\varpi - Q)}{(1+QW)^2}. \tag{G.54}$$

Indeed $g\ell_0 P'(u) - f(u, g\ell_0 P(u)) = -\ell_0 uB_+ K_l(a)/(1+\ell_0 Q_+)$; division by the actual height cocycle produces the displayed logarithmic variation. Fix $v_i > 0$ so that $P \ge p_i > 0$ on $[0, 2v_i]$. For $\xi \ge v_i$ throughout the original raw chart,

$$gH_+ = \frac{g(6 - \pi/\xi + 3k\xi)}{B} \le C(g+t), \qquad |\varpi| \le C_{v_i}\{(g+t)^3 + g^3 k\}.$$

The fixed chart and lip choices therefore give $Q - \varpi > 0$ and the same transverse margin at every running point. The positive action clock $dq = -\xi B\, d\xi/(gP)$ gives

$$W_q = W\left[-\frac{1+\varpi W}{1+QW} + gH_+\right], \qquad W(0) = \ell_0.$$

The braces increase with W; at ℓ_0 they equal $-K_l/(1+\ell_0 Q) \le -c_1 < 0$. Thus $W \le \ell_0 e^{-c_1 q}$ down to $\xi = v_i$. In raw coordinates this also gives $z = M(t)W \le \ell_0 M(t) \le C\ell_0\delta^3$, keeping the

entire noncentral inward arc in the original chart. The noncentral integral in (G.54) is exactly $\int W(Q-\varpi)/(1+QW)^2\,dq$, and is bounded.

Every $a \geq a_b$ has $u > 2v_i$ after independent coefficient shrinks. The fixed segment $[v_i, 2v_i]$ gives $A_{\mathrm{in}} = \inf \int_{v_i}^{2v_i} \xi B/P\,d\xi > 0$. Consequently its entering height satisfies $\eta_i \leq Cge^{-c_1 A_{\mathrm{in}}/g}$. On the remaining slab $[0, v_i]$, the linear coefficient $\xi B/(gP)$ is nonnegative, and the graph equation has $f \geq -Ckg\,\eta^2$. In the decreasing ξ-direction a first-exit comparison therefore keeps $0 < \eta \leq 2\eta_i$. The denominator remains at least $gp_i/2$, so the actual arc reaches $\xi = 0$ in finite positive time. The regular formula

$$f_\eta - f/\eta = \frac{\eta(H_1 gP - \xi BQ)}{(gP + Q\eta)^2}$$

bounds the remaining logarithmic variational integral by $C\eta_i/g^2$, which is exponentially small. This uses only the original central passage, with no additional observer or endpoint. Consequently

$$0 < c \leq \Lambda_+, \Lambda_+^{-1} \leq C, \qquad -\partial_a \log \eta_0^+ = \frac{aB_0(a)}{M(a)}\Lambda_+ > 0. \tag{G.55}$$

This proves monotonicity and absolute first variation for every actual raw-tail input, with no restriction $a < d_0$.

The negative inward supplier is already on its entire strict interval. Write $R = kr$,

$$P_-(r) = 1 + \pi r + 3r^2 - kr^3, \qquad \widehat{P}(R) = k^2 P_-(R/k), \qquad B_- = 1 + g(2 - R).$$

Corollary J.4 constructs the actual original-center height $\eta_0^-(r) = gZ_-(r)$ and retains $\log(\ell_0 P_-) - A_-/g + \mathcal{R}_-$. Its exact derivative is

$$\begin{gathered} -\partial_r \log \eta_0^- = \frac{rB_-}{gP_-}\Lambda_-, \qquad \Lambda_- = 1 - gH_- - g\mathcal{D}_-\mathcal{R}_-, \qquad 1/2 \leq \Lambda_- \leq 3/2, \\ \mathcal{D}_- = \frac{P_-}{rB_-}\partial_r, \qquad r \geq r_c, \quad 0 < R < \mathfrak{c}. \end{gathered} \tag{G.56}$$

This includes the arbitrarily close negative root, its actual endpoint power and all central continuation factors. In particular $\eta_0^- \downarrow 0$ at $R \uparrow \mathfrak{c}$.

Matching does not assume an image intersection or an action estimate outside its original domain. The original paired comparison has an actual negative match $r_* \geq r_c$ at u_*. Put $u_b = a_b/(gk)$. The native separate inward estimates give $\eta_0^+(u_b) < \eta_0^+(u_*) = \eta_0^-(r_*)$, since their entire segment has $a \leq a_b < d_0$. Negative (G.56) and its zero endpoint therefore give a unique $r_b \geq r_*$. This is a match of the separate actual centre maps, before any raw observer is attached. It assigns no observer value to a possibly nonadmitted auxiliary port image at a_b. Set $R_b = kr_b > 0$. Positive (G.55) and negative (G.56) monotonicity then give a unique actual negative match for every $a \geq a_b$:

$$\eta_0^+(a) = \eta_0^-(r(a)), \qquad R_b \leq R(a) < \mathfrak{c}. \tag{G.57}$$

The order of the actual centre images therefore extends matching beyond $gku < d_0$. The absolute output estimate below holds on all $r \geq r_c$, without requiring the stronger bound $R \geq R_b$. For an input of the original I_θ, the through trajectory already reaches both the centre and Σ_-. Equality at that same centre and scalar uniqueness identify the matched negative graph with this trajectory. Its output-window and closing inverse-image admission are inherited before an observer is evaluated. The auxiliary formal shots do not enlarge those observer domains.

We now bound the actual negative output gain at the fixed raw cut $t = d_q$, followed by its original regular section bridges. Since $R = kr$, the negative spatial radius is gR. In

$R = kr, W = k^2\eta$, with raw height $z = g^2W$, the exact positive-time field, after a positive constant time change, is

$$R' = g\widehat{P}(R) + Q_-(R)W, \qquad W' = W\{RB_-(R) - Cg^2(1-R)W\}, \qquad Q_-(R) = \ell(g(1-R)).$$

This retains the independent k-dependence of $\widehat{P} = k^2 + \pi kR + 3R^2 - R^3$. For the actual raw negative graph let $t = g - v > 0$, $\mathcal{W} = z/t^2$, $b = gR$, and

$$p = 3 + \pi/r + r^{-2} - R = \widehat{P}(R)/R^2, \quad m_-(t) = 3g - t + \pi kg^2/t + k^2g^3/t^2,$$
$$Q_-(t) = \ell(g-t), \quad N_-(t,\mathcal{W}) = 1 + 2g - t - Ct(g-t)\mathcal{W}, \quad D_- = m_- + Q_-\mathcal{W}.$$

Its original lip is $\mathcal{W}_l = \ell_0 gp$. Corollary J.3 proves actual passage and original-buffer admission on all $r \geq r_c, 0 < R < \mathfrak{c}$, including $p \to 0$, by its inverse graph and strip climb.

Set $\nu = 6g/(1+2g)$, $\lambda_g = 2\nu$. The exact density (J.10) on this negative graph is

$$\begin{aligned} I_- &= \int_{\log z_l}^{\log z_T} J_-\, d\log z, \\ J_- &= -\frac{t}{N_-}\left[3 - \nu + \{-A + (3-\nu)C(g-t)\}\mathcal{W} - \frac{\pi kg^2}{t^2}\right]. \end{aligned} \tag{G.58}$$

The output gain contains e^{I_-}, so an upper bound for I_- suffices. Choose $0 < \rho_0 < \inf \mathfrak{c}/4$. For $R \leq \rho_0$, p has bounds and reciprocal bounds on $r \geq r_c$. Its weak $\log z$-length is $O(1 + |\log g|)$, and $|J_-| \leq Cg$ there. This part contributes $O(g(1+|\log g|))$. For $R \geq \rho_0$, choose a fixed small auxiliary height $\omega > 0$ with $\sup|-A + (3-\nu)C(g-t)|\omega \leq 1/4$, then a fixed

$$k_{\text{tail}} \leq \frac{\rho_0^2}{4\Pi}.$$

It is a separate quadratic-chart bound, fixed before any final raw coefficient shrink. While $\mathcal{W} \leq \omega$, $t \geq b = gR$ gives $\pi kg^2/t^2 \leq \Pi k/\rho_0^2 \leq 1/4$. After choosing g_{tail} with $\nu < 1/4$, the bracket in (G.58) is at least 2. Hence $J_- \leq 0$ throughout the arbitrarily long root dwell. No logarithm of p is estimated by a coefficient bound. Choose a fixed inner cut $d_* \ll \omega$ and then shrink g; on the strong part up to d_*, $\mathcal{W} \geq \omega$ and $D_- \geq c\mathcal{W}$. Thus $d\log z \leq C\, d\log t$. The remaining density has majorant $C(t + kg^2/t)$, with integral at most $C(d_* + kg^2/b) \leq C(d_* + g/r_c)$. The compact strong continuation from this auxiliary d_* to the original fixed raw cut d_q contributes a fixed bound. Thus on the whole original negative interval

$$I_- \leq C. \tag{G.59}$$

The singular root dwell is controlled by its sign; this estimate imposes no condition on $g\log k$.

Here T denotes $t = d_q$. Let ψ_- include the original regular bridge from this cut to Σ_- and the divided closing, and choose $c_- = \psi_-'(z_{\text{ref}}) > 0$ at a coefficient-only buffered reference. It and its reciprocal are observer units. The $s = -1$ specialization of (J.11), valid algebraically on this whole negative domain, and the exact label change give

$$\Gamma'_{-,b_R} = c_- g^2 \ell_0^{-1-\nu} g^{-\nu} (gR)^{-\lambda_g} p^{-\nu} \mathcal{K}_-(r), \qquad b_R = -R^2/2, \quad 0 < \mathcal{K}_- \leq C. \tag{G.60}$$

Here $\mathcal{K}_-$ is the product U_s of that formula, without its initial $p^{-\nu}$; its upper bound follows from (G.59) and the remaining unit factors. Here $\mathscr{B}_-$ is the original moving-initial transverse factor of (J.9), a negative unit. The terminal factors are evaluated on their admitted original images. The power $p^{-\nu}$, the exact λ_g, and the absolute gain are all retained; $\mathcal{K}_-^{-1}$ is not needed. Changing back to $b_r = -r^2/2$ gives exactly $\Gamma'_{-,b_r} = k^2\Gamma'_{-,b_R}$; no $k^{-\lambda_g}$ has been declared a unit. Using the actual center variation (G.56),

$$0 < \frac{d\Gamma_-}{d\log \eta_0^-} = \frac{g\widehat{P}(R)}{B_-\Lambda_-}\Gamma'_{-,b_R} \leq Cc_- g^3 g^{-\nu-\lambda_g} R^{2-\lambda_g} p^{1-\nu} \leq Cc_- g^3. \tag{G.61}$$

Indeed $0 < p \le C$ on $r \ge r_c$, $R < \mathfrak{c}$, and $2 - \lambda_g > 1, 1 - \nu > 0$. These positive powers pay both $R \to 0$ and $p \to 0$. Also $g^{-\nu-\lambda_g}$ is a unit because $\nu + \lambda_g = O(g)$, independently of k. Thus differentiation with respect to the actual centre height absorbs the possibly singular root factor in the output gain.

Choose $c_+ > 0$ as the analogous ordinary input observer slope unit. Equations (G.52) and (G.55) give

$$\frac{d\Gamma_+}{d\log\eta_0^+} = \frac{-\Gamma'_{+,a}M(a)}{aB_0(a)\Lambda_+(a)} \ge c(d_0)c_+a^3.$$

On the same actual matched orbit the complete physical derivative is therefore the exact ratio

$$0 < \Pi'(x) = \frac{d\Gamma_-/d\log\eta_0^-}{d\Gamma_+/d\log\eta_0^+} \le C\frac{c_-}{c_+}\frac{g^3}{a^3} \le C(d_0)g^3.$$

Only the ratio of the original regular observer units is bounded here; the full exterior gains and their endpoint powers remained in (G.52) and (G.60). The input center subtraction a_θ has zero phase derivative. The ordinary C^3 divided-closing chain rule of Lemma A.2 is uniform through $m = 0$ on these original buffered images. All coefficients remain frozen, independently of g, k, the ghost gap and the phase scale. □

Corollary G.7 (The bounded positive quadratic carrier)**.** *Fix $\Pi_0 < \infty$, the original nested raw-section windows, and their ordinary C^3 regular observers and divided finite closing. The passive compact (A, C)-set is contained in their prescribed common coefficient neighbourhood, with the original buffer and observer margins. Use fibres with $K_{\mathrm{fin}} > 0$. For the field* (G.31) *there are independent upper bounds for $g, k > 0$, one common multiplier bound $0 < m \le m_* \le m_0$, and a fixed raw central-height cap $H_{\mathrm{cap}} > 0$ with the following property. On each retained real, fold or adjacent ghost fibre, $0 < \pi \le \Pi_0$, the original enlarged windows admit one connected physical carrier below H_{cap}. Its lower interval has at most one curvature zero, which is simple and crosses downward in the increasing physical input. The actual $F(x) = x - m\Pi_\theta(x)$ has at most two zeros of F', counted with multiplicity, on the entire carrier. Its overlapping upper complement has $F' > 0$. The ghost raw tail $gku \ge d_0$ also has $F' > 1/2$; it is retained on the original carrier together with every intervening input. On the remaining root-free fibres, $F' > 0$.*

Proof. We combine the low-height curvature orientation with the paired upper bound and the raw-tail contraction, all in the original affine input. The strict overlaps below ensure that these estimates control every point of the connected carrier before Rolle's theorem is applied.

Use the original affine input and divided output of Lemma 2.2 and (3.2). Make the comparison and reserve choices of Proposition G.5: $K_0, r_c, X, \Delta_0, \pi_\flat$, then δ_q, δ, M, with $M\delta \gg \Pi_0$. Prepare the bounded supplier on $u < 4M$ and retain $u \le 3M$. Choose d_0 for the small error in (G.45), with $0 < d_{\mathrm{in}} < d_0 < \min(a_0, d_q/2)$. Fix $0 < H_{\mathrm{cap}} < z_*$ before coefficient and multiplier bounds, and apply Proposition G.6 with $a_b = d_0/2$, $h_c = H_{\mathrm{cap}}/2$; include $g_{\mathrm{tail}}, k_{\mathrm{tail}}$ in the final rectangle.

The same-centre suppliers give $N_\pm \ge 1/2$, and

$$Z_-(r) \downarrow 0 \quad (kr \uparrow \mathfrak{c}), \qquad Z_-([r_c, \mathfrak{c}/k)) = (0, Z_-(r_c)].$$

The comparison action margin places $Z_+(u_{\mathrm{th}})$ inside this image and, by its bounded action, above $Z_-(r_{\max})$ for a fixed $r_{\max} > r_c$. Strict monotonicity therefore gives every retained lip its unique actual match,

$$Z_+(u) = Z_-(r), \qquad R_u = \frac{uB_+\widehat{P}(R)}{RB_-P(u)}\frac{N_+}{N_-} > 0.$$

Real/fold inputs have $u_{\rm th} \le u < a$, with only $u < d$ at the fold; ghost curvature inputs use the original enlarged raw-window preimage with $u \ge u_{\rm th}, gku < d_0$. The separate outward suppliers admit both raw images before observers. Scalar order and Lemma 2.3 make the window preimages and original speed, height and stop clips intervals.

The comparison is admitted to the mother interval: its central raw height is below h_c, and its central speed is below the original margin after shrinking coefficients. Below it the positive port has $Y/P > \ell_0$ and the centre has $Y/P < \ell_0$. Ghost positivity gives an intermediate lip hit. At every hit the action-clock derivative is strictly negative, so the hit is unique. Its radial derivative has the opposite sign, whereas scalar first variation in the original input is positive. Implicit differentiation therefore gives $a_x < 0$ in that same affine input. This first-hit construction and the centre-image identification above preserve both original output and closing clips. Allocate

$$I_{\rm tail} = \{x \in I_\theta : x \le x_{\rm comp},\ \Delta < 0,\ a(x) \ge d_0\}, \qquad I_{\rm low} = \{x \in I_\theta : x \le x_{\rm comp}\} \setminus I_{\rm tail}.$$

The tail is an initial interval or empty. Its entry bound $H \le (gk)^2 C g e^{-c/g} < h_c$ puts its whole admitted part in Proposition G.6, including $a = d_0$. That estimate strictly overlaps the native curvature interval on $a_b \le a < d_0$; the connected $I_{\rm low}$ retains the strict hypothesis $a < d_0$.

On the bounded range, the two action-three supplier bounds and bounded rows of $H_\pm$ give, by the separated quotient rule,

$$\sum_{i+j\le 2} |\mathcal{D}_+^i \mathcal{D}_-^j (n-1)| \le C_M g, \qquad \mathcal{D}_+\mathcal{D}_- n = -\frac{(\mathcal{D}_+ N_+)(\mathcal{D}_- N_-)}{N_-^2}. \tag{G.62}$$

Thus (G.46) consumes the full product and section rows. Proposition G.5 gives the downward crossings on the bounded lip, fixed-normal and joint-growing packages. The exact resection (J.53) and composition (J.54) identify the same map on their overlaps, retaining coefficient memory and both gains.

For $\sigma_1 = \pi_\flat/(24M)$, use the positive bounds $R_0(\delta), R_0(\sigma_1)$ from (G.40), and choose

$$k_0 < \frac{\min\{R_0(\delta), R_0(\sigma_1)\}}{4M}, \qquad k_0 \le \min\{\delta/(2M), k_{\rm tail}\}.$$

These enforce the proposition's mixed-range exclusion; the remaining $R \ge \delta, S \le \delta$ and $S \ge \delta$ packages have positive curvature. The closed alternatives cover $r \ge 2M, u \ge 2M$, within the enlarged domains $S, R < \delta_q$ and $S > \delta/2, a < 2d_0$; the bounded range $u < 4M$ overlaps at $2M \le u \le 3M$. Every equality face is retained. Hence every curvature zero on the connected lower interval is a simple downward crossing, so there is at most one. Rolle applied to $F'' = -m\Pi_\theta''$ gives at most two zeros of F', with multiplicity; a zero at the curvature crossing is double.

The paired lemma gives $0 < \Pi_\theta' \le C_*$ on the original buffer interval $[\eta_{\rm comp}, \eta_{\max}]$, with $\eta_{\rm comp} = gZ_+(u_*) < gZ_+(u_{\rm th}) < \eta_{\max}$. The signed high package applies for $\eta_0 > gK_H(1 + g\pi^2)$, $H = (gk)^2\eta_0 < z_*$. Shrink independently so

$$gK_H(1 + g\Pi_0^2) < \eta_{\max}/2, \qquad (gk)^2\eta_{\max} < H_{\rm cap}/2.$$

The paired and high packages strictly overlap, retain the high equality in the paired package and cover every intervening upper input under the original clips. Finally take $m_* \max(C_*, C_H, C(d_0)g_0^3, C_{\rm contraction}) < 1/2$. The tail and upper complement have $F' > 1/2$, adding no critical point to the same physical F. For $\pi \le \pi_\flat$, Lemma H.15 gives $F' > 0$. □

H Bounded action, degree loss, and remote roots

At degree loss, the incident saddles have different small exponents, so one limiting action does not describe the whole return interval. We first connect bounded action to the central source passage, then construct the growing-action and fixed-exit maps on the same physical sections. Comparing their actual input and output derivatives gives at most two zeros of the complete displacement on the entire interval.

H.1 Bounded negative action and the central source interval

The bounded-footpoint passage and the central source passage use different sections and phase labels. We connect them by following the same orbits between those sections. This gives the overlap needed to combine a negative-curvature estimate with a bounded-derivative estimate.

Let $\tau > 0$, $\sigma > 0$, and

$$\begin{aligned} P_\sigma(\xi) &= (1+\sigma\xi)(1-\xi/x_*)(1-\sigma j\xi) = 1-\Lambda_1\xi+3\Lambda_3\xi^2+\Lambda_2\xi^3, \\ \Lambda_1 &= x_*^{-1}-\sigma+\sigma j, \qquad \Lambda_2 = \sigma^2 j/x_*, \qquad 3\Lambda_3 = \sigma(j-1)/x_* - \sigma^2 j. \end{aligned} \tag{H.1}$$

Here x_* ranges in a positive compact interval and j in a fixed compact subset of $[0,\infty)$. Physical raw sections require $j > 0$. The value $j = 0$ denotes only a one-sided normalized limit. We take either a positive compact j-interval, or a sufficiently small interval $0 \le j \le j_0 < 1$. In the latter case $\Lambda_3 < 0$. In both cases, after decreasing σ,

$$\Lambda_1 \ge \lambda_1 > 0, \qquad 0 \le \Lambda_2 \le C\sigma^2, \qquad |\Lambda_3| \le C\sigma.$$

No comparison between τ and σ, or between either of them and j, is imposed.

The source parameters and physical coordinates are

$$\begin{aligned} (a, p_{\mathrm{s}}, \gamma_{\mathrm{s}}, s_{\mathrm{s}}) &= (-1, \sqrt{\tau}\Lambda_1, \tau\Lambda_3, \tau^{3/2}\Lambda_2), \\ \xi = \sqrt{\tau}X, \quad \eta = \tau Y, \qquad v - \gamma_{\mathrm{s}} &= e\xi, \quad z = e^2\eta, \quad e = \tau\Lambda_2. \end{aligned} \tag{H.2}$$

For a source quartet, denote its positive graph denominator by

$$\mathcal{D}_{a,p,\gamma,s}(X,Y) = \{1 - A(\gamma+sX) + C(\gamma+sX)^2\}Y - a - pX + 3\gamma X^2 + sX^3.$$

We write $\mathcal{D}_{\mathrm{s}}$ for the quartet in (H.2); barred subscripts below mean the same formula with the displayed barred quartet. These denominators determine where an orbit can be written as a graph. Writing

$$B_\tau = 1 + \tau(2\Lambda_3 + \Lambda_2\xi), \quad L_\tau = \Lambda_3 + \Lambda_2\xi, \quad Q_\tau = 1 - A\tau L_\tau + C\tau^2 L_\tau^2,$$

the exact graph equation is

$$\eta_\xi = \frac{\eta\{\xi B_\tau + C\tau^2\Lambda_2 L_\tau \eta\}}{Q_\tau\eta + \tau P_\sigma}. \tag{H.3}$$

Fix $L > 4(1 + \sup x_*)$. On the actual incoming section $\xi = L$ use the increasing coordinates

$$h_L = \eta + \tau\log(2\eta) - L^2/2, \qquad \beta_L = h_L/\tau - \log(2\tau). \tag{H.4}$$

These formulas prescribe initial points; neither expression is asserted to be constant along a perturbed orbit. In particular, $h_L = 0$, or $\beta_L = b_\tau := -\log(2\tau)$, is an actual interior source orbit.

Lemma H.1 (Source ranges on the actual incoming section)**.** *There is a fixed $\delta_\sharp > 0$ such that the following constructions are uniform on $-\delta_\sharp \le h_L \le 0$, with enlarged input and output intervals.*

1. *For $B_0 \le \beta_L \le b_\tau$, where B_0 is fixed large, the source passage exists and its actual central height obeys*

$$|Y(0) + \log Y(0) - \beta_L| \le C_L, \qquad Y(0) = \beta_L - \log \beta_L + O_L(1). \tag{H.5}$$

2. *Every fixed compact β_L-interval reaches a fixed compact strict source family: on fixed internal source intervals,*

$$0 < m_B \le Y(0) \le M_B, \qquad Y > 0, \qquad \mathcal{D}_{\mathrm{s}} \ge c_B(Y+1) > 0. \tag{H.6}$$

3. *At a negative base $\beta_L = -B$, $B \ge B_-$, freeze*

$$q = B^{-1/2}, \quad \rho = \sqrt{\tau B}, \quad (x,y) = (\xi/\rho, \eta/\rho^2), \quad (\lambda_1, \lambda_2, k) = (\rho\Lambda_1, \rho^3\Lambda_2, \rho^2\Lambda_3). \tag{H.7}$$

For fixed cuts $x = \pm L_{\mathrm{f}}$, $L_{\mathrm{f}} = 8$, this orbit belongs to a common enlarged negative-action family with core $|x| \le R_{\mathrm{f}} = 5$ and $P_\sigma(\rho x) \ge 3/4$. The input and output energy windows can be taken respectively inside $[-7/4, -3/8]$ and $[-2, -1/4]$, with strict margins.

All these same-orbit passages reach $\xi = \pm L$ with

$$\eta(\pm L) = L^2/2 + O_L(\delta_\sharp + \tau|\log\tau|). \tag{H.8}$$

Their outward continuations to the fixed physical sections have positive graph denominators and retain the actual regular maps.

Proof. For the first assertion, freeze $B = \beta_L \ge B_0$, put $q = B^{-1/2}$, $\rho = \sqrt{\tau B}$, and use the source scaling $(\xi, \eta) = (\rho x, \rho^2 y)$. Its coefficients are

$$(\bar a, \bar p, \bar\gamma, \bar s) = (-q^2, \sqrt{\tau}\Lambda_1 q, \tau\Lambda_3, \tau^{3/2}\Lambda_2/q).$$

Set $H = y + q^2\log(2y) - x^2/2$ and $R = L/\rho$. On a provisional quadratic corridor $y \asymp 1 + x^2$, direct subtraction in the exact graph equation gives

$$\begin{aligned} |H_x| &\le C_L\{|\bar p| + |\bar\gamma|(1+|x|) + \bar s(1+x^2)\}, \\ |\partial_H H_x| &\le C_L\left\{\frac{|\bar p|}{1+x^2} + \frac{|\bar\gamma|}{1+|x|} + \bar s\right\}. \end{aligned} \tag{H.9}$$

The terms with $\bar p = \bar\gamma = \bar s = 0$ cancel exactly. The first integral on $[-R, R]$ is $O_L(q^2)$, since its three largest products are $L|\Lambda_1|q^2, L^2|\Lambda_3|q^2, L^3\Lambda_2 q^2$. The second integral tends to zero uniformly for $B_0 \le B \le b_\tau$. Start at $x = 0$ with $H(0,y) \in [1/4, 3]$. The monotonicity of $y + q^2\log(2y)$ improves the bounds at any first contact with the corridor boundary:

$$(1+x^2)/16 < y < 4(1+x^2), \qquad \mathcal{D}_{\mathrm{bar}} \ge cy.$$

Thus both cuts are reached. The scalar energy variation is a positive exponential with exponent bounded by the second integral in (H.9); hence the incoming energy map has an actual inverse on a common enlarged interval. Its prescribed value at $x = R$ is exactly $1 + B^{-1}\log(2/B)$. Inverting and undoing the scale gives (H.5). This also constructs the orbit $h_L = 0$ and a real larger source chord through it.

For compact β_L, work instead in (X, Y) and put $H_0 = Y + \log(2Y) - X^2/2$, $R_0 = L/\sqrt{\tau}$. At the incoming cut, $H_0 = \beta_L + \log 2$. On a provisional bounded energy region the denominator is at least $c(Y+1)$: near $\xi = 0$, $P_\sigma \ge 1/2$, while away from that fixed small interval bounded energy forces $Y \ge X^2/4$, which dominates the bounded polynomial. The exact energy subtraction gives

$$|H_0'| \le C_L\{p_{\mathrm{s}} + |\gamma_{\mathrm{s}}|(1+|X|) + s_{\mathrm{s}}(1+X^2)\}(1+|H_0|).$$

Its coefficient has a bounded integral because $p_{\mathrm{s}}R_0 = L\Lambda_1$, $|\gamma_{\mathrm{s}}|R_0^2 = L^2|\Lambda_3|$, and $s_{\mathrm{s}}R_0^3 = L^3\Lambda_2$. Gronwall's inequality improves a fixed sufficiently large energy bound, and proves continuation. Inverting $Y + \log(2Y)$ at the centre proves (H.6). Choose larger compact incoming labels first to obtain strict image margins. The resulting internal energy coordinate may retain a bounded translation; no identity with $\beta_L + \log 2$ on another section is used.

For the negative base in (H.7), the energy $H_{\mathrm{f}} = y + q^2\log(2y) - x^2/2$ at $x = L/\rho$ is

$$b_B(\beta) = q^2\beta + q^2\log(2q^2), \qquad b_B(-B) = -1 - \frac{\log(B/2)}{B}. \tag{H.10}$$

Also $\rho^2 = -h_L + \tau\log(2\tau) \le -h_L$. Choose $\delta_\sharp$ so that $P_{\mathrm{f}}(x) := P_\sigma(\rho x) \ge 3/4$ on $[-8, 8]$. On the positive quadratic tail the exact subtraction gives

$$|(H_{\mathrm{f}})_x| \le Cq^2\{\rho\Lambda_1 + \rho^2|\Lambda_3||x| + \rho^3\Lambda_2 x^2\}.$$

Its integral to L/ρ is $O_L(q^2)$. A first-contact argument in a quadratic corridor therefore reaches $x = 8$, with the actual energy in the interior of $[-3/2, -1/2]$, after increasing B_-. This places the orbit in the negative-action family. Its relative curvature estimate will be proved below.

To obtain phase derivatives on a neighbourhood of these orbits, we construct the full negative core on the larger input window $[-7/4, -3/8]$. Fix the finite derivative budget m and an integer $M > 2m + 8$. Follow the positive fast arc to $y = q^{M+2}$. The logarithmic-height equation has $dH_{\mathrm{f}}/d\log y = O_{C^m}(q^2\rho)$; its clock length is $O(|\log q|)$. It cannot reach $x = 1/2$ first, since that would imply $H_{\mathrm{f}} \ge -1/8 + o(1)$. Hence its actual lip coordinate satisfies

$$p = \sqrt{-2b} + O_{C^m}(q^2|\log q|), \qquad 1/2 < p < 2.$$

In the thin core put $\mathfrak{q} = q^2\log(y/q^2)$ and

$$Q_{\mathrm{f}}(x) = 1 - Aq^2(k + \lambda_2 x) + Cq^4(k + \lambda_2 x)^2.$$

Until its first return to the same lip height, its exact equation is

$$\mathfrak{q}_x = \frac{x\{1 + q^2(2k + \lambda_2 x)\} + Cq^6\lambda_2(k + \lambda_2 x)e^{\mathfrak{q}/q^2}}{P_{\mathrm{f}}(x) + Q_{\mathrm{f}}(x)e^{\mathfrak{q}/q^2}} = \frac{x}{P_{\mathrm{f}}(x)} + O(q^2\rho) + O(q^M). \tag{H.11}$$

The actions $\int_0^p v/P_{\mathrm{f}}(v)\,dv$ and $\int_0^r v/P_{\mathrm{f}}(-v)\,dv$ have opposite comparison differences at $r = p/2$ and $r = 2p$, with fixed margins. The logarithmic height first decreases, stays a fixed amount below the lip near zero, then increases strictly on the negative half. It therefore has one first exit before $|x| = 5$, with

$$r = p + O(\rho + q^2|\log q|) > 1/4.$$

The denominator in (H.11) is at least $3/4$. Each exponential derivative costs at most two powers of q^{-1}; the choice of M and the transverse exit equation give bounded finite phase derivatives on nested windows. The negative fast arc then reaches $x = -8$, with $b_- = b + O_{C^m}(\rho + q^2|\log q|)$ strictly inside $[-2, -1/4]$. The exact independent energy coordinates of Theorem D.1 differ from H_{f} by $O_{C^m}(|k|)$ at these fixed positive-height cuts. Slightly larger windows thus contain those coordinates and their images as well.

The same energy estimates on both outward tails prove (H.8). In raw variables, the quadratic corridor of (C.9) has a denominator comparable to the squared radial variable. Choose L large and $\delta_\sharp$ small first. All the enlarged chords above enter its interior, so the first-contact argument continues them to the fixed raw sections, across any intervening axis root at positive height. The fixed regular transitions are then included. These are actual first hits of the original field throughout. □

Lemma H.2 (The bounded annulus)**.** *Fix positive constants $p_-, r_{\mathrm{a}}, R_{\mathrm{BD}}$, with $p_- < \inf x_*$. Consider one connected family with positive and negative strong-axis footpoints*

$$p_- \le p < x_*, \qquad r_{\mathrm{a}}/2 \le r \le 2R_{\mathrm{BD}}, \qquad b_+ = -p^2/2, \quad b_- = -r^2/2.$$

The projections are synchronized to fixed weak sections, and their outward continuations use the fixed physical sections of Lemma 2.2. On slightly larger such domains, normalize each actual embedding by its own exact value and positive first derivative. The resulting complete map $\Pi = \phi_- \circ S \circ \phi_+^{-1}$ satisfies

$$\phi_+' \frac{P_\sigma(p)}{p+r} \frac{\Pi''}{\Pi'} = 3\Lambda_2 - \frac{\Lambda_1}{pr} + E, \qquad \|E\|_{C^1_{b_+}} \longrightarrow 0. \tag{H.12}$$

The derivative of the error follows the actual matched family. That family has a uniform positive gap from the positive root, as proved below; convergence includes the reciprocal coefficient face $j = 0$. For sufficiently small coefficients this map is strictly concave and its displacement has at most two zeros.

Proof. The curvature formula is uniform only after the incoming footpoint has been separated from the positive root. We first obtain this separation by matching the two normal heights. On fixed small strong lips the normal rates are positive units, including at $p = x_*$, since

$$-P_\sigma'(x_*) \ge c_*, \qquad P_\sigma(-r) \ge c_-, \qquad rB_\tau(-r) \ge c_-.$$

The strong projection and its integrating normal coordinate are uniform through the required finite phase order: the backward tangential Green factor is at most $Ce^{C\tau t}$, whereas the normal variations are $C(1+t)^N e^{-ct}$. Their integral therefore defines the projection and the coordinate satisfying exactly $\dot z = -pB_\tau(p)z$ on the positive side, with the corresponding equation in reversed time on the negative side. Use the original fixed weak cuts $c_+ < p_-$ and $c_- < r_{\mathrm{a}}/2$. Integration along the projected axis gives their actual normal heights

$$w_+(p) = u_+(p)e^{-J_+(p)/\tau}, \qquad w_-(r) = u_-(r)e^{-J_-(r)/\tau}, \qquad u_\pm^{\pm 1} \le C,$$

where the units and their finite phase derivatives are bounded on the fixed lips, independently of any outer cut, and

$$J_+(p) = \int_{c_+}^{p} \frac{xB_\tau(x)}{P_\sigma(x)}\,dx, \qquad J_-(r) = \int_{c_-}^{r} \frac{xB_\tau(-x)}{P_\sigma(-x)}\,dx.$$

On the intervening weak interval $P_\sigma, Q_\tau \ge c > 0$, so $-\dot\xi = \tau P_\sigma + Q_\tau\eta \ge c\tau$ and its travel time is at most C/τ. Put $H = \xi B_\tau + C\tau^2\Lambda_2 L_\tau\eta$, so $d\log\eta/dt = -H$. At a critical height, $d^2\log\eta/dt^2 = H_\xi(\tau P_\sigma + Q_\tau\eta) > 0$. The standing passive neighborhood has $C > 0$, and $H_\xi = 1 + \tau(2\Lambda_3 + 2\Lambda_2\xi) + C\tau^2\Lambda_2^2\eta > 1/2$ on this word. Thus there is no interior height maximum; the buffered weak endpoint heights bound the whole passage and its logarithmic rate. At each fixed weak section the integrating normal coordinate and η differ by a positive unit. For the actual map M_τ between those normal coordinates,

$$\left|\tau \log \frac{M_\tau(w)}{w}\right| \le K.$$

Its positive first multiplier $M_\tau'(0)$ is retained. Matching the two heights now bounds $J_+(p)$ by $\sup J_-(r) + K + C\tau$. The latter is uniform for $r \le 2R_{\mathrm{BD}}$, whereas the simple positive root gives

$$J_+(p) \ge a_* \log \frac{\ell_*}{x_* - p} - C, \qquad a_* > 0.$$

The logarithmic residue $x_* B_\tau(x_*)/[-P'_\sigma(x_*)]$ is bounded below by a_*. Consequently every actual member has $p \le x_* - d_{\mathrm{a}}$, for one fixed $d_{\mathrm{a}} > 0$. This gap depends only on the fixed lips and middle passage, so the outer cuts can now be chosen without changing it.

Enlarge the input within $p < x_* - d_{\mathrm{a}}/2$. The limiting root-free action has a fixed compact output envelope: at $\sigma = 0$, its negative action grows linearly at infinity. Choose strict lower and upper exit brackets there, then decrease σ independently so they and their polynomial margins persist. Denote an enlarged upper output bound by $R_{\mathrm{a}} \ge 2R_{\mathrm{BD}}$. We can therefore choose the annular cuts

$$L_{\mathrm{a}} > 4(1 + \sup x_* + R_{\mathrm{a}}).$$

The source cut L, its label h_L, and $\delta_\sharp(L)$ stay fixed. The positive-height arcs from the fixed lips to $\pm L_{\mathrm{a}}$ and their small enlarged chords have uniform height and denominator margins. Independent final coefficient bounds place eL_{a} inside the original raw ports; scalar uniqueness identifies both decompositions there.

Apply the compact coefficient-annulus argument in the proof of Corollary D.2, with $t^2 = \tau$, $\lambda = (\Lambda_1, \Lambda_2)$, and $k = \Lambda_3$. The root-free polynomial margin, strict exit brackets and nonzero negative exit slope established above are its inputs. The lip and first-exit estimates give absolute C^3 error $O(\tau(1 + |\log\tau|))$, including synchronized resection; no division by the reversible coefficient distance is needed. If $S_0(-p^2/2) = -r_0^2/2$ is defined by the exact limiting action,

$$\|S - S_0\|_{C^3} \longrightarrow 0, \qquad S'_0 = \frac{P_\sigma(-r_0)}{P_\sigma(p)}. \tag{H.13}$$

The exact polynomial identity therefore gives

$$\frac{P_\sigma(p)}{p + r_0}\frac{S''_0}{S'_0} = \frac{P'_\sigma(p)/p + P'_\sigma(-r_0)/r_0}{p + r_0} = 3\Lambda_2 - \frac{\Lambda_1}{pr_0}.$$

The two outer embeddings may have very different first derivatives. We estimate each embedding relative to its own exact first derivative. On a fixed small positive-height lip let $p(t)$ be the synchronized axis solution and $w = \eta - \xi^2/2 + p(t)^2/2$. The exact field gives

$$\dot w = \tau\{\xi P_\sigma(\xi) - pP_\sigma(p)\} + \xi\eta(Q_\tau - B_\tau) - C\tau^2\Lambda_2 L_\tau \eta^2.$$

Since $|\xi + p|$ is bounded below, this has the form $\dot w = \tau a(t)w + \tau\eta d(t)$, with bounded finite derivatives and exponentially decreasing η. The fixed weak section is a positive axis distance away and the axis speed is $O(\tau)$, so its remaining time is at least c/τ. Terminal hitting-time derivatives multiply the same exponentially small normal height. Variation of constants, with four phase derivatives, gives $w = O_{C^4}(\tau)$; resecting a moving lip costs $O_{C^4}(\tau|\log\tau|)$. On both fixed outer cuts, therefore,

$$\eta_\pm(\pm L_{\mathrm{a}}, b) = L_{\mathrm{a}}^2/2 + b + O_{C^4_b}(\alpha_\tau), \qquad \alpha_\tau = \tau(1 + |\log\tau|). \tag{H.14}$$

Set $s = e\xi$, $Z = e^2\eta$, $k_0 = \Lambda_3$. The exact raw tail equation is

$$Z_s = \frac{n_2 Z^2 + n_1 Z}{\ell_o Z + m_o}, \quad \begin{aligned} \ell_o &= 1 - A(\tau k_0 + s) + C(\tau k_0 + s)^2, \\ m_o &= s^3 + 3\tau k_0 s^2 - \tau\Lambda_1 es + \tau e^2, \\ n_1 &= s(1 + s + 2\tau k_0), \quad n_2 = C(\tau k_0 + s). \end{aligned}$$

The initial data (H.14) enter the invariant corridor $Z \asymp s^2$, $\ell_o Z + m_o \asymp s^2$. Division writes the right side as $F = (n_2/\ell_o)Z + q_o + \mathfrak{r}/(\ell_o Z + m_o)$, where

$$|\mathfrak{r}| \le C(|s|^4 + \tau|k_0||s|^3 + \tau e|s|^2 + \tau e^2|s|).$$

If j is bounded away from zero, $\tau|k_0||\log e| \to 0$, because $k_0 = O(\sigma)$ and $e \asymp \tau\sigma^2$. Thus the first variation $J_1 = Z_b$ is comparable to e^2. For the reciprocal range $j \to 0$, retain instead $k_0 < 0$. Direct differentiation gives the exact first cocycle

$$F_Z = \frac{n_2\ell_o Z^2 + m_o(n_1 + 2n_2 Z)}{(\ell_o Z + m_o)^2}, \qquad \operatorname{sgn}(s)(n_1 + 2n_2 Z) \geq c|s|.$$

The latter follows from $Z \asymp s^2$ and the small fixed raw radius. At the actual common denominator, the term $3\tau k_0 s^2$ in m_o contributes nonpositively to $\operatorname{sgn}(s)F_Z$ on both outward tails. The rest gives

$$\operatorname{sgn}(s)F_Z \leq C(1 + \tau e/|s|^2 + \tau e^2/|s|^3), \qquad 0 < J_1 \leq Ce^2.$$

The normalization below uses this exact positive derivative and requires only the displayed upper bound. For $2 \leq n \leq 3$, in both cases,

$$\int |F_{Z^n}| J_1^{n-1}\, d|s| \leq C(eL_{\mathrm{a}}^{3-2n} + \tau|k_0|L_{\mathrm{a}}^{2-2n} + \tau L_{\mathrm{a}}^{1-2n} + \tau L_{\mathrm{a}}^{-2n}) \leq C\tau.$$

The quantitative normalization following (A.3), through order three, now applies with initial errors from (H.14) and $\int |F_{Z^n}| J_1^{n-1} = O(\tau)$, $n = 2, 3$. Each tail and its actual-image inverse are consequently $\mathrm{id} + O_{C^3}(\alpha_\tau)$ after normalization by the retained $J_1(b_*) > 0$. At this actual image the ordinary C^3 observer adds $O_{C^3}(J_1(b_*))$ by (3.6). Thus $\phi_\pm = \mathrm{id} + o_{C^3}(1)$, including when their individual gains are much smaller than e^2.

Finally the exact chain rule is

$$\frac{\Pi''}{\Pi'} = \frac{1}{\phi_+'} \left\{ \frac{S''}{S'} + \frac{\phi_-''(S)}{\phi_-'(S)} S' - \frac{\phi_+''}{\phi_+'} \right\}.$$

On the compact family, S, its first three derivatives and its positive first inverse derivative are bounded. Multiply by $P_\sigma(p)/(p + r)$ and use $r - r_0 \to 0$ in phase C^3. Both exterior corrections are $o_{C^1_{b_+}}(1)$. This proves (H.12). Its principal term is at most $3C\sigma^2 - \lambda_1/(2(\sup x_*)R_{\mathrm{BD}}) < 0$. Positive affine changes preserve this sign in the actual input. Thus $x - m\Pi(x)$ is strictly convex, proving the two-zero bound. □

Proposition H.3 (Bounded-footpoint domains and derivatives)**.** *On one ordered through interval ending at the actual cut $h_L = 0$, suppose the two fixed comparisons described below and the enlarged annular domains of Lemma H.2 use the same actual physical sections. The union of the central source interval $-\delta_2 \leq h_L \leq 0$ and the part with negative strong footpoint $0 < r \leq 2R_{\mathrm{BD}}$ has an overlapping cover by the bounded annulus and the negative, compact, and positive source-label intervals constructed below. All endpoints are fixed before choosing a displacement zero.*

The completed map has strictly negative curvature on the annulus and on the entire negative source interval. On the compact and positive source intervals its derivative in the actual affine input is bounded by one constant. After one common multiplier shrink, the displacement has positive derivative on these latter intervals, including the cut $h_L = 0$. The same assertions hold on the enlarged intervals used for the overlaps. The degree-loss passage below verifies all domain and overlap hypotheses of this proposition.

Proof. Fix the strict source domain and the constants for its finite phase derivatives. Choose B_- large, then $\delta_\sharp$ small enough for the relative negative-curvature estimate below. Choose a fixed $0 < \delta_1 \leq \delta_\sharp/8$, set $p_0 = \sqrt{2\delta_1}$, and require $P_\sigma \geq 1/2$ on $[-2p_0, 2p_0]$. The orbit prescribed by $h_L = -\delta_1$ is constructed by a compact fast arc to a small lip, followed by the core first-exit

argument in (H.11). Its limiting incoming footpoint is p_0, and the limiting outgoing footpoint is determined by

$$\int_0^r \frac{v}{P_\sigma(-v)}\,dv = \int_0^{p_0} \frac{v}{P_\sigma(v)}\,dv.$$

The comparison values $r = p_0/2, 2p_0$ have strict opposite action differences. Thus the actual two footpoints lie in $(p_0/2, 2p_0)$ after shrinking τ. Choose $r_{\mathrm{a}} < p_0/4$. A second fixed comparison $h_L = -p_{\mathrm{a}}^2/2$, with $0 < p_{\mathrm{a}} < r_{\mathrm{a}}/8$, has $r < r_{\mathrm{a}}/4$ and $p > p_{\mathrm{a}}/2$. Scalar order and the actual synchronized projections imply that every $r \geq r_{\mathrm{a}}/2$ in the bounded annulus has $p \geq p_{\mathrm{a}}/2$, while every $r < r_{\mathrm{a}}$ lies in $-\delta_1 < h_L \leq 0$. Both comparisons have strict margins, so these ranges have an open overlap. The comparison uses order along the same orbits; h_L need not equal the footpoint energy on the other section.

Fix $\delta_1 < \delta_2 < \delta_\sharp$. On $-\delta_2 \leq h_L \leq 0$, divide the actual incoming labels into

$$[\beta_{\min}, -B_-], \quad [-B_-, B_+], \quad [B_+, b_\tau], \qquad \beta_{\min} = -\delta_2/\tau - \log(2\tau).$$

Each interval is used with slightly larger fixed phase thresholds. For a negative base, Lemma H.1 supplies the independent enlarged windows $\widehat{J}_+ = [-7/4, -3/8]$ and $\widehat{J}_- = [-2, -1/4]$, frozen cuts $L_f = 8$ and core $R_f = 5$. The actual incoming frozen energy is strictly inside $[-3/2, -1/2]$; its separate ideal coordinate differs by $O(|k|)$. Choose a fixed $J \Subset \widehat{J}_+$ containing these values with a strict buffer. With the frozen dictionary (H.7), put

$$\rho_{\mathrm{id}} = (\lambda_1^2 + \lambda_2^2)^{1/2}.$$

Use the tail part of the proof of Theorem D.1 with $t = q$, $\lambda = (\lambda_1, \lambda_2)$ and the frozen k. Its coefficients are exactly $(a, p, \gamma, s) = (-q^2, q^2\lambda_1, q^2 k, q^2\lambda_2)$, and its auxiliary cuts are $v = q^2 k \pm d_H$, with fixed d_H. The enlarged windows above supply its quadratic corridors; at $L_f = 8$ its fourth-order kernel majorants become small after a q shrink. The same sequential intrinsic p- and s-secants, with the two reference labels fixed, give

$$\|\phi_\pm^{\mathrm{int}} - \mathrm{id}\|_{C^3} + \|(\phi_\pm^{\mathrm{int}})^{-1} - \mathrm{id}\|_{C^3} \leq C q^2 \rho_{\mathrm{id}}. \tag{H.15}$$

For the actual ordinary C^3 observers, including divided closing, the separate raw gains satisfy, for fixed $0 < \nu < 1$,

$$0 < d \leq C s_f^{2-\nu} \leq C s_f = C q^2 \lambda_2 \leq C q^2 \rho_{\mathrm{id}}, \qquad s_f = q^2\lambda_2 = e\rho. \tag{H.16}$$

Equation (3.6) on the fixed buffered cut-label chord therefore preserves (H.15) for the observed tails and their actual-image inverses. The fourth phase order is used only in the intrinsic polynomial kernel, before composing with these observers.

On the negative source interval the limiting curvature tends to zero with the reversible coefficient distance. We therefore compare the actual core S_q with its limiting action S_0 before dividing by that distance. The exact reversible energy gives $S_q(0, k) = S_0(0, k) = \mathrm{id}$. Integrating the coefficient variation in (D.19) before taking any limit yields

$$\|S_q - S_0\|_{C^2} \leq C \rho_{\mathrm{id}} q^2 (1 + |\log q|).$$

One intrinsic coefficient secant with two phase derivatives produces the factor ρ_{id}. This makes the error small relative to the core curvature even as that curvature vanishes. For $b \in J$, put $p = \sqrt{-2b}$, and let r_0 be its limiting action exit. Equation (D.22) gives

$$\frac{S_0''(b)}{S_0'(b)} = \frac{p + r_0}{P_{\mathrm{f}}(p)} \left(3\lambda_2 - \frac{\lambda_1}{p r_0}\right).$$

The even coefficient k cancels exactly. Here $\rho^2 = \tau B \le -h_L$, $\rho_{\mathrm{id}} \asymp \rho$, $\lambda_1/\rho_{\mathrm{id}} \ge c_1 > 0$, and $\lambda_2/\rho_{\mathrm{id}} = O(\rho^2\sigma^2)$. The compact root-free margins and positive first variation thus give $S_0''/\rho_{\mathrm{id}} \le -c_0 < 0$, without a Taylor error in k.

The complete curvature also contains the two section terms:

$$\Pi''(\phi_+(b)) = \frac{\phi_-'(S_q(b))}{\phi_+'(b)^2}\left\{S_q'' + (S_q')^2\frac{\phi_-''(S_q)}{\phi_-'(S_q)} - S_q'\frac{\phi_+''}{\phi_+'}\right\}.$$

The separate tail bounds above make the two section terms $O(q^2\rho_{\mathrm{id}})$, while the prefactor is a positive unit and differs from one by that amount. Consequently, at the actual normalized input $u_* = \phi_+(b)$,

$$\frac{\Pi''(u_*)}{\rho_{\mathrm{id}}} = \frac{S_0''(b)}{\rho_{\mathrm{id}}} + O(q^2(1 + |\log q|)) < -c. \tag{H.17}$$

Choose B_- large enough to absorb this error, then $\delta_\sharp$ small enough for the root-free domain and the fixed relative coefficient margins. No comparison of q with ρ is used. The incoming resection identifies the point u_* at which this sign is evaluated. For the two retained positive first gains,

$$(\Pi^{\mathrm{raw}})'' = \frac{g_-}{g_+^2}\Pi'', \qquad g_+, g_- > 0. \tag{H.18}$$

Each negative base is fixed while differentiating, so this is one strict sign on the whole negative interval, including nonroot inputs.

On the compact middle interval, (H.6) and the larger label windows give a compact strict core and positive variation units $A_\pm$. Before normalization its two auxiliary raw variations are

$$\frac{dz_\pm}{dY_0} = s_s^{2-\lambda_\gamma}(d_{H,\pm}/L_t)^{\lambda_\gamma}A_\pm U_\pm, \quad s_s = \tau^{3/2}\Lambda_2, \quad \lambda_\gamma = \frac{12\gamma_s}{1+2\gamma_s}, \quad A_\pm^{\pm1}, U_\pm^{\pm1} \le C. \tag{H.19}$$

The common singular power cancels in dz_-/dz_+; distinct fixed port radii have a bounded ratio. The actual regular bridges and divided closing have positive derivative units, so $0 < \Pi' \le C_B$. This cancellation gives the derivative bound without a lower bound on a raw gain or a bound on $\gamma_s \log s_s$. On the positive interval, (H.5) gives, after choosing B_+ large,

$$Y(0) > K(1 + p_{\mathrm{s}}^2), \qquad 0 < s_{\mathrm{s}}^2 Y(0) \le C_L\tau^3\sigma^4 j^2|\log\tau| < z_*.$$

Since $\alpha_0 = -s_{\mathrm{s}}^2$ and $\alpha_1 = s_{\mathrm{s}}p_{\mathrm{s}}$, these are the hypotheses of Corollary 5.1 on the actual central section. Its complete derivative bound holds on this entire interval, including $h_L = 0$ and the enlarged source chord through that orbit. Take the maximum of the compact and high-height derivative constants, then choose m so that $\Delta' > 1/2$ on both intervals. The annular curvature comes from Lemma H.2, on its open overlap with the source interval constructed by the two fixed comparisons. All assertions concern restrictions of one physical map and every input in those overlaps. Boundary coefficients denote normalized one-sided limits; no orbit at a collapsed raw section is asserted. □

H.2 Degree-loss and remote-root compactifications

At degree loss the three equatorial roots occupy two scales, and the two incident saddle exponents vanish at different rates. We construct one through interval before dividing it according to accumulated action. This ensures that the local estimates can later be combined for the complete physical displacement.

Proposition H.4 (Degree-loss passage preparation)**.** *Let $\delta = \kappa\tau$, with $0 < \kappa, \tau \ll 1$, and assume*

$$a = \kappa A_{\mathrm{d}}, \qquad \mathcal{R}ac = B\kappa, \qquad a + c = 1 - 3\nu, \\ \mathcal{R}(1 - 3\nu) = \omega + B\kappa. \tag{H.20}$$

The quantities

$$B,\ \omega,\ \mathcal{R},\ A_{\mathrm{d}},\ c,\ 1-3\nu,\ c-a$$

and their reciprocals are required to lie in prescribed compact intervals. Consider the source coefficients

$$(a_s, p_s, \gamma_s, s_s) = \left(-1, \sqrt{\tau}\frac{\omega}{\sqrt{B}}, \tau\kappa\mathcal{R}\nu, \tau^{3/2}\kappa^2\mathcal{R}\sqrt{B}\right). \tag{H.21}$$

On the fixed physical passage, below the common raw-height bound, the ordered through interval is covered by the bounded-source, growing-action, finite-middle-height, and fixed-exit ranges constructed below, together with their high-height continuation. The actual matching maps, derivative estimates, and strict overlaps are uniform on the displayed compact coefficient family for positive κ, τ. At $\kappa\tau = 0$, only the one-sided limits of normalized estimates are asserted. The finite-V and fixed-exit ranges retain the original section maps and finite closing on their actual buffered domains.

Proof. Set

$$r_s = \delta\mathcal{R}, \qquad v - \gamma_s = r_s u, \qquad z = r_s^2 w, \qquad \gamma_s = r_s\nu.$$

The full two-dimensional system, after a positive change of time, is

$$\dot{u} = \delta p_d(u) - K_d(u)w, \qquad \dot{w} = -w\{L_d(u) + E_d(u)w\}, \tag{H.22}$$

where $p_d = p_{\mathrm{d}}$ from (E.1) and $(K_d, L_d, E_d) = (K, L, E)$ from (E.3). Thus the saddle rates and exponents are exactly those in (E.5). On the present coefficient family,

$$\chi = \frac{\delta\ell_+}{b_+} = \tau X_d, \qquad \beta = \frac{\delta\ell_-}{b_-} = \kappa\tau B_d, \qquad \frac{\beta}{\chi} = \kappa Q_d, \tag{H.23}$$

with X_d, B_d, Q_d positive units. These identities provide the uniform spectral gaps used in the two local saddle charts.

Fix $0 < \eta_\pm < 1$, set $u_+ = a(1-\eta_+)$, $u_- = -1+\eta_-$, and on $[u_-, u_+]$ write

$$h = -\delta p_d = \delta\mathcal{R}(u+1)(a-u)(c-u) > 0, \qquad A_m = \frac{L_d}{h}.$$

The scalar equation and its partial fractions are (E.7) and (E.16). For $J(u) = \exp\int_{u_+}^{u} A_m$ and $J_- = J(u_-)$,

$$J(u) = \left(\frac{1+u_+}{1+u}\right)^{1/\beta}\left(\frac{a-u_+}{a-u}\right)^{1/\chi}\left(\frac{c-u}{c-u_+}\right)^{1+1/\beta+1/\chi}. \tag{H.24}$$

Writing

$$\Lambda_\beta = \log\frac{(1+u_+)(c-u_-)}{(1+u_-)(c-u_+)}, \qquad \Lambda_\chi = \log a + O(1),$$

we obtain

$$\log J_- = \frac{\Lambda_\beta + (\beta/\chi)\Lambda_\chi}{\beta} + O(1), \qquad J_- \geq e^{c_0/\beta}. \tag{H.25}$$

Indeed $\Lambda_\beta \geq c_0 > 0$, while $(\beta/\chi)|\log a| = O(\kappa|\log\kappa|) = o(1)$.

To measure how the middle passage concentrates near its negative end, use the normalized action $\rho = h(u_-)J/(J_- h)$ of (E.15), so $\rho(u_-) = 1$. If $D_m = L_d - h'$, direct differentiation gives

$$\frac{d\rho}{\rho} = \frac{D_m}{h}\,du, \qquad D_m'(u) = 1 - 4\delta\mathcal{R}(u+\nu) > 0.$$

For one fixed $u_0 \in (u_-, 0)$, this implies $\rho(u_0) \leq e^{-c/\beta}$ and $\operatorname{Var}\rho \leq 2$. Since

$$\frac{h'}{h} = \frac{1}{u+1} - \frac{1}{a-u} - \frac{1}{c-u},$$

the explicit expression (H.24) also yields

$$\int_{u_-}^{u_+} \rho \left(|E_d| + \left| \frac{h'}{h} \right| \right) du = o(1). \tag{H.26}$$

The only noncompact term is bounded by $C\beta + C(1 + |\log a|)e^{-c/\beta}/a$. Thus no quotient ν/κ has entered the estimate.

We next place this middle transition inside a single interval of trajectories joining the fixed physical sections. Put

$$s_0 = \kappa\sqrt{B}, \qquad x_* = \frac{a}{s_0} = \frac{A_d}{\sqrt{B}}, \qquad u = s_0\xi, \qquad w = s_0^2\tau y. \tag{H.27}$$

With

$$\Lambda_1 = \frac{\omega}{\sqrt{B}}, \qquad \Lambda_2 = \kappa^2\mathcal{R}\sqrt{B}, \qquad \Lambda_3 = \kappa\mathcal{R}\nu,$$

the exact axis polynomial is

$$P_s(\xi) = (1 + s_0\xi)\left(1 - \frac{\xi}{x_*}\right)\left(1 - \frac{s_0\xi}{c}\right), \tag{H.28}$$

and the full source system has the form

$$\dot{\xi} = -\tau\{P_s + Q_\tau y\}, \qquad \dot{y} = -y\{\xi B_\tau + C\tau^3\Lambda_2(\Lambda_3 + \Lambda_2\xi)y\}, \tag{H.29}$$

where

$$B_\tau = 1 + \tau(2\Lambda_3 + \Lambda_2\xi), \qquad Q_\tau = 1 - A\tau(\Lambda_3 + \Lambda_2\xi) + C\tau^2(\Lambda_3 + \Lambda_2\xi)^2.$$

We construct two boundaries of the relevant through interval on the same fixed physical sections. For the upper comparison, prescribe the actual point $h_L = 0$ of (H.4), with $\eta = \tau y$. Put $b_\tau = -\log(2\tau)$, $q_\tau = b_\tau^{-1/2}$. Freezing the source height scale changes the active quartet to

$$\left(-q_\tau^2, \sqrt{\tau}\Lambda_1 q_\tau, \tau\Lambda_3, \frac{\tau^{3/2}\Lambda_2}{q_\tau}\right) \longrightarrow 0. \tag{H.30}$$

Lemma H.1 constructs this orbit by an actual incoming-to-central first hit, together with a real larger source chord, using (H.3) with $P_\sigma = P_s$. Choose the fixed $L > 4(1 + \sup x_*)$ of that lemma. Its energy comparison gives, uniformly also on the larger source chord,

$$\eta(\pm L) = \frac{L^2}{2} + O_L(\tau b_\tau), \qquad Q_\tau\eta + \tau P_s \geq c\eta. \tag{H.31}$$

The same proof gives $\eta \asymp \tau b_\tau + \xi^2$ and the denominator margin throughout the central source chord. Thus $h_L = 0$ is an interior physical orbit, with neighbouring through trajectories on both sides.

To continue to the fixed raw sections, put $W = w/u^2$ and $r = |u|$. Direct substitution in (H.22) gives

$$rW_r = \frac{WA_\pm(r, W)}{K_d(\pm r)W - \delta p_d(\pm r)/r^2} - 2W, \tag{H.32}$$

where

$$A_+ = \frac{L_d(r) + E_d(r)r^2W}{r}, \qquad A_- = -\frac{L_d(-r) + E_d(-r)r^2W}{r}.$$

After choosing a fixed $U > 2\max(1, c, |\nu|) + 2$, then L, the strip

$$\frac{1}{3} < W < \frac{3}{4}, \qquad K_d(\pm r)W - \delta p_d(\pm r)/r^2 > c_1 \tag{H.33}$$

is invariant from $s_0 L$ to U. Equation (H.31) enters its interior. The same reciprocal equation in raw coordinates continues it to the fixed sections and crosses each intervening equatorial root at positive height.

The lower boundary is the broken trajectory

$$W^s_+(a,0) \cup \{(u,0) : -1 \le u \le a\} \cup W^u_+(-1,0). \tag{H.34}$$

The invariant-manifold theorem in the two fixed weighted half-boxes constructs the upper stable and unstable branches. Their graph denominators have a positive margin after division by the local transverse scale. To attach them to the fixed sections, first leave the positive box at $\xi = x_* + \ell_*$, with $\ell_* > 0$ fixed. Its height and denominator are bounded below by $c\ell_*$. The graph equation therefore reaches $\xi = L$, with

$$\eta^{\mathrm{st}}_+(L) = (L^2 - x_*^2)/2 + o(1).$$

On the negative side leave the weighted box at $u = -R$, with $R = 1 + \rho_*$, $\rho_* > 0$ fixed. Its outgoing graph has

$$w^{\mathrm{st}}_-(-R) = (R^2 - 1)/2 + o(1), \qquad K_d w^{\mathrm{st}}_- - \delta p_d \ge c(R^2 - 1).$$

The barriers $2(r^2 - 1)/5$ and $3(r^2 - 1)/5$ continue this graph until it enters (H.33). Both branches then reach $u = \pm U$, and afterwards the fixed raw sections. The source cuts $u = \pm s_0 L$ occur before all three equatorial roots have been passed. The intermediate fixed cuts complete that continuation before the raw tails are attached.

Between the saddle boxes $p_d < 0$, so $\dot u = -(h + K_d w) < 0$ for every $w \ge 0$. On either exterior side the stopped branch has $\dot u < 0$ away from its saddle and $\partial_w \dot u = -K_d < 0$. Scalar order thus keeps every graph between the stopped and upper comparison graphs in the same positive denominator domain. The inclination lemma in the two saddle boxes, together with Lemma 2.3, identifies the stable branch as the lower endpoint of the through interval. Write the two actual positive-section traces as $h^{\mathrm{st}}_{\mathrm{in}} < h^0_{\mathrm{in}}$, where h^0_{in} is the trace of $h_L = 0$. There is no cycle at or below $h^{\mathrm{st}}_{\mathrm{in}}$.

The quadratic outer equation contracts towards its organizing corridor in logarithmic radius. More explicitly, for a fixed small raw cut v_0, its deviation at $\pm v_0$ is bounded by

$$C(r_s U/v_0)^\alpha + C r_s/v_0 + C(|A - 1| + |C - 1|), \qquad 1 < \alpha < 2.$$

This follows from $\mu_i = O(r_s^{4-i})$, $i = 1, 2, 3$, and variation of constants after $u = \pm U$, retaining the passive coefficient differences from the organizing equation. The fixed regular maps then put both boundary traces, and by scalar order the whole interval $(h^{\mathrm{st}}_{\mathrm{in}}, h^0_{\mathrm{in}})$, in the prescribed raw section windows. This scalar order locates the orbits in the raw windows. We estimate the derivatives separately on the enlarged chords transported into those windows.

The two fixed comparisons of Proposition H.3 supply the overlapping source interval $-\delta_2 \le h_L \le 0$ and bounded strong-footpoint annulus. Their enlarged independent windows reach the same raw sections by (H.8) and the outer bridges above.

Let r_b be the negative strong footpoint where that section is met. On the lower part of the through interval let r_ℓ be the first point on the actual curve $y/P_s = \ell_0$. Choose a fixed $r_c > 2$. A fixed real comparison through orbit has a positive ratio at $-r_c$; choose ℓ_0 below that ratio. Scalar order gives a unique lower interval on which the initial ratio is smaller than ℓ_0. Before its first hit, directly from (H.29),

$$\frac{d}{dr} \log \frac{y(-r)}{P_s(-r)} \ge \frac{c}{\tau(1 - s_0 r)} \tag{H.35}$$

on the full interval $r_c \le r < 1/s_0$. A strict through graph has positive height at the negative root while $P_s(-r) \to 0$. Hence the hit exists. Its uniqueness and the positive first variation at a fixed section give

$$\partial_{h_{\rm in}} r_\ell < 0, \qquad \partial_{h_{\rm in}} r_b < 0, \qquad \frac{dr_\ell}{dr_b} > 0 \tag{H.36}$$

on the common domain, with $h_{\rm in}$ the actual raw input height. No equality between the two footpoint coordinates has been used.

We need a coefficient-independent threshold at which a first ratio hit has sufficiently large action. Fix a positive weak-section input P_0 in the positive saddle box. Its local Dulac formula and $\chi \asymp \tau$ give $-\tau \log z_+ \le C(P_0)$ for inputs $P \ge P_0$. The exact core equation consequently gives $\log(y/P_s)(-r_c) \ge -C(P_0)/\tau$, unless the first hit has already occurred. If there were no hit before R, (H.35) would imply

$$\log(y/P_s)(-R) \ge \{-C(P_0) + c(R - r_c)\}/\tau.$$

Choosing $R = R(P_0)$ large is a contradiction. Thus $r_\ell \to \infty$ forces $P \to 0$ uniformly. Choose P_0 so that its positive action exceeds the fixed middle variational exponent by two; then fix $r_{\rm ge}^{\rm out} > R(P_0)$ and $r_{\rm ge} > r_{\rm ge}^{\rm out}$. The growing-action estimates below therefore start with a strict lower action margin.

At $r_\ell = r_{\rm ge}$, the actual strong projection has a compact image. Choose an independent $R_{\rm BD}$ so that this image lies strictly below $2R_{\rm BD}$. The transverse inverse and (H.36) then give a fixed $r_{\rm ge}^\sharp > r_{\rm ge}$ whose entire closed interval $[r_{\rm ge}, r_{\rm ge}^\sharp]$ lies inside the enlarged annular resection. In particular,

$$r_b > 2R_{\rm BD} \implies r_\ell > r_{\rm ge}.$$

Only now decrease the upper s_0-bound so that this whole fixed overlap lies in the growing-action product range. Fix $V_1 > 1$ and $0 < \sigma_1 < \sigma_2 < \sigma_0 < 1/4$, with strictly enlarged ranges at every endpoint. The four passage regions will be restrictions of this ordered through interval.

Choose one coefficient-only real action offset $A_{c,0}(\theta)$, held fixed by every phase derivative, and put

$$\mathcal{P}(r) = P_s(-r)/B_\tau(-r), \qquad \mathcal{D} = (\mathcal{P}(r)/r)\partial_r, \qquad \mathcal{A}_-(r) = \int_{r_c}^{r} \frac{vB_\tau(-v)}{P_s(-v)}\, dv - A_{c,0}.$$

Lemma H.8 constructs the two centre maps independently, retaining the exact initial polynomial and both core multipliers. Its product rows (H.114) and matched identity (H.115) give the actual graph, with $|\zeta| + |X\partial_X\zeta| \le C\tau$ on its enlarged image.

Construct one negative section map for both the growing and fixed-exit ranges. Fix $u_c \in (-1, u_-)$ with $|u_c| > 2\sigma_0$, retain its enlarged product buffer $\sigma_0 \le \bar\sigma < |u_c|/2$, and choose $\Delta > 0$ below one third of the minimum of the gap (E.27) on $[\sigma_1, \sigma_2]$, also with $\Delta < \frac{1}{4}\inf_c \Gamma_{-,c}$. Enlarge the intervals of (E.20) once more, with uniform strict margins:

$$\mathcal{J}_c \Subset \mathcal{J}_c^\sharp \Subset \mathcal{J}_c^{\sharp\sharp} \Subset (0, u_c^2/2).$$

Apply Lemma H.6 on the original enlarged lip range, with its fixed lower threshold and product buffer. Equations (H.51) and (H.52) give the actual landing value, its strict derivative and its positive first-variation unit. The action supremum in (H.53) is $O(\tau(1 + |\log\tau|))$, uniformly in independent κ, j. Since $s_0 r \le \sigma_0$, these heights lie strictly inside

$$\mathcal{J}_{\rm gr} = [u_c^2/4, 3u_c^2/4] \Subset \mathcal{J}_{\rm gr}^\sharp = (u_c^2/8, 7u_c^2/8).$$

Choose a fixed compact positive-height interval $\mathcal{I}_{\rm out}$ containing the closures of $\mathcal{J}_{\rm gr}^\sharp$ and $\mathcal{J}_c^{\sharp\sharp}$ with a strict buffer, uniformly in c. Thus it contains $u_c^2/2$, while the fixed-exit chord stays strictly below that height.

For every $W \in \mathcal{I}_{\mathrm{out}}$, the limiting graph $w(u) = W + (u^2 - u_c^2)/2$ has a uniform positive height from u_c to a fixed large negative cut $-U_-$. Ordinary ODE dependence gives a positive exact denominator, including at $u = -1$, and uniform C^5 bounds for this finite passage and its inverse. Choose U_- to enter (H.33). The tail estimates (C.15)–(C.20), with the exact first cocycle extracted through three variations, give

$$z_-(W) = z_-^{\mathrm{st}} + \Omega Z_-(W), \qquad \Omega = r_s^{2-\lambda_\gamma}, \quad \lambda_\gamma = \frac{12\gamma_s}{1+2\gamma_s},$$
$$Z_-' > c, \qquad \|Z_-\|_{C^3} + \|Z_-^{-1}\|_{C^3} \le C.$$

Since $\Omega \to 0$ and the stopped trace lies inside its fixed raw window, the whole chord is in the domain of d_θ after shrinking the coefficients. Compose with that map, put $t_b = d_\theta(z_-^{\mathrm{st}})$, and fix an interior reference $W_\circ$. With $U_{\mathrm{out}} = \Omega^{-1}(d_\theta \circ z_-)'(W_\circ)$, this defines a single actual map

$$\begin{aligned} &\Xi(W) = t_b + \Omega U_{\mathrm{out}} D^n(W), \qquad U_{\mathrm{out}}^{\pm 1} \le C, \\ &(D^n)' > c, \qquad \|D^n\|_{C^3} + \|(D^n)^{-1}\|_{C^3} \le C. \end{aligned} \tag{H.37}$$

For compact ν, $|\lambda_\gamma \log r_s| \to 0$, so $\Omega \asymp r_s^2$; the exact power is retained for the remote-root face. In particular $\Xi' \asymp \Omega$ and $|\Xi''/\Xi'| \le C$ on the entire common chord.

Fix $r_\circ$ inside the enlarged growing domain and define

$$a_{\mathrm{grow}} = \Omega s_0^2 U_{\mathrm{out}}, \qquad t_\dagger = \Xi(W_\theta(r_\circ)) + a_{\mathrm{grow}} r_\circ^2/2,$$
$$T(r) = \Xi(W_\theta(r)) = t_\dagger - a_{\mathrm{grow}} \mathcal{B}(r).$$

The fixed-endpoint variation in Lemma H.6 gives

$$\mathcal{B}'(r) = rG(r), \qquad G(r) = \Gamma_\theta(r) \frac{\Xi'(W_\theta(r))}{\Omega U_{\mathrm{out}}} > 0. \tag{H.38}$$

Its exact chain rule is

$$\mathcal{D} \log G = \mathcal{D} \log \Gamma_\theta - s_0^2 \mathcal{P}(r) \Gamma_\theta(r) \frac{\Xi''(W_\theta(r))}{\Xi'(W_\theta(r))}. \tag{H.39}$$

Estimates (H.60) and (H.120) control the two terms that enter the complete displacement after finite closing.

On the positive side use the fixed local cuts and stopped outer chord of Lemma E.1. Its backward-integral and tail construction holds through five phase derivatives. Thus the natural raw input is

$$P = \frac{x_{\mathrm{raw}} - x_b}{s_{\mathrm{in}}}, \qquad s_{\mathrm{in}} = \Omega a^{2-\lambda_\gamma} U_{\mathrm{in}}, \qquad U_{\mathrm{in}}^{\pm 1} \le C, \tag{H.40}$$

and the normalized map from the fixed outer chord and its inverse have uniform C^5 bounds. In the positive saddle coordinates $x = (u-a)/a, y = w/a^2$, write the invariant restrictions on $x = -\eta_+$ as

$$A_+(y) = -\Xi_{+,m}(y), \quad A_0 = A_+(0) > 0, \qquad Z_{+,m}(y) = z_+ y g_+(y), \quad g_+(0) = 1.$$

Here A_0, z_+ are positive units and A_+, g_+ have uniform finite derivative bounds. On the outer chord the invariant coordinate constructed in that lemma is $\mathscr{L}_\theta(P) = -I(P)/A_0 = \Phi_+(P)/A_0$; hence

$$\mathscr{L}_\theta(0) = 0, \qquad \mathscr{L}_\theta' > c > 0, \qquad \|\mathscr{L}_\theta\|_{C^5} + \|\mathscr{L}_\theta^{-1}\|_{C^5} \le C.$$

Choose $p_0 > 0$ small, independently of the coefficients, so that $\mathscr{L}_\theta(P) \le \vartheta < 1$ on $[0, p_0]$. Equality of the two invariants is exactly

$$\mathscr{L}_\theta(P) = \frac{A_+(y)}{A_0} \{z_+ y g_+(y)\}^\chi. \tag{H.41}$$

This equation has a unique small nonnegative solution $y = y(P)$. Indeed, on $0 < y \le \chi^2$ the logarithmic derivative of its right side, multiplied by y, is

$$B_\chi(y) = \chi + y\frac{A_+'(y)}{A_+(y)} + \chi y\frac{g_+'(y)}{g_+(y)} = \chi + O(y) \ge \chi/2.$$

The right side of (H.41) tends uniformly to one at $y = \chi^2$, while its limit at zero is zero. Thus it covers $[0, \vartheta]$ for all sufficiently small χ. Taking logarithms gives

$$y = z_+^{-1}\mathscr{L}_\theta(P)^{1/\chi}\exp O(y/\chi + y).$$

In particular the exponential factor is a positive unit. From $dy/d\mathscr{L} = y/(\mathscr{L}B_\chi(y))$, differentiation twice more and the bounded derivatives of A_+, g_+ give

$$|\partial_P^j y(P)| \le C_j\chi^{-j}\mathscr{L}_\theta(P)^{1/\chi - j}, \qquad 0 \le j \le 3, \quad P > 0. \tag{H.42}$$

For $\chi < 1/4$ these derivatives have zero limits at $P = 0$. Their suprema on the fixed interval tend to zero, since $\chi^{-j}\vartheta^{1/\chi - j} \to 0$.

Define, with its continuous value at zero,

$$\Phi_\theta(P) = \{z_+ y(P)\}^\chi = \mathscr{L}_\theta(P)\frac{A_0}{A_+(y(P))}g_+(y(P))^{-\chi}. \tag{H.43}$$

The second expression and (H.42) show that $\Phi_\theta - \mathscr{L}_\theta \to 0$ in C^3. Consequently $\Phi_\theta' > c > 0$, its C^3 norm and inverse norm are uniform, and $\Phi_\theta(P) = PU_\theta(P)$ with U_θ a positive unit on $[0, p_0]$. Only phase derivatives are used here.

Use $\lambda = \ell_0$ in the normalized middle action $w = \ell_0 h\rho V$ of (E.8). The exact reference value follows directly from that raw height. Put

$$\varepsilon_+ = \frac{\lambda h_-}{a^2 J_-}, \qquad \Lambda_+^m = (z_+\beta\varepsilon_+)^\chi, \qquad \Phi_\theta(a_{\mathrm{in}}) = \Lambda_+^m.$$

Write $q_s = \beta/\chi$, $j = 1/c$, $d = \mathcal{R}c$, and $\mathscr{R} = (1 - ju_-)/(1 - ju_+)$. The logarithms in (H.24) are exactly

$$\Lambda_\beta = \log\frac{(1+u_+)\mathscr{R}}{1+u_-}, \qquad \Lambda_\chi = \log\frac{(a-u_+)\mathscr{R}}{a-u_-}.$$

The reference telescope is

$$\begin{gathered}\beta\log(z_+\beta\varepsilon_+) = -\Lambda_\beta - q_s\Lambda_\chi + \beta\log(\tau^2 U_{\mathrm{tel}}),\\ U_{\mathrm{tel}} = \frac{z_+\ell_0\beta\delta d}{a^2\tau^2}(1+u_-)(a-u_-)(1-ju_+), \qquad U_{\mathrm{tel}}^{\pm 1} \le C.\end{gathered} \tag{H.44}$$

Here $\delta = \kappa\tau$, $a = \kappa A_d$, and $\beta = \kappa\tau B_d$; hence all factors of κ cancel in U_{tel}. Since $\Lambda_\beta = \Gamma_{-,c} + O(a)$ and $\Lambda_\chi = \log a + O(1)$, the error is bounded by $C\{a + q_s(1 + |\log a|) + \beta(1 + |\log\tau|)\}$. It tends to zero under independent coefficient bounds, giving

$$q_s\log\Phi_\theta(a_{\mathrm{in}}) = \beta\log(z_+\beta\varepsilon_+) = -\Gamma_{-,c} + o(1). \tag{H.45}$$

Thus $\Lambda_+^m \to 0$ and the reference $a_{\mathrm{in}} \in (0, p_0)$ exists uniquely. It corresponds to $y = \beta\varepsilon_+$, hence to the raw middle input $V = \beta$. For every orbit in this local input interval, the exact height dictionary is

$$\begin{gathered}y = \varepsilon_+ V, \qquad V = \beta\left(\frac{\Phi_\theta(P)}{\Phi_\theta(a_{\mathrm{in}})}\right)^{1/\chi},\\ \frac{w(u_+)}{h(u_+)} = \lambda\rho(u_+)V.\end{gathered} \tag{H.46}$$

We have constructed the through interval, its first ratio hits and its input, output and centre-height maps. Proposition H.10 uses them to cover every input, including truncated intervals and boundaries between the four ranges. Proposition H.13 constructs the last two transitions on the same shooting chord and buffered closing image. □

The remote positive root requires a different coordinate at infinity. Its reciprocal keeps the finite passage coefficients regular while the physical tail scales are retained exactly.

Proposition H.5 (The remote-root face)**.** *Let d, A_d range in positive compact intervals and put*

$$\begin{aligned} &0 \le j \le j_0, \qquad a = \kappa A_d, \qquad \mathcal{R} = dj, \qquad B = dA_d, \\ &\omega = d(1-a+ja), \qquad j\nu = \frac{j-ja-1}{3}, \qquad \delta = \kappa\tau. \end{aligned} \tag{H.47}$$

For $j > 0$ this is (H.20) *with $c = j^{-1}$. For $j > 0$, the conclusion of Proposition H.4 extends uniformly as $j \downarrow 0$. At $j = 0$ this denotes only the one-sided closure of the analytic estimates, not an additional physical coefficient fibre or an asserted inverse raw gate.*

Proof. The factorization and translated root are

$$p_d(u) = d(u+1)(u-a)(1-ju), \qquad \gamma_s = \frac{\kappa\tau d}{3}(j-ja-1). \tag{H.48}$$

All coefficients of the exact field on bounded u-intervals are analytic at $j = 0$. In the middle calculation the apparently singular quantities reduce to

$$\frac{1-ju}{1+j}, \qquad \frac{j}{1-ju}, \qquad D'_m = 1 - 4\gamma_s - 4\kappa\tau dju.$$

The limiting functions needed in the accumulated-action and fixed-exit arguments are

$$H_{0,j}(u) = -\frac{u(1+u)(1-ju)}{1+j}, \qquad \Gamma_j(u) = \log\frac{(1+u)(1-ju_-)}{(1+u_-)(1-ju)},$$

and

$$\Gamma_{-,j} - \Gamma_j(-\sigma) = \log\frac{1+j\sigma}{1-\sigma} > 0.$$

Thus the small-mass, Lambert, and fixed-exit estimates above extend continuously to $j = 0$.

The bounded-action domains also have regular coordinates. Put $\sigma = s_0 = \kappa\sqrt{B}$, $x_* = A_d/\sqrt{B}$. Then

$$P_\sigma(\xi) = (1+\sigma\xi)(1-\xi/x_*)(1-\sigma j\xi), \qquad \Lambda_2 = \sigma^2 j/x_*, \quad 3\Lambda_3 = \sigma(j-1)/x_* - \sigma^2 j < 0.$$

The positive root remains simple and the negative compact-output interval has a positive polynomial margin. The two fixed phase comparisons, the compact central-height estimate, and the negative frozen first-exit construction in Lemma H.1 are consequently uniform on the compact j-block. Lemma H.2 uses the sign $\Lambda_3 < 0$ on this face: its actual first tail cocycle is only bounded above by Ce^2, with $e = \tau\Lambda_2$, and is retained exactly in the relative normalization. Thus it requires no bound on $\tau|\Lambda_3||\log e|$, even when j is arbitrarily small. The bounded-action estimate is therefore uniform even as the third root escapes the compact root chart.

It remains to check the unbounded tails. Their exact source coefficients are

	$a_\pm$	$p_\pm$	$\gamma_\pm$	$s_\pm$
$+$	$-\tau d/A_d$	$\tau d(1-a+ja)/A_d$	γ_s	$r_s a$
$-$	$-\kappa^2\tau dA_d$	$\kappa\tau d(1-a+ja)$	γ_s	r_s,

$r_s = \kappa\tau dj.$

After the transverse blow-down, the two maps to the fixed raw sections have first-cocycle scales

$$r_s^{2-\lambda_\gamma}U_-, \qquad r_s^{2-\lambda_\gamma}a^{\,2-\lambda_\gamma}U_+, \qquad \lambda_\gamma = \frac{12\gamma_s}{1+2\gamma_s}, \tag{H.49}$$

with normalized phase derivatives bounded. The common factor $r_s^{2-\lambda_\gamma}$ cancels in the complete transition, while $|\lambda_\gamma \log a| = o(1)$. With $q_s = \beta/\chi$, the exact telescope (H.44) gives

$$q_s \log \Lambda_+^m = -\Lambda_\beta + o(1), \qquad e^{-C/q_s} \le \Lambda_+^m \le e^{-c/q_s},$$

and contains no term $\beta \log j$. The common section scales are $s_{\mathrm{in}} = \Omega a^{2-\lambda_\gamma} U_{\mathrm{in}}$ and $a_{\mathrm{out}} = \Omega U_{\mathrm{out}}$, with $\Omega = r_s^{2-\lambda_\gamma}$.

The local boxes and normalized outer maps have uniform finite phase bounds on the compact j-block, so the weak-coordinate construction (H.41)–(H.42) has exact reference

$$\Phi_\theta(a_{\mathrm{in}}) = (z_+\beta\varepsilon_+)^\chi, \qquad q_s \log \Phi_\theta(a_{\mathrm{in}}) = -\Gamma_{-,j} + o(1).$$

Proposition H.13 uses these regular functions $H_{0,j}, \Gamma_j$, their strict action gap and the exact raw scales to extend the last two passage estimates to this face.

For the outer geometry no fixed cut beyond $|\nu|$ is available. Choose instead the positive source cut $\xi = x_* L_t$, so that $u/a = L_t$ exactly. The actual source construction gives $1/3 < \eta/\xi^2 < 3/4$ and a positive divided denominator there. On the negative side continue from $u = -s_0 L$ only to $u = -L_t$. On this interval the regular factorization gives

$$|p_d(-r)| \le Cr, \qquad |\delta p_d(-r)|/r^2 \le C\tau/L, \qquad |K_d - 1| + |-L_d(-r)/r - 1| \le C\kappa\tau.$$

The equation for w/u^2 therefore has the same invariant strip and positive denominator. These are the two finite entry chords for the bounded source quartets in the table. Their normalized tail equations (C.9) continue to the fixed raw sections; the positive tail crosses $u = c = j^{-1}$, if that root lies before the section, at positive height. The stopped branches are attached by the same weighted local graphs and finite bridges. Since $p_d < 0$ on $-1 < u < a$, and $\partial_w \dot{u} = -K_d < 0$, every intermediate graph remains on the through side of these branches. This constructs the entire ordered interval before its zero bounds are applied.

The same ratio and strong-projection orders retain the bounded/growing overlap. The positive Γ_j gap retains the fixed-exit overlap. On the real enlarged source chord through the actual cut $h_L = 0$, (H.5) again gives $Y_0 \asymp b_\tau \to \infty$, while $s_s^2 Y_0 \le C\tau^3\kappa^4 j^2 b_\tau \to 0$. Hence this chord supplies the overlap of the actual high interval with the low carrier through $h_L = 0$. All four domains and estimates are uniform for $j > 0$; $j = 0$ retains only the normalized one-sided meaning. □

H.3 Direct negative landing and its first action variation

A negative ratio hit must be continued to a fixed section before it can be used in the physical return. The next lemma constructs this landing and estimates its first action derivative. The terminal section is kept fixed while the starting lip moves.

Lemma H.6. *Use the exact field* (H.22) *on either the compact coefficient graph* (H.20) *or the reciprocal graph* (H.47)*, with physical $j > 0$. Keep the original sections, compact passive coefficients, and coefficient-only scales. Put*

$$\epsilon_d = \kappa\tau, \qquad s_0 = \kappa\sqrt{B}, \qquad x_c = -u_c, \qquad \sigma = s_0 r.$$

Choose the original enlarged product buffer $0 < \sigma_0 \le \bar{\sigma} < x_c/2 < 1/2$, and fix $R_0 > r_c$. On the original enlarged range

$$R \le r \le \bar{\sigma}/s_0, \qquad R \ge R_0, \qquad q_\ell = \frac{\epsilon_d}{\sigma} = \frac{\tau}{\sqrt{B}\,r}, \qquad L_\ell = 1 + |\log q_\ell|, \tag{H.50}$$

the graph from the original lip $w = \ell_0 h$ reaches u_c in positive time, and its inverse reaches the original centre $u = 0$. Both graphs have positive height and denominator.

Write $W_\theta(r) = w(u_c; r)$. There is an exact positive unit $\Gamma_\theta(r)$ such that

$$\left| W_\theta(r) - \frac{x_c^2 - \sigma^2}{2} \right| \le C\epsilon_d\{1 + \sigma L_\ell\}, \qquad W_\theta'(r) = -s_0^2 r \Gamma_\theta(r), \tag{H.51}$$

and

$$c \le \Gamma_\theta \le C, \qquad |\Gamma_\theta - 1| \le C(q_\ell L_\ell + \epsilon_d). \tag{H.52}$$

If $\Gamma(\sigma) = \Gamma_\theta(\sigma/s_0)$, then, with all coefficients frozen,

$$\begin{aligned} |\partial_\sigma \log \Gamma| &\le C\left\{ \frac{q_\ell}{\sigma} L_\ell + \epsilon_d \right\}, \\ |\mathcal{D} \log \Gamma_\theta| &\le C\left\{ \frac{q_\ell}{r} L_\ell + s_0 \epsilon_d \right\}, \qquad \mathcal{D} = \frac{\mathcal{P}(r)}{r}\partial_r. \end{aligned} \tag{H.53}$$

Constants are fixed by the original compact ranges, R_0, r_c, ℓ_0 and the section buffers, independently of the later R, κ, τ, j. In particular the supremum of the action row is $O(\tau(1 + |\log \tau|))$ as $\tau \to 0$, uniformly in independent κ, j. At fixed $r, \tau > 0$, sending only $\kappa \to 0$ does not assert that this row tends to zero or that $\Gamma_\theta \to 1$.

Proof. Put $x = -u$, $H(x) = h(-x)$, $k(x) = K(-x)$, $b(x) = -L(-x)$, and $e(x) = -E(-x)$. The exact graph is

$$w_x = F(x, w) = \frac{w(b + ew)}{H + kw}, \qquad w(\sigma; \sigma) = \ell_0 H(\sigma). \tag{H.54}$$

Indeed the original dictionary gives $s_0^2 \tau P_s(-r) = h(-s_0 r)$ exactly. In the compact and reciprocal graphs, respectively,

$$H = \epsilon_d \mathcal{R}(1 - x)(a + x)(c + x), \qquad H = \epsilon_d d(1 - x)(a + x)(1 + jx).$$

On $[\sigma, x_c]$, $a/\sigma \le C/R_0$. The exact coefficients (E.3) therefore give

$$\begin{aligned} H \asymp \epsilon_d x, \quad |H'| + |H''| \le C\epsilon_d, \quad |H'/H| \le C/x, \qquad k \asymp 1, \\ |k - 1| + |k'| \le C\epsilon_d, \qquad |b - x| \le C\epsilon_d x, \\ |(b/x)'| \le C\epsilon_d, \quad |b'/b| \le C/x, \qquad |e| + |e'| \le C\epsilon_d^2. \end{aligned} \tag{H.55}$$

At the reciprocal face these estimates use $\epsilon_d \mathcal{R} = \epsilon_d dj$ and $\epsilon_d \mathcal{R}\nu = \gamma_s$; no bound on ν or $\log j$ is required. A common small upper bound on τ makes q_ℓ small without comparing any radial parameters.

First admit the graph. On a provisional band $0 < w < Mx^2$, the denominator and $b + ew$ are positive, and

$$0 < F \le Cx + C\epsilon_d^2 w, \qquad F \ge \frac{cxw}{C\epsilon_d x + Cw}.$$

A fixed large M prevents first contact with Mx^2; the lip is below it since $H(\sigma)/\sigma^2 = O(q_\ell)$. Thus the graph stays positive and reaches x_c. On $[\sigma, 2\sigma]$, height is increasing and

$$\frac{dx}{dw} \le C\frac{\epsilon_d \sigma + w}{\sigma w}.$$

For fixed small $m > 0$, the time to height $m\sigma^2$ is at most $C\epsilon_d L_\ell + Cm\sigma < \sigma$, after a common coefficient shrink. A smaller quadratic lower barrier then gives

$$0 < w \le Cx^2, \quad b + ew \asymp x, \qquad w \asymp x^2 \quad (2\sigma \le x \le x_c). \tag{H.56}$$

For the inverse to the centre, the same exact coefficients give, on $[s_0 r_c, \sigma]$ while $w/H \le \ell_0$,

$$\frac{d}{dx}\log(w/H) = \frac{b+ew}{H+kw} - \frac{H'}{H} \ge c/\epsilon_d - C/x > 0.$$

Here $\epsilon_d/x \le C\tau/r_c$. Backward continuation remains below this lip, whose hit at σ is therefore the first hit after the inner cut. At that cut its height w_b satisfies $w_b \le C\epsilon_d(a + s_0 r_c) \le C\kappa^2\tau$. On $[0, s_0 r_c]$, $H \ge c\epsilon_d a > 0$, and for $0 < w \le 2w_b$ the exact derivative satisfies

$$F \ge -C\frac{\epsilon_d^2}{\epsilon_d a}w^2 \ge -C\tau w^2.$$

In the inverse direction this prevents first contact with $2w_b$. Positive denominator and scalar uniqueness give a positive centre height. This argument permits either sign of $e(0)w$.

Write $D = H + kw$ and

$$N = \frac{H' + (k'+e)w}{D}, \qquad I_x = \int_\sigma^x N(t, w(t;\sigma))\,dt, \qquad I = I_{x_c}.$$

Direct differentiation along the graph gives $F_w = (\log(w/D))_x + N$. The near height change and the quadratic corridor imply

$$\int_\sigma^{2\sigma} \frac{\epsilon_d}{D}\,dx \le Cq_\ell L_\ell, \qquad \int_{2\sigma}^{x_c} \frac{\epsilon_d}{D}\,dx \le Cq_\ell.$$

The remaining part of N has integral $O(\epsilon_d)$, so $|I_x| \le C(q_\ell L_\ell + \epsilon_d)$. Keep the complete moving initial factor

$$C_\sigma = (1 + \ell_0 k(\sigma))H'(\sigma) - b(\sigma) - \ell_0 e(\sigma)H(\sigma). \tag{H.57}$$

The moving initial point contributes the intrinsic derivative $\ell_0 H'(\sigma) - F(\sigma, \ell_0 H(\sigma))$. Combining this with the Jacobian to the fixed terminal section gives exactly

$$w_\sigma(x;\sigma) = \frac{w(x;\sigma)}{D(x;\sigma)}C_\sigma e^{I_x}, \qquad \Gamma(\sigma) = \frac{w(x_c;\sigma)}{D(x_c;\sigma)}\frac{-C_\sigma}{\sigma}e^I. \tag{H.58}$$

Here $-C_\sigma/\sigma = 1 + O(q_\ell + \epsilon_d)$, and the terminal ratio is $1 + O(\epsilon_d)$. These identities prove the strict sign, the unit estimate, and $|w_\sigma(x;\sigma)| \le C\sigma$. The value estimate follows by integrating the exact residual

$$F - x = \frac{w(b - xk) + ew^2 - xH}{D}.$$

The first two terms cost $O(\epsilon_d)$; the near xH/D term costs $O(\epsilon_d \sigma L_\ell)$, and the far term $O(\epsilon_d)$. Include the original initial value $\ell_0 H(\sigma)$ to obtain (H.51).

Apply Lemma A.6 to this same intrinsic landing graph, with starting abscissa σ, terminal cut x_c, and identity observer. Its constant coefficients are

$$t = -r_s, \qquad b_{\mathrm{k}} = 2\gamma_s, \qquad v_{\mathrm{k}} = \gamma_s, \qquad (\mu_2, \mu_1, \mu_0) = (3\gamma_s, \epsilon_d\omega, \epsilon_d\kappa B), \qquad r_s = \epsilon_d\mathcal{R}.$$

Here $b_{\mathrm{k}}, v_{\mathrm{k}}$ denote the lemma's constant coefficients. The two exact coefficient graphs give

$$H(x) = -r_s x^3 + 3\gamma_s x^2 + \epsilon_d\omega x + \epsilon_d\kappa B, \qquad b(x) = x(1 + 2\gamma_s - r_s x),$$

so its M, Q, ζ are precisely H, k, e in (H.54). On the original enlarged range, $2\sigma < x_c$, and (H.55) gives

$$H(x)/x^2 \asymp q_\ell \quad (\sigma \le x \le 2\sigma), \qquad |\gamma_s| + r_s \le C\epsilon_d, \qquad \kappa/\sigma \le C/R_0.$$

Thus the lemma's mass bounds hold with $\rho = q_\ell$, its $\Theta \le C q_\ell$, and, since the observer is the identity, $\mathcal{T} \le C\epsilon_d$. Its smallness conditions follow from one independent upper bound on τ and the original coefficient bounds, uniformly on the compact and reciprocal graphs. The actual corridor and terminal height are those already constructed above.

Take the lemma's frozen quadratic-label scale equal to one, so that $\beta_\ell = -\sigma^2/2$. By (H.58), the terminal slope in this label is exactly $\Gamma(\sigma) = -w_\sigma(x_c;\sigma)/\sigma$. With $E_\sigma = \sigma\partial_\sigma$, (A.15) gives

$$|E_\sigma \log\Gamma + \lambda_\gamma| \le C\{q_\ell L_\ell + \epsilon_d(\sigma/x_c)^e\}, \qquad \lambda_\gamma = \frac{12\gamma_s}{1+2\gamma_s}, \quad e = 2-\lambda_\gamma > 3/2.$$

The spectral power $\sigma^{-\lambda_\gamma}$ has its exact exponent held fixed in this phase derivative. Since $|\lambda_\gamma| \le C\epsilon_d$, $\sigma < 1$, and $e > 1$, division by σ yields

$$|\partial_\sigma \log\Gamma| \le C\{(q_\ell/\sigma)L_\ell + \epsilon_d\}.$$

The original action identity $\mathcal{D} = s_0(\mathcal{P}/r)\partial_\sigma$, with $\mathcal{P}/r \asymp 1$, gives the second row of (H.53). For $r \ge R_0$, $q_\ell = \tau/(\sqrt{B}\,r)$ and $s_0\epsilon_d = O(\kappa^2\tau)$ give its supremum $C\tau(1+|\log\tau|)$, independently of the later R and of the reciprocal parameter. □

Corollary H.7. *On a fixed original band $0 < \sigma_1 \le \sigma \le \sigma_2 \le \bar\sigma$,*

$$\left\| W_\theta(\sigma/s_0) - \frac{u_c^2 - \sigma^2}{2} \right\|_{C^2_\sigma[\sigma_1,\sigma_2]} \le C\epsilon_d(1+|\log\epsilon_d|) \longrightarrow 0 \quad \text{as } \epsilon_d \to 0. \tag{H.59}$$

If, independently, the actual scalar inverse phase $\Psi_\theta \to \Psi_0$ in C^2 on a fixed buffered interval containing this band's height image, then $\Psi_\theta(W_\theta(\sigma/s_0))$ converges in C^2_σ to $\Psi_0((u_c^2-\sigma^2)/2)$, under the hypotheses of that independent convergence.

Proof. On the fixed band, (H.52) and (H.53) give $\Gamma - 1, \Gamma_\sigma = O(\epsilon_d(1+|\log\epsilon_d|))$. Apply these to $W_\sigma = -\sigma\Gamma$, then use the value row (H.51). The composition claim is the ordinary C^2 chain rule on the assumed buffered image; for the inverse phase use Proposition E.5 under its own limiting hypotheses. □

In coordinates scaled by the starting lip, the fixed terminal section has a moving coordinate. Its contribution remains in the derivative: if $x = \sigma t$, $w = \sigma^2 v$, then

$$\frac{d}{d\sigma}W_\theta(\sigma/s_0) = 2\sigma v(x_c/\sigma;\sigma) + \sigma^2 v_\sigma(x_c/\sigma;\sigma) - x_c v_t(x_c/\sigma;\sigma).$$

The limiting heights lie inside the original buffered $\mathcal{I}_{\text{out}}$, including $u_c^2/2$; the value estimate places the whole enlarged landing image there after a common shrink. With the actual raw map (H.37),

$$T(r) = \Xi(W_\theta(r)), \qquad T'(r) = -s_0^2 r\Gamma_\theta(r)\Xi'(W_\theta(r)).$$

Its raw endpoint, exact power $\Omega = r_s^{2-\lambda_\gamma}$, and ordinary C^3 normalized observer bounds are retained. No coefficient derivative of a moving raw endpoint is asserted. For the positive unit $G = \Gamma_\theta\Xi'/(\Omega U_{\text{out}})$, the original chain rule gives the explicit bound

$$|\mathcal{D}\log G| \le C\left\{\frac{q_\ell}{r}L_\ell + s_0\epsilon_d + s_0^2\mathcal{P}(r)\right\}. \tag{H.60}$$

The last term comes from the bounded logarithmic derivative Ξ''/Ξ' of the independently constructed output map; the exact common tail power cancels. The error in the Γ_θ derivative tends to zero with τ; the output-composition error also requires $s_0 \to 0$. We now construct the matching independently of this landing estimate.

H.4 Separate source heights in the growing range

The positive and negative half-orbits reach the common centre with exponentially small heights. We construct these two height maps separately, retaining their linear multipliers and initial polynomial factors. Equality of their logarithms then gives the matched action derivative while allowing both centre multipliers to tend to zero.

Lemma H.8 (Actual source matching at degree loss)**.** *Fix the original compact or reciprocal coefficient ranges, sections, passive coefficient neighbourhood and buffered positive weak chord of* (H.43)*. For one fixed $R_0 > r_c$, the separate centre-height maps on $0 < P < p_0$ and $R_0 \le r \le \bar\sigma/s_0$ have the pure rows* (H.93) *and* (H.107)*. Their product remainder has the total-order-two rows* (H.114) *before matching. There is a fixed $R = r_{\mathrm{ge}}^{\mathrm{out}} > R_0$, chosen before the independent coefficient bounds, such that every negative lip with $R \le r \le \bar\sigma/s_0$ matches one actual input in the original strict positive chord. On this entire enlarged graph,*

$$-X\partial_X r = e_s\frac{\mathcal{P}(r)}{r}(1+\zeta), \qquad |\zeta| + |X\partial_X\zeta| \le C\tau, \qquad e_s = \tau/\chi \asymp 1.$$

All phase derivatives freeze the coefficients and retain the original centre, action constants, physical scales and raw section windows.

Proof. The two centre-height maps. Use the fixed coefficient field and positive weak chord, with scales

$$\epsilon = \kappa\tau, \qquad s_0 = \kappa\sqrt{B}, \qquad a = \kappa A_d, \qquad x_* = A_d/\sqrt{B}, \qquad r_s = \epsilon\mathcal{R}, \qquad \gamma_s = \epsilon\mathcal{R}\nu. \tag{H.61}$$

Distinguish the positive unit B from $\mathscr{B} = B_\tau(-r)$. For the original inner cut $r_c > 2$ and fixed $R_0 > r_c$, use

$$R \le r \le \bar\sigma/s_0, \qquad R \ge R_0, \qquad 0 < \bar\sigma < 1/2. \tag{H.62}$$

Constants are independent of the later R. The positive domain is $0 \le P \le p_0$ with its original larger chord buffer.

The negative source equation.

Write $x = -u = s_0 r$, $w = s_0^2\tau y$, and

$$P_-(r) = P_s(-r), \qquad K_-(r) = K(-s_0 r), \qquad e_-(r) = -E(-s_0 r). \tag{H.63}$$

The exact polynomial is

$$P_-(r) = \frac{\mathcal{R}}{\sqrt{B}}(1 - s_0 r)(x_* + r)(c + s_0 r) \quad \text{on the compact graph,} \tag{H.64}$$

and

$$P_-(r) = \frac{d}{\sqrt{B}}(1 - s_0 r)(x_* + r)(1 + j s_0 r) \quad \text{on the reciprocal graph.} \tag{H.65}$$

The identities $\mathcal{R}A_d c = B$ and $B = dA_d$, respectively, give exactly the original $P_s(-r)$. Set

$$\mathscr{B}(r) = 1 + 2\gamma_s - \epsilon\mathcal{R}s_0 r, \quad \mathfrak{d}(r) = \frac{P_-(r)}{r\mathscr{B}(r)}, \quad \mathcal{D} = \mathfrak{d}(r)\partial_r, \quad h_1(r) = \frac{P_-'(r)}{r\mathscr{B}(r)}. \tag{H.66}$$

Use the chosen coefficient-only real action offset $A_{c,0}(\theta)$ and define

$$\mathcal{A}_-(r) = \int_{r_c}^{r}\frac{dt}{\mathfrak{d}(t)} - A_{c,0}, \qquad \mathcal{D}\mathcal{A}_- = 1. \tag{H.67}$$

Direct differentiation of (H.64), (H.65), and (H.66) gives

$$\begin{aligned} P_- \asymp 1+r, \quad \mathscr{B} \asymp 1, \quad \mathfrak{d} \asymp 1, \quad |\mathcal{D}^m \mathfrak{d}| \le C_m, \quad |\mathcal{D}^m h_1| \le C_m/r \quad (0 \le m \le 3), \\ |K_- - 1| \le C\epsilon, \quad |\mathcal{D}^m K_-| \le C_m \tau \quad (1 \le m \le 3). \end{aligned} \tag{H.68}$$

For the polynomial monotonicity used below, put $\sigma = s_0 r$. On the compact graph,

$$P_-' = \frac{\mathcal{R}}{\sqrt{B}}\{c(1-2\sigma) + 2\sigma - 3\sigma^2 + a(1-c-2\sigma)\},$$

whereas on the reciprocal graph

$$P_-' = \frac{d}{\sqrt{B}}\{1 - 2\sigma + j\sigma(2-3\sigma) + a(j(1-2\sigma)-1)\}.$$

The fixed bound $\bar{\sigma} < 1/2$ and a common a-shrink give $P_-' \ge c$ in either graph. Since $\mathscr{B}' = -\epsilon \mathcal{R} s_0 \le 0$, it follows that

$$\mathcal{P} := P_-/\mathscr{B} \asymp 1 + r, \qquad \mathcal{P}' \ge c. \tag{H.69}$$

Indeed, P_-', P_-'', P_-''' are bounded by C, Cs_0, Cs_0^2, with $s_0 \le \bar{\sigma}/r$. Also

$$K_- = 1 - A(\gamma_s - \epsilon \mathcal{R} s_0 r) + C(\gamma_s - \epsilon \mathcal{R} s_0 r)^2. \tag{H.70}$$

Its positive-order action rows are $O(\epsilon s_0) = O(\tau)$. On the reciprocal graph, $\mathcal{R} = dj$ and $\gamma_s = \epsilon d(j - ja - 1)/3$ are regular at $j = 0$.

From the original rational graph equation one obtains exactly

$$y_r = \frac{y\{r\mathscr{B} + e_- s_0 \tau y\}}{\tau(P_- + K_- y)}. \tag{H.71}$$

Along the inverse passage introduce accumulated action q by $r_q = -\tau \mathfrak{d}(r)$, starting at r, and put $W = y/P_-$. The source lip is exactly $W(0) = \ell_0$. Then

$$(\log W)_q = -\frac{1 + \varpi(r) W}{1 + K_-(r) W} + \tau h_1(r) = -1 + \tau h_1(r) + G(r, W), \tag{H.72}$$

where

$$\varpi = e_- s_0 \tau \mathfrak{d}, \qquad G(r, W) = \frac{(K_- - \varpi) W}{1 + K_- W}. \tag{H.73}$$

Here $|e_-| \le C\epsilon^2$; direct differentiation gives

$$|\mathcal{D}^m \varpi| \le C_m \tau, \qquad 0 \le m \le 3. \tag{H.74}$$

These derivative bounds apply to the exact nonlinear coefficient.

Negative source variations.

The endpoint at the original inner cut is

$$T(r) = \frac{1}{\tau} \int_{r_c}^{r} \frac{dt}{\mathfrak{d}(t)} = \frac{\mathcal{A}_-(r) + A_{c,0}}{\tau}. \tag{H.75}$$

It satisfies

$$T \ge c(r - r_c)/\tau, \qquad \mathcal{D}T = 1/\tau, \qquad \mathcal{D}^2 T = \mathcal{D}^3 T = 0. \tag{H.76}$$

Apply Lemma A.3 with $(\kappa, H, Q, \omega, \rho) = (\tau, h_1, K_-, \varpi, \tau)$ and density $L = G$. Its phase derivative is the initial action derivative δ at fixed q:

$$\delta f(r(q; r)) = (\mathcal{D} f)(r(q; r)), \qquad \partial_q f = -\tau \mathcal{D} f, \tag{H.77}$$

because $\mathcal{A}_-(r(q;r)) = \mathcal{A}_-(r) - \tau q$. In particular δ commutes with ∂_q. The polynomial factors remain positive on the whole route $r_c \le v \le r$, with $s_0 v \le \bar{\sigma} < 1/2$, and on a fixed enlargement at r_c. Equations (H.68) and (H.74) give the required phase and mixed rows with $\alpha = 0$. Since K_- is close to one and $\varpi, \tau h_1$ are small, both $\mathcal{F}/W$ and $\mathcal{F}_W$ have a fixed negative margin on $0 \le W \le \ell_0$. The lemma therefore gives

$$0 < W(q;r) \le \ell_0 e^{-cq}, \qquad 0 \le q \le T(r), \tag{H.78}$$

and the fixed-clock rows

$$|\delta^i G| \le C\tau(1+q)^N e^{-cq} \quad (i = 1, 2), \qquad |G| + |G_q| \le C(1+q)^N e^{-cq}. \tag{H.79}$$

The inverse passage thus reaches the inner cut positively, with positive denominator, and its lip is the first ratio hit. Define on this actual passage

$$J(r) = \int_0^{T(r)} G(r(q;r), W(q;r))\, dq. \tag{H.80}$$

The ratio equation gives exactly $G = -K_- W_q + (\tau K_- h_1 - \varpi)W$. Integration by parts therefore isolates the nonlinear mass:

$$J = \ell_0 K_-(r) - K_-(r_c) W(T;r) + \int_0^T QW\, dq, \qquad Q = \tau(K_- h_1 - \mathcal{D}K_-) - \varpi. \tag{H.81}$$

Here $\mathcal{D}^i Q = O(\tau)$, $0 \le i \le 2$, using $\mathcal{D}^3 K_-$ in the second row. The same fixed-clock estimates give an integrable $C\tau(1+q)^N e^{-cq}$ bound for QW and its phase rows through order two. Apply the endpoint formulas (A.10) to this density and the terminal chain rules of Lemma A.3 to $K_-(r_c)W(T;r)$, with $T_1 = 1/\tau$, $T_2 = 0$. The fixed gap $R_0 - r_c > 0$ makes all endpoint terms $\tau^{-N}(1+T)^N e^{-cT}$ flat. Together with $K_-(r) - 1 = O(\epsilon) = O(\tau)$, this proves

$$|J - \ell_0| + |\mathcal{D}J| + |\mathcal{D}^2 J| \le C\tau. \tag{H.82}$$

The constants depend on $R_0 - r_c > 0$, not on the later R.

The polynomial-memory identity (A.11) retains the initial factor:

$$\int_0^T \tau h_1(r(q;r))\, dq = \log P_-(r) - \log P_-(r_c). \tag{H.83}$$

Consequently the actual height at the inner cut is

$$y_c(r) = \ell_0 P_-(r) e^{-T(r)} e^{J(r)}. \tag{H.84}$$

Uniformly for $r \ge R_0$,

$$0 < y_c(r) \le C(1+r)e^{-c(r-r_c)/\tau} \le C_N \tau^N \quad \text{for every fixed } N. \tag{H.85}$$

Its first two logarithmic rows are

$$\mathcal{D}\log y_c = h_1 - 1/\tau + \mathcal{D}J, \qquad \mathcal{D}^2 \log y_c = \mathcal{D}h_1 + \mathcal{D}^2 J. \tag{H.86}$$

We now continue to the original common centre, without renormalizing it to an image-dependent section. On $0 \le v \le r_c$, $P_-(v)$ and $\mathscr{B}(v)$ are positive units. The exact linear multiplier is

$$L_-(v) = \exp\left\{-\frac{1}{\tau}\int_v^{r_c} \frac{t\mathscr{B}(t)}{P_-(t)}\, dt\right\} \le 1, \qquad A_{\mathrm{core},-} = \int_0^{r_c} \frac{t\mathscr{B}(t)}{P_-(t)}\, dt. \tag{H.87}$$

The exact equation on this fixed compact segment is $y_v = A_- y + \mathcal{K}_- y^2$, where

$$A_- = \frac{v\mathscr{B}}{\tau P_-}, \qquad \mathcal{K}_-(v,y) = \frac{e_- s_0}{P_- + K_- y} - \frac{v\mathscr{B}K_-}{\tau P_-(P_- + K_- y)}. \tag{H.88}$$

On the original fixed small height buffer, all normal derivatives through order three of $\mathcal{K}_-$ are bounded by C/τ. Its inward linear propagator is exactly $L_- \le 1$, and (H.85) gives $y_c/\tau \ll 1$. Lemma A.5, with $\kappa_c = \tau$, gives an actual positive centre hit and $y(0; y_c) = L_-(0) y_c \mathcal{U}_-(y_c)$, where

$$|\log \mathcal{U}_-| + \sum_{i=1}^{2} |(y_c \partial_{y_c})^i \log \mathcal{U}_-| \le C y_c/\tau. \tag{H.89}$$

This uses the full multiplier, with no reciprocal bound on it, and holds for either sign of $e_-(0)$.

Use the same centre-height scale as on the positive side:

$$Z_-(r) = \frac{w(0;r)}{s_0^2 \tau} = \ell_0 P_-(r) \exp\left\{ -\frac{\mathcal{A}_-(r) + A_{c,0} + A_{\mathrm{core},-}}{\tau} \right\} e^{J(r)} \mathcal{U}_-(y_c(r)). \tag{H.90}$$

Choose the coefficient-only logarithmic constant

$$c_- = \log \ell_0 - (A_{c,0} + A_{\mathrm{core},-})/\tau. \tag{H.91}$$

The separate negative remainder is

$$S_-(r) = \log Z_-(r) + \mathcal{A}_-(r)/\tau - \log P_-(r) - c_- = J(r) + \log \mathcal{U}_-(y_c(r)). \tag{H.92}$$

The chain rule, (H.85)–(H.86), and (H.89) bound the two core contributions by $C y_c/\tau^2$ and $C y_c/\tau^3$. Combining with (H.82) gives

$$|S_-| \le C, \qquad |\mathcal{D} S_-| + |\mathcal{D}^2 S_-| \le C\tau. \tag{H.93}$$

The physical centre height is therefore

$$z_-(r) = r_s^2 s_0^2 \tau e^{c_-} P_-(r) e^{-\mathcal{A}_-(r)/\tau} e^{S_-(r)}. \tag{H.94}$$

The factor e^{S_-}, its inverse and its first two action rows are uniformly bounded.

Positive source variations.

The positive weak coordinate from (H.43) satisfies

$$\Phi(P) = \{z_+ y(P)\}^\chi, \qquad \chi = \tau X_d, \qquad z_+^{\pm 1} + X_d^{\pm 1} \le C. \tag{H.95}$$

The local invariant gives

$$0 \le y(P) \le C\vartheta^{1/\chi}, \quad 0 < \vartheta < 1, \quad 0 \le P \le p_0, \tag{H.96}$$

and $\Phi' > c > 0$ with its stated uniform phase bounds. Fix $P_\dagger \in (0, p_0)$, independently of coefficients, and set $X = \Phi(P)/\Phi(P_\dagger)$. On $P > 0$,

$$\frac{y(P)}{y(P_\dagger)} = X^{1/\chi}, \qquad \mathcal{E} := X\partial_X = \chi^{-1} y \partial_y. \tag{H.97}$$

Use this exact logarithmic derivative.

Construct $Z_+(P)$ independently by following positive time from $u_+ = a(1 - \eta_+)$, with $w = a^2 y(P)$, to $u = 0$. In $v = u/a$, $v_+ = 1 - \eta_+$, put

$$h(av)/a^2 = \tau H_+(v), \qquad L(av)/a = v\mathscr{B}_+(v), \qquad K_+(v) = K(av). \tag{H.98}$$

The original exact functions are

$$H_+(v) = \frac{\mathcal{R}}{A_d}(1+av)(1-v)(c-av) \tag{H.99}$$

on the compact graph, and

$$H_+(v) = \frac{d}{A_d}(1+av)(1-v)(1-jav) \tag{H.100}$$

on the reciprocal graph. On $0 \le v \le v_+$, $H_+, \mathscr{B}_+, K_+$ are positive units, with uniform finite coefficient-function bounds. In particular the positive core has no inverse κ scale. Its exact scalar equation is

$$y_v = \frac{y\{v\mathscr{B}_+ + aE(av)y\}}{\tau H_+ + K_+ y}. \tag{H.101}$$

Its linear multiplier and positive core action are

$$L_+(v) = \exp\left\{-\frac{1}{\tau}\int_v^{v_+} \frac{t\mathscr{B}_+(t)}{H_+(t)}\,dt\right\} \le 1, \qquad A_{\mathrm{core},+} = \int_0^{v_+} \frac{t\mathscr{B}_+(t)}{H_+(t)}\,dt. \tag{H.102}$$

Freeze the coefficients and set $y(v) = \tau^2\eta(v)$, with initial height $\eta_{\mathrm{in}} = y(P)/\tau^2$ at the same cut v_+. Exact division gives

$$\eta_v = \frac{v\mathscr{B}_+}{\tau H_+}\eta + \widetilde{\mathcal{K}}_+(v,\eta)\eta^2, \qquad \widetilde{\mathcal{K}}_+ = \frac{\tau aE(av)}{H_+ + \tau K_+\eta} - \frac{v\mathscr{B}_+K_+}{H_+(H_+ + \tau K_+\eta)}. \tag{H.103}$$

Choose a fixed small buffer $0 \le \eta \le \eta_b < \inf X_d^2$. The nonlinear coefficient and its first three ordinary height derivatives are bounded uniformly. Indeed the displayed denominators are positive units and each such derivative introduces a bounded factor τK_+. Then $\tau^2\eta_b < \chi^2$ lies inside the original height range; (H.96) puts every original $0 < P \le p_0$ in a smaller initial subbuffer after a common τ-shrink. The inward propagator remains exactly $L_+ \le 1$, without a reciprocal reserve. Thus Lemma A.5, with $\kappa_c = 1$, supplies the actual positive centre hit and its Euler-height rows. Since $\eta_{\mathrm{in}}\partial_{\eta_{\mathrm{in}}} = y_+\partial_{y_+}$ for $y_+ = y(P)$, these give

$$|\log \mathcal{U}_+(y_+)| + \sum_{i=1}^{2} |(y_+\partial_{y_+})^i \log \mathcal{U}_+(y_+)| \le Cy_+/\tau^2, \tag{H.104}$$

where $\mathcal{U}_+$ is the normalized factor at $v = 0$. The actual central height is

$$Z_+(P) = C_+(\theta)y(P)\mathcal{U}_+(y(P)), \qquad C_+(\theta) = \frac{a^2}{s_0^2\tau}e^{-A_{\mathrm{core},+}/\tau}. \tag{H.105}$$

This retains exactly $a^2/(s_0^2\tau) = A_d^2/(B\tau)$. The prefactor of the exponential is a coefficient-only unit times τ^{-1}. Scalar first variation gives $Z_+'(P) > 0$ for $P > 0$.

Define on the original independent positive domain

$$S_+(P) = \log Z_+(P) - \log Z_+(P_\dagger) - \chi^{-1}\log X = \log \mathcal{U}_+(y(P)) - \log \mathcal{U}_+(y(P_\dagger)). \tag{H.106}$$

The coefficient-only multiplier cancels as an exact reference difference. Equations (H.97) and (H.104) give

$$|\mathcal{E}^i S_+| \le C\chi^{-i}\tau^{-2}y(P) \le C\tau^{-i-2}\vartheta^{1/\chi}, \qquad i = 1, 2. \tag{H.107}$$

The rows vanish continuously at $P = 0$. This positive map remains independent of the negative image.

Admission to the positive input chord.

Choose any fixed $p_* \in (0, p_0)$ inside the original counted weak chord, with its larger original raw chord buffer. Put $j(P) = \Phi(P)/\Phi'(P)$. Its uniform phase bounds give $j'(P) = 1 + O(P)$; choose p_* small enough that $j'(P) > 3/4$ on $0 \le P \le p_*$. The endpoint is fixed before R. By $\Phi' > c$, $\Phi(p_*)$ is bounded below by a fixed positive constant. Equations (H.95), (H.102), and (H.105) give

$$-\tau \log Z_+(p_*) \le C_*(p_*), \tag{H.108}$$

uniformly in κ, τ, j after a common small bound. Indeed, $-\tau \log y(p_*) = -(\tau/\chi) \log \Phi(p_*) + \tau \log z_+$ and $A_{\mathrm{core},+}$ are bounded; the remaining scale costs $O(\tau(1 + |\log \tau|))$.

On the negative side, (H.90) gives

$$-\tau \log Z_-(R) \ge c(R - r_c) - \tau \log\{\ell_0 P_-(R)\} - C\tau. \tag{H.109}$$

First choose a fixed $R = r_{\mathrm{ge}}^{\mathrm{out}} > R_0$, larger than any already prescribed original growing threshold, so that $c(R - r_c) > C_*(p_*) + 2d_*$ for a fixed $d_* > 0$. Only afterwards decrease the coefficient bounds so that $\tau \log(1 + R)$ and the fixed terms in (H.109) are below d_*. This preserves the full independent coefficient family. Choose the counted threshold $r_{\mathrm{ge}} > R$ with a fixed positive buffer before decreasing the coefficients. It follows that

$$0 < Z_-(R) < e^{-d_*/\tau} Z_+(p_*). \tag{H.110}$$

Furthermore (H.92)–(H.93) imply

$$\mathcal{D} \log Z_- = -1/\tau + h_1 + \mathcal{D}S_- < 0. \tag{H.111}$$

Thus every actual growing lip in (H.62) has centre height strictly between zero and $Z_+(p_*)$. The continuous strictly increasing positive map Z_+ maps $[0, p_*]$ onto $[0, Z_+(p_*)]$. There is a unique $P(r) \in (0, p_*)$ with $Z_+(P(r)) = Z_-(r)$. Scalar uniqueness at the common centre identifies the two actual graph segments with the same orbit.

The local weak chord is the one parameterized by the original raw input

$$P = (x_{\mathrm{raw}} - x_b)/s_{\mathrm{in}}, \qquad s_{\mathrm{in}} = \Omega a^{2-\lambda_\gamma} U_{\mathrm{in}}, \qquad \Omega = r_s^{2-\lambda_\gamma}, \quad U_{\mathrm{in}}^{\pm 1} \le C. \tag{H.112}$$

The independently constructed stopped outer chord and its local invariant give the actual continuation to this raw section for all $0 \le P \le p_0$, with its original buffers. The matched inputs lie in the strict interior $P < p_*$. Both local points are on the same invariant level in the original coordinate box; the monotone positive invariant coordinate gives the intervening orbit segment there. Thus every lip has the actual positive-time continuation from raw input through the positive middle cut and centre to the negative lip. The separate annular overlap remains the earlier construction.

The matched action derivative.

On the separate original product domain define

$$R_{\mathrm{prod}}(P, r) = \log P_-(r) + S_-(r) - S_+(P). \tag{H.113}$$

It remains defined whether or not the two images intersect. Its phase rows satisfy

$$\begin{aligned} |\mathcal{E} R_{\mathrm{prod}}| + |\mathcal{E}^2 R_{\mathrm{prod}}| &\le C\tau^{-4}\vartheta^{1/\chi}, \\ |\mathcal{D} R_{\mathrm{prod}}| + |\mathcal{D}^2 R_{\mathrm{prod}}| &\le C, \qquad \mathcal{E}\mathcal{D} R_{\mathrm{prod}} = 0. \end{aligned} \tag{H.114}$$

Although $\log P_-(r)$ can grow, its two action derivatives have the bounds needed here. On the actual matched graph the exact identity is

$$\mathcal{A}_-(r) = \tau\{c_- - \log Z_+(P_\dagger)\} - e_s \log X + \tau R_{\mathrm{prod}}(P, r), \qquad e_s = \tau/\chi = 1/X_d. \tag{H.115}$$

The constant retains all coefficient-only terms; e_s is a positive unit.

Define separate product functions

$$N_- = 1 - \tau\{h_1 + \mathcal{D}S_-\}, \qquad N_+ = 1 + \chi\mathcal{E}S_+, \qquad n(P,r) = N_+(P)/N_-(r). \tag{H.116}$$

For a common small τ bound,

$$N_\pm \geq 1/2, \quad |N_- - 1| + |\mathcal{D}N_-| \leq C\tau, \quad |N_+ - 1| + |\mathcal{E}N_+| \leq C\tau^{-3}\vartheta^{1/\chi}. \tag{H.117}$$

Differentiate equality of the two actual centre heights: $\mathcal{E}\log Z_+ = N_+/\chi$ and $\mathcal{D}\log Z_- = -N_-/\tau$. It gives exactly

$$-\mathcal{E}r = e_s\mathfrak{d}(r)n(P,r) = e_s\frac{\mathcal{P}(r)}{r}(1+\zeta), \qquad \zeta = n - 1. \tag{H.118}$$

The matched graph is consequently strictly decreasing. To differentiate the quotient along this graph, both separate variables must be varied:

$$\mathcal{E}_{\text{graph}}\zeta = \mathcal{E}_+ n - e_s n\mathcal{D}_- n. \tag{H.119}$$

Here $\mathcal{E}_+$ and $\mathcal{D}_-$ act on the separate positive and negative variables of the product. The second pure derivatives in (H.93) and (H.107) bound precisely the two derivatives in (H.119). Thus

$$|\zeta| + |\mathcal{E}_{\text{graph}}\zeta| \leq C\{\tau + \tau^{-3}\vartheta^{1/\chi}\} \leq C\tau. \tag{H.120}$$

This is uniform on the whole enlarged graph with independent κ, τ, j; the error tends to zero with τ.

The normalized source heights retain the physical conversion $z(\gamma_s) = r_s^2 s_0^2 \tau Z$, the exact raw power Ω, the positive and negative tail quartets on the reciprocal graph

$$\left(-\frac{\tau d}{A_d}, \frac{\tau d(1-a+ja)}{A_d}, \gamma_s, r_s a\right), \qquad \left(-\kappa^2\tau d A_d, \kappa\tau d(1-a+ja), \gamma_s, r_s\right), \tag{H.121}$$

and the passive coefficient restrictions used to construct the raw chords. No estimate of $\gamma_s \log r_s$, no derivative with respect to a coefficient, and no frozen moving raw endpoint is used. At $j = 0$ only the one-sided normalized estimates are intended; actual raw sections are asserted for $j > 0$. □

H.5 One physical interval and four passage regions

The preceding constructions use different phase coordinates. We now show that their domains cover one connected interval of actual through trajectories, with open overlaps. Their derivative estimates can then be combined for the same affine displacement.

Lemma H.9 (A common entrance and gate choice)**.** *One auxiliary choice for the four passage regions, followed by independent small coefficient bounds, has the following properties. The incoming raw height h is x_{raw} in* (H.40) *and x_{in} in Lemma E.1; its stopped value is x_b. Thus the canonical weak input is exactly*

$$P = q(h) = \frac{h - x_b}{s_{\text{in}}}, \qquad s_{\text{in}} = \Omega a^{2-\lambda_\gamma} U_{\text{in}} > 0, \qquad \Omega = r_s^{2-\lambda_\gamma}, \quad U_{\text{in}}^{\pm 1} \leq C. \tag{H.122}$$

This is the affine coordinate on the original raw section, distinct from the displacement coordinate $x = h - a_\theta$; no new input observer is inserted.

Let $\widehat{p}_0 = p_0$ be a fixed buffered raw cutoff in (H.41)*, and choose $0 < \widehat{p}_* < \widehat{p}_0$ so that $\mathfrak{j}' = (\Phi_\theta/\Phi_\theta')' > 3/4$ on $[0, \widehat{p}_*]$. Set $q_{\text{ref}} = \widehat{p}_*/4$ and use $P_\dagger = q_{\text{ref}}$ in* (H.97)*. The final negative*

thresholds are denoted by $\widehat{r}_{\mathrm{ge}}^{\mathrm{out}} = r_{\mathrm{ge}}^{\mathrm{out}}$ *and* $\widehat{r}_{\mathrm{ge}} = r_{\mathrm{ge}}$*, with a fixed positive gap. Every actual first-lip input with* $r_\ell \geq r_{\mathrm{ge}}$ *satisfies*

$$0 < P < q_{\mathrm{ref}}/2 < q_{\mathrm{ref}} < \widehat{p}_* < \widehat{p}_0, \tag{H.123}$$

including first hits outside the growing product range.

The source constant $\delta_\sharp(L)$ *is selected once for the full bounded-source domain and relative negative-curvature estimate. Its gates are those below in Lemma H.11. The final* r_{ge} *precedes* R_{BD} *and its annular overlap; all fixed gates and phase buffers precede the final coefficient bounds. No comparison of the independent scales is imposed.*

Proof. Fix the original raw windows and a finite $V_1 > 1$, say $V_1 = 2$. Choose the local cuts and ordinary C^3 section maps for this value in Lemma E.1, retaining its enlarged interval $0 \leq V \leq V_1 + 1$. Keep these choices throughout. Use the strict source block and the two comparison orbits of Proposition H.3: fix its derivative constants, then B_-, then $\delta_\sharp(L)$ for both the domain and relative estimate, and finally δ_1, δ_2 and the gates below. Fix $r_c > 2$ and choose ℓ_0 below the ratio of a fixed real comparison orbit at $-r_c$, as in (H.35).

Fix a source envelope $\bar{\sigma}_{\mathrm{src}} = 1/3$ for the separate centre-map estimates on $r_c \leq r \leq \bar{\sigma}_{\mathrm{src}}/s_0$ in Lemma H.8; its derivative constants are fixed before the positive cutoff and negative thresholds below. The growing and output intervals will be chosen inside this envelope. The independent stopped outer chord and local invariant give the positive input map on $[0, p_0]$ in (H.40)–(H.43). Choose the smaller fixed $\widehat{p}_*$ above so that the positive action at $P_0 = q_{\mathrm{ref}}/2$ also exceeds the fixed middle variational exponent by two. This is possible since $\Phi_\theta(P) = PU_\theta(P)$ with uniform positive units and τ/χ a positive unit. These choices precede the final negative threshold.

The full first-hit cutoff following (H.36) provides a fixed $R(P_0)$. Choose $R = r_{\mathrm{ge}}^{\mathrm{out}}$ larger than this and every earlier growing threshold, and so that $c(R - r_c) > C_*(P_0) + 2d_*$ for a fixed $d_* > 0$ in (H.108). Choose $r_{\mathrm{ge}} > R$ with a fixed positive buffer. The separate centre maps give (H.110) with endpoint P_0 instead of p_* after independent coefficient shrink. Their monotonicity and scalar uniqueness therefore construct the actual product reference $r_\ell = r_{\mathrm{ge}}$ with raw input $0 < P_{\mathrm{ge}} < P_0$, in the original buffered chord. The local invariant supplies its positive-time continuation to the raw section, as in (H.112).

On the full first-hit domain of the original through interval, (H.36) gives $r_{\ell,h} < 0$. Thus $r_\ell(h) \geq r_{\mathrm{ge}}$ implies $h \leq h_{\mathrm{ge}} = x_b + s_{\mathrm{in}} P_{\mathrm{ge}}$, and the same positive affine input gives $0 < P(h) \leq P_{\mathrm{ge}} < P_0$. This proves (H.123) throughout the first-hit domain, including inputs beyond the product range. The extension follows from order on the full domain; the product derivative estimates remain restricted to their original range.

Only now choose R_{BD} so the compact strong projection of $r_\ell = r_{\mathrm{ge}}$ lies strictly below $2R_{\mathrm{BD}}$. The transverse inverse and first-hit order give a fixed $r_{\mathrm{ge}}^\sharp > r_{\mathrm{ge}}$ in the enlarged annular overlap. The annular proof now fixes its own outer cuts L_{a}, after this output bound and its limiting-action envelope. The earlier source cut L, h_L, $\delta_\sharp(L)$, and literal gates stay fixed; both passages continue to the same raw ports. The original weak cuts satisfy $c_+ = p_{\mathrm{a}}/8 < p_- = p_{\mathrm{a}}/2$ and $c_- = p_{\mathrm{a}}/8 < r_{\mathrm{a}}/2$, so the annular root-gap construction uses this same auxiliary tuple. Only final independent coefficient bounds are reduced to admit the new positive-height tail cuts. Choose $0 < \sigma_1 < \sigma_2 < \sigma_0 < 1/4$, the common negative section u_c, its larger $\sigma_0 < \bar{\sigma} \leq \bar{\sigma}_{\mathrm{src}}$ buffer, with $\bar{\sigma} < |u_c|/2$, and strict action gap Δ as in the construction after (H.36). After these fixed choices, shrink the independent coefficients so the fixed annular overlap lies in the counted product range and $s_0 r_{\mathrm{ge}}^\sharp < \sigma_1$. All source, raw and shooting buffers retain strict margins. All these restrictions are independent upper bounds on the coefficients. After they have been imposed, the finite multiplier will be chosen. □

Proposition H.10 (One actual carrier for the four passage regions)**.** *Use the common choice of Lemma H.9, the original field, through word, sections, raw windows, L, h_L and closing maps. The source constant $\delta_\sharp(L)$ is the once-selected constant for the full original bounded-source domain and its relative sign; δ_1, δ_2 keep their literal values from that tuple. Let I_θ be the connected interval of actual through inputs in this word, within the original buffered windows and positive graph denominator domain, with*

$$0 < H_\theta(h) := z(\gamma_s; h) < z_{\mathrm{cap}}.$$

Here z_{cap} is the original common fixed raw-height cap, chosen within the standing high-height bound. The stopped trace x_b is excluded. Write h^0 for the actual trace of $h_L = 0$ and $I_\theta^- = I_\theta \cap \{h \le h^0\}$.

1. *The original through interval (x_b, h^0) has the strictly increasing shooting coordinate $W_\theta(h) = w(u_c; h)$ with image $(0, W_\theta(h^0))$, where $W_\theta(h^0) \to u_c^2/2$ uniformly. Thus every point of the full original enlarged chord $\overline{\mathcal{J}_c^{\sharp\sharp}}$ corresponds to an actual through orbit, with a strict height buffer. Its backward shot and both raw continuations are that same orbit.*
2. *Let h_S be the trace of $h_L = -\delta_2$ and let h_A, h_B be the gate traces of $r_b = r_a/2, 2R_{\mathrm{BD}}$, respectively. Put*

$$B_\theta = I_\theta^- \cap ([h_S, h^0] \cup [h_B, h_A]), \qquad h_T = \min\{h_S, h_B\}. \tag{PCov1}$$

Let $I_{\ell,\theta}$ be the original first-ratio-hit domain. On it put $\sigma(h) = s_0 r_\ell(h)$. Wherever $r_\ell \ge r_{\mathrm{ge}}$ use the original raw middle input

$$V(h) = \frac{w(u_+; h)}{\ell_0 h_{\mathrm{ax}}(u_+)\rho(u_+)} = \frac{y(u_+; h)}{\varepsilon_+} > 0, \quad h_{\mathrm{ax}} = -\delta p_d, \quad \varepsilon_+ = \frac{\ell_0 h_{\mathrm{ax}}(u_-)}{a^2 J_-}. \tag{PCov2}$$

All factors in this definition are actual and positive for a physical coefficient fibre. Define the following subsets of that same carrier:

$$\begin{aligned} G_\theta &= I_\theta^- \cap I_{\ell,\theta} \cap \{r_\ell \ge r_{\mathrm{ge}},\ \sigma \le \sigma_0\}, \\ D_\theta &= I_\theta^- \cap I_{\ell,\theta} \cap \{\sigma \ge \sigma_1,\ 0 < V \le V_1\}, \\ E_\theta &= I_\theta^- \cap I_{\ell,\theta} \cap \{\sigma \ge \sigma_1,\ V \ge 1\}. \end{aligned} \tag{PCov3}$$

The V coordinate in the last two lines is well defined because $\sigma_1/s_0 > r_{\mathrm{ge}}$. These are connected, relatively closed intervals, possibly empty, and

$$I_\theta^- = B_\theta \cup G_\theta \cup D_\theta \cup E_\theta. \tag{PCov4}$$

Each of $G_\theta, D_\theta, E_\theta$ has the exact raw input buffer

$$0 < q(h) := \frac{h - x_b}{s_{\mathrm{in}}} < q_{\mathrm{ref}} < \widehat{p}_* < \widehat{p}_0, \qquad s_{\mathrm{in}} = \Omega a^{2-\lambda_\gamma} U_{\mathrm{in}} > 0. \tag{PCov5}$$

The compact annular/growing overlap and the original product and V overlaps are on these actual inputs, with their equality dividers included. Intersecting with any original closed raw-window or height truncation preserves the cover; no divider is required to occur.

3. *G_θ lies in the growing product domain, and D_θ, E_θ lie in the enlarged domains of the finite-V and fixed-exit maps constructed in Proposition H.13. The high-height interval is $I_\theta^+ = I_\theta \cap \{h \ge h^0\}$. Together with (PCov4) it covers the whole I_θ, including every non-root through input. On all these intervals the physical equation is*

$$F(h - a_\theta) = h - a_\theta - f_\theta(m\mathfrak{T}_\theta(h)), \qquad \mathfrak{T}_\theta = d_\theta \circ T_\theta^\infty. \tag{PCov6}$$

The statements about actual raw maps concern $j > 0$; the reciprocal face $j = 0$ has only the uniform one-sided normalized interpretation.

Proof. The independently constructed stopped and $h_L = 0$ graphs bound the through interval. Every intermediate graph reaches $u_c \in (-1, u_-)$ with positive denominator. Scalar first variation makes W_θ strictly increasing, with stopped limit zero. For the upper comparison graph set $r_0 = s_0 L$, $r = -u$ and $\mathscr{W} = w/r^2$. At its initial cut

$$\mathscr{W}(r_0) = \tfrac{1}{2} + O_L(\tau b_\tau), \qquad b_\tau = -\log(2\tau).$$

On $r_0 \le r \le |u_c|$, direct substitution of the original polynomial in the graph equation gives, on $1/4 \le \mathscr{W} \le 3/4$,

$$r\mathscr{W}_r = 1 - 2\mathscr{W} + \mathscr{E}, \qquad |\mathscr{E}| \le C\{\kappa\tau + \kappa\tau/r\} \le C\{\kappa\tau + \tau/L\}. \tag{PCov7}$$

Indeed $|\delta p_d(-r)| \le C\kappa\tau(r + a)$ and $a/r \le x_{\max}/L$, while $K_d = 1 + O(\kappa\tau)$ and $-L_d(-r)/r = 1 + O(\kappa\tau)$; the $E_d w$ term is smaller on this corridor. The regular reciprocal factorization gives the same bounds uniformly as $j \downarrow 0$. Both boundary vectors point inward. The logarithmic-radius equation contracts, so variation of constants gives $\mathscr{W}(|u_c|) \to 1/2$. Hence $W_\theta(h^0) \to u_c^2/2$. The uniform endpoint margins put $\overline{\mathcal{J}_c^{\sharp\sharp}}$ strictly inside this shooting image after shrink. Strict first variation gives a unique through input for each W; backward uniqueness identifies its shot and the two already constructed raw continuations. This proves (i) independently of the fixed-exit phase.

The same incoming first variation preserves the ordering of the actual h_L labels. Thus the central source interval is $[h_S, h^0]$. The exact gate inverse and centre matching give the annular interval $[h_B, h_A]$. The fixed orbit $h_L = -\delta_1$ has trace h_C strictly inside both intervals, because $-\delta_2 < -\delta_1 < 0$ and its actual gate labels lie strictly inside the annular bands. Consequently

$$[h_S, h^0] \cup [h_B, h_A] = [h_T, h^0]. \tag{PCov8}$$

The first fixed comparison alone supplies this overlap. The compact overlap also supplies an actual reference with $r_\ell = r_{\mathrm{ge}}$ and $r_b < 2R_{\mathrm{BD}}$; its trace h_{ge} satisfies $h_B < h_{\mathrm{ge}}$. Every input failing (PCov8) is below $h_T \le h_B$, so the supplied full original first-hit order gives $r_\ell > r_{\mathrm{ge}}$. The unified entrance then gives (PCov5) on all inputs with $r_\ell \ge r_{\mathrm{ge}}$, including first hits beyond the product range. Their derivative estimates are applied only within the growing product domain.

For this actual local input q, the original positive cut is reached on the stopped outer chord's local invariant. Its height is positive, finite and strictly increasing in h. Formula (PCov2) is exactly the original normalization $w = \ell_0 h_{\mathrm{ax}} \rho V$, so $V_h > 0$, while the full first-hit order gives $\sigma_h < 0$. All sets in (PCov3) are therefore intervals and include their finite level boundaries. If an input outside B_θ has $\sigma \le \sigma_0$, it is in G_θ. Otherwise $\sigma > \sigma_0 > \sigma_1$ and, because $V_1 > 1$, its positive V satisfies $V \le V_1$ or $V \ge 1$. It is in D_θ or E_θ. This proves (PCov4) for every through input, whether or not F vanishes.

The compact band $r_\ell \in [r_{\mathrm{ge}}, r_{\mathrm{ge}}^\sharp]$ is strictly inside the annular interval and, after the prescribed s_0 bound, the growing product domain. The inequalities $\sigma_1 < \sigma_2 < \sigma_0 \le \bar\sigma < |u_c|/2$ retain the actual product overlap. At $\sigma = \sigma_0$ the other passage region has a strict σ_1 buffer, and at $V = 1, V_1$ the last two regions overlap because $1 < V_1$. If a level is absent, the corresponding truncated interval is empty. The bounded endpoints and annular/growing overlap are included in these intervals.

On G_θ, $r_\ell \ge r_{\mathrm{ge}} > r_{\mathrm{ge}}^{\mathrm{out}}$ and $\sigma \le \sigma_0 \le \bar\sigma$, so the independently constructed negative lip and positive chord identify the actual growing graph. On $D_\theta \cup E_\theta$ the same raw buffer (PCov5) supplies the original positive local cuts. The construction in Proposition H.13, using (i) as its

independent shooting geometry, identifies the bounded-V early-cut passage and the fixed-exit shooting map with precisely these orbits. In particular its action bound places the whole E_θ inside the enlarged inverse image, with $\Theta = 0$ buffered. This last step is the only use of that proposition; the shooting geometry in part (i) was constructed independently.

Finally the h^0 comparison graph has original source height $Y_0 \asymp b_\tau \to \infty$ and physical height $H_\theta(h^0) = s_s^2 Y_0 \to 0$. The original source dictionary gives $|\alpha_0| + \alpha_1^2 = s_s^2(1 + p_s^2)$, with $p_s^2 = O(\tau)$. It follows that

$$K(|\alpha_0| + \alpha_1^2) < H_\theta(h^0) < z_{\mathrm{cap}}.$$

Scalar height order puts every retained input at or above h^0 in that same original high-height interval. The real enlarged $h_L = 0$ source chord supplies its open overlap with the low carrier. Thus I_θ^+ includes the equality h^0 and all its non-root inputs; the upper cap is common and fixed, rather than selected at a zero. Original-time uniqueness gives (PCov6), with no raw gain inverted in the domain construction. □

H.6 Bounded source

Lemma H.11 (Physical derivative estimates on the bounded-source intervals)**.** *Keep the original field, L, h_L, raw ports and enlarged windows, regular maps and ordinary C^3 divided closing. Keep the unified tuple's literal*

$$\begin{gathered}\delta_1 = \min\{\delta_\sharp(L)/8, x_{\min}^2/8192, 1/8192\},\\ \delta_2 = 2\delta_1, \quad p_{\mathrm{cmp}} = \sqrt{2\delta_1}, \quad r_a = p_{\mathrm{cmp}}/8, \quad p_a = r_a/16,\\ p_- = p_a/2, \quad c_\pm = p_a/8.\end{gathered}$$

The constant $\delta_\sharp(L)$ was fixed once, sufficiently small for the bounded-source domain and relative estimate, before these gate constants were defined. On the actual source word put

$$\beta = h_L/\tau - \log(2\tau), \quad b_\tau = -\log(2\tau), \quad \beta_{\min} = -\delta_2/\tau - \log(2\tau).$$

There are fixed thresholds B_-, B_+ and constants C_B, C_H, uniform in the original compact and reciprocal coefficient families, such that:

1. *On every actual input with $\beta_{\min} \le \beta \le -B_-$, the complete divided transition Π has $\Pi'' < 0$ in the original affine raw input, hence $F'' = -m\Pi'' > 0$ for $m > 0$.*
2. *On $-B_- \le \beta \le B_+$, one has $0 < \Pi' \le C_B$.*
3. *On $B_+ \le \beta \le b_\tau$, including $h_L = 0$, and on a real enlarged source chord through that orbit, one has $0 < \Pi' \le C_H$. The same bound holds on its admitted height-dominant continuation.*

All assertions hold on enlarged overlap intervals and at every nonclosed through input. One final choice $m \max\{C_B, C_H\} < 1/2$ gives $F' > 1/2$ on the last two pieces. The input is always $x = h - a_\theta$, and

$$\Pi(x) = \mathcal{C}_{\theta,m}(T_\theta(a_\theta + x)), \qquad F(x) = x - m\Pi(x) = x - f_\theta(md_\theta(T_\theta(a_\theta + x))). \tag{BS1}$$

Proof. Apply Proposition H.3 to the same sections and divided closing. Its estimates (H.17)–(H.18) give the first assertion, and (H.19) gives the second. The independent frozen windows, $L_f = 8$, $R_f = 5$ and polynomial margin 3/4 are supplied by Lemma H.1; both exact raw gains and their common-power cancellation are retained in (BS1). It remains to check the literal gates and the upper chord.

The final $\delta_\sharp(L)$ is chosen there for both the domain and relative curvature estimate, after B_- and before the displayed literal δ_1, δ_2. Since $\delta_2 \le \delta_\sharp/4$, the central block has enlarged lower chords down to $h_L = -3\delta_1 > -\delta_\sharp$. Compact labels $[-B_- - 1, B_+ + 1]$ and positive

labels beginning below B_+ give actual overlaps by monotone incoming first variation. A final τ shrink makes $\beta_{\min} < -B_-$ and $b_\tau > B_+ + 2$.

For positive labels choose the signed high-height constants K, z_*, C_H first, then B_+ large. Equation (H.5) gives, through $\beta = b_\tau$ and the enlarged chord below,

$$Y_0 > K(1 + p_s^2), \qquad H = s_s^2 Y_0 \le C_L \tau^3 \sigma^4 j^2 (1 + b_\tau) < z_*. \tag{BS10}$$

To realize the upper chord, freeze $B = b_\tau$ in the original positive-energy construction. For $\beta/B \in [3/4, 3/2]$, its incoming energy $\beta/B + B^{-1}\log(2/B)$ stays strictly inside $[1/4, 3]$. The energy defect is $O_L(B^{-1})$ and its variation integral tends uniformly to zero; the same first-contact proof and positive inverse therefore construct this entire chord. It has $Y_0 = \beta - \log\beta + O_L(1)$ and $\eta(\pm L) = L^2/2 + O_L(\tau b_\tau)$ at the unchanged cuts. The original quadratic corridor and raw-window margins continue both sides. In particular $0 \le h_L \le \tau b_\tau/2$ is a real enlargement above the $h_L = 0$ orbit.

By (BS10), the signed high-height bound used in Proposition H.3, namely Corollary 5.1, applies throughout this chord and its continuation $K(|\alpha_0| + \alpha_1^2) \le H < z_*$, including the equality face. It gives the third assertion, with the height scale frozen at each base point.

The full transported output, including reference values, has a common bound M_{out} on the original section windows. Choose m last with $m \le m_0$, $mM_{\text{out}} < a_{\text{cl}}/2$ and $m\max(C_B, C_H) < 1/2$. Then (BS1) gives $F' > 1/2$ on the compact, positive and high pieces, while the negative curvature gives $F'' > 0$ for $m > 0$. The enlarged estimates include threshold equalities and every nonclosed input. At $m = 0$, $F(x) = x$; collapsed sections denote only normalized one-sided limits. □

H.7 Growing exit

The growing displacement and its curvature. The action derivative of the matched map is known, but the finite closing is evaluated at its full translated output. We first bound that output and its first two action derivatives, then use the logarithmic slope to prove convexity in the original affine input. All phase derivatives fix every coefficient. Use the already constructed joint auxiliary tuple, the actual first lip $r = r_\ell$, and its original enlarged domain

$$\widehat{r}_{\text{ge}}^{\text{out}} \le r \le \bar\sigma/s_0, \qquad s_0 = \kappa\sqrt{B}, \qquad 0 < \sigma_0 < \bar\sigma < 1/2. \tag{GC1}$$

The counted interval has $r \ge \widehat{r}_{\text{ge}}$ and $s_0 r \le \sigma_0$. Write the actual affine raw input as

$$h = x_b + s_{\text{in}} P, \qquad s_{\text{in}} = \Omega a^{2-\lambda_\gamma} U_{\text{in}} > 0, \qquad \Omega = r_s^{2-\lambda_\gamma}, \qquad U_{\text{in}}^{\pm 1} \le C. \tag{GC2}$$

The joint entrance construction supplies $0 < P < \widehat{p}_*/2$. Set

$$X = \Phi_\theta(P)/\Phi_\theta(q_{\text{ref}}), \qquad \mathcal{E} = X\partial_X = \mathfrak{j}(P)\partial_P, \qquad \mathfrak{j} = \Phi_\theta/\Phi_\theta' > 0, \qquad e_s = \tau/\chi, \tag{GC3}$$

with the denominator held fixed. Let

$$\mathcal{P}(r) = P_s(-r)/B_\tau(-r), \qquad \mathfrak{d} = \mathcal{P}/r, \qquad \mathcal{D} = \mathfrak{d}\partial_r.$$

The polynomial estimates and the matched action derivative give

$$\begin{gathered} \mathcal{P} \asymp 1 + r, \quad \mathcal{P}' > 0, \quad \mathfrak{d}^{\pm 1} \le C, \quad |\mathcal{D}\log\mathfrak{d}| \le C, \quad e_s^{\pm 1} \le C, \quad \mathfrak{j}' > 3/4, \\ -\mathcal{E}r = e_s \mathfrak{d}\, n, \qquad n = 1 + \zeta, \qquad n \ge 1/2, \qquad |\zeta| + |\mathcal{E}\zeta| \le C\tau. \end{gathered} \tag{GC4}$$

Here $\mathcal{E}$ acts on the actual graph, including the negative variable in the complete quotient row.

Retain the full actual output

$$\begin{aligned} T(r) = \Xi(W_\theta(r)) = t_\dagger - a_{\text{grow}}\mathcal{B}(r), \qquad a_{\text{grow}} = \Omega s_0^2 U_{\text{out}} > 0, \qquad \mathcal{B}' = rG, \quad G > 0, \\ W_\theta' = -s_0^2 r\Gamma_\theta, \qquad G = \Gamma_\theta\, \Xi'(W_\theta)/(\Omega U_{\text{out}}). \end{aligned} \tag{GC5}$$

Thus T includes its coefficient-dependent translation, original moving endpoint, tails and regular output section map. The actual output lies in a fixed compact physical section window J_T. The first action derivative of the landing, composed with this same Ξ, satisfies

$$\begin{aligned}|\mathcal{D}\log G| &\le C\left\{\frac{\tau}{\sqrt{B}\,r^2}\left(1+\left|\log\frac{\tau}{\sqrt{B}\,r}\right|\right)+s_0\kappa\tau+s_0^2\mathcal{P}(r)\right\}\\ &\le C\{\tau(1+|\log\tau|)+s_0\}.\end{aligned}\tag{GC6}$$

The ordinary finite closing has $f_\theta(0)=0$, $f'_\theta>c_f>0$, and uniform ordinary C^3 bounds on a buffered interval containing mJ_T for $0\le m\le\bar m$. It is the original closing after the output map d_θ already retained in T. The physical affine displacement is exactly

$$F_\theta(h)=h-a_\theta-f_\theta(mT_\theta(h)),\qquad T_\theta(h)=T(r(P(h))).\tag{GC7}$$

Proposition H.12 (Convexity and zero count in the growing range)**.** *On the whole counted growing interval, there are common constants C_T, C_{cl} and a multiplier bound $m_0>0$, selected after the independent coefficient bounds, such that*

$$|T|\le C_T,\qquad 0<\mathcal{E}T\le C_T,\qquad |\mathcal{E}^2T|\le C_T.\tag{GC8}$$

For every actual through input in that interval, whether or not $F_\theta=0$,

$$\begin{aligned}\left|\mathcal{E}^j\{f_\theta(mT)-f_\theta(0)\}\right|&\le C_{\mathrm{cl}}m,\\ \left|\mathcal{E}^j\{\log f'_\theta(mT)-\log f'_\theta(0)\}\right|&\le C_{\mathrm{cl}}m,\qquad j=0,1,2,\quad 0\le m\le m_0.\end{aligned}\tag{GC9}$$

For $0<m\le m_0$, the complete physical displacement is strictly convex:

$$F''_\theta(h)>0\qquad \textit{at every point of the growing interval.}\tag{GC10}$$

Consequently it has at most one critical point, that point is simple and upward, and it has at most two isolated zeros counted with multiplicity. At $m=0$, $F'_\theta(h)=1$. The constants are uniform on the full prescribed passive coefficient family and both coefficient graphs, with the reciprocal estimates uniform for physical $j>0$ approaching its one-sided boundary.

Proof. Bounds on the full output and its action derivatives. Put $L(r)=-T_r=a_{\mathrm{grow}}rG(r)>0$ and

$$d_r=\min\{1,(\widehat r_{\mathrm{ge}}-\widehat r_{\mathrm{ge}}^{\mathrm{out}})/2\}>0.$$

The original lower buffer places every $[r-d_r,r]$ in (GC1). Since $\mathfrak{d}^{-1}$ is bounded, (GC6) gives

$$|\partial_r\log L|\le r^{-1}+\mathfrak{d}^{-1}|\mathcal{D}\log G|\le C_L.$$

Let D_T be the diameter of J_T. Using the actual output values,

$$D_T\ge T(r-d_r)-T(r)=\int_{r-d_r}^{r}L(v)\,dv\ge d_re^{-C_Ld_r}L(r).\tag{GC11}$$

Thus $L\le C$ without dividing by a_{grow}, Ω or a raw gain. Equation (GC4) now gives

$$\mathcal{E}T=e_s\mathfrak{d}\,Ln>0,\qquad \mathcal{E}\log(\mathcal{E}T)=-e_sn\{\mathcal{D}\log\mathfrak{d}+\mathcal{D}\log L\}+\frac{\mathcal{E}\zeta}{n},\qquad \mathcal{D}\log L=\mathfrak{d}/r+\mathcal{D}\log G.\tag{GC12}$$

Every term on the right is bounded. Hence $\mathcal{E}T\le C$ and $|\mathcal{E}^2T|\le C$; $|T|$ is bounded by the fixed compact J_T. The reference value of the output is included in this bound, so (GC8) applies at the actual argument of the closing map.

Composition with the finite closing. The full translated T and its first two action rows satisfy (GC8) on the buffered actual image. The ordinary C^3 closing has positive derivative units there. Lemma 3.3, specifically (3.4), therefore gives (GC9) for all through inputs, with constants fixed before m.

Strict convexity in the actual affine input. Before finite closing, the exact logarithmic slope is

$$\begin{aligned} H_0 &:= \mathcal{E}\log\frac{\mathfrak{j}(P)}{e_s G(r)\mathcal{P}(r)n(P,r)} \\ &= \mathfrak{j}'(P) + e_s n\left\{\frac{\mathcal{P}'(r)}{r} + \mathcal{D}\log G(r)\right\} - \frac{\mathcal{E}\zeta}{n}. \end{aligned} \tag{GC14}$$

This identity follows from $\mathcal{E}r = -e_s\mathfrak{d}n$; the derivative of the matched quotient is included. By (GC4), (GC6) and $\mathcal{P}' > 0$,

$$H_0 \geq 3/4 - C\{\tau(1 + |\log\tau|) + s_0\} > 1/2 \tag{GC15}$$

after common independent τ, κ bounds. Here $s_0 = \kappa\sqrt{B}$, with B a positive unit. The original passive coefficient set and joint auxiliary tuple are unchanged.

Since $\mathcal{E}T = \mathfrak{j}T_P$ and (GC5) gives $T_P = a_{\mathrm{grow}} e_s G\mathcal{P}n/\mathfrak{j} > 0$, define

$$\mathscr{R} = \frac{s_{\mathrm{in}}}{m f'_\theta(mT)T_P} > 0.$$

Every factor in its numerator, a_{grow} and e_s is coefficient-only. Its complete logarithmic row is therefore

$$H_m := \mathcal{E}\log\mathscr{R} = H_0 - m\frac{f''_\theta(mT)}{f'_\theta(mT)}\mathcal{E}T. \tag{GC16}$$

The absolute output estimate (GC8) makes the closing term at most $C_m m$, with a common C_m fixed before the multiplier. Choose last $m_0 \leq \min\{\bar{m}, 1, 1/[4(1 + C_m)]\}$. Then $H_m > 1/4$.

Restore the physical input using (GC2). Direct differentiation of the same displacement (GC7) gives at every through point

$$F'_\theta(h) = 1 - \frac{1}{\mathscr{R}}, \qquad F''_\theta(h) = \frac{\mathscr{R}_h}{\mathscr{R}^2} = \frac{H_m}{s_{\mathrm{in}}\mathfrak{j}(P)\mathscr{R}} > 0. \tag{GC17}$$

The two raw gains remain in this expression, so its sign holds even when either gain is arbitrarily small. Translation to $h - a_\theta$ leaves these derivatives unchanged.

Lemma 3.1 now gives the critical and zero counts, with multiplicity; (GC17) excludes an identity interval.

The bound (GC6) uses $r \geq R_0$ and $s_0^2\mathcal{P}(r) \leq Cs_0$; its logarithm contains no j. The bounds on κ, τ are independent, uniformly on the prescribed compact j ranges. □

H.8 Finite height and fixed exit

H.8.1 The last two passage regions and their physical curvature

Bounded amplified height is described by two small-power section maps. For larger height we instead use the action coordinate on a fixed negative section. We construct both maps on overlapping domains before comparing their derivatives in the physical input.

Retain the single joint auxiliary choice, its original physical sections and nested windows, the weak cuts, and the original coefficient family. Let I_ℓ be the positive-height first-lip domain in the original through interval. Write

$$P = \frac{h - x_b}{s_{\mathrm{in}}}, \qquad s_{\mathrm{in}} = \Omega a^{2-\lambda_\gamma} U_{\mathrm{in}} > 0, \qquad \Omega = r_s^{2-\lambda_\gamma}, \qquad q_s = \frac{\beta}{\chi} = \kappa Q_d, \tag{FVFE1}$$

where h is actual incoming height, Q_d, U_{in} are positive units, and no positive lower bound for either raw first gain is imposed. The variable P is the canonical raw input of the joint entrance construction; it is distinct from the saddle ratio q_s. Use the unchanged raw middle coordinate

$$V = \frac{y_+(h)}{\varepsilon_+}, \qquad \varepsilon_+ = \frac{\ell_0 h_-}{a^2 J_m(u_-)}, \qquad \sigma = s_0 r_\ell. \tag{FVFE2}$$

Here y_+ is the actual height w/a^2 at $u = u_+$. Every phase derivative fixes all coefficients and all gains.

Proposition H.13 (Physical passage and convexity in the last two ranges)**.** *For the joint auxiliary choice, followed by independent active coefficient bounds, put*

$$J_{34} = I_\ell \cap \{\sigma \geq \sigma_1\}, \qquad D = J_{34} \cap \{0 < V \leq V_1\}, \qquad E = J_{34} \cap \{V \geq 1\}, \qquad V_1 > 1. \tag{FVFE3}$$

These are intervals in the same actual incoming height, with every displayed divider equality included. Each lies in the prescribed enlarged domain of its corresponding actual transition, and $D \cup E = J_{34}$. Every input in these sets, whether closed or nonclosed, has its original raw trace in $0 < P < q_{\mathrm{ref}} < \widehat{p}_* < \widehat{p}_0$. The finite-$V$ map uses the original early negative cut on the fixed enlarged interval $0 < V \leq V_1 + 1$. For larger amplified heights, the independently constructed fixed-exit map uses the buffered shooting chord on $u = u_c$. The two constructions agree on $1 \leq V \leq V_1$ by original-time uniqueness, so their union is the same physical transition on J_{34}.

For the original return write

$$\mathcal{F}(h) := F(h - a_\theta) = h - a_\theta - f_\theta(mT(h)). \tag{FVFE4}$$

The map T includes the actual negative output adapter. For every $0 < m \leq m_0$,

$$\mathcal{F}''(h) > 0 \qquad (h \in J_{34}). \tag{FVFE5}$$

Thus the whole combined interval has at most one simple upward critical point and at most two isolated zeros with multiplicity. At $m = 0$, $\mathcal{F}' = 1$. The whole carrier is counted separately in Proposition H.14.

Proof. The shooting chord and its phase coordinate. Part (i) of Proposition H.10 realizes $\overline{\mathcal{J}_c^{\sharp\sharp}} \Subset (0, u_c^2/2)$ as unique through orbits, identified with the backward shots by uniqueness. Proposition E.5 applies to this entire buffered chord: its scalar first-exit construction is independent of the positive raw-section dictionary. Its extracted phase $\Psi_\theta(W)$ converges in C^2 to $\Gamma_c \circ \mathcal{W}^{-1}$, whose derivative is $1/H_{0,c}(r) > 0$. Hence its increasing inverse $W = \mathcal{H}_\theta(\Theta)$ has bounded first two derivatives and a positive derivative unit on

$$\Theta_- \leq \Theta \leq \Gamma_- - \Delta, \qquad \Theta_- < 0, \tag{FVFE8}$$

strictly inside its inverse image. The negative buffer comes from the original enlarged chord around u_-; the reciprocal functions give the same uniform gaps.

Continuation of the shooting chord to the raw input. The exact positive invariant, including its local unit functions, is

$$\Phi_\theta(P) = \{z_+ y_+(P)\}^\chi, \qquad \Phi_\theta(P) = PU_\theta(P), \qquad \Phi_\theta' > c, \qquad U_\theta^{\pm 1} \leq C. \tag{FVFE9}$$

Use the reference $P_{\mathrm{ref}} = a_{\mathrm{in}}$ of (H.45):

$$\Phi_\theta(P_{\mathrm{ref}}) = (z_+ \beta \varepsilon_+)^\chi, \qquad q_s \log \Phi_\theta(P_{\mathrm{ref}}) = -\Gamma_- + o(1). \tag{FVFE10}$$

The exact identity (H.44) proves this limit on both graphs: on the reciprocal graph use its prescribed d, j in place of $\mathcal{R}c, 1/c$. All factors remain regular at $j = 0$, so its error bound is unchanged and no $\beta \log j$ occurs. This reference corresponds to $V = \beta$. On the compact shooting chord the limiting gap from Γ_- has a fixed positive minimum d_{fx}. The scalar shooting estimate and (FVFE10) give, after shrink,

$$\beta \log V \le \Gamma_- - d_{\mathrm{fx}}/2, \qquad \beta \log y_+ \le -d_{\mathrm{fx}}/3. \tag{FVFE11}$$

Since $\beta \le C\chi$, this implies $y_+ < \chi^2$. Thus (FVFE9) can be inverted on the original invariant chord, and

$$\Phi_\theta(P) \le Ce^{-d_{\mathrm{fx}}/(3q_s)}, \qquad 0 < P \le e^{-d_{\mathrm{fx}}/(4q_s)} < \widehat{p}_0 \tag{FVFE12}$$

after a further independent κ bound. In particular the physical distance from the stopped trace satisfies $\sup(h - x_b) \le Ce^{-d_{\mathrm{fx}}/(4q_s)} \to 0$ on the entire shooting chord, since $s_{\mathrm{in}} \le C$. The two local points have the same invariant in the same larger coordinate box. Positivity and monotonicity along that invariant level give the intervening orbit segment; the independently actual raw continuation identifies its raw trace with the shot through orbit. This places the full chord within the independently selected fixed cutoff $\widehat{p}_0$, before any zero selection.

The bounded-V passage. Lemma E.1 independently continues the entire larger interval $0 \le V \le V_1 + 1$ to the early cut $u_0 = -1 + \eta_0 > u_-$. Equations (E.13) and (E.14) give its positive invariant at most $Ce^{-c_0/(2q_s)}$. The inversion used in (FVFE12) places the whole interval strictly inside the raw cutoff. The assigned set D is its restriction to $0 < V \le V_1$. On the negative early cut, the lemma's exact invariant is

$$I_-(w) = \alpha_\theta R, \qquad R = \beta^{-\beta} C(v)^\beta, \qquad v = \widehat{V} = V + o_{C^3}(\chi). \tag{FVFE14}$$

Its fixed cut choice puts the full $R \le 1 + o(1)$ interval strictly inside the outer positive-height chord. At $R = 0$ the output is the stopped height $w_- > 0$. Thus both actual tails, ordinary C^3 observers and finite closing are on their original buffered domains, with their exact values and positive first gains retained.

The domains of the two passage maps. The joint entrance supplies the actual product reference $r_\ell = \sigma_1/s_0$ after the active shrink. Direct landing followed by the independent inverse shooting phase gives

$$\Theta(h_{\sigma_1}) \longrightarrow \Gamma_c(-\sigma_1), \qquad \Gamma_- - \Gamma_c(-\sigma) = \log \frac{c + \sigma}{c(1 - \sigma)}. \tag{FVFE15}$$

On the reciprocal chart the gap is $\log((1 + j\sigma)/(1 - \sigma))$, uniformly positive on the fixed product buffer. The retained choice of Δ and its strict buffer imply

$$0 < g_0 \le \Theta(h_{\sigma_1}) \le \Gamma_- - 2\Delta. \tag{FVFE16}$$

The actual orders $r_{\ell,h} < 0$ and $V_h > 0$ imply that every member of E has $0 \le \Theta = \beta \log V \le \Theta(h_{\sigma_1})$. Thus the entire fourth range lies in the domain (FVFE8) of the inverse phase map, including inputs whose orbits are not closed. The equality $V = 1$ has the negative-action buffer there. Since $\mathcal{H}_\theta([0, \Gamma_- - 2\Delta]) \Subset \mathcal{J}_c^\sharp \Subset \mathcal{J}_c^{\sharp\sharp}$ uniformly after shrink, its whole closed output-height subset has a strict actual shooting buffer. Decrease the independent coefficient bounds further so that $\beta \log(V_1 + 1) < g_0/2$. Then $V(h_{\sigma_1}) = \exp(\Theta(h_{\sigma_1})/\beta) > V_1 + 1$, so the third range retains the full buffered finite-V passage. The two constructions identify the same raw inputs and outputs on $1 \le V \le V_1$ by original-time uniqueness. No physical input is relabelled at $V = 1$. Their actual $\sigma \ge \sigma_1$ domain has $r_\ell \ge \sigma_1/s_0 > r_{\mathrm{ge}}$, so the joint entrance's full first-hit order supplies the sharper raw bound $P < q_{\mathrm{ref}} < \widehat{p}_* < \widehat{p}_0$.

Physical convexity for bounded V. For $m > 0$, the actual-section construction gives exactly

$$\begin{aligned} h &= x_b + P_* K(v), \qquad f_\theta(mT(h)) = b_* + a_* H(v), \\ K(v) &= \Psi(v^\chi), \quad H(v) = D(\beta^{-\beta} C(v)^\beta), \qquad P_*, a_* > 0. \end{aligned} \tag{FVFE17}$$

Here all regular tails and finite closing are already included. Put $e_C = vC'/C > 0$. Lemma E.1 supplies the stronger normalized estimates below, with Σ_+ the coefficient-only positive invariant scale in its construction:

$$\begin{aligned} e_C &= 1 + O(\epsilon_\theta), \qquad \left| \frac{ve_C'}{e_C} \right| \le C\epsilon_\theta, \qquad |D''/D'| \le C, \\ \sup_t |t\Psi''(t)/\Psi'(t)| &\le C\Sigma_+, \qquad \epsilon_\theta \le C\beta^{-N} e^{-c_0/(3\beta)}, \qquad \Sigma_+ \le Ce^{-c_0/(2q_s)}. \end{aligned}$$

The exact Wronskian row, with $R = \beta^{-\beta} C(v)^\beta$, is

$$\begin{aligned} \mathcal{Q}(v) &= v\left(\frac{K''}{K'} - \frac{H''}{H'} \right) \\ &= \chi - \beta e_C - \frac{ve_C'}{e_C} + \chi v^\chi \frac{\Psi''}{\Psi'} - \beta e_C R \frac{D''}{D'} \ge \chi/2 > 0. \end{aligned} \tag{FVFE18}$$

On this interval $R \le 1 + o(1)$. With $q_s = \beta/\chi = \kappa Q_d \longrightarrow 0$, the displayed estimates give the relative bound

$$\frac{\mathcal{Q}(v)}{\chi} \ge 1 - Cq_s - C\frac{\epsilon_\theta}{\chi} - C\Sigma_+ \ge \frac{1}{2}.$$

Indeed, $\beta = q_s\chi$ implies

$$\frac{\epsilon_\theta}{\chi} \le Cq_s \beta^{-N-1} e^{-c_0/(3\beta)} \longrightarrow 0.$$

The input-section error has weight $\chi\Sigma_+$, whereas the bounded output and closing error has weight β. Thus independent bounds on κ, τ give this sign for every $v > 0$, without extending the small-power compositions in ordinary C^3 to the excluded stopped endpoint. Since $dh/dv = P_* K' > 0$, differentiation of the exact physical displacement, without multiplying it by a nonconstant factor, yields

$$\mathcal{F}''(h) = \frac{a_*}{P_*^2} \frac{H'}{(K')^2} \frac{\mathcal{Q}(v)}{v} > 0. \tag{FVFE19}$$

Physical convexity on the fixed-exit interval. The assigned set E has $0 \le \Theta \le \Theta(h_{\sigma_1})$, strictly inside the larger buffered action interval (FVFE8). Use the common actual map (H.37), $\Xi(W) = t_b + a_{\text{out}} D^n(W)$, $a_{\text{out}} = \Omega U_{\text{out}} > 0$, on its whole buffered height chord. Its ordinary C^3 normalized bounds and $0 < \Omega \le C$ retain the full output in the original raw and closing windows. For the fixed closing buffer $a_{\text{cl}} > 0$, impose on the whole chord

$$m_0 \left\{ |t_b| + a_{\text{out}} \sup_{\mathcal{J}_c^{\sharp\sharp}} |D^n| \right\} < a_{\text{cl}}/2. \tag{FVFE20a}$$

The braces are uniformly bounded on the original output window, so this selects m independently of the first gain.

Put $\widehat{H} = D^n \circ \mathcal{H}_\theta$. The exact raw formulas are

$$T(h) = t_b + a_{\text{out}} \widehat{H}(\Theta(P)), \qquad \Theta(P) = \beta \log \beta + q_s \log \frac{\Phi_\theta(P)}{\Phi_\theta(P_{\text{ref}})}. \tag{FVFE21}$$

Writing $p = h - x_b = s_{\mathrm{in}}P$, retain the raw gain in the two rows

$$0 < pT_h = a_{\mathrm{out}}\widehat{H}' q_s \frac{P\Phi'_\theta}{\Phi_\theta} \le C a_{\mathrm{out}} q_s, \tag{FVFE22}$$

$$p\frac{T_{hh}}{T_h} = P\frac{\Phi''_\theta}{\Phi'_\theta} - P\frac{\Phi'_\theta}{\Phi_\theta} + q_s P\frac{\Phi'_\theta}{\Phi_\theta}\frac{\widehat{H}''}{\widehat{H}'} = -1 + O(P + q_s) \le -\tfrac{1}{2}. \tag{FVFE23}$$

The ordinary C^3 closing has derivative units and bounded pre-Schwarzian on the whole sampled interval. (FVFE22)–(FVFE23) give

$$p\left\{\frac{T_{hh}}{T_h} + m\frac{f''_\theta(mT)}{f'_\theta(mT)}T_h\right\} \le -\tfrac{1}{2} + Cma_{\mathrm{out}}q_s < -\tfrac{1}{4}. \tag{FVFE24}$$

Therefore

$$\mathcal{F}''(h) = -mf'_\theta(mT)T_h\left\{\frac{T_{hh}}{T_h} + m\frac{f''_\theta(mT)}{f'_\theta(mT)}T_h\right\} > 0. \tag{FVFE25}$$

No lower bound for s_{in}, Ω, or T_h was used.

The positive input order makes J_{34} one interval. The two convexity estimates concern the identical physical F on an overlapping cover, so (FVFE5) holds on their union. Lemma 3.1 gives the critical and zero counts with multiplicity, also for every finite collection approaching the excluded stopped boundary. The endpoints at the phase boundaries lie inside the enlarged domains of their respective passage maps. At $m = 0$ the original completed displacement has derivative one. The combined reference logarithms, normalized section maps and first-cocycle powers above are regular at the reciprocal face; none introduces a $\beta \log j$ or $\tau \log j$ restriction. Actual physical maps are asserted for $j > 0$, with $j = 0$ a one-sided normalized limit. □

H.9 Counting zeros on the complete degree-loss interval

Each local calculation either gives strict convexity of the physical displacement or excludes critical points by a positive derivative. Because the local maps agree on overlapping intervals, the same alternative holds at every point of the connected through interval.

Proposition H.14 (Physical counting on the degree-loss carrier)**.** *Fix the compact positive-unit ranges of the degree-loss graph, or the reciprocal graph with $0 < j \le j_0$. Use the joint auxiliary choice of the growing entrance and its actual gates. There are independent active coefficient bounds, a common raw-height cap and a last finite-closing multiplier bound with the following property. On the resulting connected interval I_θ of actual through inputs on the original section, the complete affine displacement*

$$F_\theta(x) = x - f_\theta\big(m\mathfrak{T}_\theta(a_\theta + x)\big), \qquad x = h - a_\theta, \tag{PC1}$$

has at most one critical point. Every critical point is a simple upward crossing of F'_θ. Consequently it has at most two isolated zeros, counted with multiplicity, and no identity interval. The interval I_θ consists of the same through word in the original nested windows, with positive passage denominator and $0 < z(\gamma_s) < z_{\mathrm{cap}}$ for one fixed physical height cap. The assertion holds at every actual input of I_θ, without assuming its orbit is closed. The original field, marked sections, nested raw windows, L, h_L and closing map define this same return. Here $\mathfrak{T}_\theta = d_\theta \circ T_\theta^{\infty}$ is the original transported output and $f_\theta = C_\theta - C_\theta(0)$. Thus $f_\theta(0) = 0$ and at $m = 0$ the displacement is $F_\theta(x) = x$. No physical passage at $j = 0$ or at a stopped boundary is asserted.

Proof. Proposition H.10 supplies the closed low cover $B_\theta \cup G_\theta \cup D_\theta \cup E_\theta$ and its overlapping high continuation on the connected actual interval I_θ. It includes every divider equality and every nonroot input; only the stopped endpoint is excluded. All restrictions have the same affine displacement (PC1).

On the annular part of B_θ, Lemma H.2 gives $\Pi'' < 0$ after the ordinary C^3 divided closing is included. Its principal margin is $\lambda_1/(2x_{\max}R_{\mathrm{BD}})$; the independent active bounds make the errors smaller than half this margin. Lemma H.11 gives the same sign on negative source inputs and $F' > 1/2$ on the compact, positive and high source pieces. Propositions H.12 and H.13 give $F'' > 0$ on G_θ and on the whole combined $D_\theta \cup E_\theta$. These signs are in the original affine input and include both exact raw gains and the translated output inside the closing map.

Take the finite intersection of the independent active bounds, then one multiplier bound for all derivative estimates and the full absolute output envelope in the original closing buffer. Every critical point of (PC1) consequently has $F'' > 0$, including those on the dividers. Lemma 3.1 gives at most one simple upward critical point and the two-zero bound with multiplicity. An identity interval would contain a critical point with $F'' = 0$. Section 5 assembles the coefficient cover and Hausdorff capture. □

The growing-action estimates can now be applied on the physical interval. They control the matched quotient, which determines curvature before finite closing:

$$\mathcal{E} \log \frac{\mathrm{j}(P)}{e_s G(r)\mathcal{P}(r)n(P,r)} > \frac{1}{2}, \qquad \mathcal{E} = X\partial_X. \tag{H.124}$$

This is (GC14)–(GC15). To compose with the finite closing, we also need the full output and its first two action derivatives:

$$|T| + |\mathcal{E}T| + |\mathcal{E}^2 T| \le C, \qquad \mathcal{E} = X\partial_X, \quad \mathcal{E}T > 0. \tag{H.125}$$

Estimate (GC8) gives this bound on the buffered growing image, and (GC9) transfers it through the finite closing.

H.10 A direct Jacobian estimate

In the following two sign ranges, the graph Jacobian itself controls the complete derivative. An exact logarithmic derivative separates the endpoint factor from an interior term whose integral has a uniform upper bound, so no action compactification is needed.

Lemma H.15. *Assume either $\alpha_0 \le 0, \alpha_1 \le 0$, or $\alpha_0 = -s^2 < 0, \alpha_1 = sp > 0, \gamma > 0$ and*

$$\frac{s}{\gamma^{3/2}} \le k_0, \qquad \frac{p}{\sqrt{\gamma}} \le \pi_0 < \sqrt{12}.$$

On the physical input interval selected by Lemma 2.5, the central transition has a uniformly bounded positive derivative. After the final choice of the finite saddle–node multiplier, the complete return is a contraction.

Proof. Put $\xi = v - \gamma$, the local spatial coordinate of (4.2), and set

$$M = -P(\gamma + \xi) = -\alpha_0 - \alpha_1\xi + 3\gamma\xi^2 + \xi^3, \qquad \mathscr{D} = M + \ell z = -\dot{v}.$$

Along a through orbit, regarded as a graph $z = z(v)$, write

$$f = \frac{z\{q_\gamma\xi + \xi^2 + Cvz\}}{\mathscr{D}}, \qquad \nu = \frac{6\gamma}{q_\gamma}, \qquad N = -\alpha_1 + (3-\nu)\xi^2.$$

Direct differentiation gives

$$f_z = \frac{d}{dv}\log\frac{z^{1+\nu}}{\mathscr{D}} + \frac{N + \{\ell'(v) + (1-\nu)Cv\}z}{\mathscr{D}}. \tag{H.126}$$

If $\alpha_1 \le 0$, then $N \ge |\alpha_1| + 2\xi^2$ after shrinking $|\gamma|$. Let $\kappa = \frac{1}{2}\inf \ell$. On $\mathscr{D} \ge \kappa z$, the last quotient in (H.126) is bounded below. On the complement, $M < 0$,

$$z < \frac{|M|}{\ell - \kappa}, \qquad |M| \le C(\delta + |\gamma|)N.$$

Choosing the fixed raw width δ and then the coefficient neighbourhood makes the numerator at least $N/2$ there. Thus its integral in the physical direction from the positive to the negative section is bounded above.

In the second case put $\xi = (s/\sqrt{\gamma})\widehat{x}$, with $\pi = p/\sqrt{\gamma}$ and $k = s/\gamma^{3/2}$. Then

$$M = s^2\{1 - \pi\widehat{x} + 3\widehat{x}^2 + k\widehat{x}^3\}.$$

Where $N < 0$, the quadratic identity

$$1 - \pi\widehat{x} + 3\widehat{x}^2 = 3\left(\widehat{x} - \frac{\pi}{6}\right)^2 + 1 - \frac{\pi^2}{12}$$

gives a positive multiple of $1 + \widehat{x}^2$. On $N < 0$ one has $\widehat{x}^2 \le C\pi/k$, hence $|k\widehat{x}| \le C\sqrt{\pi_0 k_0}$. The cubic term is therefore absorbed by decreasing k_0. Since $-N \le sp$, the change of variable cancels the prefactor $sp\,s/(s^2\sqrt{\gamma}) = \pi \le \pi_0$, and hence

$$\int_{\{N<0\}} \frac{-N}{\mathscr{D}}\,d\xi \le C\int_{\mathbb{R}} \frac{d\widehat{x}}{1+\widehat{x}^2} < \infty.$$

Where $N \ge 0$, the preceding two-denominator argument applies. Indeed $M < 0$ can then occur only at $\xi = -u$, $u > 3\gamma$, and $N \ge cu^2$, $|M| \le u^3 \le \delta u^2$.

Integration of (H.126) yields

$$0 < T'_\theta \le C\frac{(z^{1+\nu}/\mathscr{D})_-}{(z^{1+\nu}/\mathscr{D})_+} \le C_0.$$

The two endpoints lie in the fixed positive-height windows, which bound the quotient in this estimate. The estimate holds on the full interval $\mathcal{T}_\theta \cap \operatorname{int} K_+^\sharp \cap T_\theta^{-1}(\operatorname{int} K_-^\sharp)$, including all intervening captured inputs. The bounded divided closing of Lemma 3.3, followed by a final multiplier shrink, gives the contraction. □

I Nonnegative opening: determinant and trace faces

The high-height estimate reduces the remaining problem to bounded rescaled heights. At their lower boundary, the determinant chart has a hyperbolic saddle, while the trace chart includes its saddle–node limit. The determinant count requires the absolute first variation, including both tail gains. The trace count also requires the additional outgoing saddle. We treat both passages in the affine input and with the divided closing of Lemma 2.2.

Lemma I.1 (Finite coefficient–phase atlas)**.** *Fix the signed high-height constant K_H and put $K = 2K_H$. For $\alpha_0 \ge 0$, below $z(\gamma) \le K(\alpha_0 + \alpha_1^2)$, the nonzero coefficients have the overlapping descriptions*

$$\alpha_0 = r_D^2 > 0, \qquad \alpha_1 = r_D A_D, \qquad |A_D| \le P_D, \tag{I.1}$$

$$\alpha_0 = r_L^2 a_L, \qquad \alpha_1 = \sigma r_L, \qquad 0 \le a_L \le a_*, \quad \sigma \in \{-1, 1\}, \tag{I.2}$$

where $P_D > 2/\sqrt{a_*}$. The determinant and signed-trace receivers below cover every low-height through input, including regular continuations, by overlapping stopped and compact strict intervals. Their cuts precede a displacement zero. If $\alpha_0 = \alpha_1 = 0$, every positive-height through input is in the signed high-height range.

Proof. Use (I.1) when $|\alpha_1| \le P_D\sqrt{\alpha_0}$; otherwise set $r_L = |\alpha_1|$, so $a_L < a_*/4$. The face $\alpha_0 = 0$, $\alpha_1 \neq 0$, has $a_L = 0$. These are whole-coefficient alternatives with a strict overlap and no restriction on either γ/r_D or γ/r_L. Their bounded normalized heights are respectively $K(1 + P_D^2)$ and $K(1 + a_*)$. Their physical phase covers are constructed in Lemma I.2 and Proposition I.3. For the trace limiting field $X' = a_L + \sigma X - Y$, $Y' = -XY$, at $a_L = 0$ its centre manifold has $Y = \sigma X + O(X^2)$, hence $\dot{Y} = -\sigma X^2 + O(X^3)$; its weak quadratic coefficient is nonzero. The double-zero coefficient face has no positive low-height input, and is covered by Corollary 5.1. □

I.1 Preparation on the actual determinant sections

Lemma I.2 (Determinant section preparation)**.** *Fix $P_D < \infty$, the passive compact coefficient block, and the original buffered windows. For independent sufficiently small $r = r_D > 0$ and $|\gamma|$, $\alpha_0 = r^2$, $\alpha_1 = rp$, $|p| \le P_D$, choose a fixed positive unscaled raw-height cap z_*. On the actual interval*

$$\mathcal{I}_\theta = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < H_\theta(x) < z_*\}, \qquad H_\theta(x) = z_\theta(\gamma; a_\theta + x), \tag{I.3}$$

the stopped transition has the exact composition (I.15)*, with the two absolute positive gains* (I.14) *and actual cut gap* (I.5)*. The normalized end maps and their inverses have common ordinary C^3 bounds and derivative units. The incoming and outgoing local chords are independent. The compact-spectrum saddle estimate* (I.16) *and the complete absolute variation* (I.17) *hold, including saddle ratio one. Every phase derivative fixes all coefficients and the actual m.*

The low stopped and compact strict receivers, their full regular continuations, and the signed high receiver cover $\mathcal{I}_\theta$ as in (I.18)*. On the strict and high pieces the full Π', including divided closing, is uniformly bounded. Retain intervening inputs, equality dividers and all original clips; exclude natural stopped endpoints. The parameters m, r, γ and the two local chord lengths are independent.*

Proof. We construct the two section maps with their absolute gains before estimating the stopped derivative. Keep $x = h - a_\theta$, $\Pi = \mathcal{C}_{\theta,m} \circ T_\theta(a_\theta + \cdot)$, and all original finite-side factors. The original domain is

$$J_\theta^{\mathrm{out}} = \mathcal{T}_\theta \cap \operatorname{int} K_+^\sharp \cap T_\theta^{-1}(\operatorname{int} K_-^\sharp) \cap \{\sigma_\theta < d_{\mathrm{out}}\}.$$

Order of the raw passage and central height makes (I.3) an interval, before capture is invoked. With $v = \gamma + rX$, $z = r^2 Y$, a positive time scaling gives

$$\begin{aligned} X' &= 1 + pX - 3\gamma X^2 - rX^3 - \ell(\gamma + rX)Y, \\ Y' &= -Y\{q_\gamma X + rX^2 + Cr(\gamma + rX)Y\}, \qquad q_\gamma = 1 + 2\gamma. \end{aligned} \tag{I.4}$$

At $r = \gamma = 0$, the saddle $(0, 1)$ has eigenvalues $(p \pm \sqrt{p^2 + 4})/2$. Their uniform nonzero moduli give common saddle boxes, transverse cuts $X = \pm L_c$, and persistent branch intersections for independent small r, γ. The incoming gap is exactly

$$\xi_D = Y(L_c) - Y_b^+(L_c) > 0. \tag{I.5}$$

For a limiting through graph starting at $Y(0) > 1$, put $D = Y - 1 - pX$. The identity $\dot{D}|_{D=0} = -XY$ excludes a first exit from $D > 0$ on the negative half-axis; reverse time

treats the positive half-axis. Thus every such graph is strict with quadratic tails. The gap is continuous, strictly increasing, and tends uniformly to zero at the stopped boundary. A fixed larger strict graph bounds the cut heights by order. On any bounded central-height range, the complement $\xi_D \geq \xi_*/2$ therefore has a finite compact strict cover with positive finite-cut height and speed margins.

Choose enlarged tail cuts $\pm L_t$ and buffered auxiliary ports with $0 < d_\epsilon < d_+$, joined exactly to the original sections. For $X = \epsilon R$, $\epsilon = \pm 1$, the outward equation is exactly (C.9) with $(a, s) = (1, r)$:

$$Y_R = F^\epsilon(R, Y) := F_\epsilon(R, Y)|_{a=1,s=r}. \tag{I.6}$$

The enlarged corridor has Y and its denominator comparable to R^2 up to $d_\epsilon/r < d_+/r$. Choose $|\lambda_\gamma| < \nu_0 < 1$. The intrinsic tail estimates with reference graph Y_* and first variation j give

$$F_Y = \lambda_\gamma/R + O(r + |p|R^{-2} + R^{-3+\nu_0}), \qquad j = (R/L_t)^{\lambda_\gamma} U_\epsilon, \quad U_\epsilon^{\pm 1} \leq C, \tag{I.7}$$

$$|F_{Y^h}| \leq C_h\{rR^{-2h+2} + |\gamma|R^{-2h+1} + |p|R^{-2h} + R^{-2h-1}\}, \qquad h = 2, 3. \tag{I.8}$$

The first remainder has bounded integral to the actual endpoint. Use the normalization of Lemma A.1,

$$\eta = (Y - Y_*)/j, \qquad \eta_R = N = \{F(R, Y_* + j\eta) - F(R, Y_*) - jF_Y(R, Y_*)\eta\}/j. \tag{I.9}$$

Its changed kernel weight is

$$\|N\|_{C^3} \leq C\{rR^{-2+\nu_0} + |\gamma|R^{-3+\nu_0} + P_D R^{-4+\nu_0} + R^{-5+\nu_0}\}. \tag{I.10}$$

Choose L_t large, then shrink $r, |\gamma|$ independently. The lemma with $k = 2$ supplies enlarged chords and smaller inverse images, remaining in the corridor, with

$$\|\mathcal{T}_\epsilon\|_{C^3} + \|\mathcal{T}_\epsilon^{-1}\|_{C^3} \leq C, \qquad 0 < c \leq \mathcal{T}_\epsilon' \leq C. \tag{I.11}$$

At the actual port the raw centre and gain are $z_{\mathrm{ref}} = r^2 Y_*(d_\epsilon/r)$, $d = r^2 j(d_\epsilon/r) > 0$, with $d \leq Cr^{2-\nu_0}$. The fixed-port proof of Lemma F.9 admits the whole raw images: choose the auxiliary abscissa ratio from the window reserve, then shrink the coefficients on the remaining compact bridges. That proof uses only the common quadratic corridors just established. On these images Lemma 3.3 gives the exact ordinary C^3 normalization

$$\widehat{G}(u) = \frac{G(z_{\mathrm{ref}} + du) - G(z_{\mathrm{ref}})}{G'(z_{\mathrm{ref}})d} = u + O_{C^3}(d), \tag{I.12}$$

with common inverse bounds. Its divided-closing rows (3.5) give, including $m = 0$,

$$\|\mathcal{C}_{\theta,m}\|_{C^3} \leq C, \qquad 0 < c \leq \mathcal{C}_{\theta,m}' \leq C. \tag{I.13}$$

Finite composition, with the positive-tail inverse on the incoming side, now gives the two exact absolute embedding gains

$$\sigma_\pm = r^{2-\lambda_\gamma} V_\pm, \qquad \lambda_\gamma = \frac{12\gamma}{1 + 2\gamma}, \qquad 0 < c \leq V_\pm \leq C, \tag{I.14}$$

and the complete stopped transition

$$x = x_b + \sigma_+\delta, \qquad \Pi(x) = y_b + \sigma_- O_\theta(D_\theta(I_\theta(\delta))), \quad I_\theta(\delta) = \xi_D, \quad 0 < \delta \leq \delta_* < 1. \tag{I.15}$$

The exact centres, cut factors, bridges and divided closing remain in these maps. I, O vanish at zero, have derivative units, and are ordinary C^3 with bounded inverses. The incoming and

outgoing saddle chords have separate buffers. Their common power makes the quotient of the two absolute gains a unit; the individual gains need no positive lower bound.

The saddle of (I.4) belongs to a compact analytic family: its two rates have uniform positive moduli and $\wp \in [\wp_0, \wp_1] \Subset (0, \infty)$. Reduce the independent incoming and outgoing chords separately to common compatible local sections; their actual gap holonomies are ordinary C^3 with positive derivative units. Lemma A.8 therefore gives a local map $cz^{\wp}e^{\mathcal{R}}$, with two Euler remainder rows uniform through $\wp = 1$. Differentiation replaces its remainder by $\Psi = \mathcal{R} + \log(1 + z\mathcal{R}'/\wp)$. The actual holonomies add $O(z) + O(z^{\wp_0})$ to its value and first Euler row. Thus, for fixed $0 < \alpha < \min\{1, \wp_0\}$, the actual gap-to-gap map satisfies

$$D'_\theta(z) = C_\theta \wp z^{\wp-1} e^{\Psi_\theta(z)}, \qquad C_\theta^{\pm 1} \le C, \qquad |\Psi_\theta| + |z\Psi'_\theta| \le C z^\alpha (1 + |\log z|). \tag{I.16}$$

The estimate remains uniform at ratio one because it uses no division by $\wp - 1$.

Put $i_0 = I'(0) > 0$, $o_0 = O'(0) > 0$. Differentiating the complete composition in its original affine input gives exactly

$$\begin{aligned} \Pi'(x_b + \sigma_+\delta) &= \frac{\sigma_-}{\sigma_+} O'(D(I(\delta)))D'(I(\delta))I'(\delta) = \mathcal{A}_\theta \delta^{\wp-1} e^{R_\theta(\delta)}, \\ \mathcal{A}_\theta &= \frac{\sigma_-}{\sigma_+} o_0 C_\theta \wp i_0^{\wp}, \quad 0 < c \le \mathcal{A}_\theta \le C, \qquad |R_\theta| + |\delta R'_\theta| \le C\delta^\alpha (1 + |\log \delta|). \end{aligned} \tag{I.17}$$

Here the full remainder is

$$R = \Psi(I(\delta)) + (\wp - 1)\log\frac{I(\delta)}{i_0\delta} + \log\frac{I'(\delta)}{i_0} + \log\frac{O'(D(I(\delta)))}{o_0}.$$

The ordinary C^3 bounds give $I/(i_0\delta) = 1 + O(\delta)$ in the required Euler rows, while $D(I)$ and its first Euler row are $O(\delta^{\wp_0})$. Thus every term has the stated bound. The common power in (I.14) cancels exactly, without restricting $\gamma \log r$; at $\wp = 1$ this full remainder is still used.

Finally put $S_D = r^2(1 + p^2)$. The actual set cover is

$$\begin{aligned} \mathcal{I}_{\mathrm{lo}} &= \{x \in \mathcal{I}_\theta : H_\theta \le 2K_H S_D\}, \qquad \mathcal{I}_{\mathrm{hi}} = \{x \in \mathcal{I}_\theta : H_\theta > K_H S_D\}, \\ \mathcal{I}_\theta &= \mathcal{I}_{\mathrm{lo}} \cup \mathcal{I}_{\mathrm{hi}}, \qquad \mathcal{I}_{\mathrm{lo}} \subset \{0 < \xi_D < \xi_*\} \cup \{\xi_D \ge \xi_*/2 \text{ in compact strict receivers}\}. \end{aligned} \tag{I.18}$$

The low bound $Y(0) \le 2K_H(1 + P_D^2)$ gives the finite strict cover above; shrink ξ_* so its stopped part lies in (I.15). On strict cores the finite-cut variations are units, and the common raw power (I.14) cancels before the actual bridges and divided closing bound Π'. The signed high-height corollary bounds the same derivative above.

The physical overlaps are $\xi_*/2 < \xi_D < \xi_*$ and $K_H S_D < H_\theta < 2K_H S_D$. Gap equality faces belong to the adjoining receiver, and both height equality faces are low. Window, speed and cap clips preserve the cover and all intervening inputs; a removed overlap places the connected retained interval in one adjoining receiver. All these choices are made before selecting a displacement root. □

I.1.1 The stopped trace passage

The trace chart contains a central saddle–node and an additional equatorial saddle in the same shrinking neighbourhood. The central normal form alone does not describe the return: its outgoing height must also pass through this saddle. We include both passages in the order of Figure 5.

Proposition I.3 (Uniform stopped-trace estimate)**.** *Fix $0 < p_0 < p_1 < \infty$, the passive compact coefficient block, and the original buffered raw windows. There are independent bounds*

$r_*, \gamma_* > 0$, an opening bound $a_* > 0$, a fixed positive unscaled raw-height cap h_*, and a final multiplier bound m_* such that, for

$$\alpha_0 = r^2 a_L, \qquad \alpha_1 = r p_L, \qquad 0 < r < r_*, \quad |\gamma| < \gamma_*, \quad |a_L| \le a_*, \quad p_0 \le |p_L| \le p_1,$$

the single original receiver

$$\mathcal{I}_\theta = \mathcal{I}_\theta(h_*) = \{x : a_\theta + x \in J_\theta^{\mathrm{out}}, \quad 0 < H_\theta(x) < h_*\}, \qquad H_\theta(x) = z_\theta(\gamma; a_\theta + x), \tag{I.19}$$

has $F' > 1/2$ everywhere for $p_L < 0$, and $F' = 0 \Rightarrow F'' > 0$ for $p_L > 0$, where $F(x) = x - m\Pi(x)$, $0 < m \le m_*$. It consequently has at most one or two isolated zeros, respectively, counted with total multiplicity. These conclusions apply to every retained interval with the original window, speed and height clips, including all intervening nonroot inputs and artificial equality dividers. Both signs of a_L, $a_L = 0$, the closed trace endpoints and empty receivers are included; natural stopped endpoints are excluded. The positive trace has a concave terminal receiver overlapping its compact strict complement. No relation among m, r, γ is imposed.

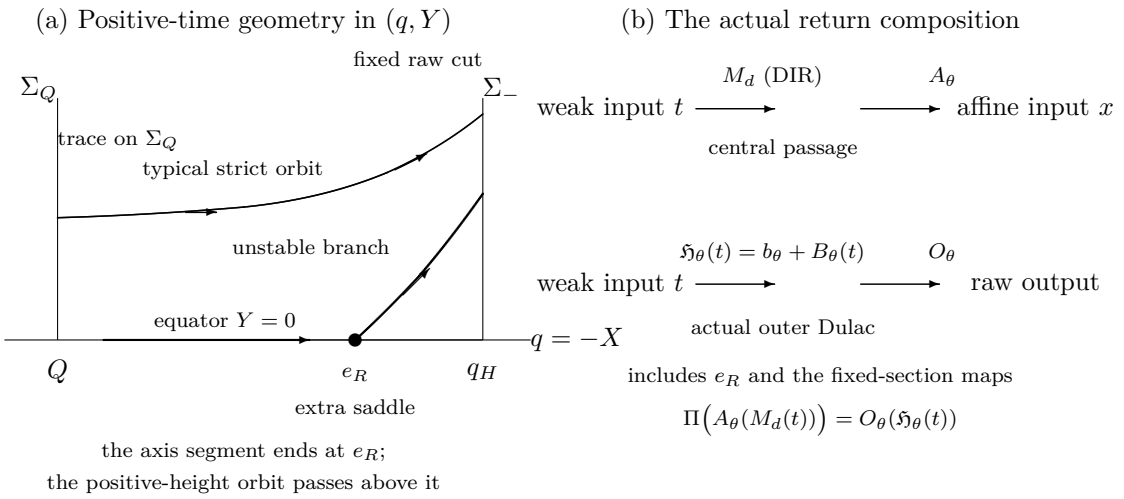


Figure 5: The stopped trace and the map used in its zero estimate. In panel (a), $q = -X$ increases in positive time. The broken limiting connection consists of the equatorial segment from the trace section Σ_Q to the additional equatorial saddle e_R, followed by its unstable upper-half-plane branch toward the fixed raw section Σ_-. The thinner curve is a representative strict positive-height orbit; it remains above the unstable branch on their common q-interval. The label *trace on* Σ_Q marks the incoming height supplied by the central saddle–node passage. All distances and slopes are schematic. Panel (b) records the two sides of the physical composition: the central saddle–node passage is the inverse Dumortier–Ilyashenko–Rousseau (DIR) normal-form map M_d, while the outgoing height has the retained offset b_θ and is then sent through the actual outer Dulac map O_θ, which includes the additional saddle and the fixed-section connectors. Arrows in panel (b) denote maps, not flow trajectories.

Proof. For positive trace, we first estimate the outgoing equatorial passage in its actual height coordinate. We then join it to the inverse central passage and the incoming section map. Their logarithmic derivatives, together with the regular connectors, give concavity near the stopped

orbit. A compact strict comparison and the high-height estimate extend the count over the whole receiver. For negative trace, a direct first-variation identity suffices.

Put $v = \gamma + rX$, $z = r^2 Y$, and $\ell(v) = 1 - Av + Cv^2$, $q_\gamma = 1 + 2\gamma$. A positive time change gives

$$\begin{aligned} X' &= a_L + p_L X - 3\gamma X^2 - rX^3 - \ell(\gamma + rX)Y, \\ Y' &= -Y\{(1 + 2\gamma)X + rX^2 + Cr(\gamma + rX)Y\}. \end{aligned} \tag{I.20}$$

Keep the original affine $x = h - a_\theta$ and $\Pi = \mathcal{C}_{\theta,m} \circ T_\theta(a_\theta + \cdot)$. Lemma 3.3 bounds divided closing through ordinary order three, including $m = 0$. Phase derivatives fix all coefficients and the actual multiplier. The two window admissions, height and speed clips in (2.22) make (I.19) an interval before capture.

Choose the finite normal-form order and an initial $|a_L| \le a_0$. Take the signed high-height constant, increase K_H so that $|\bar{a}| \le a_0/K_H$, $|\bar{p}| \le p_1/\sqrt{K_H}$, $\bar{Y}(0) = 1$ lies inside its compact source block, and put $K_L = 2K_H$. Fix $X = L_c$ beyond the central equilibria. The barrier estimate (I.35) below fixes a bounded actual gap range before the terminal chords are selected. Retain each auxiliary cut's exact continuation, centre and gain to the original port. First take $p_L \in [p_0, p_1]$.

The outgoing equatorial saddle. Set $q = -X$. The exact equations are

$$\begin{aligned} q' &= M(q) + L_-(q)Y, & M(q) &= -a_L + p_L q + 3\gamma q^2 - rq^3, \\ Y' &= Y\{qB(q) + \zeta(q)Y\}, & B(q) &= 1 + 2\gamma - rq, \\ L_-(q) &= \ell(\gamma - rq), & \zeta(q) &= Cr(rq - \gamma). \end{aligned} \tag{I.21}$$

Fix $w_0 = 1/8$, $w_1 = 3/4$, and an auxiliary raw width δ inside the original ports, sufficiently small for this fixed band. Choose a small a_0 and a large Q; then shrink $r, |\gamma|$, initially with $|a_L| \le a_0$. There is exactly one root $R > Q$ of M: $M(Q) > 0$, $M'(Q) > 0$, and M' has exactly one positive root, beyond which M decreases to $-\infty$. At R,

$$rR^2 = p_L + 3\gamma R - a_L/R, \qquad \kappa = -M'(R) = p_L + rR^2 - 2a_L/R \ge p_0/2.$$

Evaluating M on fixed compact intervals shows that $R \to \infty$. The identity $rR = 3\gamma + p_L/R - a_L/R^2$ then gives $rR \to 0$, uniformly in the retained parameters. The tangent and normal eigenvalues are $-\kappa$ and $RB(R) > 0$. Thus the root is a saddle, its incoming equatorial branch points from Q towards R, and its characteristic number satisfies

$$\beta_R = \frac{\kappa}{RB(R)} \longrightarrow 0.$$

Put

$$\begin{aligned} h(q) &= \frac{M(q)}{q^2 B(q)}, \quad b(q) = qh(q), \quad \mathcal{A}(s) = \int_Q^s \frac{dq}{b(q)}, \\ \mathcal{D} &= b(s)\partial_s, \qquad \epsilon = |a_L|/Q^2 + p_1/Q + |\gamma| + rR. \end{aligned}$$

This is the outgoing construction of Lemmas F.3 and F.4, with

$$(s, p, 1) \quad \text{in their axis polynomial replaced by} \quad (r, p_L, -a_L).$$

We verify the changed inputs. The first five Euler derivatives of $h = (-a_L/q^2 + p_L/q + 3\gamma - rq)/B(q)$ are $O(\epsilon)$ on $Q/2 \le q \le 2R$ and its fixed logarithmic collars; the same holds for $L_-/B - 1$ and $\zeta q/B$. Moreover $b > 0$ up to R, $|b'| \le C\epsilon$, and $\mathcal{A} \to \infty$ at R. On the fixed entrance interval,

$$M(q)/q \ge p_0 - 2a_0/Q - 3|\gamma|Q - rQ^2 \ge p_0/2 \qquad (Q/2 \le q \le Q).$$

After choosing $R_0 > 2Q$, the bound $b \le Cp_1$ on $[Q, R_0]$ follows by an independent shrink of $r, |\gamma|$.

For clarity, the two equations used in that construction are

$$\begin{aligned} u' &= h + cw, \qquad w' = w\{1 - 2h + (k - 2c)w\}, \\ u &= \log q, \quad w = Y/q^2, \quad c = L_-/B, \quad k = \zeta q/B, \end{aligned} \tag{I.22}$$

in positive normal time $d\tau_n = qB(q)\,dT$, and

$$\frac{dw}{d\log q} = F(\log q, w) = w\left\{\frac{1 + 2\gamma - rq + Crq(rq - \gamma)w}{L_-(q)w + 3\gamma - rq + p_L/q - a_L/q^2} - 2\right\}. \tag{I.23}$$

The first is unchanged in form, so Lemmas B.1 and B.2 give the strong footpoint and its actual fixed-cut height on $R_0 \le s < R$:

$$\log y = -\mathcal{A}(s) + 2\log s + \log U(s) + E_Q(y), \qquad 0 < y \le e_0 = C_Q R_0^2 e^{-c(R_0 - Q)/p_1}. \tag{I.24}$$

Here U and its reciprocal and logarithmic derivatives are bounded, $E_Q(0) = 0$, $\|E_Q\|_{C^3} \le C_Q$, and

$$\mathcal{D}\log y = -K, \qquad |K - 1| + |\mathcal{D}K| \le C\epsilon + C_Q e_0, \qquad K \ge \tfrac{1}{2}. \tag{I.25}$$

The passage stays in $Q/2 < q < 2R$, and $y \downarrow 0$ at R.

For the continuation, fix $|\gamma| < \delta/2$. The strong-section abscissa is $q_s(s) = sU_s(s)$, with $q_s' \asymp 1$ and three bounded logarithmic derivatives of U_s. On $w_0 \le w \le w_1$, the denominator in (I.23) is bounded below, $F(w_0) > c$, $F(w_1) < -c$, $F_w \le -1$, and its required state and logarithmic-radius derivatives are bounded. Since $q_s(R) < q_H = (\delta + \gamma)/r$, the same invariant-strip and moving-entrance argument as in (F.25)–(F.28) gives a unique raw exit and bounded logarithmic phase derivatives. Only their first row is needed here. The new term $-a_L/q^2$ has the same integrable weight as the former q^{-2}, with bounded coefficient; no reciprocal β_R is introduced.

The full raw image is admitted before composing the closing. Indeed, choose $0 < d_{\mathrm{r}} \ll \delta$, then require $2rR < d_{\mathrm{r}}$. Comparison from $rq = d_{\mathrm{r}}$ to the raw exit with $w_\Gamma(\xi) = (\sqrt{1 + \xi^2} + 1)^{-1}$ gives, uniformly in s,

$$|W_H(s) - w_\Gamma(\delta + \gamma)| \le Cd_{\mathrm{r}}/\delta + C_{d_{\mathrm{r}},\delta}(|A - 1| + |C - 1| + |\gamma| + rp_L + |a_L|r^2).$$

This is the raw comparison in the negative-gate proof with the same changed polynomial. Choose d_{r}/δ from the original window reserve, then shrink the coefficients; the raw height is $(\delta + \gamma)^2 W_H$, with organizing-height translation $O(\delta|\gamma|)$. The compact bridge and divided closing are therefore defined on the whole image. Their ordinary C^3 bounds give the completed output $\Gamma_-(s)$ with

$$\Gamma_-' < 0, \qquad |s\partial_s \log(-\Gamma_-')| \le C, \qquad |\mathcal{D}\log(-\Gamma_-')| \le C\epsilon.$$

The two physical radii $(\delta \pm \gamma)/r$ remain distinct in their absolute gains.

Define $O(y(s)) = \Gamma_-(s)$. The same logarithmic chain rule as in Proposition F.8 now gives

$$\begin{aligned} O'(y) &= \frac{(-\Gamma_-')b}{yK} > 0, \\ y\frac{O''}{O'} &= -1 - \frac{\mathcal{D}\log(-\Gamma_-') + b' - \mathcal{D}\log K}{K} = -1 + O(\epsilon + C_Q e_0) < -\tfrac{1}{2}. \end{aligned} \tag{I.26}$$

It holds on the actual height interval $0 < y < y(R_0)$, uniformly as $\beta_R \to 0$; no derivative at the excluded endpoint is used.

The central passage and its connectors. At $a_L = r = \gamma = 0$ the central eigenvalues are $p_L, 0$, with the nonzero weak quadratic coefficient of Lemma I.1. Apply the finite-smooth multiparameter orbital normal form of [3, Theorem 3 and formula (2.2), October 2000 author copy p. 6] to the reversed field, at the preselected joint order $N \geq 12$. Its positive orbital factor gives the inverse physical passage, with no scalar-opening rank condition.

The axis branch has $X_{\mathrm{ax}} = -a_L/p_L + O(a_L^2)$, $Y = 0$, and the interior branch satisfies

$$\partial_{a_L} Y_{\mathrm{int}}|_{a_L=0} = \frac{1+2\gamma}{\ell(\gamma)(1+2\gamma) + p_L C r\gamma} > c > 0.$$

Their weak separation is a_L times a positive unit; the strong rate stays nonzero. A finite compact cover gives common boxes and phase bounds across $a_L = 0$. The physical boundary is the interior saddle for $a_L > 0$, the axis saddle for $a_L < 0$, and their common root at zero. Center this weak root at $x_+(d)$, with the other at $x_+(d) - d$, $d \asymp |a_L|$. Only signed-side phase derivatives are needed.

In reversed normal-form time take $x' = g_d(x)$, $\eta' = -\eta$, a fixed strong entrance $\eta = \eta_0 > 0$, and a fixed weak exit $x = \rho$. The orbit from $(x_+(d) + t, \eta_0)$ reaches it at height $\eta_0 M_d(t)$. With bounded normal-form coefficient $\mathfrak{a}$, the two weak factors give

$$\begin{gathered} G_d(t) = \frac{t(t+d)}{1 + \mathfrak{a}(x_+(d)+t)}, \qquad g_d(x_+(d)+t) = G_d(t), \\ M_d(t) = \exp\left[-\int_{x_+(d)+t}^{\rho} \frac{dx}{g_d(x)} \right]. \end{gathered} \tag{I.27}$$

For small $d \geq 0$, $t > 0$,

$$M_d' = \frac{M_d}{G_d}, \qquad \frac{M_d''}{M_d'} = \frac{1 - G_d'}{G_d}, \qquad 1 - G_d' \geq \delta_G > 0, \quad G_d \asymp t(t+d). \tag{I.28}$$

The quantities G_d and M_d are uniformly small on a short terminal interval: $G_d \leq Ct(t + d_*)$ for $0 \leq d \leq d_*$, while the lower action bound $c\int_t^{\rho/2} du/[u(u+d_*)]$, with fixed small $d_* > 0$, diverges at zero. Keeping the two coalescing factors together makes this estimate uniform as their separation vanishes.

Absorb the fixed factor η_0 into the incoming section map. From the central strong exit to $q = Q$, the flight has fixed finite length and $q' \geq cq > 0$. Ordinary graph variations give

$$\begin{gathered} \mathfrak{H}_\theta(t) = b_\theta + B_\theta(t), \qquad B_\theta(0) = 0, \\ 0 < c_Q \leq B_\theta' \leq C_Q, \qquad |\mathfrak{H}_\theta''/\mathfrak{H}_\theta'| \leq C_Q, \qquad 0 \leq b_\theta \leq C_Q |a_L|. \end{gathered} \tag{I.29}$$

Here $b_\theta = 0$ for $a_L \leq 0$. For $a_L > 0$, the interior saddle's strong branch has a uniform spectral gap; its smooth finite flight to Q gives the positive offset. Retain this offset in O, using phase regularity on each signed side.

After Q, R_0 are fixed, $y_\theta(R_0)$ has a uniform positive lower bound. At $r = \gamma = 0$ it is the height of a finite regular passage on a compact parameter block, and continuity preserves positivity. Choose $y_* > 0$ below this bound. Only now shrink a_*, t_* so that

$$\mathfrak{H}_\theta([0, 2t_*]) < y_*/2 < y_\theta(R_0). \tag{I.30}$$

Thus every strict terminal orbit, including those with positive offset, lies in the domain of (I.26).

For the incoming connector, at $r = \gamma = 0$ the boundary branch beyond the positive local cut satisfies $X, Y > 0$ and $D = Y - a_L - p_L X > 0$. At $D = 0$, reverse time points into $D > 0$, since $\dot{D} = p_L D - XY$. Also $Y_X \leq 2X$ for large Y on bounded X-intervals. These facts

continue the branch to every fixed positive cut without loss of its graph or finite-X blow-up. At large fixed X,

$$\frac{dW}{d\log X} = \frac{W}{W - p_L/X - a_L/X^2} - 2W, \qquad W = Y/X^2.$$

Choose the cut so that $0 < p_L/X + a_L/X^2 < 1/4$. The right side is at least $1/2$ for $0 < W \le 1/4$ with $D > 0$, and is negative for $W \ge 1$. The compact family enters a common quadratic corridor at a fixed cut L_t, as does a small chord of strict incoming orbits for nearby coefficients. For nearby coefficients the rational tail equation has $F_W \le -1$ and bounded F_{WW}. Its first variation decays exponentially, so the relative-jet argument (A.3) bounds the pre-Schwarzian uniformly. The same fixed raw comparison as on the negative side admits the whole chord into the positive window. After the finite connector and original bridges, the actual input adapter therefore satisfies

$$x = A_\theta(M_d(t)), \qquad A'_\theta > 0, \qquad |A''_\theta/A'_\theta| \le C_Q. \tag{I.31}$$

Its absolute first derivative needs no lower bound.

Curvature and the overlap of actual phase intervals. The three constructed maps refer to the same orbit by flow uniqueness: A_θ continues to the actual input, $\mathfrak{H}_\theta$ to $q = Q$, and O_θ through the extra saddle and closing. Thus

$$x = A_\theta(M_d(t)), \qquad \Pi(x) = O_\theta(\mathfrak{H}_\theta(t)).$$

All factors are increasing. Put $H = \mathfrak{H}_\theta(t)$. Differentiation in t, keeping the affine input x in the displacement, gives

$$\begin{aligned} G_d \frac{d}{dt} \log \Pi'(x(t)) = G_d \frac{O''(H)}{O'(H)} H' + G_d \frac{H''}{H'} \\ - M_d \frac{A''(M_d)}{A'(M_d)} - (1 - G'_d). \end{aligned} \tag{I.32}$$

The first term is nonpositive by (I.26) and (I.30). The next two are bounded by $C_Q(G_d + M_d)$, whereas the last is at most $-\delta_G$. Shorten the terminal interval so that the whole expression is at most $-\delta_G/2$. Since $G_d > 0$ and $x'(t) > 0$, this proves $\Pi'' < 0$ in the original affine input.

Fix a positive local cut $X = L_c$ beyond the nearby central equilibria, and let $Y_b^+(L_c)$ be the incoming boundary height. The normalized actual gap is

$$\xi = Y(L_c) - Y_b^+(L_c) = \frac{z(\gamma + rL_c) - r^2 Y_b^+(L_c)}{r^2}. \tag{I.33}$$

Thus $r^2\xi$ is the actual raw gap on this section, with the same zero, ordering and orientation as ξ. Order and the compact local connector make $\xi(t)$ strictly increasing. More precisely, the connector on the fixed local cuts gives $\xi(t) = \psi_\theta(M_d(t)) - \psi_\theta(0)$, with $0 < c \le \psi'_\theta \le C$. Since $G_d(u) \ge cu^2$, the action in (I.27) at $t = t_*/2$ is at most C/t_*. Consequently $\xi_\theta(t_*/2) \ge c\exp(-C/t_*) > 0$, uniformly on the completed coefficient block, including $d = 0$. Choose

$$0 < 2\xi_0 < \inf_\theta \xi_\theta(t_*/2). \tag{I.34}$$

Then $0 < \xi \le 2\xi_0$ lies in the terminal interval. The whole lower range $Y_0 = H_\theta(x)/r^2 \le K_L$ reaches this same cut. Indeed its exact scalar graph is

$$Y_X = \frac{Y\{(1 + 2\gamma)X + rX^2 + Cr(\gamma + rX)Y\}}{\ell(\gamma + rX)Y - a_L - p_L X + 3\gamma X^2 + rX^3}. \tag{I.35}$$

The through denominator is positive. Above a fixed $M_0 > K_L + 1$ it is at least $\ell_0 Y/2$, and on $0 \le X \le L_c$ $Y_X \le C_{L_c} + Y$. Comparison after the first crossing of M_0 and Gronwall give the previously fixed bound

$$0 < \xi = Y(L_c) - Y_b^+(L_c) \le B. \tag{I.36}$$

Only original through admission is used here. The divider height is nonnegative and is subtracted on the same section.

On $\xi \ge \xi_0$ the limiting passage is uniformly separated from its stopped graph. To obtain this separation before constructing a strict cover, let t_{ref} be the unique phase with $\xi = \xi_0$. The uniform vanishing of $M_d(t)$ bounds t_{ref} away from zero. The standing normal form gives a nonbroken physical passage at every such positive phase: for $a_L > 0$ it crosses above the interior saddle with $Y_0 > a_L$, and for $a_L \le 0$ with $Y_0 > 0$. Take the positive minimum over the compact reference-phase and coefficient block at $r = \gamma = 0$; scalar order then gives

$$Y_0 - \max\{a_L, 0\} \ge \eta_0 > 0 \qquad (\xi \ge \xi_0).$$

For $D = Y - a_L - p_L X$, $D(0) > 0$ and $\dot D = p_L D - XY$ excludes a first zero in either relevant half-axis orientation. The bound $Y_X \le C + Y$ excludes finite-X blow-up. Thus the bounded complement has a compact strict-family cover with positive core height and speed margins. At large $|X|$, its $W = Y/X^2$ equation points upward below $1/4$ and downward above 1, so both half-tails enter quadratic corridors at a common fixed cut. Their actual outward continuations and fixed raw comparison give both original-window margins. In particular $\xi_0 \le \xi \le 2\xi_0$ is a physical terminal/strict overlap.

For a strict limiting graph, Proposition C.2 with $a = a_L$, $p = p_L$, $b = B_+$, and $\sigma = \partial_b Y > 0$ gives

$$\frac{S''}{S'} = -2\int_{\mathbb{R}} \frac{X(a_L + p_L X)\sigma}{D^3}\, dX = -p_L \int_{\mathbb{R}} \frac{\sigma}{D^2}\, dX < 0. \tag{I.37}$$

The compact strict cover bounds σ below and D above on a common core; $p_L \ge p_0$ gives a uniform negative margin.

To transfer this margin, use the exact normalization (I.9). The changed coefficient a_L gives the weight

$$\|N\|_{C^3} \le C\{rR^{-2+\nu_0} + |\gamma|R^{-3+\nu_0} + |p_L|R^{-4+\nu_0} + |a_L|R^{-5+\nu_0}\}, \qquad 0 < \nu_0 < 1.$$

The argument of Proposition C.3, with this integrable weight and the admitted enlarged chords, gives phase C^3 convergence independently in r, γ. Exterior factors normalized by their exact value and slope converge to the identity by (I.12)–(I.13). Use the actual maps $h(b), k(b)$ and their normalization (C.7), with $c_h = h'(b_*) > 0$, $c_k = k'(b_*) > 0$. Let $\beta = b_* + (x + a_\theta - h(b_*))/c_h$. As in (F.40),

$$\Pi''(x) = \frac{c_k}{c_h^2}\Phi''(\beta) < 0. \tag{I.38}$$

Both raw first variations are $r^{2-\lambda_\gamma}$ times positive units. Their quotient, the actual bridges and divided closing bound Π' on the strict pieces and their regular finite-flight ends; the individual gains may vanish.

At every actual point with $Y_0 > K_H$, freeze the dilation at that point before a phase derivative. The exact normalized parameters are

$$(\bar a, \bar p, \bar\gamma, \bar s) = (a_L/Y_0, p_L/\sqrt{Y_0}, \gamma, r\sqrt{Y_0}), \qquad \bar Y(0) = 1. \tag{I.39}$$

The fixed unscaled cap h_* gives $\bar s < \sqrt{h_*}$. The signed compact source block of Corollary 5.1 receives these parameters. Its two absolute original-port variations have the same $\bar s^{2-\lambda_\gamma}$ power, including every bridge and (I.13). Their quotient bounds the derivative of this same physical Π at every high input, including nonroots.

Thus the original set cover is

$$\mathcal{I}_\theta = \{Y_0 \le K_L\} \cup \{Y_0 > K_H\}, \qquad \{Y_0 \le K_L\} \subset \{0 < \xi \le 2\xi_0\} \cup \{\xi \ge \xi_0\}.$$

Each set is restricted to $\mathcal{I}_\theta$. The physical overlaps are $K_H < Y_0 < K_L$ and $\xi_0 < \xi < 2\xi_0$. Enlarged chords receive both positive gap equality faces; the height equality faces are low. Original window, speed or height clipping preserves this cover, including all intervening inputs, and excludes the broken $t = 0$. On the terminal and strict pieces $\Pi'' < 0$; on the high and regular pieces $\Pi' \le C$. Choose m_* last so that the latter have $F' > 1/2$. Every zero of F' has $F'' = -m\Pi'' > 0$. Lemma 3.1 gives the whole positive-trace bound.

The negative trace. Now take $\alpha_1 = -rp$, $p \in [p_0, p_1]$, and $\alpha_0 = r^2 a_L$, $|a_L| \le a_*$. Let δ_{raw} bound $|v - \gamma|$ on the original small raw chart. In the raw coordinate $u = v - \gamma$, put

$$\mathcal{M} = -r^2 a_L + rpu + 3\gamma u^2 + u^3, \qquad \mathscr{D} = \mathcal{M} + \ell(v)z > 0,$$
$$\nu = \frac{6\gamma}{1 + 2\gamma}, \qquad N = rp + (3 - \nu)u^2 \ge rp + 2u^2.$$

The graph equation $z_v = f = z\{(1 + 2\gamma)u + u^2 + Cvz\}/\mathscr{D}$ satisfies (H.126) with these $\mathcal{M}, N$.

$$f_z = \frac{d}{dv} \log \frac{z^{1+\nu}}{\mathscr{D}} + \frac{N + \{\ell'(v) + (1 - \nu)Cv\}z}{\mathscr{D}}. \tag{I.40}$$

On $\mathcal{M} < 0$,

$$|\mathcal{M}| \le a_* r^2 + rp|u| + 3|\gamma|u^2 + |u|^3 \le C(a_* r + \delta_{\mathrm{raw}} + |\gamma|)N.$$

Let $\kappa_0 = \frac{1}{2} \inf \ell > 0$. Where $\mathscr{D} \ge \kappa_0 z$, the last quotient in (I.40) is uniformly bounded below. On the complement, $\mathcal{M} < 0$ and $z \le |\mathcal{M}|/(\ell - \kappa_0)$; the displayed estimate makes its numerator at least $N/2$ on the standing small raw chart after the independent coefficient shrink. It is therefore nonnegative there. Integrating from the positive raw section to the negative one reverses the orientation of the interval and gives

$$0 < T' \le C \frac{(z^{1+\nu}/\mathscr{D})_-}{(z^{1+\nu}/\mathscr{D})_+} \le C_0. \tag{I.41}$$

The endpoint ratio is bounded on the fixed actually admitted raw windows: endpoint heights are positive units and graph speeds have fixed positive lower bounds. No lower central speed is used. The estimate holds on the full receiver (I.19), including its speed clip and every intervening nonroot input. The ordinary divided closing has bounded positive derivative, so $0 < \Pi' \le C_{\mathrm{neg}}$. Add $m_* C_{\mathrm{neg}} < 1/2$ to the final multiplier choice; then $F' > 1/2$ throughout, giving at most one simple zero. The positive-phase estimates remain uniform at $a_L = d = 0$. □

Lemma I.4 (Nonnegative-opening domains)**.** *On every coefficient fibre with $\alpha_0 \ge 0$, the coefficient alternatives and phase intervals of Lemma I.1 cover the connected captured physical interval, with their regular continuations and all points between captured roots. The compact strict and regular portions have bounded complete raw derivative. The common high-height interval is contractive after the multiplier shrink of Corollary 5.1, including the positive-height equality face $\alpha_0 = \alpha_1 = 0$. The only remaining stopped portions are the determinant passage of Lemma I.2 and the signed trace passages of Proposition I.3.*

Proof. For each whole coefficient fibre use the alternative of Lemma I.1. The actual determinant cover (I.18) and signed trace receiver (I.19) retain its stopped, strict, regular and high pieces, including the original clips and every intervening input. Their two absolute tail gains have a common power, which cancels before the complete raw derivative is bounded on the strict and regular pieces. The signed high-height estimate bounds that same derivative above, including when both opening coefficients vanish. Choosing the finite multiplier last makes these bounded pieces contractive. The gap-zero completion is excluded. The whole-fibre sharp counts are Propositions 5.6 and 5.7; Corollary 5.8 takes the maximum across coefficient alternatives. □

J Quadratic inward and outward section estimates

We estimate the actual quadratic passages on their original sections, keeping the coefficient-dependent first multipliers. The inward maps provide heights on one common centre section; their equality defines the match. The outward maps carry the two lips to the original raw ports. Together these estimates control the section terms in physical curvature. The final paired estimate bounds the complete derivative above a comparison orbit, where the separate first variations may be large. We use the scalar variation and continuation formulas of the physical section toolkit throughout.

J.1 Quadratic radial embeddings on common raw sections

For the quadratic passage estimates below, fix the original raw-section and ordinary C^3 observer reserves and the passive compact set. Phase derivatives freeze g, k, π, A, C, m, with $g, k > 0$ independently small. In the positive or reflected negative radial coordinate, use

$$\begin{aligned} P_s(v) &= 1 - s\pi v + 3v^2 + skv^3, & B_s(v) &= 1 + g(2 + skv), \\ Q_s(v) &= 1 - Ag(1 + skv) + Cg^2(1 + skv)^2, & \zeta_s(v) &= sCkg^3(1 + skv), \qquad s = \pm 1. \end{aligned} \tag{J.1}$$

The exact scalar graph is

$$Y_v = \frac{Y(vB_s + \zeta_s Y)}{g(P_s + Q_s Y)}, \qquad x = sgkv, \quad z = g^3k^2Y. \tag{J.2}$$

Decreasing v follows positive time for $s = +1$ and its inverse for $s = -1$. On each domain where the actual lip continuation exists, denote its original-centre height by $Z_s(x_s)$ and define

$$\begin{aligned} A_s(x_s) = \int_0^{x_s} \frac{vB_s}{P_s}\, dv, \quad \mathcal{D}_s = \frac{P_s(x_s)}{x_s B_s(x_s)}\partial_{x_s}, \\ \log Z_s = \log\{\ell_0 P_s(x_s)\} - A_s(x_s)/g + \mathcal{R}_s, \quad H_s = P_s'/(vB_s), \quad N_s = -g\mathcal{D}_s \log Z_s. \end{aligned} \tag{J.3}$$

The polynomial factor and action in this formula retain the entrance height and the full linear contraction to the centre. Coefficient-only memory remains in $\mathcal{R}_s$; the estimates below concern its positive-order rows. Write $J = A_+$ and omit $s = +1$ subscripts when only the positive graph is used.

On a fixed compact segment $[0, b]$ with $P_s \geq p_b > 0$, the scalar graph has the common preparation

$$\begin{aligned} Y_v &= \frac{vB_s}{gP_s}Y + K_s(v, Y)Y^2, \qquad K_s = \frac{\zeta_s P_s - vB_s Q_s}{gP_s(P_s + Q_s Y)}, \\ m_s(v) &= \exp\left[-\frac{1}{g}\int_v^b \frac{wB_s(w)}{P_s(w)}\, dw\right] \leq 1, \qquad m_s(b) = 1. \end{aligned} \tag{J.4}$$

Fix a prescribed height buffer $0 \leq Y \leq h_b$ before shrinking g, k. The positive units B_s, Q_s, the margin p_b, and rational differentiation give $|\partial_Y^j K_s| \leq C_b/g$, $0 \leq j \leq 3$, on that buffer. Lemma A.5 therefore applies with $\kappa_c = g$ whenever the incoming height satisfies $0 < y \leq h_b/2$ and y/g is small. Each regional proof supplies these input bounds and the logarithmic phase rows of y; the continuation keeps the same original centre and uses no reciprocal bound on $m_s(0)$.

Lemma J.1. *Fix the raw section buffers, then a compact neighborhood of the passive coefficients A, C, and $\pi \in [\pi_\flat, \Pi]$. There are fixed $M, L, d_q, \delta_q, g_0, k_0 > 0$ such that the following holds for independently small $0 < g < g_0$, $0 < k < k_0$. For either $s = \pm 1$, let*

$$x_0 \geq 2M, \quad q = x_0^{-1}, \quad S = kx_0 \leq \delta_q, \quad a = gkx_0, \quad b_0 = -x_0^2/2.$$

The orbit from the lip at $v = g + sa$ reaches $v = g + sd_q$ with positive height. Its raw embedding $H_s(b_0)$, and any increasing observer $\Phi(H_s(b_0))$ on the prescribed buffered raw image for which $\|\partial_z \log \Phi'\|_{C^1} \le C$, have positive first derivative. Put

$$E = x_0\partial_{x_0} = a\partial_a, \qquad h_s = \frac{3 - s\pi q + q^2 + sS}{1 + 2g + sgS}, \quad D_s = h_s E, \quad \lambda = \frac{12g}{1+2g}, \quad L_g = 1 + |\log g|.$$

After separate affine normalizations of the embeddings, their slopes obey

$$D_s \log \phi_s' = -\lambda h_s + G_s, \qquad |G_s| + |D_s G_s| \le C g L_g (q + S). \tag{J.5}$$

Primes use the lip label b_0; g, k, π, A, C are held fixed. The two raw passages exist separately. The conclusion does not require their inward central images to intersect.

Proof. The common spectral estimate separates the exact homogeneous power from its logarithmic residual. We first check outward existence and admission to the raw buffers, then apply its derivative bounds with the original moving initial data and terminal observer.

Write $x = |v - g|$, $W = z/x^2$, $\tau = \log(x/a)$, and set

$$\begin{aligned} m_s(x) &= 3g + sx - s\pi k g^2/x + k^2 g^3/x^2, \\ Q_s(x) &= 1 - A(g + sx) + C(g + sx)^2, \\ N_s(x, W) &= 1 + 2g + sx + sCx(g + sx)W, \qquad D = m_s + Q_s W. \end{aligned}$$

Division of the exact raw field gives

$$W_\tau = F_s(x, W) := W(N_s/D - 2), \qquad W(0) = \ell_0 g p, \quad p = 3 - s\pi q + q^2 + sS. \tag{J.6}$$

Use the exact signed dictionary for Lemma A.6,

$$t = s, \quad b = 2g, \quad v = g, \quad (\mu_2, \mu_1, \mu_0) = (3g, -s\pi k g^2, k^2 g^3), \quad a = gS = gkx_0, \quad \sigma = gk.$$

Choose M large and δ_q small. For $S \le \delta_q$, $m_s(x) \asymp g$ on $[a, 2a]$, for both signs, and

$$|b| + |v| + |\mu_2| + a + |\mu_1|/a + |\mu_0|/a^2 \le Cg, \qquad \Theta = g(S + \pi q + q^2) \le Cg(S + q).$$

Thus the common lemma's intrinsic construction applies with $\rho \asymp g$. For the negative sign it gives the original root-free outward passage and invariant strip even when m_- changes sign beyond $2a$.

Common positive admission. For $s = +1$, the same construction applies on the larger domain

$$x_0 \ge 2M, \qquad a = gkx_0 \le a_0, \qquad 0 < g < g_0, \quad 0 < k < k_0, \tag{J.7}$$

without a bound on S. Indeed $kg/x \le q$ for $x \ge a$, so the choice of M gives

$$x + 2g \le m_+(x) \le x + 4g, \qquad 6g + a + \pi g q + g q^2 \le C(g + a).$$

These are the two exterior hypotheses with $\rho = g + a$. Choose d_q within that lemma's fixed cut allowance and $a_0 < d_q/L$, $L \ge 2$, with $g_0 + a_0 < \min(\rho_0, 1)$. Shrink g_0 so that $g_0\delta_q < a_0$. Its construction gives, for both signs on their respective domains,

$$D \ge c(g + W), \quad 0 < W \le w_1, \qquad W \ge w_0 \quad (x \ge La). \tag{J.8}$$

For the positive sign, in fact $D \ge c(g + x + W)$. The limiting graph in (A.16) keeps $t = s$, hence $m_s^*(x) = sx$. Its positive window is

$$|W(d_q) - W_+^*(d_q)| \le C(g + a) + C(a/d_q)^c.$$

The original regular-section reserves give common neighbourhoods of the two limiting terminal families whose continuations lie strictly inside the prescribed raw buffers. Choose a_0, then g_0, inside the positive reserve. For the negative sign the same window with $\rho \asymp g$ and $a \le g\delta_q$ gives its original buffered image after shrinking g_0. Thus both regular continuations are evaluated on admitted images. The positive graph is the inverse of the actual incoming passage; the negative graph follows positive time, and $z_x > 0$.

For the logarithmic rows on $S \le \delta_q$, use $\rho \asymp g$ for both signs. The coefficient bounds hold on a slightly enlarged open phase interval, with all coefficients fixed under E and $\beta = b_0$. The admitted regular continuation followed by the observer is the χ of the common lemma; its prescribed ordinary C^3 logarithmic-slope reserve gives $\mathcal{T} \le C$.

Write $W_l = \ell_0 gp$, $\eta = \log z$, and $\nu = 6g/(1+2g) = \lambda/2$. The exact moving factor specializes to

$$\mathscr{B}_s = EW_l - F_s(a, W_l) = \ell_0 \left[g(6 - s\pi q + 3sS) - \frac{1 + 2g + sa + sCa(g + sa)W_l}{1 + \ell_0 Q_s(a)} \right]. \tag{J.9}$$

It is a negative unit. With $\mathscr{L}_s = [2m_s + xm_s' + (xQ_s' + sCx(g + sx))W]/D$, the exact spectral density is

$$\mathscr{L}_s \, d\tau = \nu \, d\eta + J_s \, d\eta, \qquad J_s = \frac{s\{(3-\nu)x - \pi k g^2/x + x[-A + (3-\nu)C(g+sx)]W\}}{N_s}. \tag{J.10}$$

Put $I(a) = \int_{\eta_l(a)}^{\eta_T(a)} J_s \, d\eta$. For an observer choose a fixed reference z_* in its buffered image, put $c_\theta = \Phi'(z_*) > 0$, and $\widehat{\Phi}' = \Phi'/c_\theta$. The logarithmic derivative assumption makes $\widehat{\Phi}'$ a positive unit there. For the raw embedding take $c_\theta = 1$. The common exact slope, using $b_0 = -a^2/(2g^2k^2)$, is

$$\begin{aligned} (\Phi \circ H_s)'(b_0) &= d_\theta a^{-\lambda} U_s, \quad d_\theta = c_\theta g^2 k^2 \ell_0^{-1-\nu} g^{-\nu}, \\ U_s &= (-\mathscr{B}_s)(1 + \ell_0 Q_s(a)) p^{-\nu} \frac{W_T}{D_T} z_T^\nu \widehat{\Phi}'(z_T) e^I. \end{aligned} \tag{J.11}$$

By (A.15), its first two logarithmic residual rows satisfy

$$|E \log U_s| + |E^2 \log U_s| \le C\{gL_g(q+S) + a^{2-\lambda}\} \le CgL_g(q+S),$$

since $2 - \lambda > 1$ and $a = gS$. The spectral factor retains $\lambda = 12g/(1+2g)$. With $R_s = E \log U_s$ and $G_s = h_s R_s$,

$$D_s \log \phi_s' = -\lambda h_s + G_s, \qquad D_s G_s = h_s (Eh_s) R_s + h_s^2 E R_s.$$

The bounded h_s, Eh_s give (J.5). Separate affine value/slope calibrations are constant under E and do not change these rows. □

Corollary J.2. *Keep the original positive raw section and observer buffers of Lemma J.1, and fix $\delta > 0$. There are $a_0, g_0, k_0 > 0$ such that its positive outward passage and observer are defined on the whole range*

$$x_0 \ge 2M, \qquad S = kx_0 \ge \delta, \qquad a = gS \le a_0, \qquad 0 < g < g_0, \quad 0 < k < k_0.$$

With the same fixed-coefficient $E = x_0 \partial_{x_0}$, quadratic label and coefficient-only slope normalization, one has

$$|E \log \phi_+'| + |E^2 \log \phi_+'| \le C_\delta a(1 + |\log a|). \tag{J.12}$$

There is no upper bound on S independent of g, and no comparison of g with k. The observer has exactly the same stated ordinary C^3 logarithmic-slope input as in the lemma.

Proof. Use the common positive admission (J.7) and its original buffered continuation. Apply Lemma A.6 with the same positive dictionary, $\rho = g + a$, $\sigma = gk$ and terminal cut d_q. Its coefficient and mass hypotheses were checked before imposing $S \geq \delta$; the admitted ordinary C^3 observer again gives $\mathcal{T} \leq C$. Now

$$g \leq a/\delta, \quad \Theta = a + \pi gq + gq^2 \leq C_\delta a, \quad L_\rho \leq 1 + |\log a|, \quad \lambda = \frac{12g}{1+2g} \leq 12a/\delta.$$

With $e = 2 - \lambda > 3/2$, the spectral rows give

$$|E \log \phi'_+ + \lambda| + |E^2 \log \phi'_+| \leq C_\delta\{a(1 + |\log a|) + (a/d_q)^e\} \leq C_\delta a(1 + |\log a|).$$

Absorbing $\lambda = O_\delta(a)$ proves (J.12). The exact quadratic label, coefficient-only $(gk)^2$ gain and $a^{-\lambda}$ are retained. □

Corollary J.3. *Keep the original negative raw section and buffered observers, and fix $r_c > 0$. For independently small $g, k > 0$, the actual positive-time negative outward passage is defined for every*

$$r \geq r_c, \qquad 0 < R = kr < \mathfrak{c}, \qquad p = 3 + \pi/r + r^{-2} - R > 0.$$

Here $\mathfrak{c}$ is the negative simple-root endpoint already defined in the quadratic chart. Put

$$E = r\partial_r = R\partial_R, \qquad B = 1 + g(2 - R), \qquad h = p/B, \qquad \mathcal{A} = hE = \mathcal{D}_-.$$

With primes in either quadratic label $-r^2/2$ or $-R^2/2$,

$$|\mathcal{A} \log \phi'_-| + |\mathcal{A}^2 \log \phi'_-| \leq Cg(1 + |\log g|). \tag{J.13}$$

The ordinary C^3 buffered observer input is the same as in the lemma; it includes the actual divided finite closing on its original windows.

Proof. Use the exact negative equations of the lemma, with $a = gR$, $q = 1/r$, $W_l = \ell_0 gp$. Here p may tend to zero at the simple root, so the weak height interval can be arbitrarily long. The moving initial variation has a factor p, which will absorb this dwell in the differentiated integrals. We first construct the graph in logarithmic height. On the stated range p is bounded above, R is bounded, $a \leq Cg$, and $m_-(a) = gp$. Also $m'_-(x) < 0$. To establish the actual graph without presupposing a positive raw radial denominator, use $\eta = \log z$, $\chi = \log x$, and

$$\chi_\eta = V = (m_- + Q_- W)/N_-, \qquad W = e^{\eta - 2\chi}.$$

This equation is regular while N_- is a positive unit. Initially $D_l = gp(1 + \ell_0 Q_-(a)) > 0$. At a first zero of $D = m_- + Q_- W$, its derivative along this graph would be $D_\eta = Q_- W > 0$. Hence $D > 0$ throughout the graph. Below a smaller auxiliary positive strip threshold, $D \leq gp + Q_- W$ and $V < 1/4$; thus $W_\eta > 0$ and $W \geq \ell_0 gp$. Consequently

$$W_\tau = WN_-/D - 2W \geq \frac{\inf N_-}{\ell_0^{-1} + \sup Q_-} - 2W \geq c_0 > 0.$$

A fixed logarithmic length therefore reaches that threshold; the fixed positive margin supplies the remaining climb to the original w_0. Thus $x_1 \leq La$, uniformly even when p tends to zero. Choose g_0 so that $La < d_q$. Thereafter the original strip is invariant and $D \geq cW$. The limiting strong graph and contraction estimate of the lemma give

$$|W(d_q) - W^*_-(d_q)| \leq Cg + C(La/d_q)^c \longrightarrow 0.$$

This proves first passage and admission to the original observer buffer before derivative estimates. Its initial height length satisfies

$$\eta_1 - \eta_l = \log \frac{W_1}{\ell_0 g p} + 2 \log \frac{x_1}{a} \le C(1 + |\log g| + |\log p|).$$

Write $Q_- = q_0 + q_1 x + C x^2$, where $q_0 = 1 - Ag + Cg^2$, $q_1 = A - 2Cg$. At fixed η the exact partial derivatives are

$$D_\chi = -x - \pi k g^2/x - 2k^2 g^3/x^2 - (2q_0 + q_1 x)W, \qquad N_{-,\chi} = -x + Cgx W.$$

On a sufficiently small fixed inner raw interval they imply

$$V_\chi \le -c\Xi, \qquad |V_{\chi\chi}| \le C\Xi, \qquad \Xi = x + kg^2/x + k^2 g^3/x^2 + W. \tag{J.14}$$

Indeed $D \le gp + Q_- W \le Cg + CW$; the correction $-DN_{-,\chi}/N_-^2$ is absorbed by the negative terms in D_χ/N_-. The next derivative has the same density.

Let $X_j = \mathcal{A}^j \chi|_\eta$, $l_j = \mathcal{A}^j \eta_l$, and $t_j = \mathcal{A}^j \eta_T$. The full moving initial rows are

$$\begin{aligned} l_1 &= (2p + Ep)/B, \qquad l_2 = O(p), \\ X_{1,l} &= h - V_l l_1 = pK_l > 0, \\ X_{2,l} &= \mathcal{A} X_{1,l} - V_{\chi,l} X_{1,l} l_1 = O(p). \end{aligned}$$

with K_l a positive unit. Here l_1, Eh and El_1 are bounded. The two inverse equations are (A.7). Up to that inner cut, (J.14) gives $0 < X_1 \le Cp$, $\int \Xi X_1 \, d\eta \le Cp$, and $|X_2| \le C X_1$, by division by the positive first variation. On the strong part choose an auxiliary fixed inner cut $d_* \le d_q$ so that $V_\chi/V \le -3/2$, then shrink g_0 to ensure $La < d_*$. The decay follows from $D \ge cW$, $W \ge w_0$, and $V_\chi/V \le -2 + Cg/w_0 + Cx$. Thus $X_1 \le Cp(a/x)^{3/2}$ up to that cut. The remaining compact strong-graph segment to the original d_q has bounded coefficients and finite logarithmic length, and preserves this bound up to a fixed factor. At the original terminal cut, (A.7) gives $|t_1| + |t_2| \le Cpa^{3/2}$. Use the exact density $J_- = J$ from (J.10), with $I = \int_{\eta_l}^{\eta_T} J \, d\eta$. Its two complete rows are (A.8), with $D = \mathcal{A}$, $r = \chi$, and $Dr_l = h$. On the initial part $|\partial_\chi^j J| \le Cg$, $j = 0, 1, 2$, since $x \le La \le Cg$ and $kg^2/x \le g/r$. Its differentiated integrals are at most $Cgp(1 + |\log g| + |\log p|) \le Cg(1 + |\log g|)$. On the strong part $d\eta \le C \, d\log x$, and the integrals are bounded by

$$Cp \int_{La}^{d_*} (x + kg^2/x)(a/x)^{3/2} \, d\log x \le Cp(a + kg^2/a) \le Cg.$$

If the actual entry is below La, the fixed logarithmic segment from x_1 to La has the same Cg bound. The compact final segment contributes $Cpa^{3/2}$. The displayed initial terms are $O(g)$, and all terminal partials are bounded. Therefore $|\mathcal{A}I| + |\mathcal{A}^2 I| \le CgL_g$.

Finally use the exact complete slope factor of the lemma. Its moving-initial and $1 + \ell_0 Q_-(a)$ logarithmic rows are $O(g)$. Although $p^{-\nu}$ is unbounded as p tends to zero, its two action logarithmic rows are $O(g)$, since $\mathcal{A} \log p = Ep/B$ and $\mathcal{A}^2 \log p = O(p)$. The terminal W_T/D_T, z_T^ν and ordinary C^3 observer rows are $O(pa^{3/2})$, by the full terminal formulas. The extracted $a^{-\lambda}$ contributes $-\lambda h = O(g)$ and $-\lambda \mathcal{A} h = O(g)$. All coefficient-only slope factors are frozen, including the change between the two quadratic labels. These estimates prove (J.13) on the whole original domain, including arbitrarily close approach to its negative root endpoint. □

J.2 The complete negative inward action remainder

We next continue the negative lip back to the original centre and estimate its action remainder. This supplies one side of the later matching equation. The positive and negative heights are constructed separately; matching is imposed only on the intersection of their images.

Corollary J.4. *Use* (J.1) *with* $s=-1, x_s=r$, $\pi\in[\pi_\flat,\Pi]$, *and fixed small* ℓ_0. *Fix* $r_c>0$. *On the entire strict negative interval* $r\ge r_c$, $0<R=kr<\mathfrak{c}$, *continue the lip* $Y=\ell_0P_-(r)$ *to* $\xi=0$ *along the inverse of the actual positive-time negative branch. Its height* $Z_-(r)>0$ *is defined on this entire interval and enters the original central height window after one coefficient shrink. In* (J.3), *its remainder obeys*

$$\sum_{j=1}^{3}|\mathcal{D}_-^j\mathcal{R}_-|\le Cg. \tag{J.15}$$

Changing the central height by a coefficient-only multiplicative scale changes only a constant in this logarithmic formula. The estimate does not differentiate coefficients or bound the value of $\mathcal{R}_-$ *by* Cg. *All constants are independent of the distance to the negative simple root and of any comparison between* g *and* k.

Proof. Choose an auxiliary fixed $0<b<r_c/2$. On $[0,r_c]$, the polynomial is uniformly positive after decreasing k, and

$$A_-(r_c)-A_-(b)\ge a_*>0.$$

For $v\ge b$, $kv<\mathfrak{c}$, write

$$Q=1-Ag(1-kv)+Cg^2(1-kv)^2,\quad H=\frac{P_-'(v)}{vB_-(v)},\quad \omega=-\frac{Ckg^3(1-kv)P_-(v)}{vB_-(v)}.$$

Here Q is a positive unit. With $\mathcal{D}_v=P_-(v)\partial_v/(vB_-(v))$, direct rational differentiation in (v^{-1},kv) gives

$$\sum_{j=0}^{3}|\mathcal{D}_v^jH|\le C,\qquad \sum_{j=1}^{3}|\mathcal{D}_v^jQ|\le Cg,\qquad \sum_{j=0}^{3}|\mathcal{D}_v^j\omega|\le Cg^3. \tag{J.16}$$

The field $\mathcal{D}_v$ contains the factor $P_-(v)$; none of these bounds divides by its vanishing root factor.

Set $t=A_-(r)$, $q=(t-A_-(v))/g$, and $W=Y/P_-(v)$. The original negative scalar equation gives exactly

$$v_q=-g\frac{P_-(v)}{vB_-(v)},\qquad (\log W)_q=-\frac{1+\omega W}{1+QW}+gH,\qquad W(0)=\ell_0.$$

At fixed q, $\partial_t v=P_-(v)/(vB_-(v))$, so every ∂_t coefficient derivative is precisely $\mathcal{D}_v$. The logarithmic rate and the normal diagonal are at most $-c<0$ for the fixed small lip. Lemma A.3 applies with $D=\partial_t$, $\kappa=g$, $\rho=g$, $\alpha=0$, by (J.16); clock differentiation acts as $-g\mathcal{D}_v$. It constructs the actual positive-height inverse passage to b, with radial speed $P_-+QY>0$, and gives

$$|\partial_t^jW|+|\partial_t^jL|\le Cg(1+q)^Ne^{-cq},\quad 1\le j\le 3,\qquad L=(Q-\omega)W/(1+QW).$$

Retain the exact source polynomial:

$$T=(t-A_-(b))/g,\qquad V_b=\int_0^{T(t)}L\,dq,\qquad Y_b=\ell_0P_-(r)e^{-T+V_b}=P_-(b)W(T)\le Ce^{-cT}.$$

Here $T'=1/g, T''=T'''=0$, and $T\ge a_*/g$. The three complete rows (A.10) therefore give $|\partial_t^jV_b|\le Cg$: each boundary term is bounded by $Cg^{-N}(1+T)^Ne^{-cT}$. The retained polynomial gives

$$\mathcal{D}_-\log Y_b=H(r)-g^{-1}+\partial_tV_b,\qquad |\mathcal{D}_-^jY_b|\le Cg^{-j}Y_b,\quad j\le 3.$$

Reflect the segment from $-b$ to the original centre by $v = -\xi$. Preparation (J.4), with $s = -1$, then applies on the fixed margin above. Exponential flatness gives $Y_b \le h_b/2$ and small Y_b/g, so Lemma A.5 establishes the actual continuation, $0 < Y \le 2Y_b$, and its logarithmic nonlinear unit $\mathcal{N}$. The three composition rows of that lemma are bounded by $CY_b/g^{j+1} \le Cg$. The exact linear multiplier is $e^{-A_-(b)/g}$, so the artificial cut cancels and $\mathcal{R}_- = V_b + \mathcal{N}(Y_b)$ gives (J.15) at the original centre. □

J.3 Positive inward passage on the growing range

When $S = ku$ grows, the positive action speed is of order $1 + S$. We therefore estimate the inward remainder in the radial derivative $u\partial_u$, keeping the conversion to action derivatives explicit. This is the estimate used in (G.45).

For $s = +1$, the factorization is

$$P(v) = (\mathfrak{c} + kv)\{(v - d)^2 - \Delta\}, \quad \mathfrak{c} = 3 + 2kd, \quad \mathfrak{c}(d^2 - \Delta) = 1, \quad \pi = 2\mathfrak{c}d - k/\mathfrak{c},$$

with $d = \pi/6 + O(k)$, $\mathfrak{c} = 3 + O(k)$.

Corollary J.5 (Positive large-S inward action rows)**.** *Fix $0 < \pi_\flat \le \Pi < \infty$, a compact set of the passive coefficients A, C, and $\delta > 0$. Choose the original lip height ℓ_0 and physical collar width d_0 sufficiently small within their original buffered windows. There are fixed*

$$v_i, M, g_0, k_0 > 0, \qquad k_0 \le \delta/(2M),$$

such that the following holds for independently small $0 < g < g_0$, $0 < k < k_0$, $\pi \in [\pi_\flat, \Pi]$. On every strict root-free fibre $\Delta < 0$, every lip point

$$\xi = u, \qquad Y = \ell_0 P(u), \qquad S = ku \ge \delta, \qquad a = gS < d_0 \tag{J.17}$$

has an actual positive-height, positive-time first passage to $\xi = 0$. Its original centre height is $Y_{0,+} = Z_+ > 0$, with $\mathcal{R}_+$ defined by (J.3)*. With all coefficients fixed under $\mathcal{E} = u\partial_u$,*

$$|\mathcal{E}^j \mathcal{R}_+(u)| \le C_\delta a, \qquad j = 1, 2, 3. \tag{J.18}$$

The constants are independent of g, k and of $-\Delta > 0$. The statement can be restricted to the original strict raw-window preimages without extending any image or inverse.

On a real or double-root fibre the prescribed positive through lip ends at the first root $a_{\mathrm{root}} = d - \sqrt{\Delta}$. That endpoint is bounded uniformly in this coefficient block, and after the same k-shrink its through interval has $ku < \delta$. Thus (J.17) *is empty on that branch. No passage beyond the first root, or at the stopped $\Delta = 0$ limit, is included.*

Proof. Fix the finite differentiability budget used below first. Choose $v_i > 0$ so small that

$$P(v) \ge p_i > 0 \qquad (0 \le v \le 2v_i)$$

uniformly for the fixed π-interval and small k. Then fix $M > 2v_i$ beyond all positive roots and turning points of the retained compact positive factors, and large enough that

$$P(v) \asymp v^2(3 + kv), \qquad v \ge M. \tag{J.19}$$

The constants may depend on the fixed passive compact, $\pi_\flat, \Pi, \delta, \ell_0, v_i, M$. All calculations are also valid on a slightly enlarged phase range $S > \delta/2$, $a < 2d_0$; the choices of d_0, g_0, k_0 are made for that range before restriction to (J.17). This supplies actual phase buffers for the displayed derivatives.

Actual passage and value admission. For $\Delta < 0$, $P > 0$ on the positive half-axis. On $v_i \le v \le u$ put

$$q = \frac{J(u) - J(v)}{g}, \qquad v_q = -g\frac{P(v)}{vB(v)}, \qquad W = \frac{Y}{P(v)}, \qquad H = \frac{P'(v)}{vB(v)}, \quad \varpi = \frac{\zeta(v)P(v)}{vB(v)}. \tag{J.20}$$

Division of (J.2), retaining the polynomial factor, gives exactly

$$W_q = W\left\{-\frac{1+\varpi W}{1+QW} + gH\right\}, \qquad W(0) = \ell_0. \tag{J.21}$$

Here B, Q are positive units, $gH \le C_\delta a$, and $|\varpi| \le C_\delta a^3$, as also checked below. Choose ℓ_0, d_0, g_0 so that the logarithmic rate in (J.21) is at most $-c_0 < 0$ while $0 < W \le \ell_0$. A first-exit argument proves

$$0 < W(q) \le \ell_0 e^{-c_0 q}, \qquad 0 \le q \le T(u) := \frac{J(u) - J(v_i)}{g}. \tag{J.22}$$

Consequently $P + QY = P(1 + QW) > 0$. For each strict ghost coefficient the finite interval $[v_i, u]$ has a positive minimum of P; the time integral to v_i is finite. Height is positive by uniqueness. The same construction continues to $v_i/2$, on its fixed positive-P buffer, before imposing the cut v_i. Thus the fixed-q partials at that moving terminal are genuine nearby actual-flow derivatives.

Since $Y \le \ell_0 P(v)$, the physical height on the entire arc is at most

$$C\ell_0\{g^3k^2 + g^3k^2v + g^3k^2v^2 + g^3k^3v^3\} \le C\ell_0\{g^3k^2 + g^2ka + ga^2 + a^3\}.$$

Together with $0 < gkv \le a < d_0$, this admits the arc to a fixed small original chart. Neither a limiting axis orbit nor a normalization is used to infer this admission.

On $M \le v \le \delta/k$, $P(v) \le C_\delta v^2$ and $B \ge 1$. Since $u \ge \delta/k$, it follows that

$$T(u) \ge \frac{c_\delta \log(1/k) - C_\delta}{g}. \tag{J.23}$$

Choose k_0 so that the numerator has a fixed positive margin. Then T is uniformly large for small g, with no comparison between g and k. In particular the actual terminal height

$$y_i(u) := Y(v_i; u) = P(v_i)W(T(u), u) \le Ce^{-c_0 T(u)} \tag{J.24}$$

lies in any fixed sufficiently small compact-entry interval $0 < y_i < y_*$ after the permitted coefficient shrink. This places the input in the prescribed compact-continuation buffer before its derivatives are estimated.

On $[0, v_i]$, preparation (J.4) and exponential flatness give the prescribed height buffer and small y_i/g. Lemma A.5 therefore continues the actual graph to the original centre with $0 < Y \le 2y_i$, for either sign of C. Retain the exact memory (A.11), with $V(q, u) = \int_0^q L\, ds$, $L = (Q - \varpi)W/(1 + QW)$; its initial factor is $P(u)/P(v)$ and $|V| \le C\ell_0$.

Three transported coefficient rows. Write

$$h(v) = \frac{P(v)}{v^2 B(v)}, \qquad \mathcal{D}_v = \frac{P(v)}{vB(v)}\partial_v = h(v)v\partial_v, \qquad b(u) = \frac{1}{h(u)}.$$

At fixed elapsed action q, differentiating $J(v) = J(u) - gq$ yields

$$\mathcal{E}\log v = \frac{h(v)}{h(u)}, \qquad \mathcal{E}[F(v(q, u))] = b\mathcal{D}_v F. \tag{J.25}$$

The full second and third transported rows are

$$\begin{aligned}\mathcal{E}^2F &= b^2\mathcal{D}_v^2F + (\mathcal{E}b)\mathcal{D}_vF,\\ \mathcal{E}^3F &= b^3\mathcal{D}_v^3F + 3b(\mathcal{E}b)\mathcal{D}_v^2F + (\mathcal{E}^2b)\mathcal{D}_vF.\end{aligned} \tag{J.26}$$

The variable coefficients of $\mathcal{D}_v$ are differentiated in its powers. On the initial domain,

$$h(u) = \frac{3-\pi/u+u^{-2}+S}{1+2g+gS} \asymp_\delta 1+S, \qquad |\mathcal{E}^j h(u)| \le C_\delta h(u) \quad (j=1,2). \tag{J.27}$$

Hence

$$|\mathcal{E}b| + |\mathcal{E}^2 b| \le C_\delta b, \qquad b \le C_\delta/(1+S), \qquad gh(u) \asymp_\delta g+a.$$

For $v \ge M$, let $t = 1/v$, $s = kv$, and $\mathcal{E}_v = -t\partial_t + s\partial_s$. Then

$$h(v) = \frac{3-\pi t+t^2+s}{1+2g+gs}, \qquad H(v) = \frac{6-\pi t+3s}{1+2g+gs}, \qquad \varpi(v) = Cg^3 s(1+s)h(v). \tag{J.28}$$

For $0 \le s \le S$, $t \le 1/M$, and $gS = a < 2d_0$, these rational symbols give

$$h(v) \le C_\delta h(u), \qquad |\mathcal{E}_v^j h(v)| \le C_\delta h(v) \quad (j \le 2).$$

The first three $\mathcal{E}_v$ rows of $gH, Q-1, \varpi$ have size $C_\delta a$; the ϖ rows have the sharper size $C_\delta a^3$. Expanding $(h(v)\mathcal{E}_v)^j$ and using $bh(v) \le C_\delta$ proves the same bounds for the rows in (J.26).

On $v_i/2 \le v \le M$, H has uniform ordinary finite derivatives. The coefficients and finite ordinary derivatives of $\mathcal{D}_v = P/(vB)\partial_v$ are uniformly bounded, also as $-\Delta \downarrow 0$; they contain P rather than P^{-1} at the bottleneck. The corresponding ordinary rows of $gH, Q-1$ are $O(g)$, and those of ϖ are $O(kg^3)$. Since $g \le a/\delta$ on the retained domain, this proves

$$|\mathcal{E}^j(gH)| + |\mathcal{E}^j(Q-1)| + |\mathcal{E}^j\varpi| \le C_\delta a, \qquad 0 \le j \le 3, \tag{J.29}$$

uniformly through the arbitrary strict ghost gap. In particular the quotient in (J.25) and its first two Euler rows are bounded on both the growing and compact portions.

For the time rows, $\partial_q = -g\mathcal{D}_v$ on coefficient functions and $gh(v) \le C_\delta a$ on the growing portion. On the compact portion $g\mathcal{D}_v$ has bounded finite coefficients. The same rational calculations give bounded first and second q-rows and the mixed $q, \mathcal{E}$ rows required below. Their rows containing an $\mathcal{E}$ retain the $C_\delta a$ factor. No inverse of the bottleneck polynomial is introduced.

Normal variations and the moving terminal. The exact diagonal is

$$\mathscr{F}_W = -\frac{1+2\varpi W+\varpi QW^2}{(1+QW)^2} + gH \le -c_1 < 0. \tag{J.30}$$

Thus Lemma A.3 applies with $D = \mathcal{E}, \kappa = g, \rho = C_\delta a, \alpha = 0$, by (J.29) and the mixed clock bounds above. Its fixed-clock estimate is

$$|\mathcal{E}^j W| + |\mathcal{E}^j L| \le C_\delta a(1+q)^N e^{-cq}, \qquad 1 \le j \le 3, \tag{J.31}$$

with the mixed and pure clock rows specified there. The terminal rows are

$$T_1 = b/g, \qquad T_2 = (\mathcal{E}b)/g, \qquad T_3 = (\mathcal{E}^2 b)/g, \qquad |T_j| \le C_\delta/(gh(u)) \le C_\delta/a \le C_\delta/g. \tag{J.32}$$

By (J.23), for each fixed N,

$$g^{-N}(1+T)^N e^{-cT} \le C_{\delta,N} g \le C_{\delta,N} a. \tag{J.33}$$

Choose k_0 with $cc_\delta \log(1/k_0) - cC_\delta > 0$; then $g^{-N}e^{-\beta/g}$ absorbs every terminal loss, using independent upper bounds on g, k. The terminal formulas (A.10), for $V_i = V(T(u), u)$, give

$$|\mathcal{E}^j V_i| \le C_\delta a, \qquad 1 \le j \le 3. \tag{J.34}$$

The exact memory, including all moving terminal rows, is

$$y_i = \ell_0 P(u) e^{-T+V_i} = P(v_i)W(T), \qquad \mathcal{E}^j \log y_i = \mathcal{E}^j \log P(u) - T_j + \mathcal{E}^j V_i. \tag{J.35}$$

The rational symbols bound $|\mathcal{E}^j \log P(u)| \le C_\delta$, hence $|\mathcal{E}^j \log y_i| \le C_\delta/g$ for $1 \le j \le 3$. The value estimate uses $P(v_i)W(T)$, independently of $P(u)$. The compact continuation already constructed has exact multiplier $m(0) = e^{-J(v_i)/g}$. Its three composition rows from Lemma A.5 are at most $Cy_i/g^4 \le C_\delta a$, by (J.33). Consequently

$$\log Y_{0,+} = \log\{\ell_0 P(u)\} - J(u)/g + V_i + \mathcal{N}(y_i), \qquad \mathcal{R}_+ = V_i + \mathcal{N}(y_i).$$

This proves the three Euler bounds at the original centre. The fixed-cut action cancels exactly; an additive action normalization changes only a coefficient constant in $\mathcal{R}_+$.

Finally, $d, \mathfrak{c}$, and every nonnegative $\sqrt{\Delta}$ in the fixed π-block are bounded for small k. The root a_{root} is therefore bounded by the fixed M. Its prescribed through lip has $ku \le ka_{\mathrm{root}} < \delta$ after the k-shrink, as stated. The strict ghost proof permits arbitrarily small $-\Delta > 0$ but does not add a through interval at $\Delta = 0$. □

J.4 Bounded positive quadratic passages

On a bounded positive lip interval, both the root gap and the lip's distance to its first root can vanish. The inward calculation remains uniform in action coordinates. For the outward calculation we use logarithmic height through the weak stage; its moving initial factor absorbs the long dwell before the graph enters the parabolic strip.

Corollary J.6 (Bounded positive quadratic suppliers)**.** *Fix the original positive raw section and its buffered regular observer, with its original ordinary C^3 logarithmic-slope reserve $\|\partial_z \log \Phi'\|_{C^1} \le C$ on the buffered raw image, a compact neighborhood of the passive coefficients A, C, and $\pi \in [\pi_\flat, \Pi]$. Fix $U < \infty$ and use* (J.1)–(J.2)*, with physical coordinates $x = gkv, z = g^3k^2Y$. Retain the original actual moving-lip interval $Y = \ell_0 P(u)$, intersected with $u \le U$. Its lower threshold satisfies $u_{\mathrm{th}} \ge u_0 > 0$. On real or fold fibres this interval has $u_{\mathrm{th}} \le u < a$, where a is the first positive root. On root-free fibres it is the component of the original section-image preimage specified in the positive quadratic domain construction. In each case $P > 0$ on the actual inward axis path $[0, u]$. No right-hand axis continuation is added at the fold.*

There are $g_0, k_0, C_0 > 0$, uniform on these domains, such that the following statements hold for independently small $0 < g < g_0$, $0 < k < k_0$. Put $L_g = 1 + |\log g|$, with $J, \mathcal{D}_+$ as in (J.3)*. The inward lip orbit reaches the original centre $v = 0$, and its height $Z_+(u) = Y(0; u) > 0$ has*

$$|\mathcal{D}_+^j \mathcal{R}_+| \le C_0 g, \qquad 1 \le j \le 3. \tag{J.36}$$

The outward lip graph, which is the inverse of the physical positive incoming passage, reaches the original positive raw port at positive height and inside its original buffer. For the complete embedding followed by the original increasing ordinary C^3 observer, use the lip label $\beta = -u^2/2$ and any coefficient-only positive affine value and slope calibration ϕ_+. Then

$$\phi_+' > 0, \qquad |\mathcal{D}_+^j \log \phi_+'| \le C_0 g L_g, \quad j = 1, 2. \tag{J.37}$$

The prime denotes differentiation in β. Both estimates freeze g, k, π, A, C. They are uniform as the positive root gap and the distance of the lip to its first root tend to zero independently of g, k. They do not require an intersection of positive and negative inward images.

Proof. The exact positive quadratic factorization is

$$P(v) = (\mathfrak{c} + kv)\{(v-d)^2 - \Delta\}, \qquad \mathfrak{c} = 3 + 2kd, \quad \mathfrak{c}(d^2 - \Delta) = 1, \quad \pi = 2\mathfrak{c}d - k/\mathfrak{c}.$$

It gives uniform bounds for $d, d^{-1}, \mathfrak{c}, \mathfrak{c}^{-1}$ on the retained fold band. On a real fibre, $P(a) = 0$ implies

$$1 = \pi a - 3a^2 - ka^3 \le \Pi a, \qquad a \ge \Pi^{-1}.$$

On a root-free fibre, $\pi = 2\mathfrak{c}d - k/\mathfrak{c}$ bounds d away from zero. Thus the original fixed X_{up} can be chosen small enough that $u_{\mathrm{th}} = a - X_{\mathrm{up}}$, respectively $d - X_{\mathrm{up}}$, has the asserted uniform positive lower bound. All constants below are fixed before g, k.

The inward action integral and compact continuation. Choose fixed $b < u_0/2$ with $P \ge p_b > 0$ on $[0, b]$, and put

$$q = (J(u) - J(v))/g, \quad W = Y/P(v), \quad H = P'/(vB),$$
$$\omega = \zeta P/(vB), \qquad \mathcal{D}_v = P(v)\partial_v/(vB(v)).$$

Use the ratio equation and exact memory (A.11), with $\kappa = g$ and $V(q) = \int_0^q (Q - \omega)W/(1 + QW)\, ds$. At fixed q, $\mathcal{D}_+ v = P(v)/(vB(v))$, so its repeated coefficient rows are powers of $\mathcal{D}_v$. On $[b, U]$, this field has bounded finite derivatives, uniformly at the root/fold:

$$|\mathcal{D}_v^j(gH)| + |\mathcal{D}_v^j(Q-1)| + |\mathcal{D}_v^j\omega| \le Cg, \qquad 0 \le j \le 3. \tag{J.38}$$

Clock rows use $-g\mathcal{D}_v$, and the fixed small lip gives both negative margins. These are the inputs of the negative inward proof, now with $s = +1$, $D = \mathcal{D}_+$, $\kappa = g, \rho = g, \alpha = 0$. The terminal check is

$$q_b = (J(u) - J(b))/g \ge \eta/g, \qquad \mathcal{D}_+ q_b = 1/g, \quad \mathcal{D}_+^2 q_b = \mathcal{D}_+^3 q_b = 0;$$

the lower bound follows on a fixed subinterval of $[b, u_0]$. Lemma A.3 and its terminal formulas give $|\mathcal{D}_+^j V(q_b)| \le Cg$, $1 \le j \le 3$, and

$$\begin{aligned} y_b &= \ell_0 P(u) e^{-q_b + V(q_b)} = P(b) W(q_b) \le C e^{-c\eta/g}, \\ \mathcal{D}_+ \log y_b &= -g^{-1} + H(u) + \mathcal{D}_+ V(q_b), \qquad |\mathcal{D}_+^j y_b| \le C_j g^{-j} y_b \quad (1 \le j \le 3). \end{aligned} \tag{J.39}$$

Preparation (J.4) verifies the compact-continuation inputs: $P \ge p_b$, $|\partial_Y^j K| \le C/g$, $y_b \le h_b/2$, and small y_b/g. Lemma A.5 then gives

$$Z_+(u) = e^{-J(b)/g} y_b e^{\mathcal{N}(y_b)}, \qquad \mathcal{R}_+ = V(q_b) + \mathcal{N}(y_b).$$

Its three composition rows are $O(y_b/g^{j+1}) = O(g)$. The cut action cancels, proving (J.36).

Construction of the outward graph and buffer admission. Put

$$a = gku, \qquad p = \frac{P(u)}{u^2} > 0, \qquad E = u\partial_u, \qquad h = \frac{p}{B(u)}, \qquad \mathcal{D}_+ = hE.$$

In the raw graph set $W = z/x^2$, $r = \log x$, and $\eta = \log z$. Its exact coefficients are

$$\begin{aligned} m(x) &= 3g + x - \pi k g^2/x + k^2 g^3/x^2, \\ Q_0(x) &= 1 - A(g + x) + C(g + x)^2, \\ N(x, W) &= 1 + 2g + x + Cx(g + x)W, \qquad D_0 = m + Q_0 W. \end{aligned}$$

The initial data are the original lip data

$$x_l = a, \qquad W_l = \ell_0 g p, \qquad z_l = a^2 \ell_0 g p = g^3 k^2 \ell_0 P(u). \tag{J.40}$$

Use the inverse logarithmic-height equation first:

$$r_\eta = V = \frac{D_0}{N}, \qquad W_\eta = W(1-2V). \tag{J.41}$$

Here N, Q_0 are positive units on the original small corridor. At a possible first zero of D_0, one has $V = 0$ and

$$\frac{dD_0}{d\eta} = Q_0 W > 0.$$

Thus a graph starting with $D_0 > 0$ cannot cross to $D_0 \le 0$. Both x and z increase on this graph, including at an axis root crossed at positive height.

Fix $L = 2$, then a sufficiently small auxiliary height $w_0 > 0$. In the box $a \le x \le La$, $0 < W \le w_0$, the variable $x/(gk)$ lies in the fixed interval $[u_0, LU]$. Hence

$$|m| + |m_r| + |m_{rr}| \le Cg, \qquad 0 < V \le C(g + w_0), \qquad \tfrac{1}{2}W \le W_\eta \le W.$$

Write $s = \eta - \eta_l$. The length of this weak stage, up to either possible first exit, is bounded by

$$T \le C\{L_g + \log^+(1/p)\}. \tag{J.42}$$

Since $m(0) = gp$ and

$$|m_s| \le Cg(|m| + W),$$

Gronwall's inequality, together with the lower growth rate for W, gives

$$|m(s)| \le Cgpe^{Cgs} + CgW(s), \qquad r(s) - r_l \le Cgp(1+s)e^{Cgs} + Cw_0.$$

For every fixed j, the elementary bound

$$p(1+T)^j e^{CgT} \le C_j L_g^j, \qquad 0 < p \le p_{\max}, \tag{J.43}$$

holds after shrinking g. Indeed (J.42) reduces it to $g^{-Cg} \le C$ and the boundedness of $p^{1-Cg}\{1 + \log^+(1/p)\}^j$. Choose w_0 so $Cw_0 < \frac{1}{2}\log L$, then g so that $CgL_g < \frac{1}{2}\log L$. The preceding displacement estimate excludes an exit at $x = La$. Thus the graph reaches $W = w_0$ at a finite height time and an abscissa $a_* \in [a, La]$, for every $p > 0$.

Keep the original raw cut d_q fixed. Choose an auxiliary cut $0 < d_* < d_q$ sufficiently small compared with w_0, before shrinking g, k. After this shrink $a_* < d_*/2$. On $a_* \le x \le d_*$ use

$$W_{\log x} = F = W(N/D_0 - 2).$$

The strip $w_0 \le W \le w_1$, for a fixed $w_1 > 1/2$, is invariant, with $D_0 \ge cW$. Moreover $F_W \le -c$ and $V_r/V \le -\alpha$, where $\alpha = 3/2$, because $|m| + |m_r| \le C(g + x)$ on this segment and g, d_* are sufficiently small compared with w_0. The limiting strong graph at $g = k = 0$ has $W^*(x) = 1/2 + O(x)$: with $U_* = W^* - 1/2$ its equation is $xU_*' = -2U_* + O(x)$, whose bounded solution follows from the integral equation at zero. Since $x \ge a \ge gku_0$, the strip equation differs by $O(g)$; variation of constants gives

$$|W(d_*) - W^*(d_*)| \le Cg + C(a_*/d_*)^c.$$

On the remaining fixed segment $[d_*, d_q]$, $F = 1 - 2W + O(g + x)$ retains barriers $W = 1/4, 3/4$, with D_0, N, V positive units and bounded derivatives. Continuous dependence gives

$$|W(d_q) - W^*(d_q)| \le Cg + C(a_*/d_*)^c \longrightarrow 0 \tag{J.44}$$

uniformly in the independent scales, because $a_* \le LgkU$. The original passive-coefficient and raw-buffer margins admit these terminal values and the regular continuation to the positive port.

The initial action factor and the full terminal variations. Let $P_j = \mathcal{D}_+^j r|_\eta$, $j = 1, 2$, and $l_j = \mathcal{D}_+^j \eta_l$. From the original initial height,

$$l_1 = H(u), \qquad l_2 = \mathcal{D}_+ H(u), \qquad |l_1| \le C, \quad |l_2| \le Cp, \qquad \mathcal{D}_+ r_l = h, \quad |\mathcal{D}_+ h| \le Cp.$$

Both moving initial rows are

$$P_{1,l} = h - V_l l_1 \asymp p > 0, \qquad P_{2,l} = \mathcal{D}_+(P_{1,l}) - V_{r,l} P_{1,l} l_1 = O(p), \qquad V_l = \frac{gp(1 + \ell_0 Q_0(a))}{N_l}. \tag{J.45}$$

The derivative of $P_{1,l}$ here includes both moving initial coordinates. The inverse equations (A.7) give $(P_2/P_1)_\eta = V_{rr} P_1$. On the weak box, direct differentiation with $W_r = -2W$ gives

$$V_r \le Cg, \qquad |V_{rr}| \le C(g + W), \qquad 0 < P_1(s) \le Cpe^{Cgs}.$$

The integral controlling P_2/P_1 is at most

$$CgpTe^{CgT} + Cpe^{CgT} \int W \, d\eta \le C$$

by (J.43) and $\int W \, d\eta \le Cw_0$. Thus $|P_2| \le CP_1$ there. On $a_* \le x \le d_*$, the negative ratio V_r/V gives

$$0 < P_1(x) \le Cpe^{CgT}(a_*/x)^\alpha, \qquad |P_2(x)| \le CP_1(x). \tag{J.46}$$

The second bound follows by integrating $|V_{rr}| P_1 \, d\eta$ with V a unit. At d_* both rows are $O(a^\alpha)$. The remaining fixed segment has bounded η-length, since

$$\eta_T - \eta_* = 2\log(d_q/d_*) + \log(W_T/W_*).$$

The barriers bound the last logarithm, so the inverse equations transfer both $O(a^\alpha)$ rows to the original cut.

At the original fixed terminal $x = d_q$, both full implicit rows (A.7), with $D = \mathcal{D}_+$, give $|t_1| + |t_2| \le Ca^\alpha \le Cg$, since V_T is a unit. The initial factor p absorbs the arbitrary $|\log p|$ weak dwell. The weak-stage bound $V_r \le Cg$ suffices, so this argument also applies in the compact ghost box.

The exact moving factor and differentiated density. Use (J.9)–(J.10) with $s = +1, x_0 = u, p = P(u)/u^2$. These are exact identities of this same field; only their present bounded-domain estimates are used. Write $\mathscr{B} = \mathscr{B}_+$, $\mathscr{J} = J_+$, $\nu = 6g/(1 + 2g)$, $\lambda = 2\nu$. Direct rational differentiation gives $|\mathcal{D}_+^j \log(-\mathscr{B})| \le Cgp$, $j = 1, 2$. At fixed η, $W_r = -2W$, and direct differentiation gives

$$|\partial_r^j \mathscr{J}| \le C(x + kg^2/x), \quad 0 \le j \le 2, \qquad |\mathscr{J}_\eta| \le CxW.$$

Let

$$I(u) = \int_{\eta_l(u)}^{\eta_T(u)} \mathscr{J}(r(u, \eta), \eta) \, d\eta.$$

Its two complete action rows are (A.8), with $D = \mathcal{D}_+$, $Dr_l = h$, and the fixed terminal r_T. On the weak box, $x + kg^2/x \le Cg$. The integral bounds just established give

$$CgpTe^{CgT} \le CgL_g$$

for every integral in (A.8); here $P_1^2 \le CP_1$ and $|P_2| \le CP_1$. On $a_* \le x \le d_*$, the common bound is

$$Cpe^{CgT} \int_{a_*}^{d_*} (x + kg^2/x)(a_*/x)^\alpha \, d\log x \le Cpe^{CgT}(a_* + kg^2/a_*) \le Cg.$$

This uses $\alpha > 1$, $a_* \asymp a$, and $kg^2/a = g/u \le Cg$. On the remaining fixed strong segment the integrals are bounded by Ca^α. The initial boundary terms are $O(g)$; the full terminal terms are $O(a^\alpha)$ by (A.7). Hence

$$|\mathcal{D}_+ I| + |\mathcal{D}_+^2 I| \le CgL_g. \tag{J.47}$$

At artificial splits the full Leibniz terms cancel in this single integral.

The complete slope on the original observer image. Use (J.11) with $s = +1, x_0 = u$, including the regular continuation in Φ on the admitted image (J.44). Normalize its coefficient-only reference slope as in that formula; the original ordinary C^3 logarithmic-slope bound still holds. The initial rows are

$$\mathcal{D}_+ \log p = Ep/B = O(1), \qquad \mathcal{D}_+^2 \log p = O(p), \qquad \mathcal{D}_+ \log a = h, \quad \mathcal{D}_+^2 \log a = \mathcal{D}_+ h = O(1).$$

Thus the powers $p^{-\nu}$ and $a^{-\lambda}$ contribute $O(g)$; the rows of $\log(1 + \ell_0 Q_0(a))$ and $\log(-\mathscr{B})$ are $O(gp)$. The full terminal formulas (A.7) and the ordinary C^3 observer bound give $O(a^\alpha)$ for both logarithmic rows of the terminal factors. Combining these estimates with (J.47) proves (J.37), uniformly as $p \to 0$. The exact slope retains $d_\theta = c_\theta (gk)^2 \ell_0^{-1-\nu} g^{-\nu}$; frozen affine calibration preserves its logarithmic rows. □

J.5 Separate inward jets in the joint quadratic corner

The derivative of the curvature error involves second derivatives of the relative matching slope. To control these when both lips grow, we construct three action derivatives of each inward remainder on its own domain. Their quotient then has the required mixed derivatives on the product, before equality of the two actual central heights is imposed.

Lemma J.7. *Use the original field* (G.31)*, its moving lips $Y = \ell_0 P$, the original central section $\xi = 0$, and a fixed compact passive A, C neighborhood. Fix the original sufficiently small $\delta_q > 0$, $\pi \in [\pi_\flat, \Pi]$, and a sufficiently large fixed M. For independently small $g, k > 0$, restrict the positive side to strict root-free fibres $\Delta < 0$. On each separate original admissible lip domain with*

$$u, r \ge 2M, \qquad ku, kr \le \delta_q,$$

the positive inward passage and the inverse negative passage reach the same original central section at actual positive heights $Z_+(u)$, $Z_-(r)$. Use (J.1)–(J.3) *with $x_+ = u, x_- = r$, $\theta_s = x_s^{-1} + k x_s$. The separate central remainders satisfy*

$$\sum_{j=1}^{3} |\mathcal{D}_s^j \mathcal{R}_s| \le Cg\theta_s.$$

Their values retain any original coefficient-only logarithmic constants; no small zeroth remainder row is asserted.

On the product of these separate actual domains define

$$N_\pm = -g\mathcal{D}_\pm \log Z_\pm = 1 - gH_\pm - g\mathcal{D}_\pm \mathcal{R}_\pm, \qquad H_\pm = P_\pm'/(xB_\pm), \qquad n = N_+/N_-, \quad \varrho = n - 1.$$

Then $N_\pm \ge 1/2$ and

$$\sum_{i+j \le 2} |\mathcal{D}_+^i \mathcal{D}_-^j \varrho| \le Cg(\theta_+ + \theta_-). \tag{J.48}$$

On the original actual subset $Z_+(u) = Z_-(r)$, this n is exactly the relative variation in

$$R_u = \frac{uB_+(u)\widehat{P}(R)}{RB_-(r)P_+(u)}\, n, \qquad R = kr, \quad \widehat{P}(R) = k^2 P_-(R/k).$$

The admissible central-image intersection may be empty. All coefficients are frozen under the phase operators; constants are independent of the ghost gap and of any comparison between g and k.

Proof. Choose a fixed small $0 < b < M$, with $\Pi b < 1/4$, so $P_s \geq p_b > 0$ on $[0, b]$ after the independent k-shrink. On the positive strict ghost path and the negative joint interval, $P_s > 0$; B_s, Q_s are positive units. Decreasing v follows the positive passage or the inverse negative passage, respectively, in (J.2). For $v \geq b$, put $t = v^{-1}$, $V = kv$, $\vartheta = t + V$, and

$$D_s = \frac{3 - s\pi t + t^2 + sV}{1 + 2g + sgV}(-t\partial_t + V\partial_V),$$
$$h_s = P_s/(v^2 B_s), \quad H_0 = \frac{6}{1+2g}, \quad \ell_g = 1 - Ag + Cg^2.$$

These variables lie in a fixed compact rectangle and $h_s \leq C_b$, including at the ghost bottleneck. The exact formulas

$$H_s - H_0 = \frac{s[-\pi t + (3 - gH_0)V]}{B_s}, \qquad \omega_s = \frac{\zeta_s P_s}{vB_s} = sCg^3 V(1 + sV)h_s$$

give, without dividing by P_s,

$$\begin{aligned} |D_s^j(H_s - H_0)| &\leq C\vartheta && (0 \leq j \leq 3), \\ |D_s^j(Q_s - \ell_g)| &\leq CgV, \quad |D_s^j \omega_s| \leq Cg^3 V && (0 \leq j \leq 3), \\ |D_s^j \vartheta| &\leq C\vartheta && (1 \leq j \leq 3). \end{aligned} \tag{J.49}$$

Use the ratio equation and exact memory of Lemma A.3, with

$$\tau = (A_s(x_s) - A_s(v))/g, \quad T = (A_s(x_s) - A_s(b))/g, \quad W = Y/P_s(v), \quad W(0) = \ell_0,$$
$$f = (Q_s - \omega_s)W/(1 + Q_s W), \qquad I_b = \int_0^{T(x_s)} f\, d\tau.$$

At fixed τ, $\mathcal{D}_{x_s} v = P_s(v)/(vB_s(v))$, so phase rows transport to powers of D_s; clock rows use $-gD_s$. They commute, and $|\vartheta_\tau| \leq Cg\vartheta$ gives $\vartheta(v(\tau)) \leq \theta_s e^{Cg\tau}$. Thus (J.49) verifies all phase and mixed coefficient inputs with $D = \mathcal{D}_{x_s}$, $\kappa = g$, $\rho = g\theta_s$, $\alpha = Cg < c_0/12$. The fixed small lip gives the two negative margins. The lemma constructs the actual graph to b and bounds its fixed-clock rows by $Cg\theta_s(1+\tau)^N e^{-c\tau}$. Each strict fibre has finite action and flight time; their lengths need no uniform upper bound.

The moving clock has $\mathcal{D}T = 1/g$, higher rows zero, and

$$T \geq c_b \log(x_s/b)/g \geq a_b/g, \qquad g^{-N}(1+T)^N e^{-cT} \leq Cg/x_s$$

for every fixed N, after shrinking g. The first lower bound absorbs x_s; the second absorbs every fixed power of g^{-1}. Hence the full terminal formulas (A.10) give $\sum_{j=1}^3 |\mathcal{D}^j I_b| \leq Cg\theta_s$. Their exact entrance memory is

$$\begin{aligned} y_b = \ell_0 P_s(x_s) e^{-T + I_b} &= P_s(b) W(T) \leq Ce^{-cT}, \\ \Lambda_1 = \mathcal{D} \log y_b &= H_s(x_s) - 1/g + \mathcal{D} I_b, \\ \Lambda_j = \mathcal{D}^j \log y_b &= D_s^{j-1} H_s + \mathcal{D}^j I_b \quad (j = 2, 3). \end{aligned}$$

Thus $|\Lambda_1| \leq C/g$ and $|\Lambda_2| + |\Lambda_3| \leq C\theta_s$. On $[0, b]$, preparation (J.4) checks the fixed positive margin, the C/g height bounds, $y_b \leq h_b/2$, and small y_b/g. Lemma A.5 therefore gives the actual centre height, with multiplier $e^{-A_s(b)/g}$, and

$$\log Z_s = \log\{\ell_0 P_s(x_s)\} - A_s(x_s)/g + I_b + \mathcal{N}_s(y_b).$$

Its three composition rows are bounded by $Cy_b/g^4 \leq Cg/x_s$. This proves the weighted remainder estimate on the original centre and admits its height to the original low-height window. Concatenation at the same lips retains the original raw images.

Separated product and actual matching. For the product estimate let $N_0 = 1 - gH_0$ and $d_s = N_s - N_0$. The weighted coefficient and remainder rows give

$$\sum_{j=0}^{2} |\mathcal{D}_s^j d_s| \le Cg\theta_s, \qquad N_s \ge 1/2, \qquad \varrho = \frac{d_+ - d_-}{N_-}.$$

The common homogeneous term N_0 cancels in this quotient. The quotient rule through order two bounds all separated product rows; in particular the genuinely mixed row is

$$\mathcal{D}_+\mathcal{D}_- n = -\frac{(\mathcal{D}_+ N_+)(\mathcal{D}_- N_-)}{N_-^2}.$$

This proves (J.48) on the product before imposing a central-image intersection. Equality of the actual central heights then gives

$$r_u = \frac{uB_+P_-(r)}{rB_-P_+(u)}\,\frac{N_+}{N_-},$$

which is precisely the asserted relative variation after $R = kr$. Only the original central-image intersection is used; no actual image or physical carrier is enlarged. □

J.6 Physical resection at two simple quadratic roots

Near $b/a = 2$, the endpoint value of the curvature principal part can vanish at the same time as both action speeds. Fixed normal sections replace the lip's endpoint powers by pure action residues. This yields errors that vanish with each independent footpoint distance. The actual resection below then transfers the resulting curvature orientation back to the moving-lip description of the same physical map.

Lemma J.8 (Joint simple-root resection on the physical sections)**.** *Use the positive quadratic field and its original raw section buffers. Restrict the real-root coefficients to a fixed neighborhood of $b/a = 2$, so that $b - a$ is a positive unit. All phase derivatives freeze g, k, π, A, C. Write*

$$P(u) = (\mathfrak{c} + ku)(u - a)(u - b), \qquad \widehat{P}(R) = (\mathfrak{c} - R)(R + ka)(R + kb),$$

$$B_+(u) = 1 + g(2 + ku), \qquad B_-(R) = 1 + g(2 - R),$$

$$\mathfrak{a}_+ = \frac{aB_+(a)}{(\mathfrak{c} + ka)(b - a)}, \qquad \mathfrak{a}_- = \frac{\mathfrak{c}B_-(\mathfrak{c})}{(\mathfrak{c} + ka)(\mathfrak{c} + kb)}.$$

The residues and their reciprocals are bounded. Keep all coefficient-only constants in the original actions $A_+ = J_+$, $A_- = A_-(R)$. Their exact pole subtractions are

$$A_+(a - h) = -\mathfrak{a}_+ \log h + j_+(h), \qquad A_-(\mathfrak{c} - t) = -\mathfrak{a}_- \log t + j_-(t),$$

where $j_s - j_s(0)$ have uniform C^3 bounds.

There are fixed normal-height sections in $y_+ = gY$ and $y_- = gk^2Y$, and fixed $h_0, t_0, g_0, k_0 > 0$, with the following properties for independently small $g, k > 0$. The normal-section axis footpoints are $u = a - h$ and $R = \mathfrak{c} - t$. Each normal section has its own actual outward map to the original raw port and observer image. On the actual intersection of the matched graph with $0 < h < h_0$, $0 < t < t_0$,

$$\begin{aligned} A_+(a - h) - A_-(\mathfrak{c} - t) &= C_{g,k} + E_+(h) + E_-(t), \\ |(h\partial_h)^j E_+| \le C\varepsilon_g h, \qquad |(t\partial_t)^j E_-| &\le C\varepsilon_g t, \quad 0 \le j \le 3. \end{aligned} \tag{J.50}$$

where $E_+(0) = E_-(0) = 0$ *and* $\varepsilon_g = Cg(1 + |\log g|)^4$. *No bound on* $C_{g,k}$ *is required. The quotient* n *of the two separated central action derivatives has*

$$\begin{aligned} \sum_{i+j\le 2} |\mathcal{D}_+^i \mathcal{D}_-^j (n-1)| &\le C\varepsilon_g (h+t), \\ \sum_{Z\in\{n,\mathcal{D}_+ n,\mathcal{D}_- n\}} (|\partial_h Z| + |\partial_t Z|) &\le C\varepsilon_g. \end{aligned} \tag{J.51}$$

where

$$\mathcal{D}_+ = \frac{P(u)}{uB_+(u)}\partial_u, \qquad \mathcal{D}_- = \frac{\widehat{P}(R)}{RB_-(R)}\partial_R.$$

The normal-section embeddings use labels $\beta_+ = -u^2/2$, $\beta_- = -R^2/2$. *After separate coefficient-only positive affine value and slope calibrations, their complete maps satisfy*

$$\begin{aligned} \phi_s &= \mathrm{id} + O_{C^3}(\varepsilon_g), \\ |\mathcal{D}_+^j \log \phi_+'| \le C\varepsilon_g h, \qquad |\mathcal{D}_-^j \log \phi_-'| &\le C\varepsilon_g t, \quad j = 1, 2. \end{aligned} \tag{J.52}$$

These statements use only the original ordinary C^3 *logarithmic-slope reserve on the terminal observers, including the divided finite closing on the negative side.*

Let $x = a - u_{\mathrm{lip}}$, $y = \mathfrak{c} - R_{\mathrm{lip}}$ *on the original lips. The actual resection to the normal sections is increasing and satisfies*

$$\begin{aligned} \mathfrak{a}_+ \log h - (\mathfrak{a}_+ + g)\log x &= C_+ + O(x+h), \\ \mathfrak{a}_- \log t - (\mathfrak{a}_- + g)\log y &= C_- + O(y+t), \qquad |C_s| \le C. \end{aligned} \tag{J.53}$$

The remainders have the corresponding finite logarithmic derivative bounds. The negative constant contains no unbounded pure $\log k$. *If* F_s *denotes this resection in the internal labels, then*

$$S_{\mathrm{fp}} = F_- \circ S_{\mathrm{lip}} \circ F_+^{-1}, \qquad \phi_{\mathrm{lip},s} = \phi_{\mathrm{fp},s} \circ F_s \tag{J.54}$$

on the actual common image, up to the retained positive affine calibrations of the physical coordinates. Thus both constructions describe the same completed physical map. No affine identification of the two internal root-distance coordinates is asserted.

Proof. The normal sections and their exact invariant. On the positive side reverse time and use $y = gY$; on the negative side multiply time by k and use $y = gk^2Y$. Division by the positive axis normal rate gives, on both root boxes,

$$v' = gF_s(v) + c_s(v)y, \qquad y' = y\{1 + \kappa_s(v)y\}, \tag{J.55}$$

where

$$\begin{aligned} F_+ &= P/(vB_+), \quad c_+ = Q_+/(vB_+), \quad \kappa_+ = Ckg^2(1+kv)/(vB_+), \\ F_- &= \widehat{P}/(vB_-), \quad c_- = Q_-/(vB_-), \quad \kappa_- = -Cg^2(1-v)/(vB_-), \end{aligned}$$

and $Q_- = 1 - Ag(1-v) + Cg^2(1-v)^2$. At $r_+ = a$, $r_- = \mathfrak{c}$, the axis rates are exactly $gF_s'(r_s) = -g/\mathfrak{a}_s$. On fixed boxes bounded away from zero, gF_s has C^4 norm $O(g)$, $c_s = 1/v + O_{C^4}(g)$, and $\kappa_s = O_{C^4}(g^2)$, uniformly in k.

Apply the backward-integral construction (E.11)–(E.12) to $h = gF_s$. The coefficient bounds just established and $Y(-T) \le Cye^{-T/2}$ give differentiated kernels $C(1+T)^N e^{-(1/2-Cg)T}$ through order three. This supplies π_s and $Z_s = ye^{-\mathcal{B}_s}$, with $\pi_s' = gF_s(\pi_s)$, $(\log Z_s)' = 1$. The same estimates for the difference from $v' = y/v$, $y' = y$ give

$$\begin{aligned} \pi_s(v,y) &= \sqrt{v^2 - 2y} + O_{C^3}(g), \qquad \|\mathcal{B}_s\|_{C^3} \le Cg^2, \\ \mathcal{B}_s &= O(g^2 y), \qquad \partial_v \pi_s \ge \tfrac{1}{2}. \end{aligned} \tag{J.56}$$

Consequently $\log Z_s - A_s(\pi_s)/g$ is constant along each orbit.

Choose fixed small heights $y_{0,s} > 0$. On $y = y_{0,s}$ write $\pi_s(N_s(U), y_{0,s}) = U$. Then

$$N_s(U) = \sqrt{U^2 + 2y_{0,s}} + O_{C^3}(g), \qquad Z_{N,s}(h) = y_{0,s} e^{-\mathcal{B}_s(N_s(r_s-h),y_{0,s})},$$

$$|\partial_h^j \log Z_{N,s}(h)| \le Cg^2, \qquad 1 \le j \le 3.$$

These are the fixed normal sections and their actual axis footpoints. The physical orbit from a sufficiently small lip point reaches its normal section inside the unchanged box: the normal height grows at a fixed rate, the projected axis footpoint remains on the inner side of the root, and its physical displacement is $O(y_{0,s})$.

The local amplitudes and the middle first multipliers. Fix inner cuts $\bar{v}_s = r_s - \delta_s$ in the two boxes. The invariant gives the exact implicit relation

$$\log y_{\mathrm{in}} + e_s(y_{\mathrm{in}}) = \log Z_{N,s}(h) - \{A_s(r_s - h) - A_s(\bar{v}_s)\}/g,$$

$$e_s(y) = -\mathcal{B}_s(\bar{v}_s, y) - \{A_s(\pi_s(\bar{v}_s, y)) - A_s(\bar{v}_s)\}/g.$$

Here $e_s(0) = 0$, and its required height derivatives have polynomial g^{-1} bounds, since the inner cut is a fixed distance from its root. Solving this scalar identity gives

$$\begin{aligned} y_{\mathrm{in},s}(h) &= \alpha_{y,s} h^{\mathfrak{a}_s/g} e^{-[j_s(h)-j_s(0)]/g} u_s(h), \\ u_s(h) &= \frac{Z_{N,s}(h)}{Z_{N,s}(0)} e^{-e_s(y_{\mathrm{in},s}(h))}. \end{aligned} \tag{J.57}$$

with

$$\alpha_{y,s} = Z_{N,s}(0) e^{[A_s(\bar{v}_s)-j_s(0)]/g}, \qquad \alpha_{y,s}^{\pm 1} \le e^{K/g}.$$

The coefficient-only constants cancel in $A_s(\bar{v}_s) - j_s(0)$; thus this local bound permits a global $\log k$ term in the negative action. Passing to Y on the positive cut and $\widehat{Y} = k^2 Y$ on the negative cut adds only the coefficient-only factor g^{-1}.

Choose preliminary h_0, t_0 so the positive residue powers dominate these fixed exponential constants. Three ordinary endpoint derivatives introduce only polynomial g^{-1} factors and inverse endpoint powers. Their nonlinear terms are then bounded by arbitrarily high fixed endpoint powers after shrinking g. The remaining phase part of $\log u_s$ is $O_{C^3}(g^2)$. Consequently the first three logarithmic endpoint rows of $g\{\log u_s(h) - \log u_s(0)\}$ are bounded by $C\varepsilon_g h$.

For the negative growing bridge, take $R_0 = \bar{v}_-$, the already fixed inner cut with a margin below the root and away from zero, and decrease k_0 so that $kr_c < R_0$. Retain its exact first multiplier

$$\widehat{m} = \exp\left\{-\frac{1}{g}\int_{kr_c}^{R_0} \frac{RB_-}{\widehat{P}}\, dR\right\}, \qquad \widehat{m}/k^2 \le e^{K/g} k^{c_0/g-2} \le e^{K/g}.$$

Indeed $\widehat{P} \asymp R^2$ on $Kk \le R \le R_0$, giving the action lower bound $c_0 \log(1/k) - K$. Choose $g_0 < c_0/4$. No reciprocal bound for this collapsing multiplier is used. Put $W = \widehat{Y}/\widehat{P}$ and $q = (A_-(R_0) - A_-(R))/g$. Coefficients and both radial cuts are fixed under the incoming height variation; the flight $T = (A_-(R_0) - A_-(kr_c))/g$ may be arbitrarily long. On $kr_c \le R \le R_0$, the fixed root margin gives $\widehat{P} \asymp (R+k)^2$ and $k/R \le 1/r_c$. Hence

$$Q_- - 1 = O(g), \qquad H = \widehat{P}_R/(RB_-) = O(1), \qquad \omega = -Cg^3(1-R)\widehat{P}/(RB_-) = O(g^3).$$

The exact ratio equation has $\kappa = g$. On the unchanged fixed small buffer $0 \le W \le \ell_0$, these bounds keep $1 + Q_- W$ a positive unit and both $\mathcal{F}/W, \mathcal{F}_W \le -c_1 < 0$, after decreasing g_0. At the fixed initial cut $\widehat{P}(R_0)$ is a positive unit. The residue formula (J.57), with its factor g^{-1}, places $w(t) = \widehat{Y}_0(t)/\widehat{P}(R_0) \le \ell_0/2$ after the same independent coefficient shrink. Thus

Lemma A.4 applies on this full buffer. Retaining the terminal polynomial in its exact cocycle gives

$$\widehat{Y}_T = \widehat{Y}_0 \widehat{m} e^{\mathcal{N}_T(w)}, \qquad |\partial_w^j \mathcal{N}_T| \le C \quad (1 \le j \le 3), \qquad |(w\partial_w)^j \mathcal{N}_T| \le Cw \quad (0 \le j \le 3).$$

The normalized nonlinear unit therefore has bounded ordinary C^3 rows in the incoming height at the fixed cut, uniformly in g, k, T. Its three endpoint rows follow by the Euler composition formula of Lemma A.5, with $\Lambda_j = (t\partial_t)^j \log w$: the flat w absorbs the polynomial g^{-1} and endpoint powers from the local residue. These are fixed-route height rows; phase and moving-terminal rows on other routes remain those of Lemma A.3. Conversion to Y contributes k^{-2} only to $\widehat{m}/k^2$, with no reciprocal gain estimate.

Fix the prescribed height buffers of the final compact inward continuations and their coefficient bounds before the final choice of h_0, t_0. Enlarge K_{tot} to include the regular local action parts and the preceding bound for $\widehat{m}/k^2$. For any fixed endpoint power n and finite polynomial loss g^{-M}, choose $t_0 < 1$ so that $\mathfrak{a}_{\min} \log(1/t_0) > K_{\mathrm{tot}} + 2c_*$, with $c_* > 0$. Then decrease g_0; the residue and its three ordinary endpoint derivatives give

$$g^{-M} e^{K_{\mathrm{tot}}/g} t^{\mathfrak{a}_{\min}/g - j} \le C e^{-c_*/g} t^n, \qquad 0 < t \le t_0, \quad 0 \le j \le 3.$$

In particular the converted input satisfies $Y_T \le h_b/2$ and small Y_T/g in its already prescribed buffer h_b, uniformly in independently small k. The positive input has the same bound after the analogous final choice of h_0. Lemma A.5 now applies to the actual final inward segments: their linear propagators contract, and their nonlinear height bounds are polynomial in g^{-1}. Its three Euler composition rows and the displayed superpower bound control the endpoint rows on those original centre windows. No reciprocal linear multiplier is introduced in this admission. Equality of the two exact original central heights therefore gives (J.50); all coefficient-only first multipliers remain in $C_{g,k}$. Put $d_+ = h$, $d_- = t$. On the fixed root boxes, $\mathcal{D}_s = -d_s \gamma_s(d_s) \partial_{d_s}$, where γ_s are positive units with bounded C^2 norms. The corrected actions $A_+ - E_+$ and $A_- + E_-$ in (J.50) have action derivatives

$$\mathcal{N}_+ = 1 - \mathcal{D}_+ E_+, \qquad \mathcal{N}_- = 1 + \mathcal{D}_- E_-, \qquad n = \mathcal{N}_+ / \mathcal{N}_-.$$

The three weighted rows of E_s give

$$|\mathcal{D}_s^j (\mathcal{N}_s - 1)| \le C \varepsilon_g d_s \quad (0 \le j \le 2), \qquad |\partial_{d_s} \mathcal{N}_s| + |\partial_{d_s} (\mathcal{D}_s \mathcal{N}_s)| \le C \varepsilon_g.$$

For the ordinary derivatives, divide the second and third action rows of E_s by the single factor $d_s \gamma_s$. After shrinking the fixed boxes, $\mathcal{N}_s \ge 1/2$. The separated quotient calculation in the proof of Lemma J.7 now applies with errors $\varepsilon_g h$ and $\varepsilon_g t$; its mixed row is $O(\varepsilon_g^2 ht)$. Differentiating the quotient and its first action rows with the displayed ordinary bounds proves (J.51) on the product, before restriction to the actual match and without comparing h and t.

The actual fixed-normal exterior embeddings. Put $\sigma_+ = gk$, $\sigma_- = g$. At a normal section,

$$x_l = \sigma_s N_s(U), \qquad z_l = \sigma_s^2 y_{0,s}.$$

The ratio $W_l = y_{0,s}/N_s(U)^2$ is a positive unit, even when the normal chord lies beyond its axis root. Use the signed coefficients of (J.6), writing N_s^{raw} for its numerator and $D_s = m_s + Q_s W$. As in the bounded positive outward construction, $(D_s)_\eta = Q_s W > 0$ at $D_s = 0$, so the inverse-height graph retains its positive denominator. Choose a fixed auxiliary $d_* < d_q$ small compared with the lower bound for W_l. On $[x_l, d_*]$, the singular terms in $m_s - sx$ are $O(g)$. The same invariant-strip argument, starting now at x_l with no weak stage, gives

$$w_*/2 \le W \le w^*, \qquad D_s \ge cW, \qquad V_r/V = -2 + O((g+x)/w_*) \le -3/2,$$

$$|W(d_*) - W_s^*(d_*)| \le Cg + C(x_l/d_*)^c.$$

The fixed parabolic continuation therefore admits the graph to its original raw buffer and observer.

Here $\partial_U \eta_l = \partial_U^2 \eta_l = 0$, and the inverse initial rows are bounded with positive first row. Applying (A.7) as above gives

$$0 < P_1 \le C(x_l/x)^{3/2}, \quad |P_2| \le CP_1, \qquad |\partial_U \eta_T| + |\partial_U^2 \eta_T| \le C\sigma_s^{3/2}.$$

The exact first-variation cocycle and density split, with $\nu = 6g/(1+2g)$, $\lambda = 2\nu$, give

$$J_z = \frac{W_T/D_T}{W_l/D_l}(z_T/z_l)^\nu e^{I_s}.$$

The density's first two fixed-height radial derivatives are bounded by $C(x + kg^2/x)$. Thus (A.8), with $D = \partial_U$ and both lower clock rows zero, bounds its two complete rows by

$$C(\sigma_s + kg^2/\sigma_s) + C\sigma_s^{3/2} \le Cg.$$

Since z_l is phase-constant and $\beta = -U^2/2$, the full moving first variation is

$$\Gamma_s'(\beta) = \sigma_s^2 \frac{N_s(U)N_s'(U)}{U} \frac{W_l N_l^{\mathrm{raw}}}{D_l} J_z \Phi'(z_T).$$

Removing only the coefficient-only observer gain gives the exact form

$$\Gamma_s'(\beta) = c_s^* \sigma_s^{2-\lambda} y_{0,s}^{-\nu} \frac{N_s(U)N_s'(U)}{U} N_l^{\mathrm{raw}} \frac{W_T}{D_T} z_T^\nu \widehat{\Phi}'(z_T) e^{I_s}. \tag{J.58}$$

The initial factor equals $1 + O_{C^2}(g)$, because (J.56) gives $N_s(U)N_s'(U)/U = 1 + O_{C^2}(g)$. The terminal factors have two logarithmic rows $O(\sigma_s^{3/2})$, using just the original ordinary C^3 logarithmic-slope observer bound. Thus both ordinary logarithmic rows of the slope are $O(g)$. Subtracting its exact value and calibrating its exact positive slope at a fixed reference point gives $\phi_s = \mathrm{id} + O_{C^3}(g)$. The action fields contain the factors h, t, proving (J.52). The two physical gains $\sigma_+^{2-\lambda}$, $\sigma_-^{2-\lambda}$ are retained coefficient-only factors in (J.58); their ratio and their reciprocals are not estimated.

Resection of the lip and the same physical curvature. On either lip the local normal coordinate is

$$y_{\mathrm{lip},+} = g\ell_0 P(a - x), \qquad y_{\mathrm{lip},-} = g\ell_0 \widehat{P}(\mathfrak{c} - y).$$

It is g times a positive unit times the root distance. The projected lip distance is $\widehat{x} = x(1+O(g))$, respectively $\widehat{y} = y(1 + O(g))$. Applying the exact invariant between the lip and its normal-section intersection, and subtracting the two pole expansions, gives (J.53). The coefficient terms are $g \log g$, logarithms of local positive units, and the bounded projection correction $\mathfrak{a}_s \log(\widehat{x}/x)$. The negative comparison is in $gk^2 Y$, so it has no pure $\log k$. The common growing-bridge multiplier cancels from this local comparison. The finite logarithmic derivative estimates follow by differentiating this invariant on the positive interior of the lip. In particular

$$h = A_+ x^{1+g/\mathfrak{a}_+} e^{O(x+h)}, \qquad t = A_- y^{1+g/\mathfrak{a}_-} e^{O(y+t)}, \qquad A_s^{\pm 1} \le C.$$

The increasing maps give uniform inclusions between the nested endpoint neighborhoods; the lip exponents are $\mathfrak{a}_s/g + 1$, whereas the normal residue quotient has no added g.

Both resection coordinates belong to the same orbit, so (J.54) identifies their completed maps after the retained positive affine physical calibrations. The normal physical input is increasing in h, since $d\{-(a-h)^2/2\}/dh = a - h > 0$; the lip physical input is increasing in x, and the resection is increasing. Consequently the same completed physical map has the same curvature zeros and crossing directions in the two descriptions. □

J.7 Paired first variation on positive quadratic fibres

Above a comparison orbit the complete derivative is bounded through the ratio of two variations. Their separate core factors need not be units, and both exterior gains contain a small common power of gk. The next lemma retains these factors until their cancellation in the physical derivative.

Lemma J.9 (Paired first variation above a quadratic comparison)**.** *Fix the original raw sections, nested buffers, ordinary C^3 regular observers and the divided finite closing of Lemma 2.2. Use the polynomial and physical coordinates of* (G.31)*, and write $a = d - \sqrt{\Delta}$ when $\Delta \geq 0$. The passive compact (A, C)-set lies inside their prescribed common coefficient neighbourhood, with the original buffer and observer margins. Fix a finite coefficient bound Π. In the original domain construction choose X and the retained ghost width jointly for the prescribed finite comparison action threshold, and then fix $\pi_\flat$. For $\pi \in [\pi_\flat, \Pi]$, there are fixed $\eta_{\max}, C_* > 0$ and independent upper bounds for g, k with the following property on the retained real, fold and ghost fibres. Use coefficient fibres with the actual $K_{\mathrm{fin}} > 0$ and the original $0 < m \leq m_0$ finite-closing admission. Choose the comparison lip at*

$$u_* = a - X/2 \quad (\Delta \geq 0), \qquad u_* = d - X/2 \quad (\Delta < 0), \qquad \eta_{\mathrm{comp}} = gZ_+(u_*).$$

The comparison lies strictly inside the enlarged low lip interval beginning at $a - X$, respectively $d - X$. The actual completed transition in the original affine input satisfies

$$0 < \Pi' \leq C_* \qquad \textit{throughout} \qquad \eta_{\mathrm{comp}} \leq \eta_0 = gY(0) \leq \eta_{\max}.$$

This entire interval is admitted to the original section buffers. The constants are independent of the root gap and of any relation between g and k. After the common choice $mC_* < 1/2$, $F' = 1 - m\Pi' > 1/2$ on it.

Proof. Write

$$\eta = gY, \qquad \epsilon_0 = gk, \qquad D = gP + Q\eta,$$
$$Q = 1 - Ag(1 + k\xi) + Cg^2(1 + k\xi)^2, \qquad B = 1 + g(2 + k\xi).$$

The exact field is

$$\dot{\xi} = -D, \qquad \dot{\eta} = -\eta\{\xi B + H_1\eta\}, \qquad H_1 = Ckg^2(1 + k\xi), \qquad z = \epsilon_0^2\eta.$$

By (C.1), this is the source field with coefficient quartet $(-g, g\pi, g, \epsilon_0)$; ϵ_0 is held fixed under every phase derivative.

The actual comparison and the common cuts. Use the jointly chosen fixed $X > 0$. The abscissa u_* ranges over a fixed positive compact interval, and there is a fixed $\epsilon > 0$ such that

$$P \geq p_* > 0 \qquad \text{on } [0, u_* + \epsilon].$$

The factorization gives this margin. The comparison action choice of Proposition G.5 first fixes X by $J_+(a - X) \geq c \log(\delta_c/X)$, then the ghost width; the fold has the stronger inverse-distance divergence.

Corollaries J.6 and J.4 give the original inward cocycles, without changing the centre,

$$\log Z_+ = \log(\ell_0 P) - J_+/g + \mathcal{R}_+, \qquad \log Z_- = \log(\ell_0 P_-) - A_-/g + \mathcal{R}_-.$$

Their zeroth rows on the bounded comparison and the entire strict negative side are bounded. Indeed their ratio equations give $W \leq \ell_0 e^{-c\tau}$, hence $|\int (Q - \omega)W/(1 + QW)\, d\tau| \leq C\ell_0$, and the original compact continuation adds $O(Y_b/g)$. The original action constants remain, $N_- \geq 1/2$,

and $Z_- \downarrow 0$ at its simple root. Choose fixed $r_{\max} > r_c$ with $A_-(r_{\max})$ above the bounded comparison action, then shrink k. Monotonicity and the zeroth cocycles give the unique actual match

$$r_* \in [r_c, r_{\max}], \qquad Z_+(u_*) = Z_-(r_*), \qquad \eta_{\mathrm{comp}} = gZ_+(u_*).$$

On either comparison lip, $z/x^2 = \ell_0 gp$, with p a positive unit: on the negative lip r_* is bounded. Use the original inverse logarithmic-height graph until it reaches a fixed small strip height w_0. Its bounded-comparison weak length is $O(1 + |\log g|)$. The positive weak displacement estimate is $O(g(1 + |\log g|)) + O(w_0)$ in $\log x$. Choose w_0 first so small that this strip is entered before $\xi = u_* + \epsilon/2$, and then decrease g. This uses a fixed positive margin of P, and is uniform through the fold. The negative weak stage reaches its fixed strip by $|\xi| \le L_1 r_*$, with L_1 fixed.

The outward constructions of Corollaries J.6 and J.3 continue both comparison traces. Choose one fixed L larger than those entrance bounds and the bounded positive factors. On the two coefficient-dependent but phase-fixed cuts $\xi = \pm L$, the comparison has

$$c_L \le \eta(\pm L) \le C_L, \qquad D(\pm L) \asymp \eta(\pm L).$$

Increasing L, the strip contraction places both values of η/L^2 inside the same small parabolic band. The original outward continuation, including its weak-stage initial factors, lands strictly inside the original raw buffers. No raw section or observer window has been replaced.

One upper reference and all intervening inputs. Fix $\eta_{\max} > 0$. At $g = k = 0$ the core graph with centre height $\eta_{\max}$ is $\eta = \eta_{\max} + \xi^2/2$, with strictly positive height and speed on every fixed core. Ordinary dependence on $[-L, L]$ constructs its actual upper reference for small independent g, k. Choose L large enough that its two values of η/L^2 belong to the same parabolic band as the comparison.

The inverse-height first-exit calculation continues this reference to an auxiliary raw cut $d_* < d_q$, where strip contraction gives

$$|W(d_*) - W^*(d_*)| \le Cg + C(\epsilon_0 L/d_*)^c \longrightarrow 0.$$

The regular segment to the original port has the comparison's limit $W^*(d_q)$; the prescribed passive and raw-buffer margins admit it.

Scalar uniqueness orders the graphs between the comparison and this upper reference. They stay between the two finite positive core graphs and between their admitted outward tails. Thus every $\eta_0 \in [\eta_{\mathrm{comp}}, \eta_{\max}]$ has an actual through graph on the entire word and remains in the original convex raw and observer windows. Its central raw height is $z(\gamma) = \epsilon_0^2 \eta_0$; after the common coefficient shrink the original low-height and speed restrictions are satisfied. The whole interval, including nonroot inputs, is consequently used.

The paired core variation. For

$$f = \frac{d\eta}{d\xi} = \frac{\eta(\xi B + H_1 \eta)}{D}$$

use (A.5), with residual $[gP' + (Q' + H_1)\eta]/D$. On $[-L, u_* + \epsilon]$, P has a fixed positive lower bound after decreasing k; on the remaining positive core, the comparison has already entered its invariant strip. Hence

$$D_{\mathrm{comp}} \ge cg \quad \text{on } [-L, L], \qquad \eta/D \le C$$

on every graph above it. In the part where P could be negative, $\eta \ge c > 0$, so $D \ge c\eta$ after decreasing g. Since P' is bounded and $Q' = O(gk)$, $H_1 = O(kg^2)$, the remaining integral in the identity is bounded in absolute value on the full core. The endpoint quotients $(\eta/D)(\pm L)$ are positive units.

Set

$$A_{\pm} = \frac{\partial \eta(\pm L; \eta_0)}{\partial \eta_0} > 0.$$

Integration gives

$$\frac{A_-}{A_+} = \frac{(\eta/D)(-L)}{(\eta/D)(L)} \exp\left\{ \int_L^{-L} \frac{gP' + (Q' + H_1)\eta}{D}\, d\xi \right\} \in [c, C]. \tag{J.59}$$

This estimate controls the ratio of the two core variations; separate unit bounds for $A_{\pm}$ are not required. For comparison with a moving lip phase u, the exact incoming factor is

$$\frac{d\eta_0}{du} = \eta_0 \left\{ \frac{P'}{P} - \frac{uB_+}{gP} + \mathcal{R}'_+ \right\} = -\frac{uB_+\eta_0}{gP} N_+.$$

The core-end derivatives are $A_{\pm}\, d\eta_0/du$, so this complete moving-source factor cancels. The common cuts are phase-fixed.

Both original tails and the physical derivative. In the source coordinates (ξ, η), with quartet $(-g, g\pi, g, \epsilon_0)$, the displayed parabolic rational calculation (C.12) gives, on either admitted tail,

$$F_{\sigma,\eta} - \lambda_g / r = O(\epsilon_0 + g\pi r^{-2} + r^{-3+\nu}), \qquad \lambda_g = \frac{12g}{1+2g}, \qquad 0 < \nu < 1.$$

Its remainder has bounded integral from L to d_σ/ϵ_0. The parabolic bands just constructed justify this application even though the centre height may be exponentially small. Consequently the two actual tail first variations are

$$J_\sigma = (d_\sigma/(\epsilon_0 L))^{\lambda_g} U_\sigma, \qquad U_\sigma, U_\sigma^{-1} \le C.$$

At the original raw ports the complete core-plus-tail variations are exactly

$$\frac{dz_\sigma}{d\eta_0} = \epsilon_0^{2-\lambda_g} (d_\sigma/L)^{\lambda_g} A_\sigma U_\sigma. \tag{J.60}$$

All terminal factors lie on the admitted original images. The original bridges and divided closing, with $K_{\mathrm{fin}} > 0$, $m \le m_0$, have positive derivative units by Lemma 2.2.

The common $\epsilon_0^{2-\lambda_g} = (gk)^{2-\lambda_g}$ factor cancels in the ratio of the two raw variations. The ratio $(d_-/d_+)^{\lambda_g}$, the two U-factors, and the paired core ratio are bounded units. Composing the original regular bridges and divided closing therefore proves $0 < \Pi' \le C_*$ in the same original affine input on the entire stated interval. Choose $mC_* < 1/2$ only after these constants and buffers have been fixed. This gives the stated strict sign of F'. □

K Coefficient dictionaries and passage domains

We express each coefficient limit in a chart where the corresponding passage estimate is uniform. The substitutions are exact, so the estimates apply to the original coefficient fibres. We then choose overlapping chart ranges in the order required by Section 5.9. All constants precede the common raw cap and final coefficient restrictions in Lemma 5.14.

K.1 Proof of Lemma 5.9

Proof. If $p \le 0$, use Lemma H.15. For $p > 0$, first fix the positive-quadratic ghost and root-free reserves, and then the independent far-root rectangles.

Fix a small number $\mathfrak{q}_{\rm far} > 0$ for which Proposition G.2 applies. On the negative γ-side, Proposition H.5 and the dictionary below provide numbers $\mathfrak{q}_- > 0$ and $\Pi_- > 1$. Put

$$\mathfrak{q}_\flat = \frac{1}{2}\min(\mathfrak{q}_{\rm far}, \mathfrak{q}_-), \qquad \Pi_{\rm far} = \max\{\lambda_0^{-1/2}, \Pi_-, 2\sqrt{12}\}, \qquad \Pi_{\max} = 2\Pi_{\rm far}, \tag{K.1}$$

where λ_0 is the constant in (G.8). Apply the whole-carrier estimate of Corollary G.7 and Proposition G.1 on $0 < \pi \le \Pi_{\max}$.

For $\mathfrak{q} = k\pi \ge \mathfrak{q}_\flat$, the degree-loss dictionary given below is uniform once $k \le k_d(\mathfrak{q}_\flat)$. Choose $k_\flat > 0$ so that $2k_\flat$ is smaller than this number and than the k-bounds in both compact-π carriers, including the whole positive-carrier bound $k_0 \le k_{\rm tail}$. Put

$$G_{\max} = k_\flat^{-2/3}. \tag{K.2}$$

For this fixed $G_{\max}$, choose U_d so that the large-U dictionary below belongs to Proposition H.4, and then choose $U_{\rm cub} > U_d + 1$. Apply Proposition 4.1 on $0 \le U \le U_{\rm cub}$, $|G| \le G_{\max}$. The bounded positive-quadratic $k \to 0$ face remains in the first compactification; it is distinct from this compact cubic rectangle.

We first check that the two degree-loss substitutions reproduce the source coefficients exactly. In the γ-dominant chart (4.4), let $\epsilon = \operatorname{sgn}\gamma$ and put $t = k\xi$. The exact polynomial is

$$F_{\epsilon,\mathfrak{q},k}(t) = t^3 + 3\epsilon t^2 - \mathfrak{q}t + k^2. \tag{K.3}$$

For $\mathfrak{q} \ge \mathfrak{q}_\flat$ and k small it has roots $-r < t_a < t_c$, all three simple. Vieta's formulas are

$$t_a + t_c = r - 3\epsilon, \qquad r t_a t_c = k^2, \qquad \mathfrak{q} = r(r - 3\epsilon) - \frac{k^2}{r}.$$

Define

$$\begin{gathered} \tau = p^2, \qquad \kappa = \frac{k^2}{\mathfrak{q}^{3/2}}, \qquad \mathcal{R} = \frac{r}{\sqrt{\mathfrak{q}}}, \qquad B = \frac{\mathfrak{q}}{r^2}, \qquad \omega = \frac{\sqrt{\mathfrak{q}}}{r}, \\ \nu = \frac{\epsilon}{r}, \qquad a = \frac{t_a}{r}, \qquad c = \frac{t_c}{r}, \qquad A_d = \frac{B^{3/2}}{c}. \end{gathered} \tag{K.4}$$

These identities imply (H.20) exactly. Moreover

$$(\Lambda_1, \Lambda_2, \Lambda_3) = (1, \kappa^2, \epsilon/\pi^2),$$

so

$$(\sqrt{\tau}\Lambda_1, \tau\Lambda_3, \tau^{3/2}\Lambda_2) = (p, \gamma, s).$$

For $\mathfrak{q} \in [\mathfrak{q}_\flat, \infty]$, all positive-unit quantities in Proposition H.4 have compact closure: $r/\sqrt{\mathfrak{q}} \to 1$ and $(B, \omega, c, \nu, A_d) \to (1, 1, 1, 0, 1)$ as $\mathfrak{q} \to \infty$.

On the negative-γ far face $\mathfrak{q} \downarrow 0$, $\pi \to \infty$, use the roots of

$$F_{-,\mathfrak{q},k}(t) = t^3 - 3t^2 - \mathfrak{q}t + k^2.$$

The exact regular coordinates of Proposition H.5 are

$$\begin{gathered} a = \frac{t_a}{r}, \qquad j = \frac{r}{t_c}, \qquad \kappa = a, \qquad \mathcal{R} = j, \qquad d = B = A_d = 1, \\ \nu = -\frac{1}{r}, \qquad \omega = 1 - a + ja, \qquad \tau = \frac{g t_c}{a}. \end{gathered} \tag{K.5}$$

Vieta's formulas give

$$t_a + t_c = r + 3, \qquad r t_a t_c = k^2, \qquad \mathfrak{q} = r(r+3) - \frac{k^2}{r}, \qquad \pi^2 = \frac{t_c}{a}(1 - a + ja)^2, \qquad \tau = \frac{p^2}{\omega^2}.$$

Hence

$$r \sim \mathfrak{q}/3, \quad t_c \to 3, \quad a \sim 3/\pi^2, \quad j \sim \mathfrak{q}/9, \quad \tau \sim p^2.$$

Thus $(j, \kappa, \tau) \to 0$, the positive-unit entries stay in compact ranges, and $j\nu = -1/t_c = (j - ja - 1)/3$ stays bounded. The same source coefficients are recovered exactly. Fix $\mathfrak{q}_-$, then Π_-, so that $\omega \geq \omega_{\min} > 0$, and let τ_- be the reciprocal radial bound. The rectangle

$$0 < \mathfrak{q} \leq \mathfrak{q}_-, \qquad \pi \geq \Pi_-, \qquad p^2 \leq \tau_- \omega_{\min}^2 \tag{K.6}$$

lies within the range of Proposition H.5. The estimate therefore holds throughout this rectangle, independently of the relative rates of its parameters.

For the complementary cubic chart (4.5), the large U polynomial

$$y^3 + 3Gy^2 - Uy + 1$$

has three simple roots $-r < t_a < t_c$ when $U \geq U_d(G_{\max})$. Set

$$\begin{gathered} \tau = p^2, \qquad \kappa = U^{-3/2}, \qquad \mathcal{R} = \frac{r}{\sqrt{U}}, \qquad B = \frac{U}{r^2}, \qquad \omega = \frac{\sqrt{U}}{r}, \\ \nu = \frac{G}{r}, \qquad a = \frac{t_a}{r}, \qquad c = \frac{t_c}{r}, \qquad A_d = \frac{B^{3/2}}{c}. \end{gathered} \tag{K.7}$$

Again Vieta's formulas imply (H.20), and

$$(\Lambda_1, \Lambda_2, \Lambda_3) = (1, U^{-3}, GU^{-2})$$

recovers $(p, \gamma, s) = (tU, t^2G, t^3)$ exactly. Uniformly for $|G| \leq G_{\max}$, all positive units in (K.7) tend to one and $\nu \to 0$.

The compact-degree positive units have compact closure. Let $\tau_d > 0$ be their radial bound, and let the positive far-root bound τ_0 include its fixed core/high splice reserve. Choose once

$$0 < p_*^2 < \tfrac{1}{2} \min\{\tau_0, \tau_d, \tau_- \omega_{\min}^2\}. \tag{K.8}$$

Then the degree and positive far-root scales $\tau = p^2$, and the negative far-root scale $\tau = p^2/\omega^2$, lie in their required ranges. These radial restrictions must be chosen explicitly, since shrinking the raw coefficients alone does not control the chart ratios.

The chosen chart ranges now cover every coefficient alternative. If $\gamma \neq 0$ and $0 < k \leq 2k_\flat$, then:

- for $0 < \pi \leq \Pi_{\max}$, use Propositions G.5 and G.1, according to the sign of γ;
- for $\pi > \Pi_{\text{far}}$ and $\mathfrak{q} \geq \mathfrak{q}_\flat$, use Proposition H.4 through (K.4);
- for $\pi > \Pi_{\text{far}}$, $\mathfrak{q} < \mathfrak{q}_\flat$, and $\gamma > 0$, use Proposition G.2;
- in the same range with $\gamma < 0$, use Proposition H.5 through (K.5).

The compact and far π-ranges, and the two $\mathfrak{q}$-ranges, overlap strictly. If $\gamma = 0$, or if $k \geq k_\flat$, then

$$|G|^{3/2} = \frac{|\gamma|^{3/2}}{s} \leq k_\flat^{-1}.$$

For $U \leq U_{\text{cub}}$, use Proposition 4.1; for $U \geq U_d$, use Proposition H.4 through (K.7). These intervals also overlap.

The remaining radii $g = |\gamma|$, $t = s^{1/3}$, and s can be restricted independently. This gives eventual $s_*, \gamma_* > 0$; the common raw cap is fixed before their final reduction and the choice of m. The closed alternatives include every equality face, and the bands in $k, \pi, \mathfrak{q}, U$ overlap strictly. These choices depend only on the coefficients and are made before any phase point is selected. □

K.2 Proof of Lemma 5.10

Proof. At the central section,

$$Y_L(0) = p^{-2}Y_0 \le K_0(1+p^{-2}) \le 2K_0.$$

The graph equation obtained directly from (4.2) is

$$\frac{dY_L}{dX} = \frac{Y_L\{q_\gamma X + r_L X^2 + Cr_L(\gamma + r_L X)Y_L\}}{\ell(\gamma + r_L X)Y_L + p^{-2} - X + 3\gamma X^2 + r_L X^3}. \tag{K.9}$$

The denominator is positive on a through graph. At low height it may approach zero, so we estimate growth only after the height has crossed a fixed positive threshold. Choose $M_0 > 2K_0 + 1$ so that, whenever $Y_L \ge M_0$, the denominator is at least $\ell_0 Y_L/2$. On that region

$$Y_L' \le C_{L_c} + Cr_L(|\gamma| + r_L L_c)Y_L \le C_{L_c} + Y_L$$

after a raw coefficient shrink. Apply this inequality after the first crossing of M_0, or equivalently to $\max(M_0, Y_L)$. Gronwall's inequality gives

$$Y_L(L_c) \le (M_0 + C_{L_c}L_c)e^{L_c} =: B_c.$$

Strict scalar order gives $Y_L(L_c) > d_L$; since $d_L \ge 0$, (5.10) follows. No derivative lower bound for the map from $X = 0$ to $X = L_c$ has been used. □

K.3 Proof of Lemma 5.11

Proof. Fix the constant K_0 in Corollary 5.1. Its complete derivative bound applies on the connected high-height interval

$$Y_0 > K_0(1+p^2), \qquad 0 < s^2 Y_0 < z_*. \tag{K.10}$$

It remains to cover the lower-height complement by the stopped-trace and strict source intervals.

Use the positive signed trace chart

$$\alpha_0 = -r_L^2 U_0 d, \qquad \alpha_1 = r_L p_L, \qquad d = \frac{p_L^2}{U_0 p^2}, \qquad \frac{s^2}{r_L^2} = \frac{p_L^2}{p^2}, \tag{K.11}$$

where p_L and U_0 are positive units. In the notation of Proposition I.3, the signed trace parameter is

$$a_L^{\mathrm{tr}} := -U_0 d = -\frac{p_L^2}{p^2}, \qquad \alpha_0 = r_L^2 a_L^{\mathrm{tr}}.$$

The stopped-trace proposition therefore applies with the signed parameter a_L^{tr}, which includes the positive factor d. By increasing P_*, a_L^{tr} lies in its signed trace block. For an orbit in the complementary range $Y_0 \le K_0(1+p^2)$,

$$Y_L(0) = \frac{s^2 Y_0}{r_L^2} \le K_0 p_L^2(1+p^{-2}) \le C.$$

The height lemma also applies when p_L ranges over this compact positive interval: its coordinates are related to the $r_L = \alpha_1$ coordinates by $X = p_L X_{\mathrm{old}}$, $Y_L = p_L^2 Y_{\mathrm{old}}$. Its Gronwall bound is uniform for the corresponding cuts $X_{\mathrm{old}} = L_c/p_L$ in a fixed compact interval. Lemma 5.10 therefore gives a fixed number B_0 such that

$$0 < \frac{\mathfrak{g}_L}{r_L^2} \le B_0. \tag{K.12}$$

Take $\xi_* = \xi_0$ from (I.34). The compact annulus $\xi_* \le \mathfrak{g}_L/r_L^2 \le B_0$ is covered by finitely many strict source intervals, overlapping the terminal range $0 < \mathfrak{g}_L/r_L^2 \le 2\xi_*$. On these intervals the coefficients satisfy

$$a_L^{\mathrm{tr}} = -\frac{p_L^2}{p^2} \in [-P_*^{-2}C, 0], \qquad p_L \in [1/2, 3/2],$$

and all orbit segments retain strict positive margins. Proposition I.3, with (I.30) and (I.34), supplies their every-point physical cover. Proposition 5.7 controls the derivative on the connected union of these intervals. The choices are

$$K_0,\ L_c,\ B_c,\ B_0; \quad \text{terminal interval, } \xi_*;$$
$$P_*,\ \text{finite strict cover, } L_t; \quad \text{common raw cap, raw coefficient neighbourhood, } m.$$

The strict margins ensure that every later shrink preserves the domains and overlaps already chosen. □

L Notation and reading index

The original DRR field and graphic are F_*, Γ_{I4}; the canonical endpoint is H_0, Γ; the neighboring organizing family is H_c, Γ_c. The full quadratic germ is $\mathcal{Q}_2$, while $\theta = (A, C, \mu_1, \mu_2, \mu_3)$ denotes the five-parameter transverse slice. In the positive-infinity chart, $v = X/Y$ and $z = 1/Y$; h denotes a central height only where that local section is fixed. The exact return is written $\mathcal{P}_\theta(x) = m_\theta \Pi_\theta(x)$, where m_θ is the finite through multiplier and Π_θ is the completed divided transition; $F_\theta = x - \mathcal{P}_\theta$ is the displacement.

The local finite saddle–node through multiplier is $m_{\mathrm{fin}} = e^{-T_f}$ in (2.12), written m in the completed return (2.13). In the fixed-c lower construction, Λ_ρ is the complete external gain, including the two tail gains and the finite connection, as defined in (6.22). The action-model target is $m_\varepsilon = 2\varepsilon^{5/2}$; the two-coefficient calibration (6.24) imposes $\Lambda_\rho = m_\varepsilon$.

In the positive-infinity chart, the polynomials are

$$P(v) = \mu_1 + \mu_2 v - v^3, \qquad R(v) = \mu_3 + v + v^2$$

as in (2.5). The symbol γ denotes the small root of the transverse-rate equation $R(\gamma) = 0$, equivalently the root selected in (2.10); it is not, in general, a root of the equatorial drift P. The two symbols α_0, α_1 are the value and first derivative of that drift,

$$\alpha_0 = P(\gamma), \qquad \alpha_1 = P'(\gamma),$$

not transverse coefficients. The ratios $p, \pi, \mathfrak{q}, U, G$ are local coefficient dictionaries introduced at their first use in the coefficient-regime and degree-loss sections.

The scale $\tau_{\mathrm{cub}} = t^2$ is reserved for the cubic coefficient chart in (4.5). An unadorned τ is local and must be read from its defining display: in the coefficient charts it denotes the stated small coefficient scale (for example p^2 or gt_c/a), whereas in a variation integral or a reparametrized orbit it can be a dummy or clock variable. No global identification of these uses is intended. The source variable and the physical section coordinate are not identified unless the displayed affine adapter says so.

Finally, Π_θ in (2.13) is the completed, divided transition: the infinity map composed with the regular section adapters and the finite-side closing factor after the factor m has been extracted. It is the positive transition appearing in the exact displacement $F = x - m\Pi_\theta(x)$; it is not an unrestricted Poincare map outside the chosen physical sections.

The coefficient dictionaries define their ratios before use. The common raw receiver and capture retain the original physical coordinate throughout; positive affine changes transfer the derivative signs used for counting.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of competing interests

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data and code availability

The complete LaTeX source accompanies this manuscript. Supporting symbolic checks and independent multiprecision shooting examples retain their frozen coefficients and trajectory data in the accompanying reproduction material. The numerical examples are supplementary and are not premises of the analytic argument.

Declaration of generative AI and AI-assisted technologies

During preparation of this manuscript the author used OpenAI Codex for mathematical exploration and proof drafting, simulated technical readings, organization and language editing, LaTeX and code assistance, literature searches, and symbolic and internal-consistency checks. These tools do not constitute independent human peer review or formal verification. The author is responsible for the mathematical claims and the final manuscript.

References

[1] Peter De Maesschalck and Stephen Schecter. The entry–exit function and geometric singular perturbation theory. *Journal of Differential Equations*, 260(8):6697–6715, 2016.

[2] Freddy Dumortier, A. Guzmán, and Christiane Rousseau. Finite cyclicity of elementary graphics surrounding a focus or center in quadratic systems. *Qualitative Theory of Dynamical Systems*, 3:123–154, 2002.

[3] Freddy Dumortier, Yulij Ilyashenko, and Christiane Rousseau. Normal forms near a saddle-node and applications to finite cyclicity of graphics. *Ergodic Theory and Dynamical Systems*, 22(3):783–818, 2002. Author copy dated October 2000: `https://dms.umontreal.ca/~rousseac/DIR.pdf`.

[4] Freddy Dumortier, Robert Roussarie, and Christiane Rousseau. Hilbert's 16th problem for quadratic vector fields. *Journal of Differential Equations*, 110:86–133, 1994.

[5] Renato Huzak. Regular and slow–fast codimension 4 saddle-node bifurcations. *Journal of Differential Equations*, 262(2):1119–1154, 2017.

[6] Renato Huzak. Cyclicity of canard cycles with hyperbolic saddles located away from the critical curve. *Journal of Differential Equations*, 320:479–509, 2022.

[7] Renato Huzak and Kristian Uldall Kristiansen. On entry–exit formulas for degenerate turning point problems in planar slow–fast systems. *Nonlinearity*, 39(8):085026, 2026.

[8] Haibo Lu. Sharp local cyclicity of the I_2^1 graphic in a five-parameter quadratic family. Zenodo preprint, 2026. Version dated 29 September 2026, record 23023367.

[9] David Marín and Jordi Villadelprat. The cyclicity of hyperbolic hemicycles. *Journal of Differential Equations*, 433:113281, 2025.

[10] Robert Roussarie and Christiane Rousseau. Finite cyclicity of nilpotent graphics of pp-type surrounding a center. *Bulletin of the Belgian Mathematical Society–Simon Stevin*, 15(5):889–920, 2008.

[11] Christiane Rousseau and Huaiping Zhu. PP-graphics with a nilpotent elliptic singularity in quadratic systems and Hilbert's 16th problem. *Journal of Differential Equations*, 196(1):169–208, 2004.

[12] Huaiping Zhu and Christiane Rousseau. Finite cyclicity of graphics with a nilpotent singularity of saddle or elliptic type. *Journal of Differential Equations*, 178(2):325–436, 2002.